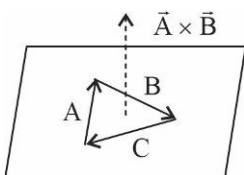


Solutions to Final Step | Part - A | Physics

Introduction to Vectors & Forces

1.(C) $\vec{A} \times \vec{B}$ lies in plane $\perp$ to $\vec{A}, \vec{B}, \vec{C}$

$(\vec{A} \times \vec{B}) \times \vec{C}$ will be in plane containing A, B and C



2.(C) $\vec{V}_T = \vec{V}_E + \vec{V}_N$ $\begin{cases} V_T = \text{actual velocity of the train} \\ V_E = \text{velocity of train along east} \\ V_N = \text{velocity of train along north} \end{cases}$

First Condition

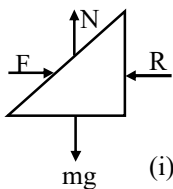
$$\frac{V_E - 10}{V_N} = 1$$

Second Condition

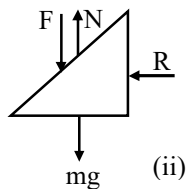
$$\frac{V_E - 30}{V_N} = 0$$

$\therefore V_E = 30, V_N = 20; V_0 (\text{my velocity}) = 50$

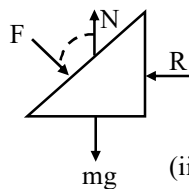
3.(B)



$$\Rightarrow R = F, N = mg$$



$$\Rightarrow R = 0, N = F + mg$$



$$\begin{aligned} \Rightarrow R &= F \sin \theta \\ N &= mg + F \cos \theta \end{aligned}$$

4.(D) $\vec{A}$ and $\vec{B}$ may be anti-parallel with magnitude of A twice of B .

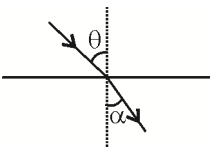
$$A^2 + B^2 + AB \cos \theta = A^2$$

$$\Rightarrow B(B + 2 \cos \theta) = 0 \quad \Rightarrow B = 0 \text{ or } \cos \theta = -\frac{B}{2A}$$

$$\therefore \vec{A} \cdot \vec{B} = AB \cos \theta = -\frac{B^2}{2} \leq 0$$

5.(C) $\vec{a}$ and $\vec{b}$ must be $\parallel$ or anti-parallel with $|\vec{a}| > |\vec{b}|$

6.(B)



$$\vec{a} \times \vec{c} = \sin \theta \hat{p}$$

$$\vec{b} \times \vec{c} = \sin \alpha \hat{p}$$

$\hat{p}$ is unit vector prep to plane of incidence (outward)

7.(C) $\sin 45^\circ = \frac{A \sin 75^\circ}{C}$ and $\sin 30^\circ = \frac{B \sin 75^\circ}{C}$, $\sin 75^\circ = \sin(30 + 45)$

$$= \sin 30^\circ \cos 45^\circ + \cos 30^\circ \sin 45^\circ = \frac{1}{2\sqrt{2}} + \frac{\sqrt{3}}{2\sqrt{2}} = \frac{\sqrt{3}+1}{2\sqrt{2}}$$

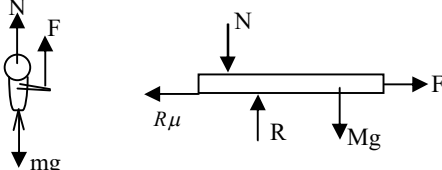
$$\Rightarrow \frac{1}{\sqrt{2}} = \frac{B(\sqrt{3}+1)}{2\sqrt{2}C} \Rightarrow B = \frac{\sqrt{2}C}{\sqrt{3}+1} = \frac{C(\sqrt{3}-1)}{\sqrt{2}}$$

8.(BD) $|\vec{A} + \vec{B}| = |\vec{A} - \vec{B}|$

$$A^2 + B^2 + 2AB \cos \theta = A^2 + B^2 - 2AB \cos \theta$$

$$AB \cos \theta = 0$$

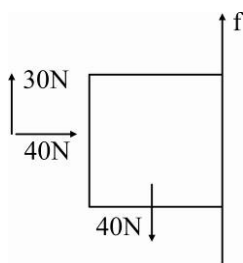
9.(BD) $\vec{a}$ is anti-parallel to $\vec{b}$ and $|\vec{a}| = |\vec{b}| \Rightarrow \vec{a} + \vec{b} = \vec{0}$

10.(A) 

$$\left. \begin{array}{l} N + F = mg \\ R = N + Mg \\ F = R\mu \end{array} \right\} \Rightarrow F = \frac{(m + M)g\mu}{1 + \mu}$$

11.(ABD) If $\mu = 0.5$

$f_{\max} = 20 \text{ N}$ which will not be needed so block will remain at rest at $f = 10 \text{ N}$



If $\mu = 0.2$, $f = 8 \text{ N}$ hence block will move down.

12.(D) $\vec{DB} = \vec{a} - \vec{b}$

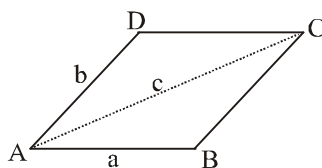
$$\vec{DB} \cdot \vec{AB} = (\vec{a} - \vec{b}) \cdot \vec{a}$$

$$= a^2 - \vec{b} \cdot \vec{a} \quad \dots \text{(i)}$$

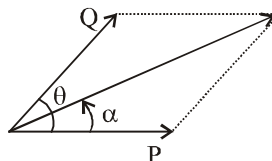
Also $\vec{a} + \vec{b} = \vec{c}$

$$a^2 + b^2 + 2\vec{a} \cdot \vec{b} = c^2 \quad \dots \text{(ii)}$$

From (i) and (ii) $\vec{DB} \cdot \vec{AB} = \frac{3a^2 + b^2 - c^2}{2}$



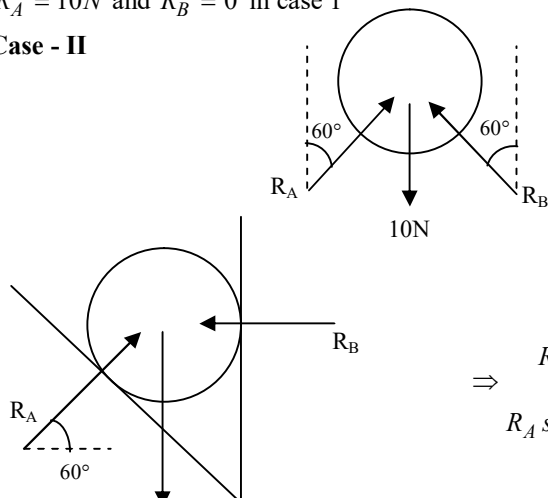
13.(D) $\tan \alpha = \frac{Q \sin \theta}{P + Q \cos \theta} \Rightarrow \frac{\sqrt{3}}{2} = \frac{2P \sin \theta}{P + 2P \cos \theta}$



14.(ABD) If $|\vec{F}_1| = 2$, $|\vec{F}_2| = 3$, and $|\vec{F}_3| = 4$ then maximum value of $\vec{F}_1 + \vec{F}_2 + \vec{F}_3$ can be 9 and minimum value can be 0 and particle can take accelerations in between $\frac{9}{m} = 4.5 \text{ m/s}^2$ and $\frac{0}{m} = 0 \text{ m/s}^2$

15.(AB) $R_A = 10N$ and $R_B = 0$ in case 1

Case - II



$$R_A \sin 60^\circ = R_B \sin 60^\circ \Rightarrow R_A = R_B = 10N$$

$$R_A \cos 60^\circ + R_B \cos 60^\circ = mg$$

$$\Rightarrow R_B = R_A \cos 60^\circ \Rightarrow R_A = \frac{2mg}{\sqrt{3}} = 20/\sqrt{3}$$

$$R_A \sin 60^\circ = mg \quad R_B = mg/\sqrt{3} = 10/\sqrt{3}$$

16.(C) $\vec{b} = x\hat{i} + y\hat{j} + z\hat{k}$

$$\vec{a} \cdot \vec{b} = 1 \Rightarrow x + y + z = 1$$

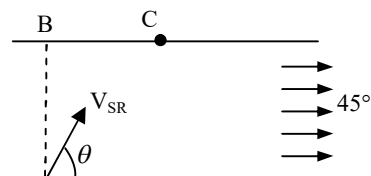
$$\vec{a} \times \vec{b} = \hat{j} - \hat{k} \Rightarrow z - y = 0 ; x - z = 1 ; y - x = -1$$

17.(AB) As $AB = BC \Rightarrow V_{SR} \cos \theta + V_{SR} \sin \theta$

$$\Rightarrow 5 \cos \theta + 1 = 5 \sin \theta \Rightarrow 5\sqrt{2} \sin(\theta - 45^\circ) = 1$$

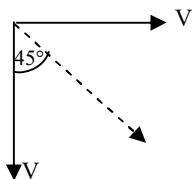
$$\Rightarrow \theta = 45^\circ + \sin^{-1} \frac{1}{5\sqrt{2}} \approx 53^\circ$$

$$\text{time} = \frac{0.4 \text{ km}}{5 \sin \theta \text{ km/hr}} = 0.1 \text{ hr} = 6 \text{ min}$$



18.(B) $\vec{b} \times \vec{c} = \vec{b} \times (-(\vec{a} + \vec{b})) = \vec{a} \times \vec{b} ; \vec{c} \times \vec{a} = (-(\vec{a} + \vec{b}) \times \vec{a}) = \vec{a} \times \vec{b}$

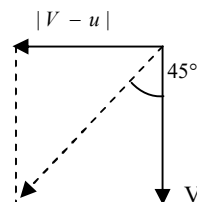
19.(CD)



Let velocity components (horizontal and vertical) be V each as for stationary observer it is falling at angle of 45° , let u ; velocity of man in the direction of horizontal component of rain.

$$\Rightarrow v = 2m/s \text{ (i.e. } 2\sqrt{2} \cos 45^\circ)$$

Now relative horizontal components relative velocity of rain in opposite direction if $u > v$ and $|v - u| = 2$ (i.e. $(2\sqrt{2} \sin 45^\circ) \Rightarrow u = 4m/s$



20.(D) $(\vec{a} + \vec{b} + \vec{c})^2 = a^2 + b^2 + c^2 + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a})$

21.(D) Magnitude of vector should not change

22.(ABCD)

(i) Maximum value of $|\vec{a} + \vec{b} + \vec{c}|$ in case 1 will be 14 and minimum value will be zero because $\vec{a}$, $\vec{b}$ and $\vec{c}$ can form a triangle.

(ii) Maximum value of $|\vec{a} + \vec{b} + \vec{c}|$ in Case II will be 12 and minimum value will be $|1 + 4 - 7| = 2$

23.(C) Sum of magnitude of any two vector is greater or equal to magnitude of the third, if three vector adding to zero.

24.(C) For Parallel

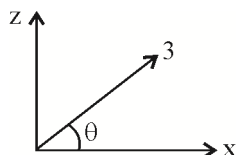
$$\frac{8t}{t^2} = \frac{-t^2}{2} \quad (t^3 = -16 \text{ (Not possible as } t \geq 0))$$

For perpendicular

$$\vec{a} \cdot \vec{b} = 0 = 8t^3 - 2t^2$$

$$t = \frac{1}{4}$$

25.(B)



26.(B) $x^2 = (1500 - 10t)^2 + (1800 - 15t)^2$, for x to be minimum, its first derivative should be zero. Thus $\frac{dx}{dt} = 0$.
 $= -20(1500 - 10t) - 30(1800 - 15t)$ or $t = 129.23s$

27.(D)

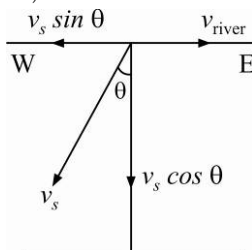
28.(A) $\cos \theta = \frac{\vec{A} \cdot \vec{B}}{|\vec{A}| |\vec{B}|} = \frac{\sqrt{3}}{2}, \theta = \frac{\pi}{6}$

29.(A) $\hat{n} = \frac{\vec{A} \times \vec{B}}{|\vec{A} \times \vec{B}|}, \vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 1 \\ 1 & -1 & 1 \end{vmatrix} = 4\hat{i} - \hat{j} - 5\hat{k}, |\vec{A} \times \vec{B}| = \sqrt{42}, \hat{n} = \frac{1}{\sqrt{42}}(4\hat{i} - \hat{j} - 5\hat{k})$

30.(C) $a_{av} = \frac{V_f - V_i}{t} = \frac{5\hat{j} - 4\hat{i}}{10}$ or $|a_{av}| = \frac{1}{\sqrt{2}}$ North west.

31.(B) $(2\hat{i} + 3\hat{j} + 8\hat{k}) \cdot (4\hat{j} - 4\hat{i} + \alpha\hat{k}) = 0$ or $-8 + 12 + 8\alpha = 0, \alpha = \frac{-1}{2}$

32.(D) $v_s \sin \theta = v_{\text{river}}$
 $\sin \theta = \frac{1}{2}, \theta = 30^\circ$



33.(D) $\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -2 & 1 \\ 4 & 3 & -2 \end{vmatrix}$

$$\hat{n} = \frac{\vec{A} \times \vec{B}}{|\vec{A} \times \vec{B}|} = \frac{\hat{i} + 10\hat{j} + 17\hat{k}}{\sqrt{390}} \quad ; \quad \vec{x} = |\vec{B}| \hat{n} = \frac{\sqrt{29}(\hat{i} + 10\hat{j} + 17\hat{k})}{\sqrt{390}}$$

34.(C) $\vec{V}_{rm} = \vec{V}_r - \vec{V}_m - 3\hat{k} - 4\hat{j}$; $\tan \beta = \frac{4}{3} \Rightarrow \beta = 53^\circ$ North vertical

35.(A) A regular hexagon will be formed if we continue.

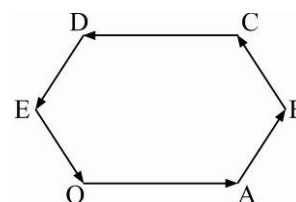
After 5th turn displacement = OE = 500 m

After 3rd turn displacement = OC = 1000 m

(OC is diameter of the circle circumscribing regular hexagon).

36.(D) Use parallelogram law.

37.(A) $\vec{V}_{cm} = (V_{rx} - V_m)\hat{i} + V_{ry}\hat{j}$



$$\text{Case 1 : } \tan 90^\circ = \frac{V_{ry}}{V_{rx} - V_m} = \frac{V_{ry}}{V_{rx} - 8} \text{ or } V_{rx} = 8 \text{ ms}^{-1}$$

$$\text{Case 2 : } \tan 30^\circ = \frac{V_{rx} - V_m}{V_{ry}} = \frac{8 - 12}{V_{ry}} \text{ or } V_{ry} = -4\sqrt{3} \text{ ms}^{-1}$$

$$V_r = 8\hat{i} - 4\sqrt{3}\hat{j} = 4\sqrt{2^2 + 3} = 4\sqrt{7} \text{ ms}^{-1}$$

$$\tan \theta \frac{V_{ry}}{V_{rx}} = \frac{4\sqrt{3}}{8} = \frac{\sqrt{3}}{2} \text{ or } \theta = \tan^{-1} \frac{\sqrt{3}}{2} \text{ with respect to road (horizontally).}$$

$$38.(C) \text{ As } 13N > 8 + 4 \text{ } [\because R_{max} = A + B]$$

$$39.(A) \quad V_{ws} = V_w - V_s \Rightarrow (V_{wx}\hat{i} + V_{wy}\hat{j}) - 10\hat{i} = 5\hat{j} \text{ or } (V_{wx}\hat{i} + V_{wy}\hat{j}) = 5\hat{j} + 10\hat{i}$$

$$\text{or } |V_w| = 5\sqrt{5} \text{ and } \tan \theta = \frac{1}{2} \text{ or } \theta = \tan^{-1} \frac{1}{2} \text{ North of east.}$$

$$40.(C) \quad \vec{d} = 5\hat{i} - 2\hat{j} + 4\hat{k}$$

$$W = \vec{F} \cdot \vec{d} = (6\hat{i} + 3\hat{j} + \hat{k}) \cdot (5\hat{i} - 2\hat{j} + 4\hat{k}) = 28J$$

$$41.(B) \quad \vec{V}_{Res} = \vec{V}_{steamer} + \vec{V}_{wind} = -5\hat{i} + 5\hat{j} + 5\hat{i} = 5\hat{j}. \text{ The flag will flutter in a direction of velocity of wind relative the velocity of steamer.}$$

$$42.(C) \quad d = 100\hat{j} + 100\hat{i} + 20\hat{j} + (-100\hat{i} - 100\hat{j}) = 20\hat{j}$$

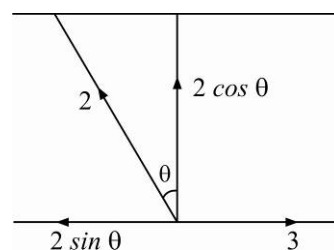
$$43.(A) \text{ Let us assume he swims at an angle } \theta \text{ with the perpendicular as shown. river is } \ell \text{ m wide time taken to cross it.}$$

$$t = \frac{\ell}{2\cos\theta}, v_x = 3 - 2\sin\theta$$

Horizontal distance covered along x direction during this period.

$$x = vt = (3 - 2\sin\theta) \frac{\ell}{2\cos\theta}$$

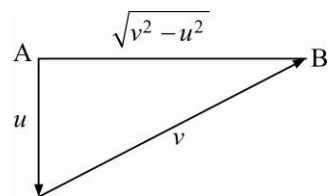
$$\text{For } t \text{ to be min } \frac{dx}{d\theta} = 0 \text{ or } \ell \left[\frac{3}{2} \sec\theta \tan\theta - \sec^2\theta \right] = 0 \text{ or } \sin\theta = \frac{2}{3}$$



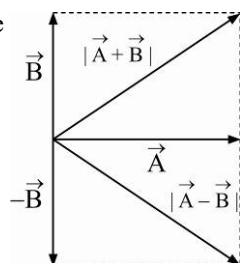
$$44.(B) \text{ See in the figure, the velocity in the direction A to B will be}$$

$$\sqrt{v^2 - u^2} \therefore t = \frac{\ell}{\sqrt{v^2 - u^2}} \text{ same time is needed to come back from B to A}$$

$$\therefore t_{net} = \frac{2\ell}{\sqrt{v^2 - u^2}}$$



$$45.(C) \text{ See figure}$$



INTEGER VALUE TYPE

$$46.(13) \quad |\vec{AB}| = |\vec{r}_2 - \vec{r}_1| = \sqrt{r_1^2 + r_2^2 - 2r_1r_2(\cos 75^\circ - \cos 15^\circ)} = \sqrt{9 + 16 - 2 \times 3 \times 4 \times \frac{1}{2}} = \sqrt{13} \therefore |\vec{AB}|^2 = 13$$

$$47.(1) \quad \vec{A} + \vec{B} = \hat{i} \Rightarrow \vec{B} = \hat{i} - \hat{i} - 2\hat{j} + 3\hat{k} = -2\hat{j} + 3\hat{k} \therefore x + y = -2 + 3 = 1$$

$$48.(2) \quad \vec{B} = \vec{C} - \vec{A} \quad \vec{A} + \vec{C} = -\vec{B}$$

$$\begin{aligned}
 |\vec{B}|^2 &= |\vec{C} - \vec{A}|^2 & \Rightarrow & |\vec{A} + \vec{C}|^2 = |\vec{B}|^2 \\
 \Rightarrow B^2 &= C^2 + A^2 - 2AC \cos \theta, & \Rightarrow & A^2 + C^2 + 2AC \cos \theta_2 = B^2 \\
 \Rightarrow A^2 &= 2A^2 - 2A^2 \cos \theta_1 & \Rightarrow & 2A^2 + 2A^2 \cos \theta_2 = A^2 \\
 \Rightarrow 2A^2 \cos \theta_1 &= A^2 & \Rightarrow & 2A^2 \cos \theta_2 = -A^2 \\
 \Rightarrow \cos \theta_1 &= \frac{1}{2} & \Rightarrow & \cos \theta_2 = -\frac{1}{2} \\
 \therefore \theta_1 &= \frac{\pi}{3} & \therefore \theta_2 &= \frac{2\pi}{3} & \therefore \frac{\theta_2}{\theta_1} &= 2
 \end{aligned}$$

49.(5) $\vec{p} = 7\hat{i} + 6\hat{j}$

$\vec{P} + \vec{Q} = 11\hat{i} + 9\hat{j} \quad \therefore \quad \vec{Q} = 4\hat{i} + 3\hat{j} \quad \therefore \quad |\vec{Q}| = \sqrt{4^2 + 3^2} = \sqrt{25} = 5$

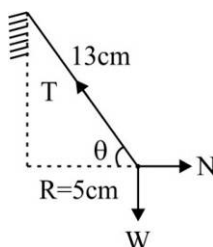
50. (150) $R = \frac{B}{2}$

$\sin \alpha = \frac{R}{B} = \frac{1}{2} \quad \therefore \quad \alpha = 30^\circ \quad \therefore \quad \theta = 180^\circ - \alpha = 150^\circ$

51. (260) $T \sin \theta = W = 240N$

$T \frac{12}{13} = W$

$\therefore T = \frac{13W}{12} = 260N$



52. (30) $30\sqrt{3}g \sin 30^\circ = mg \sin 60^\circ$

$15 = m \frac{1}{2} \quad ; \quad m = 30kg$

53. (10) $\vec{d} = 30\hat{j} + 20\hat{i} - 30\hat{i} - 30\hat{j} \quad ; \quad = -10\hat{i}$
 $= 10m \text{ west}$

54. (50) For B : $\sum F_y = 0$

$T_1 \cos \theta_1 = mg + T_2 \cos \theta_2 \quad \dots(1)$

$\sum F_x = 0$

$T_1 \sin \theta_1 = T_2 \sin \theta_2 \quad \dots(2)$

For C : $\sum F_y = 0$

$T_2 \cos \theta_2 = mg \quad \dots(3)$

$\sum F_x = 0$

$T_2 \sin \theta_2 = mg \quad \dots(4)$

By (3) and (4)

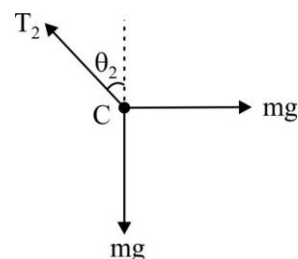
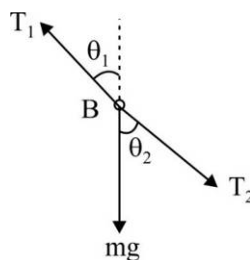
$\tan \theta_2 = 1 \quad \therefore \quad \theta_2 = \frac{\pi}{4} \text{ radian} \quad \therefore \quad T_2 = \sqrt{2}mg$

(2) divided by (1)

$\tan \theta_1 = \frac{mg}{mg + mg} = \frac{1}{2} \quad \therefore \quad T_1 = \sqrt{2}mg \cdot \frac{1}{\sqrt{2}} \sqrt{5} = \sqrt{5}mg = \sqrt{5} \cdot \sqrt{5} 10N = 50N$

55. (3) $Q = 25$

$P = 40$



$$\theta = 120^\circ$$

$$\cos 120^\circ = -\frac{1}{2}$$

$$R = \sqrt{(25)^2 + (40)^2 + 2(25)(40)\cos 120^\circ} = \sqrt{625 + 1600 - 1000} = 35$$

$$\sin \delta = \frac{Q \sin 120^\circ}{R} = \frac{25\sqrt{3}}{70} = \frac{5\sqrt{3}}{14} \quad \therefore \quad \delta = \sin^{-1}\left(\frac{5\sqrt{3}}{14}\right) \text{ with P} \quad \therefore \quad n = 3$$

56. (250) $T_1 = T_3 \cos 37^\circ$

$$T_1 = \frac{4T_3}{5} \quad \dots(i)$$

$$50 + T_3 \sin 37^\circ = 100$$

$$\frac{3T_3}{5} = 50 \Rightarrow T_3 = \frac{250}{3} N \quad \therefore \quad 3T_3 = 250 N$$

57. (1) $\widehat{PS} = \widehat{PQ}$

$$\Rightarrow \frac{\vec{S} - \vec{P}}{bR} = \frac{\vec{Q} - \vec{P}}{R} \Rightarrow \vec{S} - \vec{P} = b\vec{Q} - b\vec{P} \quad \therefore \quad \vec{S} = (1-b)\vec{P} + b\vec{Q} \quad \therefore \quad n = 1$$

58. (6) $\vec{a} = 5 \cos 37^\circ \hat{j} + 5 \sin 37^\circ \hat{k} = (0)\hat{i} + 4\hat{j} + 3\hat{k}$

$$\vec{b} = 5 \cos 37^\circ \hat{j} + 5 \sin 37^\circ \hat{i} = 3\hat{i} + 4\hat{j} + 0\hat{k}$$

$$\therefore \quad \vec{a} \cdot \vec{b} = (3)(0) + (4)(4) + 3(0) = 16$$

$$\text{But } \vec{a} \cdot \vec{b} = ab \cos \theta \quad \Rightarrow \quad 16 = (5)(5) \cos \theta \quad \therefore \quad \cos \theta = \frac{16}{25}$$

59. (90) To find minimum value of F, slipping tendency is taken down the plane. Friction acts up the plane.

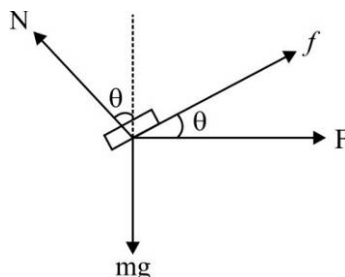
Balancing forces,

$$N = mg \cos \theta + F \sin \theta$$

$$f = mg \sin \theta - F \cos \theta$$

Putting $f = \mu N$ and solving,

$$\text{We get } F = mg \left(\frac{\tan \theta - \mu}{1 + \mu \tan \theta} \right) = 90 N$$



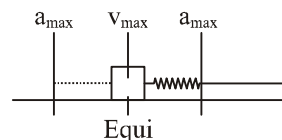
60. (8) $\vec{v}_{AB} = \vec{v}_A - \vec{v}_B = 5\hat{i} - 2\hat{j}$

$$\vec{v}_{BC} = \vec{v}_B - \vec{v}_C = 3\hat{i} + 4\hat{j}$$

$$\vec{v}_{AC} = \vec{v}_A - \vec{v}_C = (\vec{v}_A - \vec{v}_B) + (\vec{v}_B - \vec{v}_C) = \vec{v}_{AB} + \vec{v}_{BC} = 8\hat{i} + 2\hat{j}$$

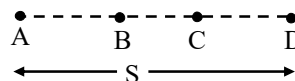
Kinematics of a Particle

1.(AC) Just after release particle accelerates till equilibrium position and beyond that its speed decreases until it reaches other extreme



2.(A) Let total time taken : t

$$\Rightarrow \quad t_{AB} = \frac{5/2}{3} = \frac{5}{6} \text{ sec}, t_{BC} = t_{CD} = \frac{t - t_{AB}}{2} = \frac{t - 5/6}{2}$$



$$BD = BC + CD \Rightarrow s/2 = \left(\frac{t-5/6}{2}\right)4.5 + \left(\frac{t-5/6}{2}\right)7.5 \Rightarrow s/t = 4 \text{ m/s}$$

3.(ABCD) Ball moving in direction of train $V_b = 10 + 1 = 11 \text{ m/s}$

Ball moving opposite to direction of train $V_b = 10 - 1 = 9 \text{ m/s}$

$$\text{Time duration between collisions} = \frac{10}{1} = 10s$$

4.(A) Slope is continuously decreasing

$$5.(C) \quad ut - \frac{1}{2}gt^2 = u(t-T) - \frac{1}{2}g(t-T)^2 = H$$

$$6.(B) \quad u \cos \theta = v \Rightarrow \theta = \cos^{-1} \frac{v}{u}$$

7.(A) Let C : highest point, A to B, $(v/2)^2 = v^2 + 2(-10)(+3)$

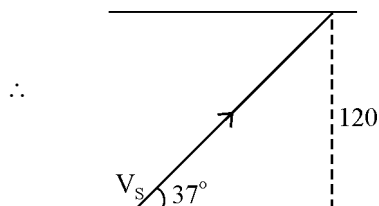
$$B \text{ to } C : 0 = \left(\frac{v}{2}\right)^2 + 2(-10)s \Rightarrow s = 1m$$

$$8.(C) \quad \vec{V_S} = \vec{V_{SR}} + \vec{V_R}$$

$$V_R > V_{SR}$$

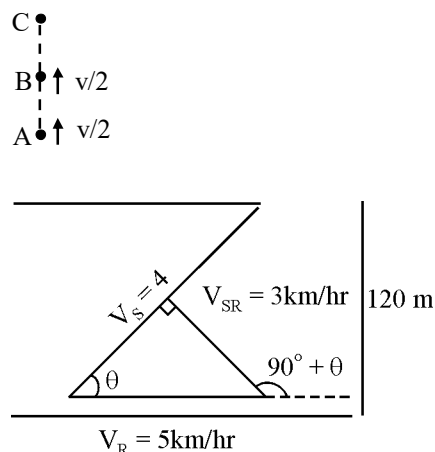
Swimmer can not cross the river along shortest path i.e., along the width of the river. Possible shortest path length will be greater than width of the river.

To travel along this path V_s must be $\perp$ to V_{SR}



$\therefore$

$$\therefore \quad V_{SR} = 4 \text{ km/hr} \\ \theta = 37^\circ$$



$$S = 120 \times \frac{5}{3} \\ = 200 \text{ m}$$

9.(C) Time taken to reach the maximum height : $t_1 = \frac{u}{g}$

If t_2 is the time taken to hit ground

$$i.e., \quad -H = ut_2 - \frac{1}{2}gt^2$$

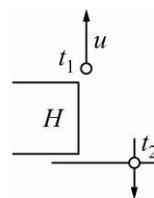
$$\text{But} \quad t_2 = nt_1$$

$$\text{So,} \quad -H = u \frac{nu}{g} - \frac{1}{2}g \frac{n^2u^2}{g^2}$$

[Given]

$$-H = \frac{nu^2}{g} - \frac{1}{2} \frac{n^2u^2}{g}$$

$$2gH = nu^2(n-2)$$



10.(BC) Component of acceleration parallel to velocity is rate of change of speed. (Perpendicular component of acceleration is responsible for changing the direction.

$$11.(B) \quad \frac{-da}{dt} = ka \Rightarrow -\int_{a_0}^a \frac{da}{a} = k \int_0^t dt \Rightarrow a = a_0 e^{-kt} \quad a = a_0 / 2 \text{ at } t = t_0 \Rightarrow \ln 2 = kt_0 \Rightarrow k = \ln 2 / t_0$$

$$\Rightarrow a = a_0 e^{-(\ln 2)t/t_0} \text{ i.e. } a = a_0 2^{-t/t_0} \Rightarrow \int_0^v dv = a_0 \int_0^\infty e^{-(\ln 2)t/t_0} dt$$

$$\Rightarrow v = a_0 \left[\frac{e^{-(\ln 2)t/t_0}}{-(\ln 2)/t_0} \right]_0^\infty = \frac{a_0 t_0}{\ln 2}$$

$$12.(ABC) \quad \frac{d\vec{v}}{dt} = \vec{a} \Rightarrow \left| \frac{d\vec{v}}{dt} \right| = |\vec{a}|, |\vec{v}| = \text{speed} \Rightarrow \frac{d|\vec{v}|}{dt} = \text{rate of change of speed}$$

i.e. magnitude of acc.

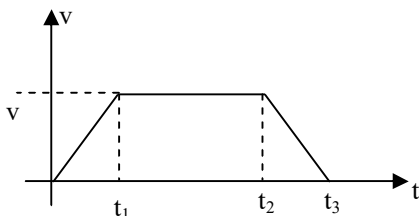
$$\text{If } \vec{a} \perp \vec{v} \text{ then } |\vec{a}| \neq 0 \text{ but } \frac{d}{dt} |\vec{v}| = 0$$

$$13.(AB) \quad v \frac{dv}{dx} = 4 - 2x \Rightarrow \int_0^v v dv = \int_0^x (4 - 2x) dx \Rightarrow \frac{v^2}{2} = 4x - x^2$$

$$\Rightarrow v = \pm \sqrt{8x - 2x^2} = 0 \Rightarrow x = 4m, v \text{ is max} = \frac{dv}{dx} = 0 \Rightarrow x = 2m$$

$$\Rightarrow \text{max velocity} = \sqrt{16 - 8} = 2\sqrt{2}m/s$$

14.(C)



Let v : maximum speed

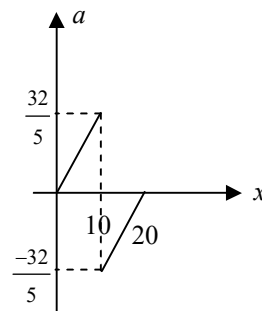
$$\Rightarrow \ell = \frac{1}{2} v t_1, 2\ell = v(t_2 - t_1), 3\ell = \frac{1}{2} v(t_3 - t_2)$$

$$\Rightarrow t_1 = 2\ell/v, t_2 = \frac{4\ell}{v}, t_3 = 10\ell/v$$

$$\Rightarrow \text{av. velocity} = \frac{\ell + 2\ell + 3\ell}{10\ell/v} = \frac{3v}{5} = \frac{\text{av. velocity}}{\text{max. velocity}} = \frac{3}{5}$$

$$15.(C) \quad a = v \frac{dv}{dx} = \begin{cases} \left(\frac{8}{10}x \right) \frac{8}{10} & \dots\dots\dots x < 10 \\ 16 \left(1 - \frac{x}{20} \right) \left(\frac{-8}{10} \right) & \dots\dots\dots 10 \leq x < 20 \end{cases}$$

$$\text{i.e. } \begin{cases} \frac{16x}{25} & \dots\dots\dots x \leq 10 \\ \frac{16}{25} (x - 20) & \dots\dots\dots 10 < x \leq 20 \end{cases}$$



16.(D) Let they cross each other after time t at $(x, 30)$

$$\Rightarrow \text{For } A : x = 3t \quad \& \quad \text{For } B : \begin{cases} x = \frac{1}{2} a \sin^2 \theta t^2 \Rightarrow x = 0.2 \sin^2 \theta t^2 \\ 30 = \frac{1}{2} a \cos^2 \theta t^2 \Rightarrow 30 = 0.2 \cos^2 \theta t^2 \end{cases} \Rightarrow t = 300 \text{ sec} = 5 \text{ min.}$$

$$17.(C) \quad a = ks \Rightarrow v \frac{dv}{ds} = ks \Rightarrow \int_0^v v dv = k \int_0^s s ds \Rightarrow v^2 = ks^2 \quad \text{i.e. } v = \sqrt{k}s$$

$$18.(A) \quad \frac{a}{10} + \frac{t}{11} = 1 \Rightarrow v \text{ is max} \Rightarrow dv/dt = 0 \text{ i.e. } a = 0 \Rightarrow t = 11 \text{ sec.}$$

$$\frac{dv}{dt} = 10(1 - t/11) \Rightarrow v_m = \int_0^{11} 10(1 - t/11) dt = \left| 10t - \frac{5t^2}{11} \right|_0^{11} = 110 - 55 = 55 \text{ m/s}$$

19.(B) $x = 9t - t^2$ (Let east is positive) and initial position is $x = 0$.

$v = 0 \Rightarrow t = 4.5 \text{ sec} \Rightarrow$ distance traveled is 5 sec.

$$= |x(4.5) - x(4)| + |x(5) - x(4.5)| \dots \dots \text{ where } x(t) \text{ distances position at time } t.$$

$$= |20.25 - 20| + |20 - 20.25| = 0.25 + 0.25 = 0.5$$

20.(C) For upward motion, net down ward $acc = \frac{W + F}{m}$; m is mass. Let h : height gain

$$\Rightarrow 0 = V^2 - 2\left(\frac{W + F}{m}\right)h \Rightarrow h = \frac{mv^2}{2(W + F)}$$

For downward motion, net downward $acc = \frac{W - F}{m}$. Let V' ; final velocity and displacement $= -h$

$$\Rightarrow V'^2 = 0^2 - 2\left\{-\frac{W - F}{m}\right\}\left\{\frac{-mv^2}{2(W + F)}\right\} \Rightarrow V' = V\sqrt{\frac{W - F}{W + F}}$$

21.(C) As the force is exponentially decreasing, so is acceleration, i.e., rate of increase of velocity will decrease with time. Thus, the graph of velocity will be an increasing curl with decreasing slope with time.

$$\therefore a = \frac{F}{M} = \frac{F_0}{m} e^{-bt} = \frac{dv}{dt}$$

$$\Rightarrow \int_0^v dv = \int_0^t \frac{F_0}{m} e^{-bt} dt \Rightarrow v = \frac{F_0}{m} \left[\left(\frac{1}{-b} \right) e^{-bt} \right]_0^t = \frac{F_0}{mb} [e^{-bt}]_0^t = \frac{F_0}{mb} (e^0 - e^{-bt}) = \frac{F_0}{mb} (1 - e^{-bt})$$

$$\Rightarrow v_{max} = \frac{F_0}{mb}$$

$$22.(B) -h_1 = -\frac{1}{2}gt_1^2, -h_2 = -\frac{1}{2}gt_2^2 \quad t_1 = 5 \text{ sec and } t_2 = 3 \text{ sec} \Rightarrow h_1 - h_2 = g/2(5^2 - 3^2) = 80 \text{ m}$$

$$23.(AC) 2x \frac{dx}{dt} = 2t \Rightarrow xv = t \quad \text{differential wrt } t \Rightarrow v^2 + xa = 1 \Rightarrow a = \frac{1 - v^2}{x} \Rightarrow a = \frac{1 - (t/x)^2}{x} = \frac{x^2 - t^2}{x^3}$$

24.(D) According to question, velocity of unit mass varies as

$$v(x) = \beta x^{-2n} \dots (i)$$

$$\frac{dv}{dx} = -2n\beta x^{-2n-1} \dots (ii)$$

$$\text{Acceleration of the particle is given by } a = \frac{dv}{dt} = \frac{dv}{dx} \times \frac{dx}{dt} = \frac{dv}{dx} \times v$$

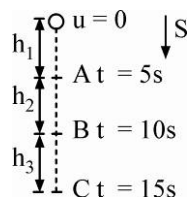
$$\text{Using equation (i) and (ii), we get : } a = (-2n\beta x^{2n-1}) \times (\beta x^{2n}) = -n\beta^2 x^{-4n-1}$$

25.(D) Distance covered by the stone in first 5 seconds (i.e. $t = 5\text{s}$) is

$$h_1 = \frac{1}{2}g(5)^2 = \frac{25}{2}g \dots (i)$$

Distance travelled by the stone in next 5 second (i.e. $t = 10\text{s}$) is

$$h_1 + h_2 = \frac{1}{2}g(10)^2 = \frac{100}{2}g \dots (ii)$$



Distance travelled by the stone in next 5 seconds (i.e. $t = 15$ s) is :

$$h_1 + h_2 + h_3 = \frac{1}{2}g(15)^2 = \frac{225}{2}g \quad \dots\dots(iii)$$

Subtract (i) from (ii), we get :

$$(h_1 + h_2) - h_1 = \frac{100}{2}g - \frac{25}{2}g = \frac{75}{2}g$$

$$h_2 = \frac{75}{2}g = 3h_1 \quad \dots\dots(iv)$$

Subtract (ii) from (iii), we get :

$$(h_1 + h_2 + h_3) - (h_2 + h_1) = \frac{225}{2}g - \frac{100}{2}g$$

$$h_3 = \frac{125}{2}g = 5h_1$$

From (i), (iv) and (v), we get : $h_1 = \frac{h_2}{3} = \frac{h_3}{5}$

26.(D) Given : $x = 8 + 12t - t^3$

$$\text{Velocity, } v = \frac{dx}{dt} = 12 - 3t^2$$

When $v = 0$, $12 - 3t^2 = 0$ or $t = 2$ s

$$a = \frac{dv}{dt} = -6t$$

$$|_{t=2s} = -12ms^{-2}$$

Retardation = $12 ms^{-2}$

27.(A) Distance, $x = (t + 5)^{-1} \quad \dots\dots(i)$

$$\text{Velocity, } v = \frac{dx}{dt} = \frac{d}{dt}(t + 5)^{-1} = -(t + 5)^{-2} \quad \dots\dots(ii)$$

$$\text{Acceleration, } a = \frac{dv}{dt} = \frac{d}{dt}[-(t + 5)^{-2}] = 2(t + 5)^{-3} \quad \dots\dots(iii)$$

$$\text{From equation (ii), we get : } v^{3/2} = -(t + 5)^{-3} \quad \dots\dots(iii)$$

Substituting this in equation (iii) we get :

$$\text{Acceleration, } a = -2v^{3/2} \quad \text{Or} \quad a \propto (\text{velocity})^{3/2}$$

$$\text{From equation (i) we get : } x^3 = (t + 5)^{-3}$$

$$\text{Substituting this in equation (iii), we get : Acceleration, } a = 2x^3$$

Hence option (A) is correct

28.(A) Let the two balls meet after t s at distance x from the platform.

For the first ball $u = 0, t = 18$ s, $g = 10$ m/s²

$$\text{Using } h = ut + \frac{1}{2}gt^2 \quad \therefore \quad x = \frac{1}{2} \times 10 \times 18^2 \quad \dots\dots(i)$$

For the second ball $u = v, t = 12$ s, $g = 10$ m/s²

$$\text{Using } h = ut + \frac{1}{2}gt^2 \quad \therefore \quad x = v \times 12 + \frac{1}{2} \times 10 \times 12^2 \quad \dots\dots(ii)$$

From equations (i) and (ii), we get : $\frac{1}{2} \times 10 \times 18^2 = 12v + \frac{1}{2} \times 10 \times (12)^2$

or $12v = \frac{1}{2} \times 10 \times [(18 + 12)(18 - 12)]$

$$12v = \frac{1}{2} \times 10 \times 30 \times 6 \quad \text{or} \quad v = \frac{1 \times 10 \times 30 \times 6}{2 \times 12} = 75 \text{ m/s}$$

29.(B) Given $u = 0$.

Distance travelled in 10s, $S_1 = \frac{1}{2} a \cdot 10^2 = 50a$ and distance travelled in 20 s, $S_2 = \frac{1}{2} a \cdot 20^2 = 200a$

$$\therefore S_2 = 4S_1$$

30.(D) Let v_x be the velocity of the scooter, the distance between the scooter and the bus = 1000 m,

The velocity of the bus = 10 ms^{-1}

Time taken to overtake = 100 s

Relative velocity of the scooter with respect to the bus = $(v_s - 10)$

$$\therefore \frac{1000}{(v_s - 10)} = 100 \text{ s} \Rightarrow v_s = 20 \text{ ms}^{-1}$$

31.(C) Given : At time $t = 0$, velocity, $v = 0$.

$$\text{Acceleration } f = f_0 \left(1 - \frac{t}{T} \right)$$

$$\text{At } f = 0, 0 = f_0 \left(1 - \frac{t}{T} \right)$$

Since f_0 is a constant,

$$\therefore \int_0^{v_2} dv = \int_{t=0}^{t=T} f dt = \int_0^T f_0 \left(1 - \frac{t}{T} \right) dt \quad \therefore v_x = \left[f_0 t - \frac{f_0 t^2}{2T} \right]_0^T = f_0 T - \frac{f_0 T^2}{2T} = \frac{1}{2} f_0 T$$

$$\text{32.(D) Average speed} = \frac{\text{total distance travelled}}{\text{total time taken}} = \frac{s + s}{t_1 + t_2} = \frac{2s}{\frac{s}{v_n} + \frac{s}{v_d}} = \frac{2v_u v_d}{v_d + v_u}$$

33.(A) Given : $x = 9t^2 - t^3$ (i)

$$\text{Speed } v = \frac{dx}{dt} = \frac{d}{dt}(9t^2 - t^3) = 18t - 3t^2 \quad \text{For maximum speed, } \frac{dx}{dt} = 0 \Rightarrow 18 - 6t = 0$$

$$\therefore t = 3 \text{ s} \quad \therefore x_{\max} = 81 \text{ m} - 27 \text{ m} = 54 \text{ m} \quad (\text{From } x = 9t^2 - t^3)$$

34.(C) Distance travelled in one rotation (lap) = $2\pi r$

$$\therefore \text{Average speed} = \frac{\text{distance}}{\text{time}} = \frac{2\pi r}{t} = \frac{2 \times 3.14 \times 100}{62.8} = 10 \text{ ms}^{-1}$$

$$\text{Net displacement in one lap} = 0. \text{ Average velocity} = \frac{\text{net displacement}}{\text{time}} = \frac{0}{t} = 0$$

35.(D) $x = ae^{-\alpha t} + be^{\beta t}$

$$\frac{dx}{dt} = -a\alpha e^{-\alpha t} + b\beta e^{\beta t}$$

$$v = -a\alpha e^{-\alpha t} + b\beta e^{\beta t}$$

For certain value of t velocity will increase.

36.(B) Let total height = H

Time of ascent = T

$$\text{So, } H = uT - \frac{1}{2}gT^2$$

Distance covered by ball in time $(T - t)$ sec.

$$y = u(T - t) - \frac{1}{2}g(T - t)^2$$

So distance covered by ball in last t sec

$$h = H - y = \left[uT - \frac{1}{2}gT^2 \right] - \left[u(T - t) - \frac{1}{2}g(T - t)^2 \right]$$

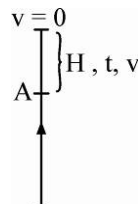
By solving and putting $T = \frac{u}{g}$ we will get $h = \frac{1}{2}gt^2$

Aliter :

Time to reach the topmost position .

$$T = u/g$$

Velocity at the top, $v = 0$



Let's consider a point A distance H below the highest point. Let it takes t second for the ball to reach the top from A. So we have to calculate H . Let's find the velocity at point A. Now the time taken to reach A is $(T - t)$

$$\therefore v_A = u - g(T - t) = u - gT - gt = u - u - gt = -gt.$$

Now consider its journey from A to the top.

$$\text{Using } v^2 = u^2 - 2gh \Rightarrow 0 = v^2 - 2gH \Rightarrow H = \frac{(-gt)^2}{2g} = \frac{1}{2}gt^2$$

$$37.(A) \text{ Initial energy equation } mgh = \frac{1}{2}mv^2 \text{ i.e. } 10 \times 5 = \frac{1}{2}v_1^2 \Rightarrow v_1 = 10 \dots (i)$$

$$\text{After one bounce, } 10 \times 1.8 = \frac{1}{2}v_2^2 \Rightarrow v_2 = 6 \dots (ii)$$

Loss its velocity on bouncing $\frac{6}{10} = \frac{3}{5}$ a factor.

$$38.(A) \text{ Initial velocity of first body } (u_1) = 0$$

$$\text{Final velocity } (v_1) = 3 \text{ m/s and initial velocity of second body } (u_2) = 4 \text{ m/s. Height } (h) = \frac{v_1^2}{2g} = \frac{(3)^2}{2 \times 9.8} = 0.46 \text{ m}$$

Therefore velocity of the second body.

$$v_2 = \sqrt{u_2^2 + 2gh} = \sqrt{(4)^2 + 2 \times 9.8 \times 0.46} = 5 \text{ m/s}$$

$$39.(D) \text{ Initial velocity } (u) = 0 ; \text{ Acceleration in the first phase } = \alpha ; \text{ Deceleration in the second phase } = \beta \text{ and total time } = t. \text{ When car is accelerating then}$$

$$\text{Final velocity } (v) = u + \alpha t = 0 + \alpha t_1 \text{ or } t_1 = \frac{v}{\alpha} \text{ and when car is decelerating.}$$

$$\text{Then final velocity } 0 = v - \beta t \text{ or } t_2 = \frac{v}{\alpha} + \frac{v}{\beta}$$

$$t = v \left(\frac{1}{\alpha} + \frac{1}{\beta} \right) = v \left(\frac{\beta + \alpha}{\alpha\beta} \right) \text{ or } v = \frac{\alpha\beta t}{\alpha + \beta}$$

$$40.(A) \text{ The velocity } (v) = \frac{ds}{dt}$$

Therefore, instantaneous velocity at point E is negative

41.(B) In one dimensional motion, the body can have at a time one value of velocity but not two values of velocities.

42.(B) Distance covered in n th second is given by $s_n = u + \frac{a}{2}(2n-1)$

Given : $u=0, a=g$ $\therefore s_4 = \frac{g}{2}(2 \times 4 - 1) = \frac{7g}{2}$; $s_5 = \frac{g}{2}(2 \times 5 - 1) = \frac{9g}{2} \therefore \frac{s_4}{s_5} = \frac{7}{9}$

43.(C) Let $PQ = s$ and L is the midpoint of PQ and v be velocity of the car at point L .

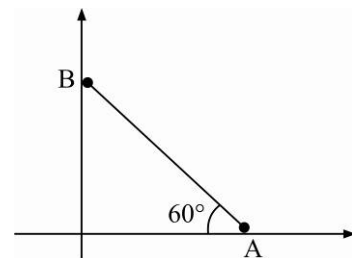
Using third equation of motion, we get

$$(40)^2 - (30)^2 = 2as$$

Or $a = \frac{(40)^2 - (30)^2}{2s} = \frac{350}{s} \dots (i)$

Also, $v^2 - (30)^2 = 2a \frac{s}{2}$

Or $v^2 - (30)^2 = 2 \times \frac{350}{s} \times \frac{s}{2}$ Or $v = 25\sqrt{2} \text{ km/hr}$



44.(C) Let both the balls are thrown from height ' h ', both will move under same acceleration i.e., acceleration due to the gravity. Let the speed of ball ' A ' when it reaches the ground be V_A .

For ball ' A ', using $v^2 = u^2 + 2gh$; $(V_A)^2 = V^2 - 2gh \Rightarrow V_A = \sqrt{V^2 - 2gh}$

Let the speed of ball ' B ' when it reaches the ground be V_B . For ball ' B ', again using $v^2 = u^2 + 2gh$

$$(V_B)^2 = (-V)^2 - 2gh \Rightarrow V_B = \sqrt{V^2 - 2gh}$$

So for both the cases, velocity will be equal.

45.(D) Assuming constant retardation :

Using $v^2 = u^2 + 2as \Rightarrow 0 = u^2 - 2as$

or $s = \frac{u^2}{2a}$ or $s \propto u^2 \Rightarrow \frac{s_1}{s_2} = \frac{u_1^2}{u_2^2} \Rightarrow \frac{s_1}{s_2} = \left(\frac{u}{4u}\right)^2 = \frac{1}{16}$

46.(C) Assuming constant retardation :

Using $v^2 = u^2 + 2as \Rightarrow 0 = u^2 - 2as$ or $s = \frac{u^2}{2a}$ or $\frac{s_1}{s_2} = \frac{u_1^2}{u_2^2}$

$$\frac{6}{s_2} = \left(\frac{50}{100}\right)^2 \Rightarrow s_2 = 24m$$

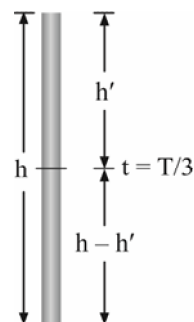
47.(A) From polygon law of vector addition we know if the vectors are arranged along the sides of a polygon the resultant of vector becomes zero. Here in this case the forces are represented by the sides of a triangle taken in same order, so their resultant is zero. It means the acceleration of the particle should be zero, hence velocity remains constant.

48.(A) $h = ut + \frac{1}{2}gt^2 \Rightarrow h = \frac{1}{2}gT^2$

After $\frac{T}{3}$ seconds, the position of ball from top.

$$h' = 0 + \frac{1}{2}g\left(\frac{T}{3}\right)^2 = \frac{1}{2} \times \frac{g}{9} \times T^2 = \frac{h}{9}$$

$\therefore$ Position of ball from ground $= h - \frac{h}{9} = \frac{8h}{9}m$.



49.(D) Assuming constant retardation :

$$\text{Using } v^2 = u^2 + 2as \Rightarrow 0 = u^2 - 2as \quad \text{or} \quad s = \frac{u^2}{2a} \quad \text{or} \quad \frac{s_1}{s_2} = \frac{u_1^2}{u_2^2}$$

$$\frac{20}{s_2} = \left(\frac{60}{120} \right)^2 \Rightarrow s_2 = 80m$$

50.(D) $\frac{ft_1}{t_2 - (t_1 + t)} = \frac{f}{2} \Rightarrow t_2 = 3t_1 + t$

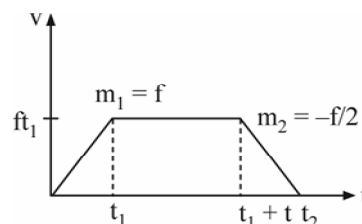
$$\frac{1}{2}(ft_1)(t_1) = S \quad \dots (i)$$

$$\frac{1}{2}ft_1(t + t_2) = 15S$$

$$\frac{1}{2}ft_1(2t_1 + 3t_1) = 15S \quad \dots (ii)$$

(i) divided (ii)

$$\frac{t_1}{2t + 3t_1} = \frac{1}{15} \Rightarrow t_1 = \frac{t}{6} ;$$



$$S = \frac{1}{2}ft_1^2 = \frac{1}{f_2}ft^2$$

51.(B) We are given the relation between time and distance as $t = ax^2 + bx$ $\dots (i)$

Differentiating equation (i) w.r.t. 'x', we get : $\frac{dt}{dx} = 2ax + b$ but $\frac{dx}{dt} = v = \frac{1}{2ax + b}$ $\dots (ii)$

Differentiating equation (ii) w.r.t 't', we get acceleration i.e.,

$$a = \frac{dv}{dt} = \frac{dv}{dx} \cdot \frac{dx}{dt} \quad \therefore \quad a = v \frac{dv}{dx} = \frac{-v \cdot 2a}{(2ax + b)^2} = -2a \cdot v \cdot v^2 = -2av^3$$

The acceleration is negative here. It means that it is the case of retardation.

52.(B) After bailing out from position A, parachutist falls freely under gravity a distance 50m. Let the velocity acquired by him at position B be 'v'.

$$v^2 = u^2 + 2as = 0 + 2 \times 9.8 \times 50 = 980 \quad [\text{As } u = 0, a = 9.8 \text{ m/s}^2, s = 50 \text{ m}]$$

At position B, parachute opens and it moves with retardation of 2m/s^2 and reach at ground (position C) with velocity of 3m/s

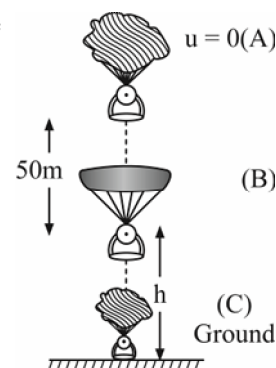
For the journey of the part 'BC' again applying the equation $v^2 = u^2 + 2as$,

where $v = 3 \text{ m/s}$, $u = \sqrt{980} \text{ m/s}$, $a = -2 \text{ m/s}^2$, $s = h$

$$\Rightarrow (3)^2 = (\sqrt{980})^2 + 2 \times (-2) \times h \Rightarrow 9 = 980 - 4h$$

$$\Rightarrow h = \frac{980 - 9}{4} = \frac{971}{4} = 242.7 \approx 243 \text{ m}$$

So, the total height by which parachutist bailed out = $50 + 243 = 293 \text{ m}$.



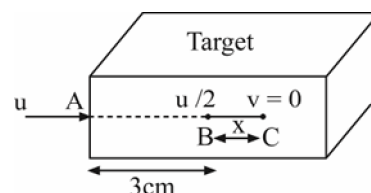
53.(A) Let initial velocity of the bullet = u

After penetrating 3 cm its velocity becomes $\frac{u}{2}$

From $v^2 = u^2 - 2as$, we get

$$\left(\frac{u}{2} \right)^2 = u^2 - 2a(3) \Rightarrow 6a = \frac{3u^2}{4} \Rightarrow a = \frac{u^2}{8}$$

Let further it will penetrate through distance x and stops at point C.



For distance BC , $v = 0, u = u/2, s = x, a = u^2/8$

$$\text{From } v^2 = u^2 - 2as \quad ; \quad 0 = \left(\frac{u}{2}\right)^2 - 2\left(\frac{u^2}{8}\right) \cdot x \quad \Rightarrow \quad x = 1 \text{ cm.}$$

54.(C) We are given $v = \alpha\sqrt{x}$

$$v = \frac{dx}{dt} = \alpha\sqrt{x} \text{ or } \frac{dx}{\sqrt{x}} = \alpha dt$$

$$\int_0^x \frac{dx}{\sqrt{x}} = \alpha \int_0^t dt = \alpha t \quad \text{or} \quad 2x^{1/2} = \alpha t \quad \text{or} \quad x = \left(\frac{\alpha}{2}\right)^2 t^2 \quad \text{or displacement is proportional to } t^2.$$

55.(A) We are given velocity $v = v_0 + gt + ft^2$.

$$\therefore \quad v = \frac{dx}{dt} \quad \text{or} \quad \int_0^x dx = \int_0^t v dt$$

$$x = \int_0^t (v_0 + gt + ft^2) dt \quad \text{or} \quad x = v_0 t + \frac{gt^2}{2} + \frac{ft^3}{3}$$

At $t = 1 \text{ sec}$

$$x = v_0 + \frac{g}{2} + \frac{f}{3}$$

56.(B) For 1st particle :

It starts moving ($u_1 = 0$) with constant acceleration.

$$\therefore \quad x_1 = x_1(t) = u_1 t + \frac{1}{2} at^2 = \frac{1}{2} at^2 \quad \dots\dots(i)$$

For 2nd particle :

It is moving with constant velocity (v)

$$\therefore \quad x_2 = x_2(t) = vt \quad \dots\dots(ii)$$

Relative position of particle 1 w.r.t. 2,

$$x_1 - x_2 = x_{1,2} = \frac{1}{2} at^2 - vt \quad \dots\dots(iii)$$

Hence graph should be parabola.

Differentiating equation (iii) w.r.t time, we get the relative velocity of particle 1 w.r.t 2

$$|v_{1,2}| = \frac{dx_{1,2}}{dt} = at - v$$

$$t = 0, v_{1,2} = -v$$

It means the slope of the parabola at $t = 0$ should be negative.

The parabola should open upwards. Hence graph 'b' fulfills the requirements.

57.(B) We are given $\frac{dv}{dt} = -2.5\sqrt{v}$ or $\frac{1}{\sqrt{v}} dv = -2.5 dt$

On integrating, within limit (at $t = 0, v_1 = 6.25 \text{ ms}^{-1}$ and at any time, $v_2 = 0$), we get

$$\int_{6.25 \text{ ms}^{-1}}^0 v^{-1/2} dv = -2.5 \int_0^t dt$$

$$2 \times \left[v^{1/2} \right]_{6.25}^0 = -(2.5)t \Rightarrow t = \frac{-2 \times (6.25)^{1/2}}{-2.5} = 2s$$

- 58.(A) When a particle is thrown vertically upwards with a speed u . at highest point the velocity of the particle will be zero.

Using $v = u + at$, we get :

$$0 = u - gt$$

$$\Rightarrow t = \frac{u}{g} \quad \dots (i)$$

The time taken by the particle, to hit the ground, is n times that taken by it to reach the highest point of its path.

Let $t_1 = nt$,

Using, $s = ut_1 + \frac{1}{2}at_1^2$, we get

$$\Rightarrow -H = u(nt) - \frac{1}{2}g(nt)^2 \Rightarrow -H = u \times n\left(\frac{u}{g}\right) - \frac{1}{2}gn^2\left(\frac{u}{g}\right)^2 \quad [\text{Using (i)}]$$

$$\Rightarrow -2gH = 2nu^2 - n^2u^2 \Rightarrow 2gH = nu^2(n-2)$$

- 59.(C) For 1st stone : $y_1 = u_1t - \frac{1}{2}gt^2$

For 2nd stone : $y_2 = u_2t - \frac{1}{2}gt^2$

Relative position of '2nd', w.r.t '1st',

$$y_{21} = y_2 - y_1 = (u_2 - u_1)t = (40 - 10)t = 30t \quad \dots (i)$$

Or $y_{21} \propto t$

First stone hits the ground after t_1

$$-240 = 10t - \frac{1}{2}(10)t^2$$

$$\Rightarrow (t^2 - 2t - 48) = 0 \Rightarrow t = 8s$$

Second stone hits the ground after t_2

$$-240 = 40t - \frac{1}{2}(10)t^2 \Rightarrow t^2 - 8t - 48 = 0 \Rightarrow t = 12s$$

It means upto the time both the stones are in the air (From $t = 0$ to $t = 8$ sec) the graph should be a straight line. When second stone hits the ground, first stone is still moving under gravitational acceleration, hence the graph should be parabolic. The speed of first stone will keep on increasing. Hence option (C) is correct.

INTEGER VALUE TYPE

60. (2) At $t = 0$

$$u = 0$$

$$t = t_1 + t_2 + t_3$$

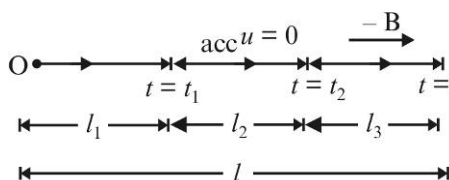
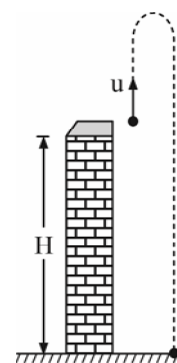
$$t_1 = \frac{v}{\alpha}, t_2 = \frac{l_2}{v}, t_3 = \frac{v}{\beta}$$

$$l_1 = \frac{v^2}{2\alpha} \text{ and } l_3 = \frac{v^2}{2\beta}$$

$$l_2 = l - \frac{v^2}{2\alpha} - \frac{v^2}{2\beta} \therefore t = \frac{l}{v} - \frac{v}{2} \left[\frac{1}{\alpha} + \frac{1}{\beta} \right]$$

For t to be minimum

$$\frac{dt}{dv} = 0 \Rightarrow v = \sqrt{\frac{2\alpha\beta l}{\alpha + \beta}}$$



61. (2) $\begin{array}{ccccccc} u = 0 & & (v = v_{max}) & & a_2 = -ve & & v_f = 0 \\ t = 0 & \xrightarrow{a_1 = ve} & t = t_1 & \xrightarrow{a_2 = -ve} & t = t_2 \end{array}$

In first case $\frac{v_1}{t_1} = a_1 \quad \dots(1)$

In second case $\frac{v_2}{t_1} = 2a_1 \quad \dots(2)$

From (1) and (2)

$$v_2 = 2v_1$$

62. (1) $v = ax + b$

where $a \longrightarrow +ve$ and $b \leq \frac{a}{2}$

$$\frac{dx}{dt} = ax + b \quad ; \quad \int \frac{dx}{ax + b} = \int dt$$

$$\frac{\log_e(ax + b)}{a} = t + c \quad \dots(1)$$

Given at $t = 0, x = 0$

$$\frac{\log_e(b)}{a} = c \quad \text{put in equation (1)}$$

$$\frac{\log_e(ax + b)}{a} = t + \frac{\log_e b}{a}$$

$$\frac{1}{a} \log_e \left(\frac{ax + b}{b} \right) = t \quad ; \quad \frac{ax + b}{b} = e^{at} \quad ; \quad x = \frac{b}{a} (e^{at} - 1)$$

63.(400) For $0 \leq t \leq 4$ second

Given $a = \frac{dv}{dt} = \sqrt{16 - t}$

$$\int_0^v dv = \int_0^t \sqrt{16 - t^2} dt$$

$$v = \frac{t}{2} \sqrt{16 - t^2} + 8 \sin^{-1} \left(\frac{t}{4} \right) \quad ; \quad a = \sqrt{16 - t^2}$$

$$v = \frac{t}{2} \sqrt{16 - t^2} + 8 \sin^{-1} \left(\frac{t}{4} \right)$$

This expression is valid upto $t = 4$ second

velocity at $t = 4$

$$v_4 = 4\pi$$

For $t > 4$, acceleration $a = 2 \text{ m/sec}^2 = \text{constant}$.

So using equation of motion for the velocity at $t = 6$ second

$$v_6 = v_4 + a(t_6 - t_4)$$

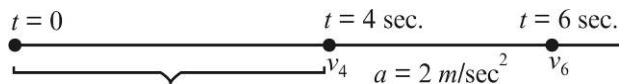
$$v_6 = 4\pi + 2 \times 2$$

$$v_6 = 4\pi + 4$$

$$\boxed{v_6 = 4(\pi + 1)} \text{ m/sec}^2$$

64.(2) Let the length of the course be 's' then for the first car

$$S = ut + \frac{1}{2} at^2$$



and for the second car $at^2 + 2ut - 2s = 0$... (1)

$$s = u't + \frac{1}{2}a't^2$$

$$a't^2 + 2u't - 2s = 0$$
 ... (2)

Solving these equation simultaneously $\frac{t^2}{-4us + 4u's} = \frac{t}{-2a's + 2as} = \frac{1}{2u'a - 2ua'}$

$$\frac{t^2}{2s(u' - u)} = \frac{t}{s(a - a')} = \frac{1}{u'a - 4a'}$$

Eliminating we get

$$s^2(a - a')^2 = 2s(u' - u)(u'a - ua')$$

$$S = \frac{2(u' - u)(u'a - ua')}{(a - a')^2}$$

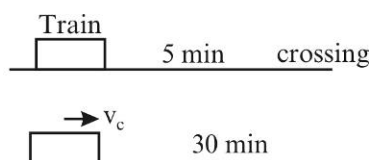
65. (4) $t_{\text{cycle}} = \frac{10\text{km}}{20\text{km/hr}} = \frac{1}{2}\text{hr} = 30\text{min}$

$$\Delta t = 5\text{min} = \frac{5}{60}\text{hr}$$

Train running as per schedule

$$v_{\text{train}} = \frac{10}{5/60} = \frac{10 \times 60}{5} = 120\text{ km/hr}$$

$$v_{\text{train}} = 30N \quad \therefore N = 4$$



66. (21) $a = \tan 45^\circ$

$$a = 1$$

$$x_A = x_B$$

$$10.5 + 10t = \frac{1}{2}at^2$$

$$t^2 - 20t - 21 = 0$$

$$t^2 - 21t + t - 21 = 0$$

$$(t - 21)(t + 1) = 0 \quad ; \quad t = 21\text{ second}$$

67. (5) If t is time taken then

$$\frac{1}{2} \times 2(t + 5)^2 = 20t$$

$$T = 5\text{ sec.}$$

68. (61) $|\vec{u}_{c/ob}| = |\vec{u}_{c/g}| - |\vec{u}_{\text{elevator}/g}| = 10 - 10$

$$u_{c/o} = 0$$

The acceleration of coin with respect to observer in the elevator

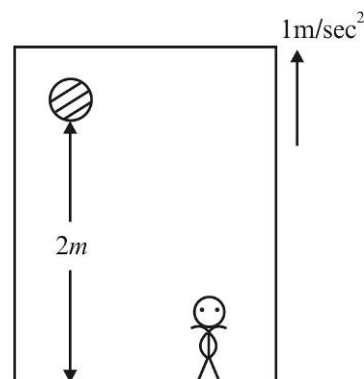
$$|\vec{a}_{c/ele}| = |\vec{a}_{\text{coin}/g}| = |\vec{a}_{\text{elevator}/g}| = g - (-a)$$

$$a_{c/ele} = g + a$$

$$S_{c/\text{elevator}} = u_{c/\text{elevator}} + \frac{1}{2}a_{c/\text{elevator}}t^2$$

$$2 = 0 + \frac{1}{2}(g + a)t^2$$

$$2 = \frac{1}{2}(9.8 + 1)t^2$$



$$t = \sqrt{\frac{2 \times 2}{10.8}}$$

$$t = 0.61 \text{ second}$$

69. (9) $V_{\text{men}} = 9 \text{ km/hr} = \frac{9 \times 5}{18} = 2.5 \text{ m/sec.}$

$$V_{\text{cyclist}} = 18 \text{ km/hr} = \frac{18 \times 5}{18} = 5 \text{ m/sec.}$$

For observer crossing men, speed = $\left(v + \frac{5}{2}\right) \text{ m/sec.}$

$$\Delta t = \frac{20}{v + \frac{5}{2}} \quad \dots(1)$$

For observer crossing cyclist, speed = $(v + 5) \text{ m/sec.}$

$$\Delta x = 30 \text{ m}$$

$$\Delta t = \frac{30}{v + 5} \quad \dots(2)$$

using equ (1) and (2)

$$\frac{20}{v + \frac{5}{2}} = \frac{30}{v + 5} \quad ; \quad U = \frac{5}{2} \text{ m/sec} = 9 \text{ km/hr}$$

70. (5) consider motion along x-axis $\frac{dx}{dt} = (3 - 5 \cos \theta)$

$$dx = (3 - 5 \cos \theta) dt$$

$$\int_0^t dx = \int_0^t 3 dt - \int_0^t 5 \cos \theta dt$$

$$5 \int_0^t \cos \theta dt = 3t \quad \dots(1)$$

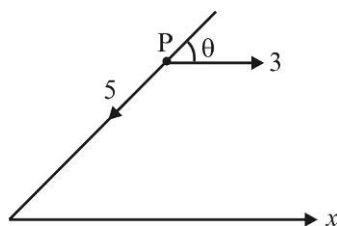
Let OP = r

$$-\frac{dr}{dt} = (5 - 3 \cos \theta)$$

$$-\int_0^t dr = \int_0^t 5 dt - 3 \int_0^t \cos \theta dt \quad \dots(2)$$

Solving equ. (1) and (2) we get

$$t = 5$$



71.(26) $a = \begin{cases} 2t + t - 2, & t > 2 \\ 2t - t + 2, & t < 2 \end{cases}$

$$\Delta v = a dt = \int_0^t (t + 2) dt = \frac{t^2}{2} + 2t$$

$$\Delta v = \left[\frac{t^2}{2} + 2t \right]_0^2 = 6$$

$$v_2 = v_0 + 6 = 4 + 6 = 10 \text{ m/sec.}$$

$$\text{Then } \Delta v = \int_2^4 (3t - 2) dt$$

$$v_4 - v_2 = \left[\frac{3t^2}{2} - 2t \right]_2^4$$

$$v_4 - 10 = 3 \left[\frac{16}{2} \right] - 2 \times 4$$

$$v_4 = 26 \text{ m/sec.}$$

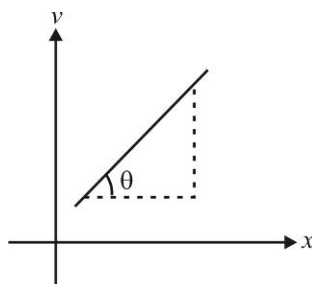
72. (600) $m = \tan \theta = \frac{3}{100}$

$$v = \frac{3}{100}x + c$$

When $x = 50$, $v = 3$

$$3 = \frac{3}{100} \times 50 + c$$

$$c = \frac{3}{2}$$



Then $v = 0 = \frac{3}{100}x + \frac{3}{2}$

$$\int_0^v v dv = \int_0^{100} \left(\frac{3}{100}x + \frac{3}{2} \right) dx$$

$$\frac{v^2}{2} = 300$$

$$v = \sqrt{600} \text{ m/sec}$$

73.(4) $v = a \sin \omega t$

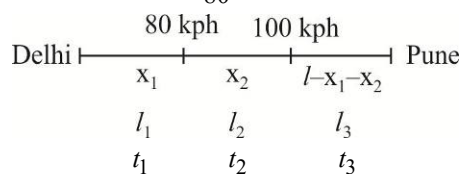
$$v = \pi \sin \pi \omega$$

$$\text{Distance} = \int_0^2 |v| dt \quad ; \quad = 2 \int_0^1 |v| dt = \frac{2\pi}{\pi} [-\cos \pi t]_0^1 = \frac{2\pi}{\pi} \times 2 = 4 \text{ meter}$$

74.(2) As time are equal $t = \sqrt{\frac{2\sqrt{2}L}{a}} = \frac{2L}{v}$

$$\frac{2\sqrt{2}L}{a} = \frac{4L^2}{v^2} \Rightarrow a = \frac{v^2}{\sqrt{2}L}$$

75.(840) If t is normal running time then $t = \frac{l}{80}$



Here $t_1 = \frac{x_1}{80} + 2$

$$t_2 = \frac{100 - 80}{20} = 1 \text{ hr}$$

$$t_3 = \frac{l - x_1 - x_3}{100}$$

Here $x_2 = 80(1) + \frac{1}{2}(20)(1)^2 = 90 \text{ km}$

Thus total time $t = t_1 + t_2 + t_3 = \frac{t}{80}$

$$\left(\frac{x_1}{80} + 2\right) + 1 + \left(\frac{l + x_1 - 90}{100}\right) = \frac{l}{80}$$

Solving it gives $l - x_1 = 840 \text{ km}$

76.(89) Stopping distance of car-1 $= \frac{(10)^2}{2 \times 2} = 25 \text{ m}$

Stopping distance of car-2 $= \frac{(12)^2}{2 \times 2} = 36 \text{ m}$

Separation between cars $= 150 - 25 - 36 = 89 \text{ m}$

Motion in Two Dimensions

1.(A) Let V : final velocity, a ; tangential acceleration

$$\Rightarrow V^2 = 2a_t(\pi R) \Rightarrow a_r = \frac{V^2}{R} = 2a_t \pi$$

Angle between velocity vector acceleration vector = Angle between tangential acceleration and total acceleration

$$u = \tan^{-1} \frac{a_r}{a_t} = \tan^{-1}(2\pi)$$

2.(C) Particle may have its velocity continuously changing $\vec{a} = \frac{d\vec{v}}{dt}$

3.(AB) $H_1 > H_2$

$$\frac{V_1^2 \sin^2 \theta_1}{2g} > \frac{V_0^2 \sin^2 \theta_2}{2g} \Rightarrow \theta_1 > \theta_2 \Rightarrow T_1 > T_2$$

4.(BCD) $\vec{r} = R(\cos \omega t \hat{i} + \sin \omega t \hat{j}) \rightarrow$ position vector $\vec{V} = \frac{d\vec{r}}{dt} = -R\omega \sin \omega t \hat{i} + R\omega \cos \omega t \hat{j} \rightarrow$ Velocity vector

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{d^2\vec{r}}{dt^2} = -R\omega^2 \cos \omega t \hat{i} - R\omega^2 \sin \omega t \hat{j} \rightarrow \text{acc. vector}$$

$$= -R\omega^2 (\cos \omega t \hat{i} + \sin \omega t \hat{j})$$

$$\vec{a} = -\omega^2 \vec{r} \rightarrow \text{Variable as } \vec{r} \text{ varies with time.}$$

Magnitude $|\vec{a}| = \omega^2 |\vec{r}| = \omega^2 R$ $|\vec{v}| = \omega R = \text{Speed const}$

$$\frac{d\vec{a}}{dt} = -\omega^2 \frac{d\vec{r}}{dt} = -\omega^2 \vec{v}$$

$$\left| \frac{d\vec{a}}{dt} \right| = \omega^2 |\vec{v}| = \omega^2 \omega R = \omega^3 R$$

$$\frac{d|\vec{v}|}{dt} = \text{Rate of change of speed} = 0 \quad (\because \text{Speed} = \text{Const} = \omega R)$$

5.(A) Final velocity vector makes angle $90 - 30 = 60^\circ$ with horizontal, when it becomes (perpendicular to initial velocity vector) Let it be $V \Rightarrow 20 \cos 30^\circ = V \cos 60^\circ = V \cos 60^\circ \Rightarrow V = 20\sqrt{3} \text{ m/s}$

$$\Rightarrow V_y = -20\sqrt{3} \sin 60^\circ = -30 \text{ m/s}, u_y = 20 \sin 30^\circ = 10 \text{ m/s} \quad (v = u + at)$$

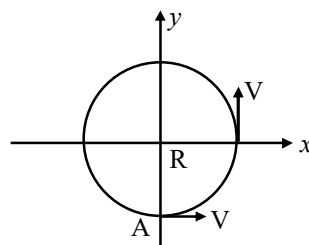
$$\Rightarrow -30 = 10 - 10t \Rightarrow t = 4 \text{ sec horizontal displacement} = u_x t = 20 \cos 30^\circ \times 4 = 40\sqrt{3} \text{ m}$$

6.(A) Let initial point A is on $-y$ -axis and at $t = T/4$, it is at $+x$ -axis.

$$\Rightarrow \text{At } t = T/4, \vec{a} = \frac{-V^2}{R} \hat{i} \quad \dots(i)$$

$$\Rightarrow \text{At } t = 0, \vec{V} = V \hat{i}; \text{ at } t = T/4, \vec{V} = V \hat{j}$$

$$\Rightarrow \Delta \vec{V} = V \hat{j} - V \hat{i} \quad \dots(ii),$$



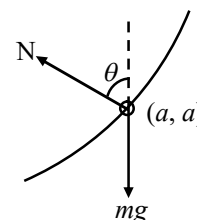
$$\text{Average velocity from } t = 0 \text{ to } t = T/4 \text{ is equal to } \frac{R\hat{i} - (-R\hat{j})}{T/4} = \frac{4R}{T} (\hat{i} + \hat{j}) \quad \dots(iii)$$

$$\Rightarrow \theta_1 = \text{angle between vectors } -\hat{i} \text{ and } \hat{i} + \hat{j} = 135^\circ$$

$$\text{and } \theta_2 = \text{angle between vector } -\hat{i} \text{ and } \hat{j} - \hat{i} = 45^\circ$$

7.(C) $\tan \theta = \frac{dy}{dx} = \frac{2x}{a} \left\{ = 2 \text{ at } (a, a) \right\}$ where θ is angle made by tangent at (a, a) with x -axis
i.e. angle made by normal with y -axis

$$\Rightarrow N \cos \theta = mg \text{ and } N \sin \theta = m\omega^2 a \Rightarrow \tan \theta = \frac{\omega^2 a}{g} \Rightarrow \omega = \sqrt{\frac{g \tan \theta}{a}} = \sqrt{\frac{2g}{a}}$$



8.(C) Energy is conserved but K.E of the particles is not zero at P ($v_x \neq 0$)

9.(C) As horizontal component of vertical remains same $\Rightarrow V \cos \phi = u \cos \theta \Rightarrow v = u \cos \theta \sec \phi$

10.(ABC) $H_P < H_A$ Acceleration is along radius vector

11.(BD) Horizontal component w.r.t. ground is along the tangent and its net initial velocity makes some angle $\theta (\neq 90)$ with horizontal $\Rightarrow$ path is parabolic.

$$12.(ABD) \frac{u^2 \sin 2\theta_1}{g} = \frac{u^2 \sin 2\theta_2}{g} \Rightarrow \sin 2\theta_1 = \sin(180 - 2\theta_2) \Rightarrow \theta_1 + \theta_2 = 90^\circ$$

$$t_1 = \frac{2u \sin \theta_1}{g} \text{ and } t_2 = \frac{2u \sin \theta_2}{g} \Rightarrow \frac{t_1}{\sin \theta_1} = \frac{t_2}{\sin \theta_2} = \frac{2u}{g}$$

$$t_1 / t_2 = \frac{\sin \theta_1}{\sin \theta_2} = \frac{\sin \theta_1}{\cos \theta_1} = \tan \theta_1 = \cot \theta_2$$

13.(A) $t = 2 \text{ sec}$

$$V \sin 45^\circ + u \sin \theta - g \cdot t \quad \dots(i)$$

$$V \cos 45^\circ = u \cos \theta \quad \dots(ii)$$

$t = 3 \text{ sec}$

$$u \sin \theta - g \cdot 3 = 0 \quad \dots(iii)$$

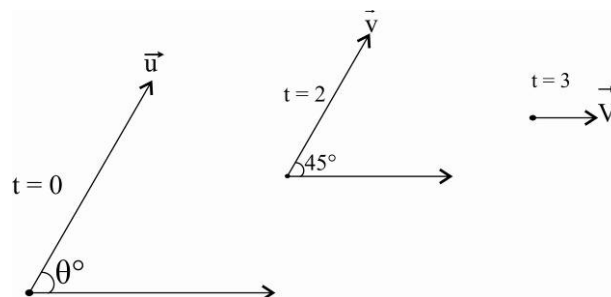
(From 1, 2 and 3)

$$V \sin 45^\circ = 10 = U \cos \theta$$

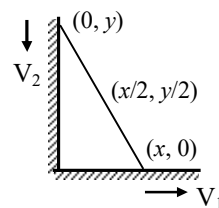
$$u \sin \theta = 30$$

$$u \cos \theta = 10$$

$$\Rightarrow \tan \theta = 3 \text{ and } u = 10\sqrt{10} \tan \theta = 3 \text{ and } u = 10\sqrt{10}$$



14.(D) $\Rightarrow \frac{dx}{dt} = V_1, \frac{dy}{dt} = V_2$ velocity of mid point $= \frac{1}{2} \frac{dx}{dt} \hat{i} + \frac{1}{2} \frac{dy}{dt} \hat{j}$
 $\Rightarrow \text{magnitude} = \frac{\sqrt{V_1^2 + V_2^2}}{2}$

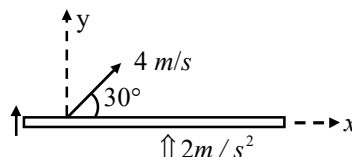


15.(B) $\Rightarrow$ w.r.t. elevator, $u_x = 4 \cos 30^\circ = 2\sqrt{3} \text{ m/s}$

$$u_y = 4 \sin 30^\circ = 2 \text{ m/s}$$

$$a_x = 0 \text{ m/s}^2 \text{ and } a_y = (-10) - (+2) = -12 \text{ m/s}^2$$

$$S_y = 0 \Rightarrow u_y t + \frac{1}{2} a_y t^2 = 0 \Rightarrow t = \frac{-2u_y}{a_y} = 1/3 = 0.33 \text{ sec}$$



16.(D) Let u : velocity of projection and θ : angle of projection

$$\Rightarrow R = \frac{u^2 \sin 2\theta}{g} \text{ and } H = \frac{u^2 \sin^2 \theta}{2g}$$

If horizontal acceleration $= g/4$ is imparted: Let T : time of

$$\text{Flight } S_y = 0 \Rightarrow 0 = u \sin \theta T - \frac{1}{2} g T^2 \Rightarrow T = \frac{2u \sin \theta}{g}$$

$$\text{New range: } S_x = u \cos \theta T + \frac{1}{2} \frac{g}{4} T^2 = \frac{2u^2 \sin \theta \cos \theta}{g} + \frac{1}{2} \frac{g}{4} \cdot \frac{4u^2 \sin^2 \theta}{g^2} = R + H$$

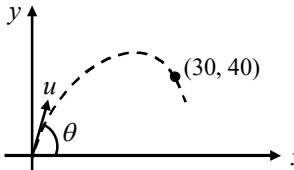
From initial point of projection to maximum height gain, $V_y = 0$

$$\Rightarrow V_y^2 = u_y^2 + 2a_y s_y \Rightarrow 0 = u^2 \sin^2 \theta - 2g s_y \Rightarrow s_{y \text{ new}} \quad \text{Height gain} = \frac{u^2 \sin^2 \theta}{2g} = H$$

17.(B) $a_r \propto t^2 \Rightarrow a_r = ct^2$ [c is constant]

$$\frac{V^2}{R} = ct^2$$

$$a_z = \frac{d|V|}{dt} = \sqrt{cR} = \text{Constant of time}$$

18.(B) 

$$30 = u \cos \theta t$$

$$40 = u \sin \theta t - \frac{10t^2}{2}$$

$$\Rightarrow 40 = 30 \tan \theta - \frac{5(30)^2}{u^2 \cos^2 \theta} \Rightarrow 4 = 3 \tan \theta - \frac{450}{u^2} (1 + \tan^2 \theta)$$

$$\Rightarrow 450 \tan^2 \theta - 3u^2 \tan \theta + 450 + 4u^2 = 0$$

$\tan \theta$ is real $\Rightarrow D \geq 0$

$$\Rightarrow 9u^4 - 1800(450 + 4u^2) \geq 0 \Rightarrow u^4 - 800u^2 - 90,000 \geq 0$$

$$\Rightarrow u^4 - 900u^2 + 100u^2 - 90,000 \geq 0 \Rightarrow u^2(u^2 - 900) + 100(u^2 - 900) \geq 0$$

$$\Rightarrow u^2 = -100 \text{ or } 900 \Rightarrow u = 30 \text{ m/s}$$

- 19.(A) Let angle made by velocity vector with horizontal is θ when speed is $\sqrt{125} \text{ m/s} \Rightarrow 10\sqrt{2} \cos 45^\circ = 5\sqrt{5} \cos \theta$
 $\Rightarrow \cos \theta = 2/\sqrt{5} \quad \Rightarrow \sin \theta = 1/\sqrt{5}$
 $\Rightarrow \text{vertical component of velocity} = \pm \sqrt{125} \sin \theta = \pm 5 \text{ m/s}$
 $\Rightarrow \text{along vertical } \pm 5 = 10\sqrt{2} \sin 45^\circ - gt \Rightarrow t_1 = 0.5 \text{ sec}, t_2 = 15 \text{ sec} \Rightarrow \text{time interval} = 1 \text{ sec.}$

- 20.(B) Along AB, relative velocity of A w.r.t. B = $u\sqrt{3} \cos 30^\circ + u \cos 60^\circ = 2u$

$$\Rightarrow \text{time} = \frac{\text{relative horizontal displacement}}{\text{relative horizontal velocity}} = \frac{x}{2u}$$

- 21.(D) For total time of flight (T), $y = 0 \Rightarrow kt(1 - \alpha t) = 0 \Rightarrow t = 0 \text{ or } 1/\alpha$

$$\text{At maximum height, } \frac{dy}{dt} = 0 \Rightarrow K(1 - 2\alpha t) = 0 \Rightarrow t = \frac{1}{2}\alpha \Rightarrow \text{maximum height} = \frac{K}{2\alpha} \left(1 - \alpha \cdot \frac{1}{2\alpha}\right) = \frac{K}{4\alpha}$$

- 22.(B) As, initial velocity $v = (i + 2j) \text{ m/s}$

For projectile : Motion along vertical horizontal direction, $x = t$

$$\text{Motion along vertical direction, } y = ut - \frac{1}{2}gt^2 \quad \dots(i)$$

$$\text{Where, } u \text{ is a vertical component of velocity } y = 2t - \frac{1}{2}10t^2 \quad \dots(ii)$$

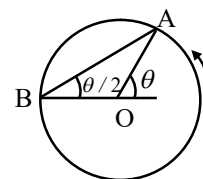
From equation (i) and (ii), we get : $y = 2x - 5x^2$

$$\begin{aligned} 23.(D) \quad \frac{u^2 \sin 2\theta}{g} - 6 &= \frac{\sqrt{3}u^2}{2g} \\ \frac{u^2 \sin 2\theta}{g} + 9 &= \frac{u^2}{g} \dots \text{Subtract.} \\ 15 &= \frac{u^2}{g} \left(1 - \frac{\sqrt{3}}{2}\right) \end{aligned}$$

$$u^2 = \frac{15g}{1 - \sqrt{3}/2}$$

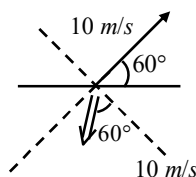
$$\frac{u^2 \sin 2\theta}{g} = \frac{15}{1 - \sqrt{3}/2} - 9 \quad \sin 2\theta = \frac{4 + 3\sqrt{3}}{10} \Rightarrow \theta = \frac{1}{2} \sin^{-1} \left(\frac{4 + 3\sqrt{3}}{10} \right)$$

- 24.(B) Let $\omega_{AO} = d\theta/dt \Rightarrow \omega_{AB} = \frac{d(\theta/2)}{dt} = \frac{1}{2} \frac{d\theta}{dt} = \frac{1}{2} \omega_{AO} \Rightarrow \frac{\omega_{AO}}{\omega_{AB}} = 2$



- 25.(D) $\Rightarrow a_{\perp} = 10 \cos 60^\circ = 5 \text{ m/s}$

$$\Rightarrow R = \frac{V^2}{a_{\perp}} = \frac{10 \times 10}{5} = 20$$



- 26.(A) Two vector $\vec{A}$ and $\vec{B}$ are orthogonal to each other, if their scalar product is zero i.e. $\vec{A} \cdot \vec{B} = 0$

$$\text{Here, } \vec{A} = \cos \omega t \hat{i} + \sin \omega t \hat{j} \quad \text{and} \quad \vec{B} = \cos \frac{\omega t}{2} \hat{i} + \sin \frac{\omega t}{2} \hat{j}$$

$$\therefore \vec{A} \cdot \vec{B} = (\cos \omega t \hat{i} + \sin \omega t \hat{j}) \cdot \left(\cos \frac{\omega t}{2} \hat{i} + \sin \frac{\omega t}{2} \hat{j} \right)$$

$$= \cos \omega t \cos \frac{\omega t}{2} + \sin \omega t \sin \frac{\omega t}{2} \quad (\because \hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = 1 \text{ and } \hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{i} = 0)$$

$$= \cos \left(\omega t - \frac{\omega t}{2} \right) \quad (\because \cos(A - B) = \cos A \cos B + \sin A \sin B)$$

But $\vec{A} \cdot \vec{B} = 0$ (as $\vec{A}$ and $\vec{B}$ are orthogonal to each other) $\therefore \cos \left(\omega t - \frac{\omega t}{2} \right) = 0$

$$\cos \left(\omega t - \frac{\omega t}{2} \right) = \cos \frac{\pi}{2} \text{ or } \omega t - \frac{\omega t}{2} = \frac{\pi}{2}$$

$$\frac{\omega t}{2} = \frac{\pi}{2} \text{ or } t = \frac{\pi}{\omega}$$

27.(A) Here $\vec{R} = 4 \sin(2\pi t)\hat{i} + 4 \cos(2\pi t)\hat{j}$

The velocity of the particle is $v = \frac{d\vec{R}}{dt} = \frac{d}{dt} [4 \sin(2\pi t)\hat{i} + 4 \cos(2\pi t)\hat{j}] = 8\pi \cos(2\pi t)\hat{i} - 8\pi \sin(2\pi t)\hat{j}$

Its magnitude is $|v| = \sqrt{(8\pi \cos(2\pi t))^2 + (-8\pi \sin(2\pi t))^2} = \sqrt{64\pi^2 \cos^2(2\pi t) + 64\pi^2 \sin^2(2\pi t)}$

$$= \sqrt{64\pi^2 [\cos^2(2\pi t) + \sin^2(2\pi t)]} = \sqrt{64\pi^2} \quad (\text{as } \sin^2 \theta + \cos^2 \theta = 1)$$

$$= 8\pi \text{ m/s}$$

28.(A) The equation of trajectory is $y = x \tan \theta - \frac{gx^2}{2u^2 \cos^2 \theta}$

Where θ is the angle of projection and u is the velocity with which projectile is projected.

For equal trajectories and for same angles of projection, $\frac{g}{u^2} = \text{constant}$

As per question, $\frac{9.8}{5^2} = \frac{g'}{3^2}$

Where g' is acceleration due to gravity on the planet.

$$g' = \frac{9.8 \times 9}{2.5} = 3.5 \text{ ms}^{-2}$$

29.(D) At time $t = 0$, the position vector of the particle is $\vec{r}_1 = 2\hat{i} + 3\hat{j}$

At time $t = 5s$ the position vector of the particle is $\vec{r}_2 = 13\hat{i} + 14\hat{j}$

Displacement from $\vec{r}_1$ to $\vec{r}_2$ is $\Delta \vec{r} = \vec{r}_2 - \vec{r}_1 = (13\hat{i} + 14\hat{j}) - (2\hat{i} + 3\hat{j}) = 11\hat{i} + 11\hat{j}$

$\therefore$ Average velocity, $v_{av} = \frac{\Delta \vec{r}}{\Delta t} = \frac{11\hat{i} + 11\hat{j}}{5 - 0} = \frac{11}{5}(\hat{i} + \hat{j})$

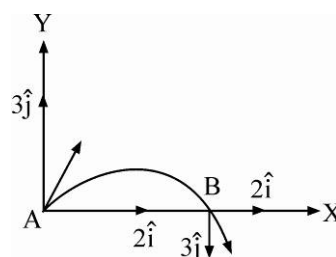
30.(A) At point B X component of velocity remains unchanged while Y component reverses its direction.

$\therefore$ The velocity of the projectile at point B is $2\hat{i} - 3\hat{j} \text{ m/s}$

31.(B) Here, $\vec{u} = 2\hat{i} + 3\hat{j}$, $a = 0.3\hat{i} + 0.2\hat{j}$, $t = 10s$

As $\vec{v} = \vec{u} + \vec{a}t$

$\therefore v = (2\hat{i} + 3\hat{j}) + (0.3\hat{i} + 0.2\hat{j})(10) = 2\hat{i} + 2\hat{j} + 2\hat{j} = 2\hat{i} + 5\hat{j}$



$$|\vec{v}| = \sqrt{(5)^2 + (5)^2} = 5\sqrt{2} \text{ units}$$

32.(D) Here, Radius, $R = 5 \text{ cm} = 5 \times 10^{-2} \text{ m}$

Time period, $T = 0.2\pi \text{ s}$

$$\text{Centripetal acceleration } a_c = \omega^2 R = \left(\frac{2\pi}{T}\right)^2 R = \left(\frac{2\pi}{0.2\pi}\right)^2 (5 \times 10^{-2}) = 5 \text{ m/s}^2$$

As particle moves with constant speed, therefore its tangential acceleration is zero. So, $a_t = 0$

The acceleration of the particle is $a = \sqrt{a_c^2 + a_t^2} = a_c = 5 \text{ m/s}^2$. It acts towards the centre of the circle.

33.(A) Here, $u = 20 \text{ m/s}$, $g = 10 \text{ m/s}^2$

For maximum range, angle of projection is $\theta = 45^\circ$

$$\begin{aligned} \therefore R_{\max} &= \frac{u^2 \sin 90^\circ}{g} = \frac{u^2}{g} \quad \left\{ \because R = \frac{u^2 \sin 2\theta}{g} \right\} \\ &= \frac{(20 \text{ m/s})^2}{(10 \text{ m/s}^2)} = 40 \text{ m} \end{aligned}$$

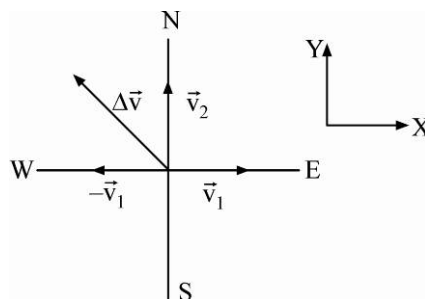
34.(D) Velocity towards east direction $\vec{v}_1 = 30\hat{i} \text{ m/s}$

Velocity towards north direction, $\vec{v}_2 = 40\hat{j} \text{ m/s}$

Change in velocity $\Delta\vec{v} = \vec{v}_2 - \vec{v}_1 = (40\hat{j} - 30\hat{i})$

$$\therefore |\Delta\vec{v}| = |40\hat{j} - 30\hat{i}| = 50 \text{ m/s}$$

$$|\vec{a}_{av}| = \frac{|\Delta\vec{v}|}{\Delta t} = \frac{50 \text{ m/s}}{10 \text{ s}} = 5 \text{ m/s}^2$$



35.(C) Let ϕ be elevation angle of the projectile at its highest point as seen from the point of projection O and θ be angle of projection with the horizontal.

$$\text{From figure, } \tan \phi = \frac{H}{R/2} \quad \dots\dots(i)$$

In case of projectile motion

$$\text{Maximum height, } H = \frac{u^2 \sin^2 \theta}{2g} \quad \text{Horizontal range, } R = \frac{u^2 \sin 2\theta}{g}$$

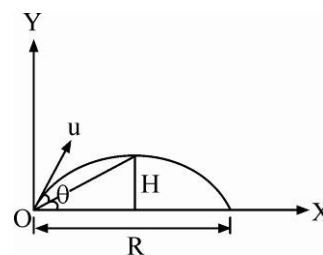
Substituting these value of H and R in (i), we get :

$$\tan \phi = \frac{\frac{2^2 \sin^2 \theta}{2g}}{\frac{u^2 \sin 2\theta}{2g}} \Rightarrow \tan \phi = \frac{\sin^2 \theta}{\sin 2\theta} = \frac{\sin^2 \theta}{2 \sin \theta \cos \theta} = \frac{1}{2} \tan \theta$$

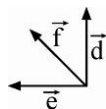
$$\tan \theta = \frac{1}{2} \tan 45^\circ = \frac{1}{2}$$

$$\text{Here, } \theta = 45^\circ \quad \therefore \tan \phi = \frac{1}{2} \tan 45^\circ = \frac{1}{2} \quad (\because \tan 45^\circ = 1)$$

$$\phi = \tan^{-1} \left(\frac{1}{2} \right)$$



36.(C) From figure4, $\vec{d} + \vec{e} = \vec{f}$



37.(A) $\Delta P = -mu \sin 45^\circ \hat{j} - mu \cos 45^\circ \hat{i}$. If it sticks to the ground.

38.(B) The centre of the tube will be at length $L/2$.

So radius $r = L/2$. The force exerted by the liquid at the other end = centrifugal force.

$$\text{Centrifugal force} = Mr\omega^2 = M\left(\frac{L}{2}\right)\omega^2 = \frac{ML\omega^2}{2}$$

39.(B) Horizontal range $R = \frac{u^2 \sin 2\theta}{g}$

For angle of projection $(45^\circ - \theta)$, the horizontal range is

$$\therefore R_1 = \frac{u^2 \sin[2(45^\circ - \theta)]}{g} = \frac{u^2 \sin(90^\circ - 2\theta)}{g} = \frac{u^2 \cos 2\theta}{g}$$

For angle of projection $(45^\circ + \theta)$, the horizontal range is :

$$R_2 = \frac{u^2 \sin[2(45^\circ + \theta)]}{g} \propto \frac{u^2 \sin(90^\circ - 2\theta)}{g} = \frac{u^2 \cos 2\theta}{g}$$

$$\therefore \frac{R_1}{R_2} = \frac{u^2 \cos 2\theta / g}{u^2 \cos 2\theta / g} = \frac{1}{1} \quad \therefore \quad \text{The range is the same}$$

40.(D) $t = \frac{a}{v'} = \frac{a}{\sqrt{v^2 - v_1^2}}$

41.(B) Mass $m = 3$ kg, force, $F = 6t^2 \hat{i} + 4t \hat{j}$

$$\therefore \text{Acceleration, } a = F/m = \frac{6t^2 \hat{i} + 4t \hat{j}}{3} = 2t^2 \hat{i} + \frac{4}{3}t \hat{j}$$

$$\text{Now, } a = \frac{dv}{dt} = 2t^2 \hat{i} + \frac{4}{3}t \hat{j}$$

$$\therefore dv = \left(2t^2 \hat{i} + \frac{4}{3}t \hat{j}\right) dt \therefore v = \int_0^3 \left(2t^2 \hat{i} + \frac{4}{3}t \hat{j}\right) dt = \frac{2}{3}t^3 \hat{i} + \frac{4}{6}t^2 \hat{j} \Big|_0^3 = 18\hat{i} + 6\hat{j}$$

42.(B) $v = \omega r = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 5 & -6 & 6 \\ 3 & -4 & 1 \end{vmatrix} = 18\hat{i} + 13\hat{j} - 2\hat{k}$

43.(D) Let particle B move upwards with velocity v . then $\tan 60^\circ = \frac{v}{10}$; $v = \sqrt{3} \times 10 = 17.3 \text{ m/s}$

44.(C) $\frac{mv^2}{r} = 25$; $v = \sqrt{\frac{2.5 \times 1.96}{0.25}} = 14 \text{ m/s}$

45.(A) Number of revolutions per minute (n) = 120. Therefore angular speed (ω)

$$= \frac{2\pi n}{60} = \frac{2\pi \times 120}{60} = 4\pi \text{ rad/s}$$

- 46.(B) The kinetic energy of the ball at point of projection is E.

$$E = \frac{1}{2}mu^2$$

At highest point velocity has its horizontal component $u \cos \theta$

Therefore kinetic energy of the ball at height point is :

$$E_H = \frac{1}{2}m(u \cos \theta)^2 = \frac{1}{2}mu^2 (\cos \theta)^2 = E \cos^2 \theta \quad \Rightarrow \quad E_H = E \cos^2 45^\circ = \frac{E}{2}$$

- 47.(B) Only horizontal component of velocity ($u \cos \theta$).

- 48.(C) $x = \alpha t^3$ and $y = \beta t^3$

$$x - \text{component of the velocity : } v_x = \frac{dx}{dt} = 3\alpha t^2 \quad \text{and } y - \text{component of the velocity : } v_y = \frac{dy}{dt} = 3\beta t^2$$

$$\text{Resultant velocity : } v = \sqrt{v_x^2 + v_y^2} = 3t^2 \sqrt{\alpha^2 + \beta^2}$$

- 49.(A) When the ball again comes to the same horizontal level it covers a distance equal to range of the projectile motion.

$$\text{Hence range, } R = \frac{u^2 \sin 2\theta}{g} = \frac{(10)^2 \sin 60^\circ}{10} = 5\sqrt{3} = 8.66m$$

- 50.(B) In circular motion we know the velocity vector is tangent to the circle. If a particle moves in a circle with constant angular speed it perform uniform circular motion. In uniform circular motion the tangential acceleration of the particle is zero. The particle moves under radial acceleration only which points to the centre of the circle. Hence the velocity and acceleration vectors are perpendicular to each other.

- 51.(C) Person will catch the ball if its velocity will be equal to horizontal component of velocity of the ball.

$$\frac{v_0}{2} = v_0 \cos \theta \quad \Rightarrow \quad \cos \theta = \frac{1}{2} \quad \Rightarrow \quad \theta = 60^\circ$$

- 52.(B) For same range, angles of projection should be θ and $(90^\circ - \theta)$.

So, time of flights.

$$t_1 = \frac{2u \sin \theta}{g} \quad \text{and} \quad t_2 = \frac{2u \sin(90^\circ - \theta)}{g} = \frac{2u \cos \theta}{g}$$

$$\therefore \quad t_1 t_2 = \frac{4u^2 \sin \theta \cos \theta}{g^2} \quad \Rightarrow \quad t_1 t_2 = \frac{2}{g} \left(\frac{u^2 \sin 2\theta}{g} \right) = \frac{2R}{g} \Rightarrow t_1 t_2 \propto R$$

- 53.(B) $v = u + at$

$$\vec{v} = (3\hat{i} + 4\hat{j}) + (0.4\hat{i} + 0.3\hat{j}) \times 10 \quad \Rightarrow \quad \vec{v} = (3 + 4)\hat{i} + (4 + 3)\hat{j}$$

$$\Rightarrow \quad |\vec{v}| = \sqrt{49 + 49} = \sqrt{98} = 7\sqrt{2} \text{ unit}$$

(This value is about 9.9 units close to 10 units. If (a) is given that is also not wrong).

- 54.(D) Here, $\vec{v} = K(y\hat{i} + x\hat{j})$

$$\vec{v} = Ky\hat{i} + Kx\hat{j} \quad \dots\dots(i)$$

$$\text{Also, } \vec{v} = \frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j} \quad \dots\dots(ii)$$

Equating equation (i) and (ii), we get :

$$\frac{dx}{dt} = Ky ; \frac{dy}{dt} = Kx \quad ; \quad \text{Now, } \frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx} \quad \therefore \quad \frac{dy}{dx} = \frac{Kx}{Ky} = \frac{x}{y}$$

Integrating both sides of the above equation, we get :

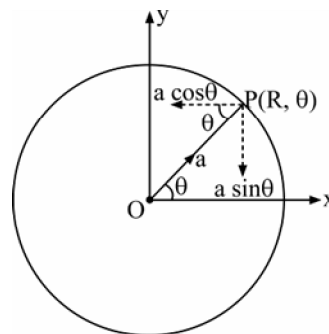
$$\int y dy = \int x dx \Rightarrow \frac{y^2}{2} = \frac{x^2}{2} + \text{constant} \quad \text{or} \quad y^2 = x^2 + \text{constant}$$

- 55.(C) For a particle in uniform circular motion, acceleration

$$a = \frac{v^2}{R} \text{ is towards the centre.}$$

From figure, we have

$$\vec{a} = -a \cos \theta \hat{i} - a \sin \theta \hat{j} \Rightarrow \vec{a} = -\frac{v^2}{R} \cos \theta \hat{i} - \frac{v^2}{R} \sin \theta \hat{j}$$



- 56.(D) Distance time relation of the particle, $s = t^3 + 5$

$$\text{speed of the particle } v = \frac{ds}{dt} = \frac{d}{dt}(t^3 + 5) = 3t^2$$

$$\text{Tangential acceleration, } a_t = \frac{dv}{dt} = \frac{d}{dt}(3t^2) = 6t$$

$$\text{At time } t = 2s \quad ; \quad \text{Speed of the particle, } v = 3(2)^2 = 12 \text{ m/s}$$

$$\text{Tangential acceleration, } a_t = 6(2) = 12 \text{ m/s}^2$$

$$\text{Centripetal acceleration, } a_c = \frac{v^2}{R} = \frac{(12)^2}{20} = \frac{144}{20} = 7.2 \text{ m/s}^2$$

$$\text{Net acceleration, } a = \sqrt{(a_c)^2 + (a_t)^2} = \sqrt{(7.2)^2 + (12)^2} = 14 \text{ m/s}^2$$

- 57.(B) Total area around the fountain that gets wet is equal to area of a circle whose radius is equal to maximum range of sprinkle water.

$$R_{\max} = \frac{v^2 \sin 90^\circ}{g} = \frac{v^2}{g} \quad ; \quad \text{Area} = \pi (R_{\max})^2 = \frac{\pi v^4}{g^2}$$

- 58.(D) Let u be the velocity of projection of the stone.

The maximum height a boy can thrown a stone.

$$H_{\max} = \frac{u^2}{2g} = 10 \text{ m} \quad \dots (i)$$

The maximum horizontal distance the boy can thrown the same stone is equal to maximum range

$$R_{\max} = \frac{u^2 \sin(2 \times 45^\circ)}{g} = \frac{u^2 \sin 90^\circ}{g} = \frac{u^2}{g} \quad ; \quad R_{\max} = \frac{u^2}{g} = 20 \text{ m}$$

- 59.(C) Centripetal acceleration $a_c = \omega^2 r$

$$\text{Angular velocity } \omega = \frac{2\pi}{T}$$

The cars makes complete circle in same time.

$$\text{As } T_1 = T_2 \Rightarrow \omega_1 = \omega_2 \quad \therefore \quad \frac{a_{c1}}{a_{c2}} = \frac{r_1}{r_2}$$

- 60.(A) Given initial velocity of the particle, $u = \hat{i} + 2\hat{j}$

$$\therefore \quad u_x = 1 \text{ m/s and } u_y = 2 \text{ m/s}$$

$$\text{Also } x = u_x t \therefore x = t \quad \dots (i)$$

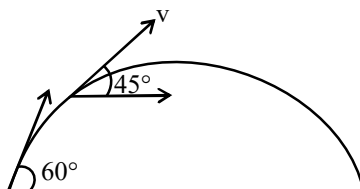
$$\text{and } y = u_y t - \frac{1}{2} g t^2 \text{ and } y = 2t - \frac{1}{2} \times 10 \times t^2 = 2t - 5t^2 \quad \dots (ii)$$

From (i) and (ii)

Equation of trajectory is $y = 2x - 5x^2$.

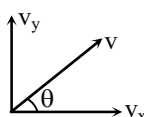
INTEGER VALUE TYPE

61.(73)



As horizontal component of velocity remains unchanged

So, $20 \cos 60^\circ = 10$



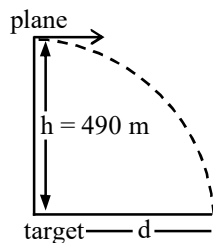
$$\tan \theta = \frac{v_y}{v_x} = \frac{u \sin 60^\circ - gt}{\theta \cos 60^\circ} = \frac{20 \times \frac{\sqrt{3}}{2} - 10 \times t}{20 - \frac{1}{2}}$$

$$\tan 45^\circ = \frac{\frac{\sqrt{3}}{2} - \frac{t}{2}}{\frac{1}{2}}$$

$$1 = \frac{\sqrt{3} - t}{1}$$

$$t = (\sqrt{3} - 1) = 0.73 \text{ sec.}$$

62.(2)



Horizontal velocity of plane = $720 \text{ km/hr} = 200 \text{ m/s}$.

Time taken by bomb to reach the surface

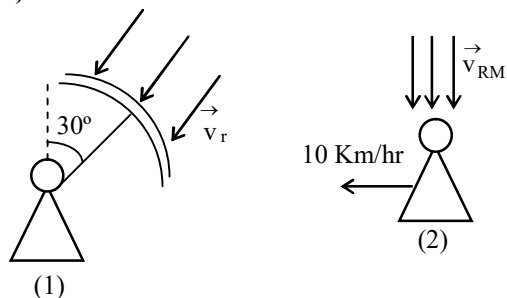
$$y = \frac{1}{2} g t^2$$

$$t^2 = \frac{2y}{g} = \frac{2 \times 490}{9.8}$$

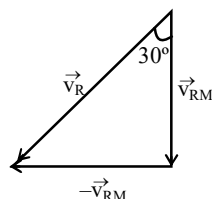
$$t^2 = 100 \text{ sec; } t = 10 \text{ sec}$$

In this time interval horizontal distance travelled by bomb = $v_x \times t = 200 \times 10 = 2000 \text{ m} = 2 \text{ km}$.

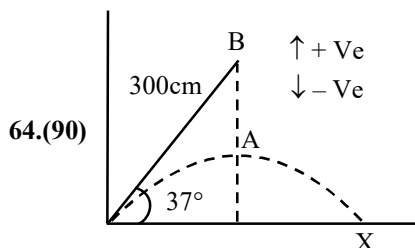
63.(20)



$\vec{v}_{RM}$ (velocity of rain with respect to man) is vertical downward $\vec{v}_{RM} = \vec{v}_R - \vec{v}_M$



$$\begin{aligned} |\vec{v}_R| \sin 30^\circ &= |\vec{v}_M| \\ \Rightarrow v_R &= \frac{10 \text{ Km/hr}}{\sin 30^\circ} = \frac{10 \text{ Km/hr}}{1/2} = 20 \text{ Km/hr} \end{aligned}$$



From $\triangle BOC$,

$$\sin 37^\circ = \frac{BC}{OB}$$

$$BC = 300 \sin 37^\circ = 300 \times \frac{3}{5} = 180 \text{ cm}$$

Let ball A and ball B collide after a time t

$\therefore$ for ball A, along vertical

$$s_y = u_y t + \frac{1}{2} a_y t^2 = 700 \sin 37^\circ t + \frac{1}{2} (-g) t^2$$

$$AC = 420 t - \frac{gt^2}{2}$$

For ball B, height fallen along vertical

$$s = ut + \frac{1}{2} a t^2 \Rightarrow -BA = (0) t + \frac{1}{2} (-g) t^2$$

$$BA = \frac{gt^2}{2} ; \quad BA + AC = BC = 180$$

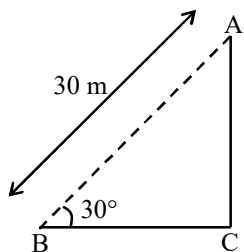
$$\frac{gt^2}{2} + 420t - \frac{gt^2}{2} = 180$$

$$420 t = 180$$

$$t = \frac{180}{420} = \frac{3}{7} \text{ sec}$$

$$\therefore BA = \frac{gt^2}{2} = \frac{980}{2} \times \frac{9}{49} = 90 \text{ cm}$$

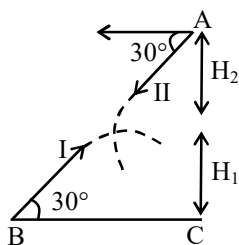
65.(46)



In $\triangle ABC$,

$$\sin 30^\circ = \frac{AC}{AB}$$

$$AC = AB \sin 30^\circ = 30 \times \frac{1}{2} = 15 \text{ m}$$



Height risen by Ist shell

$$= H_1 = 350 \sin 30^\circ t - \frac{1}{2} gt^2$$

$$= 175 t - \frac{gt^2}{2}$$

Height fallen by IInd shell

$$= H_2 = 300 \sin 30^\circ t + \frac{1}{2} gt^2$$

$$= 150 t + \frac{gt^2}{2}$$

$$H_1 + H_2 = AC = 15$$

$$\therefore 1750 t + 150 t = 15 \Rightarrow t = \frac{15}{325} = \frac{3}{65} \text{ sec} = 46.15 \text{ millisecond so answer is 46}$$

66.(80) Angle described by the point = $\frac{\pi}{10}$

Now, $v = at$

$$\text{Tangential acceleration} = a_t = \frac{dv}{dt} = a = 0.5$$

$$\therefore \text{Angular acceleration} = \alpha = \frac{a_t}{R} = \frac{0.5}{R}$$

Now, initially at $t = 0$, $v = a(0) = 0$

$$\therefore R\omega = 0 \Rightarrow \omega = 0$$

Using $\omega^2 = \omega_0^2 + 2\alpha\theta$

$$\omega^2 = 0^2 + 2\left(\frac{0.5}{R}\right)\left(\frac{\pi}{10}\right)$$

$$\omega^2 = \frac{\pi}{10R} \quad \dots(1)$$

Also, using $\omega = \omega_0 + \alpha t$

$$\omega = 0 + \frac{0.5}{R}t \Rightarrow \omega = \frac{0.5t}{R}$$

$$\therefore \left(\frac{0.5t}{R}\right)^2 = \frac{\pi}{10R} ; t = \sqrt{\frac{2\pi R}{5}}$$

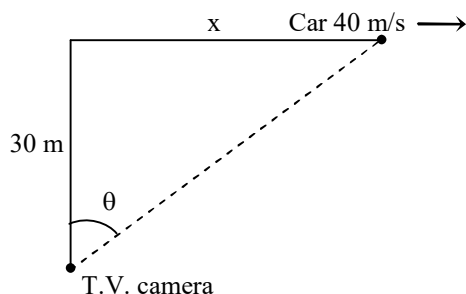
$$\therefore v = a\sqrt{\frac{2\pi R}{5}} = 0.5\sqrt{\frac{2\pi R}{5}} = \sqrt{\frac{\pi R}{5}}$$

$\therefore$ At this time,

$$\text{normal acceleration} = \frac{v^2}{R} = \frac{\pi R}{5R} = \frac{\pi}{5}$$

$$\begin{aligned} \text{Net acceleration} &= \sqrt{a_N^2 + a_t^2} = \sqrt{\frac{\pi^2}{25} + (0.25)} \\ &= 0.8 \text{ m/s}^2 = 80 \text{ cm/s}^2 \end{aligned}$$

67.(1)



We have

$$\tan \theta = \frac{x}{30} ; \frac{d}{dt} [\tan \theta] = \frac{d}{dt} \left[\frac{x}{30} \right]$$

$$\frac{d}{d\theta} (\tan \theta) \cdot \frac{d\theta}{dt} = \frac{1}{30} \frac{dx}{dt}$$

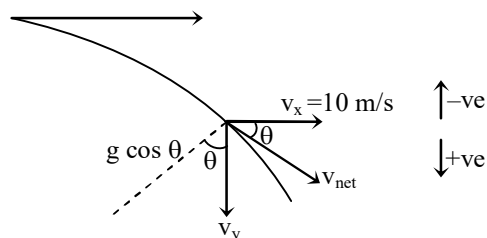
$$\sec^2 \theta \frac{d\theta}{dt} = \frac{1}{30} \times 40 \Rightarrow \frac{d\theta}{dt} = \frac{4}{3 \sec^2 \theta}$$

At the given instant, $\theta = 30^\circ$

$$\therefore \frac{d\theta}{dt} = \frac{4}{3} \cos^2 30^\circ = \frac{4}{3} \times \left(\frac{\sqrt{3}}{2}\right)^2 = \frac{4}{3} \times \frac{3}{4} = 1$$

or angular speed with which camera should be rotated = 1 rad/sec

68.(316) After 3 second, its vertical velocity will be



$$v_y = 0 + gt = 10 \times 3 = 30 \text{ m/sec}$$

$$v_x = 10 \text{ m/sec}$$

$$\Rightarrow \square\square \text{ net velocity} = v_{\text{net}} = \sqrt{v_x^2 + v_y^2}$$

$$= \sqrt{(10)^2 + (30)^2} = 10\sqrt{10} \text{ m/sec}$$

The acceleration, normal to tangential velocity

$$= g \cos \theta = g \times \frac{v_x}{v_{\text{net}}} = \frac{10 \times 10}{10\sqrt{10}} = \sqrt{10}$$

$$\text{So, radius of curvature} = \frac{v_{\text{net}}^2}{g \cos \theta} = \frac{1000}{\sqrt{10}}$$

$$= 100\sqrt{10} \text{ m}$$

- 69.(4) When we are giving an angular acceleration to the rod, the bead is also having an instantaneous acceleration $a = L \alpha$. This will happen when a force is exerted on bead by the rod. The bead has a tendency to drift more away from the centre. But due to friction between the bead and the rod this does not happen to the extent to which frictional force is capable of holding the bead. The normal reaction provides tangential force $F_t = ma = N$. The frictional force provides the necessary centripetal force. If instantaneous angular velocity is ω then

$$mL\omega^2 = \mu ma$$

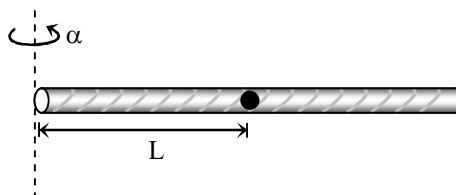
$$mL\omega^2 = \mu mLma$$

$$\Rightarrow \omega^2 = \mu\alpha$$

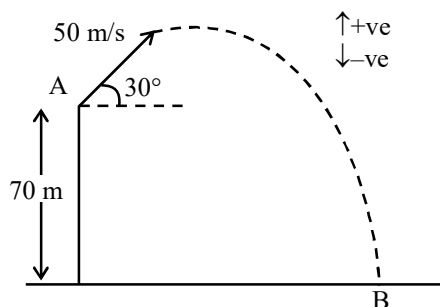
By applying, $\omega = \omega_0 + \alpha t$

$$\text{we get } \omega = \alpha t \Rightarrow \omega^2 = \alpha^2 t^2$$

$$\mu\alpha = \alpha^2 t^2 \Rightarrow t = \sqrt{\frac{\mu}{\alpha}} = 4 \text{ sec}$$



70.(7)



We have, for motion from A to B,

$$s_y = -70, a_y = -g = -10, u_y = 50 \sin 30^\circ = 25$$

$$\therefore \text{ using } s_y = u_y t + \frac{1}{2} a_y t^2$$

$$-70 = 25t + \frac{1}{2} (-10) t^2$$

$$5t^2 - 25t - 70 = 0 \Rightarrow t^2 - 5t - 14 = 0$$

$$t^2 - 7t + 2t - 14 = 0 \Rightarrow t(t-7) + 2(t-7) = 0$$

$$(t+2)(t-7) = 0$$

$$t = 7 \text{ sec}$$

71.(2) For the ball's motion along vertical direction

$$s_y = u_y t + \frac{1}{2} a_y t^2$$

$$\text{Now } \tan \alpha = \frac{5}{12}, \sin \alpha = \frac{5}{13}$$

For $s_y = 15 \text{ m}$,

$$15 = 52 \sin \alpha t + \frac{1}{2} (-g) t^2$$

$$15 = 52 \times \frac{5}{13} t - 5t^2$$

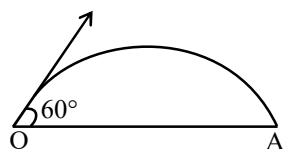
$$5t^2 - 20t + 15 = 0 \Rightarrow t^2 - 4t + 3 = 0$$

$$t = 3, t = 1 \text{ sec}$$

above are the two values of t for which ball is at a height 15 m above the ground.

So the time for which ball is at least 15 m above the ground is $= 3 - 1 = 2 \text{ sec}$

72.(50) We have, in 1st case, $R = \frac{u^2 \sin 2\theta}{g}$



$$= \frac{(40)^2 \sin 2(60^\circ)}{10} = 80\sqrt{3} \text{ m}$$

When the shell is fired with speed v then

$$R = v \cos 60^\circ \times t \Rightarrow 80\sqrt{3} = \frac{v}{2} t$$

$$\Rightarrow vt = 160\sqrt{3} \dots (1)$$

In time t , target rises by $= 9\sqrt{3} t$

$$\text{Shell rises by} = v \sin 60^\circ t - \frac{gt^2}{2} = \frac{vt\sqrt{3}}{2} - 5t^2$$

$$\therefore 9\sqrt{3} t = vt \frac{\sqrt{3}}{2} - 5t^2$$

$$9\sqrt{3} t = \frac{160\sqrt{3} \cdot \sqrt{3}}{2} - 5t^2$$

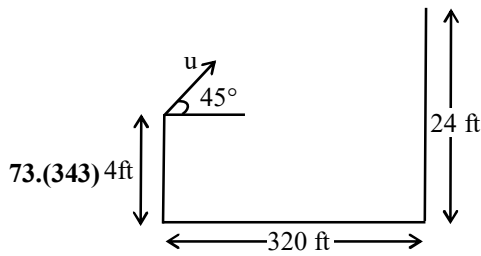
$$5t^2 + 9\sqrt{3} t - 240 = 0$$

$$t = \frac{-9\sqrt{3} + \sqrt{243 + 4800}}{10} = \frac{41\sqrt{3} - 9\sqrt{3}}{10}$$

$$= \frac{32\sqrt{3}}{10}$$

$$vt = 160\sqrt{3} \Rightarrow \frac{v \times 32\sqrt{3}}{10} = 160\sqrt{3}$$

$$v = \frac{1600}{32} = 50 \text{ m/s}$$



For motion of ball along horizontal

$$R = \frac{u^2 \sin 2(45^\circ)}{g} = 350$$

$$u^2 = 350 \times 32$$

If y is vertical displacement,

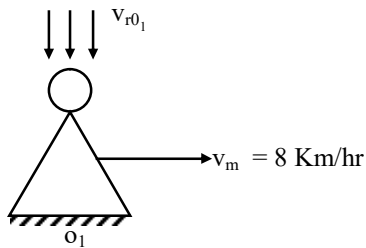
$$y = x \tan \theta - \frac{gx^2}{2u^2 \cos^2 \theta}$$

$$= 320 \tan 45^\circ - \frac{\frac{1}{2} \times 32 \times (320)^2}{350 \times 32 \times \frac{1}{2}}$$

$$y = 320 - 292.571 = 27.43 \text{ ft}$$

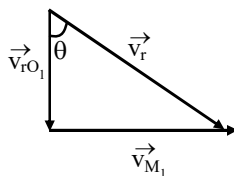
$$h = 27.43 - 24 = 3.43 \text{ ft}$$

74.(112)

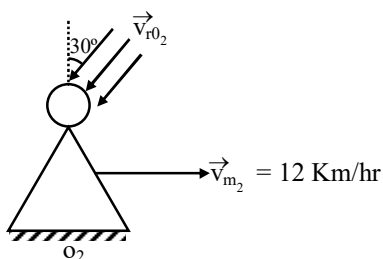


$$\vec{v}_{rO_1} = \vec{v}_r - \vec{v}_{m_1}$$

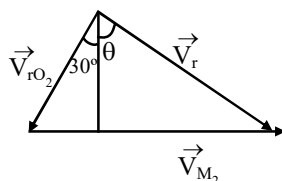
from above



$$v_r \sin \theta = v_{m_1} = 8 \text{ Km/hr}$$



$$Q_1 \quad \vec{v}_{rO_2} = \vec{v}_r - \vec{v}_{m_2}$$



$$v_{rO_2} \cos 30^\circ = v_r \cos \theta$$

$$v_{rO_2} \sin 30^\circ + v_r \sin \theta = v_{m_2}$$

$$\Rightarrow v_{rO_2} (1/2) + v_{m_1} = v_{m_2}$$

$$\text{or } v_{rO_2} = 2(12 - 8) \text{ Km/hr} = 8 \text{ Km/hr}$$

$$\Rightarrow \tan \theta = \frac{v_{m_1}}{v_{rO_2}} = \frac{8}{8} = 1 \Rightarrow \theta = 45^\circ$$

$$\Rightarrow v_r = 8\sqrt{2} \text{ Km/hr Ans.}$$

75.(388) Given $v_p = 100 \text{ km/hr}$; $v_w = 60 \text{ km/hr}$ from vector addition

$$v_r = \sqrt{v_p^2 + v_w^2}$$

$$v_r = \sqrt{100^2 + 60^2} = 10\sqrt{156}$$

$$= 116.56 \text{ km/hr}$$

resultant direction of motion will be

$$\tan \theta = \frac{v_y}{v_x} = \frac{v_w}{v_p} = \frac{60}{100} = 3/5$$

$$\theta = \tan^{-1}(3/5) \text{ west of north distance of plane after 20 min, from starting point } d = v.t.$$

$$= 116.58 = \frac{20}{60} = \frac{116.58}{3} = 38.86 \text{ km}$$

Dynamics of a Particle

- 1.(A) A block of mass m is placed on a surface with a vertical cross-section, then $\tan = \frac{dy}{dx} = \frac{d\left(\frac{x^3}{6}\right)}{dx} = \frac{x^2}{2}$

At limiting equilibrium, we get : $\mu = \tan \theta$, $0.5 = x^2 / 2$

$$\Rightarrow x^2 = 1 \quad \Rightarrow x \pm 1$$

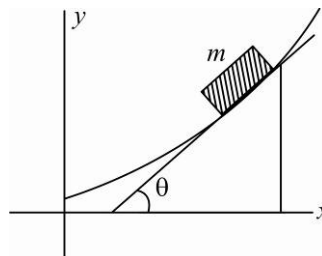
Now, putting the value of x in $y = x^3/6$, we get :

When $x = 1$

$$\therefore y = \frac{(1)^3}{6} = \frac{1}{6}$$

When $x = -1$

$$y = \frac{(-1)^3}{6} = -\frac{1}{6}$$



So, the maximum height above the ground at which the block can be placed without slipping is $\frac{1}{6}m$.

- 2.(CD) Impulse = $2mv = 0.5 \text{ Ns}$

- 3.(ABC) $T - F_2 = 0 \quad \dots \text{(i)} \quad P - T - F_1 = 0 \quad \dots \text{(ii)}$

For limiting value

$$T - F_0 = 0 \quad \dots \text{(iii)} \quad P - T - F_0 = 0 \quad \dots \text{(iv)}$$

Tension will develop when A is just about to move.

Then only friction between B and surface will develop.

The system will move when (limiting for both)

$$(T - F_0) + (P - T - F_0) = 0 \quad \Rightarrow \quad P - 2F_0 = 0 \Rightarrow P = 2F_0$$

So for $F_0 < P < 2F_0 \Rightarrow T = P - F_0$

For $P > 2F_0$, both will start moving.

$$T - F_0 = ma \quad \dots \text{(v)}$$

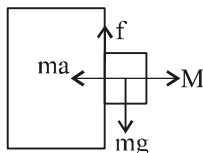
$$P - T - F_0 = ma \quad \dots \text{(vi)}$$

$$\Rightarrow a = \frac{P - 2F_0}{2m} \quad \Rightarrow \quad T = \frac{P}{2}$$

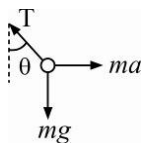
- 4.(A) w.r.t. block

$$N = ma$$

$$f_{\max} = \mu N = mg \Rightarrow \mu = \frac{g}{a}$$

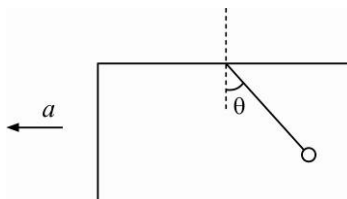


- 5.(AC) F.B.D.



$$T \cos \theta = mg \quad \dots \text{(i)}$$

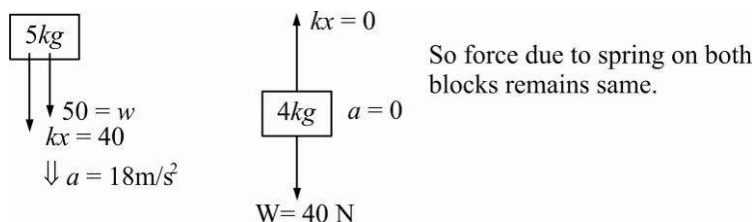
$$T \sin \theta = ma \quad \dots \text{(ii)}$$



$$\Rightarrow \tan \theta = \frac{a}{g}$$

$$\sqrt{(T \cos \theta)^2 + (T \sin \theta)^2} = m\sqrt{g^2 + a^2}$$

- 6.(C) When string is cut tension in string becomes zero. But spring doesn't change its shape just after cutting the string.



- 7.(ABD) F.B.D. of block

$$(T + N) - W = \left(\frac{W}{g}\right)a$$

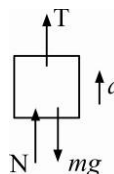
$$(W_1 + W_2) - W = \left(\frac{W}{g}\right)a$$

$$W_1 + W_2 = W + \frac{W}{g}(a)$$

if a is upwards $\rightarrow W_1 + W_2 > W$

if $a = 0 \rightarrow W_1 + W_2 = W$

if a is downwards $\rightarrow W_1 + W_2 < W$



- 8.(ABD) $v = \frac{dx}{dt} = 4\pi A \cos 4\pi t$; $a = \frac{dv}{dt} = -16\pi^2 A \sin 4\pi t$

$$F = ma \text{ and Impulse } I = m\Delta v$$

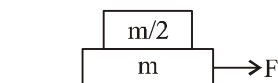
- 9.(ABDE) $f_{1\max}(B \text{ and ground}) = \frac{3mg}{2} \times 0.1 = 0.15g$

$$f_{2\max}(A \text{ and } B) = \frac{mg}{2} \times 0.2 = 0.1mg$$

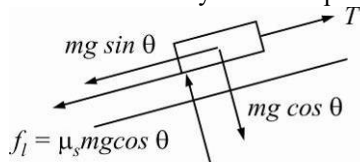
$f_{\max}$ (so that two blocks move together)

$$F_{\max} - 0.15g = \frac{3m}{2} \times 2 \Rightarrow F_{\max} = 0.45g$$

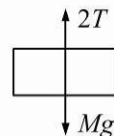
Two bodies will be at rest if $F < 0.15g$



- 10.(C) For maximum value of M limiting friction will act down the plane as m will have tendency to move up the plane.



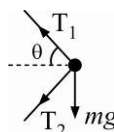
$$T = \mu_s mg \cos \theta + mg \sin \theta$$



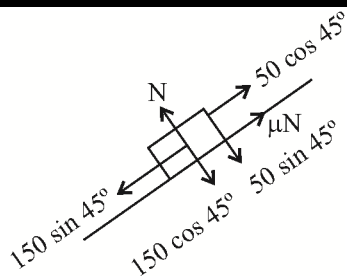
$$M = 2m(\mu_s \cos \theta + \sin \theta)$$

- 11.(A) $F_1 = F_2 = F$ for equilibrium

- 12.(B) $T_1 \cos \theta + T_2 \cos \theta = m\omega^2 r$... (i)
 $T_1 \sin \theta - T_2 \sin \theta - mg = 0$... (ii)



13.(C)



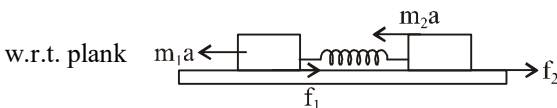
$$N = 150 \cos 45^\circ + 50 \sin 45^\circ$$

$$150 \sin 45^\circ = 50 \cos 45^\circ + \mu N$$

14.(D)

$$a_B = \frac{12t-3}{2} = \frac{dv}{dt} \Rightarrow \int_0^0 dv = \int_0^t \left(\frac{12t-3}{2} \right) dt \Rightarrow t = \frac{1}{2}$$

15.(D)



If slipping starts both the block will move with same acceleration so spring will remain in its natural length.

16.(ABCD) So as F increase $\Rightarrow f$ increase $\Rightarrow R$ increase

$$\text{When friction is limiting } R = mg\sqrt{\mu^2 + 1} \Rightarrow R \leq mg\sqrt{\mu^2 + 1} \text{ and } \tan \theta \leq \frac{\mu mg}{mg}$$

17.(B)

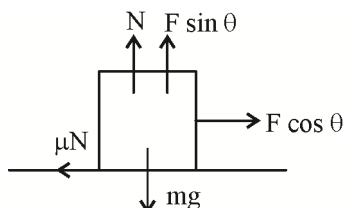
$$f_{\max} = T_{\max}$$

$$4\mu mg = mg + \frac{mv_{\max}^2}{\ell} ; v_{\max} = A\sqrt{\frac{g}{\ell}} \Rightarrow \mu = \frac{1}{4} \left(1 + \frac{A^2}{\ell^2} \right)$$

18.(AC) Just after thread is burnt : $a_{m_1} = 0$; $a_{m_2} = \frac{m_1 g - m_2 g}{m_2} = \left(\frac{m_1}{m_2} - 1 \right) g$

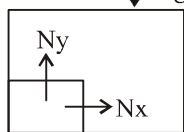
As tension in BC is zero and tension in after string remaining ' mg '.

19.(B)



$$\begin{aligned} F \cos \theta &= \mu N \\ N + F \sin \theta &= mg \\ \mu &= \tan \theta \end{aligned}$$

20.(B)



$$N_y = m a \sin 53^\circ = 40 \text{ N}$$

21.(D)

22.(C)

$$a = \frac{mg \sin \theta}{2m} = \frac{g \sin \theta}{2} \quad \leftarrow \text{m} \quad T = ma = \frac{mg \sin \theta}{2}$$

23.(AB) If they slide together $N \neq 0$

$$mg \sin \theta + N - f_1 = ma \quad \dots \text{(i)}$$

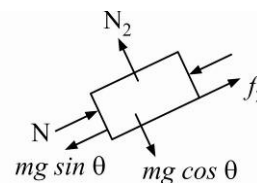
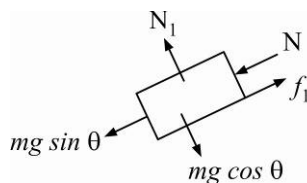
$$mg \sin \theta - N - f_2 = ma \quad \dots \text{(ii)}$$

$$\text{(i) If } \mu_1 > \mu_2 \Rightarrow f_1 > f_2$$

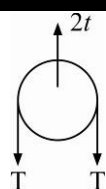
So always in contact (Block 1 slower)

$$\text{(ii) If } \mu_1 < \mu_2 \Rightarrow f_1 < f_2$$

So different acceleration (Block 1 faster)

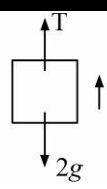


24.(B)



$$2T = 2t \Rightarrow T = t$$

$$T - g = 1a_1$$

$$t - g = a_1$$


$$T - 2g = 2a_2$$

$$t - 2g = 2a_2$$

Since $a \propto t \Rightarrow V \propto t^2$ so parabolic. A will leave the ground first.

25.(D) $\mu mg \cos \theta = mg \sin \theta = 2(mg \sin \theta - \mu mg \cos \theta)$
 $\tan \theta = 3\mu$

26.(A) $x + y + \sqrt{d^2 + x^2} = \text{constant}$

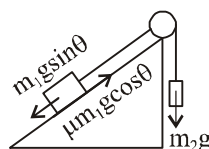
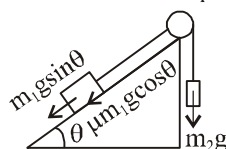
$$\frac{dx}{dt} + \frac{dy}{dt} + \frac{2x}{2\sqrt{d^2 + x^2}} \frac{dx}{dt} = 0$$

$$-v + u - v \cos \theta = 0 \Rightarrow \frac{u}{v} = 1 + \cos \theta$$

move up

move down

27.(BD)



28.(BC) A will never move up if $mg \sin \theta_2 \leq mg \sin \theta_1 + \mu mg \cos \theta_1$

29.(C) $T_{\max} = \frac{40}{\sin 30^\circ} = 80N$

$$T_{\max} - mg = ma_{\max} \Rightarrow a_{\max} = 6m/s^2$$

30.(C) Let v be tangential speed of heavier stone. Then, centripetal force experienced by lighter stone is

$$(F_C)_{\text{lighter}} = \frac{m(nv)^2}{r} \text{ and that of heavier stone is } (F_C)_{\text{heavier}} = \frac{2mv^2}{(r/2)}$$

But $(F_C)_{\text{lighter}} = (F_C)_{\text{heavier}}$ (given)

$$\therefore \frac{m(nv)^2}{r} = \frac{2mv^2}{(r/2)}$$

$$n^2 \left(\frac{mv^2}{r} \right) = 4 \left(\frac{mv^2}{r} \right)$$

$$n^2 = 4 \text{ or } n = 2$$

31.(D) Option not matching with answer

$$F = 14$$

Here, $M_A = 4kg$, $M_B = 2kg$, $M_C = 1kg$, $F = 14N$

Net mass : $M = M_A + M_B + M_C = 4 + 2 + 1 = 7kg$

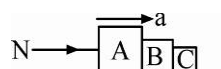
Let a be the acceleration of the system.

Using Newton's second law of motion,

$$F = Ma$$

$$14 = 7a \therefore a = 2ms^{-2}$$

Let F' be the force applied on block A by block B i.e. the contact force between A and B. Free body diagram for block A



Again using Newton's second law of motion, $F - F' = 4a$

$$14 = F' = 4 \times 2 \Rightarrow 14 - 8 = F' \therefore F' = 6N$$

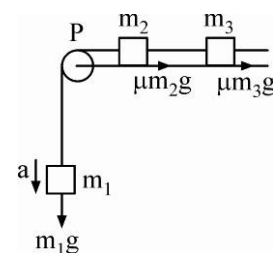
32.(A)

33.(C) Force of friction of mass $m_1 = \mu m_2 g$; Force of friction of mass $m_3 = \mu m_2 g$

Let a be common acceleration of the system :

$$\therefore a = \frac{m_1 g - \mu m_2 g - \mu m_2 g}{m_1 + m_2 + m_3} \quad \text{Here, } m_1 = m_2 = m_3 = 3 = \frac{g(1-2\mu)}{3}$$

Hence, the downward acceleration of mass m_1 is $\frac{g(1-2\mu)}{3}$

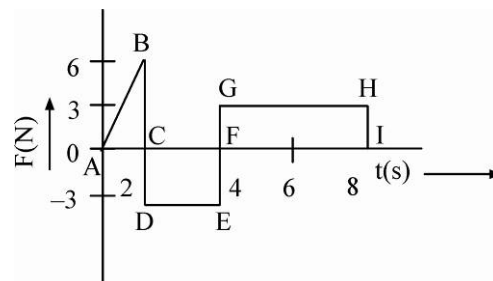


34.(C) Change in momentum = Area under $F-t$ graph in that interval

= Area of $\triangle ABC$ - Area of rectangle $CDEF$

+ Area of rectangle $FGHI$

$$= \frac{1}{2} \times 2 \times 6 - 3 \times 2 + 4 \times 3 = 12Ns$$



35.(A) Let F be the upthrust of the air. As the balloon is descending down with an acceleration a .

$$\therefore mg - F = ma \quad \dots (i)$$

Let mass m_0 be removed from the balloon so that it starts moving up with an acceleration a . Then,

$$F - (m - m_0)g = (m - m_0)a$$

$$F - mg + m_0g = ma - m_0a \quad \dots (ii)$$

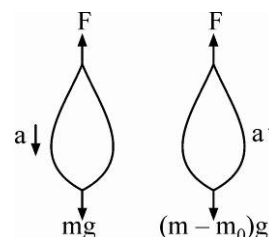
Adding eqn. (i) and eqn. (ii), we get

$$m_0g = 2ma - m_0a$$

$$m_0g + m_0a - 2ma$$

$$m_0(g + a) = 2ma$$

$$m_0 = \frac{2ma}{a + g}$$



36.(C) Let T_1 be tension in string connecting m and $2m$ T_2 be tension in string connecting $2m$ and $3m$.

Let a be common acceleration of the system.

$$\therefore a = \frac{F - (m + 2m + 3m)g}{m + 2m + 3m} = \frac{F - 6mg}{6m}$$

As the system moves with constant speed, therefore, $a = 0$

$$\therefore F - 6mg = 0 \text{ or } F = 6mg$$

The free body diagram of block m is as shown in the figure

The equation of motion of block of mass m is $F - T_1 - mg = 0$

$$6mg - T_1 - mg = 0$$

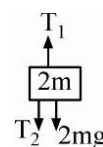
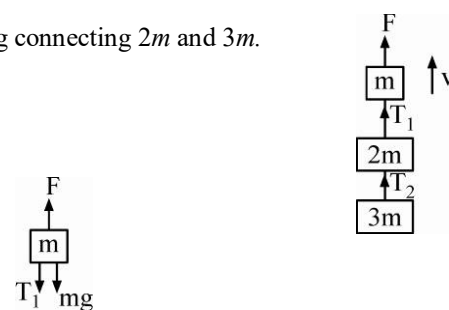
$$T_1 = 5mg$$

The free body diagram of block of mass $2m$ is as shown in the figure

The equation of motion of block of mass $2m$ is $T_1 - T_2 - 2mg = 0$

$$5mg - T_2 - 2mg = 0 \quad (\text{Using (i)})$$

$$T_2 = 3mg$$



The free body diagram of block of mass

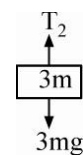
$3m$ is as shown in the figure.

The equation of motion of block of mass $3m$ is :

$$T_2 - 3mg = 0$$

$$T_2 = 3mg$$

Net force on the block of mass $2m$ is $F_{net} = T_1 - T_2 - 2mg = 5mg - 3mg - 2mg = 0$



Alternate solution

As all blocks are moving with constant speed, therefore, acceleration is zero. So net force on each block is zero.

37.(D) The situation is as shown in the figure

According to law of conservation of linear momentum

$$\vec{p}_1 + \vec{p}_2 + \vec{p}_3 = 0 \therefore \vec{p}_3 = -(\vec{p}_1 + \vec{p}_2)$$

Here $p_1 = (1\text{kg})(12\text{ms}^{-1})\hat{i} = 12\hat{i} \text{ kg ms}^{-1}$

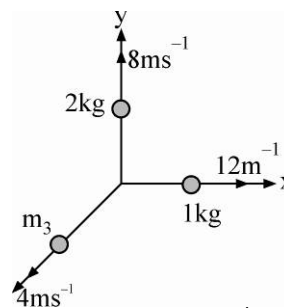
$$\vec{p}_2 = (2\text{kg})(8\text{ms}^{-1})\hat{j} = 16\hat{j} \text{ kgms}^{-1}$$

$$\therefore \vec{p}_3 = -(12\hat{i} + 16\hat{j}) \text{ kg ms}^{-1}$$

The magnitude of p_3 is :

$$p_3 = \sqrt{(12)^2 + (16)^2} = 20 \text{ kg ms}^{-1}$$

$$\therefore m_3 = \frac{p_3}{v_3} = \frac{20 \text{ kg ms}^{-1}}{4 \text{ ms}^{-1}} = 5 \text{ kg}$$



38.(A) Let m be mass of the block and L be length of the inclined plane.

According to work – energy theorem

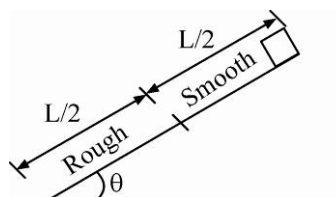
$$W = \Delta K = 0 \quad (\text{Initial and final speed are zero})$$

$$\therefore \text{Work done by friction} + \text{Work done by gravity} = 0$$

$$-\mu mg \cos \theta \frac{L}{2} + mg \sin \theta L = 0$$

$$\frac{\mu}{2} \cos \theta = \sin \theta$$

$$\mu = \frac{2 \sin \theta}{\cos \theta} = 2 \tan \theta$$



Alternate solution

For upper half smooth plane

Acceleration of the block $a = g \sin \theta$

Here, $u = 0$ (block starts from rest)

$$a = g \sin \theta, s = \frac{L}{2}$$

$$\text{Using, } v^2 - u^2 = 2as, \text{ we have } v = \sqrt{gL \sin \theta} \quad \dots (i)$$

For lower half rough plane

Acceleration of the block, $a' = g \sin \theta - \mu g \cos \theta$ where μ is the coefficient of friction between the block and lower half of the plane

$$\text{Here, } u = v = \sqrt{gL \sin \theta}$$

$$v = 0 \text{ (block comes to rest)}$$

$$a = a' = g \sin \theta - \mu g \cos \theta \quad s = \frac{L}{2}$$

Again, using $v^2 - u^2 = 2as$, we have

$$0 - (\sqrt{gL \sin \theta})^2 = 2 \times (g \sin \theta - \mu g \cos \theta) \times \frac{L}{2}$$

$$-gL \sin \theta = (g \sin \theta - \mu g \cos \theta) L$$

$$-\sin \theta = \sin \theta - \mu \cos \theta$$

$$\mu \cos \theta = 2 \sin \theta$$

$$\mu = 2 \tan \theta$$

39.(B)

40.(C) Here, Mass of person, $m = 60 \text{ kg}$

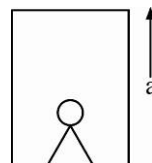
Mass of lift, $M = 940 \text{ kg}$,

$$a = 1 \text{ m/s}^2, g = 10 \text{ m/s}^2$$

Let T be the tension in the supporting cable.

$$\therefore T - (M + m)g = (M + m)a$$

$$T = (M + m)(a + g) = (940 + 60)(1 + 10), 11000 \text{ N}$$



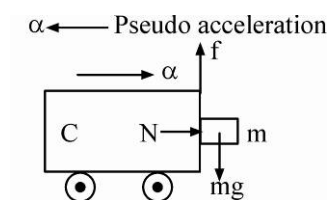
41.(C) Pseudo force or fictitious force, $F_{fle} = m\alpha$

Force of friction, $f = \mu N = \mu m\alpha$

The block of mass m will not fall as long as

$$f \geq mg$$

$$\mu m\alpha \geq mg \Rightarrow \alpha \geq \frac{g}{\mu}$$

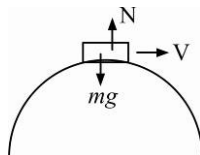


42.(C) For circular motion

$$mg - N = \frac{mV^2}{R}$$

Weight less $\Rightarrow N = 0$

$$V = \sqrt{gR} = 14.3 \text{ m/s}$$



$$43.(C) F = \frac{d}{dt}(Mv) = v \frac{dM}{dt} + M \frac{dv}{dt}$$

As v is a constant,

$$F = v \frac{dM}{dt} \quad \text{But } \frac{dM}{dt} = M \text{ kg/s}$$

$\therefore$ To keep the conveyer belt moving at $v \text{ m/s}$, force need $= mV \text{ Newton}$.

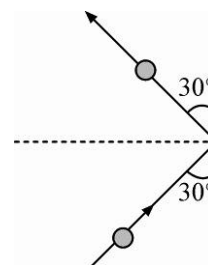
44.(C) Components of momentum parallel to the wall and each other and components of momentum perpendicular to the wall are opposite to each other. Therefore change of momentum is final momentum-initial momentum

i.e., $mv \sin \theta$ after collision

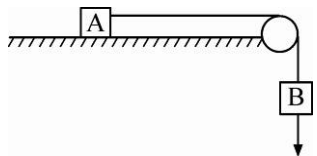
$-(-mv \sin \theta)$ Before collision

$$F \times t = \text{change in momentum} = 2mv \sin \theta$$

$$\therefore F = \frac{2mv \sin \theta}{t} = \frac{2 \times 0.5 \times 12 \times \sin 30^\circ}{0.25} = 48 \times \frac{1}{2} = 24 \text{ N}$$



45.(D)



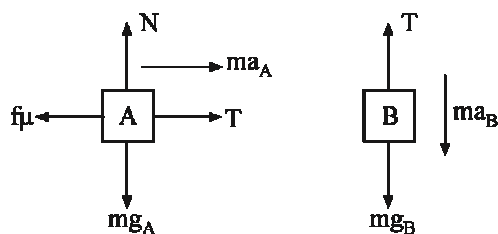
Free body diagram of two masses is

We get equations

$$T + ma = f_\mu \text{ or } T = \mu N_A \text{ (for } a = 0 \text{)}$$

and $T = ma + mg \text{ or } T = m_B g \text{ (for } a = 0 \text{)}$

$$\therefore \mu N_A = m_B g \Rightarrow m_B = \mu m_A = 0.2 \times 2 = 0.4 \text{ kg}$$



46.(D) When the lift is accelerating upwards with acceleration a , then reading on the scale.

$$R = m(g + a) = 80(10 + 5) \text{ N} = 1200 \text{ N}$$

47.(D) Let T be the tension in the rope when monkey climbs up with an acceleration a .

Then, $T - mg = ma$

$$25g - 20g = 20a \Rightarrow a = \frac{5 \times 10}{20} = 2.5 \text{ m/s}^2.$$

48.(C) Load $W = Mg = 75 \times 10 = 750 \text{ N}$

Effort (P) = 250 N $\therefore$ Mechanical advantage

$$= \frac{\text{load}}{\text{effort}} = \frac{W}{P} = \frac{750}{250} = 3$$

$$\text{Velocity ratio} = \frac{\text{distance travelled by effort}}{\text{distance travelled by load}} = \frac{12}{3} = 3$$

$$\text{Efficiency, } \eta = \frac{\text{Mechanical advantage}}{\text{Velocity ratio}} = (3/2) \times 100 = 75\%$$

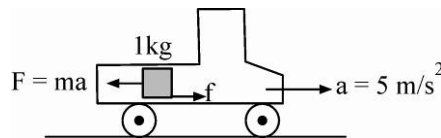
49.(A) $f_{rL} = \mu_s N = \mu_s \times mg = 0.6 \times 1 \times 10 = 6 \text{ N}$

Where f_{rL} is the force of limiting friction.

Pseudo force = $ma = 1 \times 5$; $F = 5 \text{ N}$

If $F < f_{rL}$ block does not move. So static friction is present.

Static friction = applied force $\therefore f_r = 5 \text{ N}$



50.(A) Change in momentum = $2 \times 3 \times 10 \times \sin 60^\circ$

$$= 60 \times \frac{\sqrt{3}}{2}$$

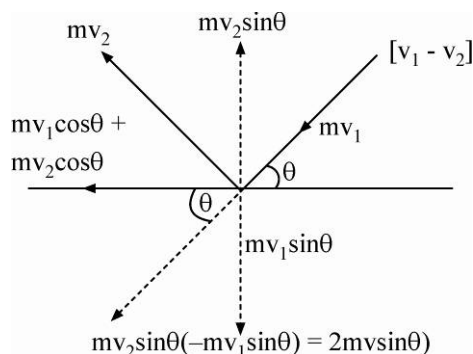
Force = Change in momentum / Impact time

$$= \frac{30\sqrt{3}}{0.2} = 150\sqrt{3} \text{ N}$$

51.(D) When $F = 0$, $600 - 2 \times 10^5 t = 0$

$$\therefore t = \frac{600}{2 \times 10^5} = 3 \times 10^{-3} \text{ s}$$

Now, impulse, $I = \int_0^t F dt = \int_0^t (600 - 2 \times 10^5 t) dt$



$$600t - 2 \times 10^5 \frac{t^2}{2} = 600 \times 10^3 - 10^5 \times (3 \times 10^{-3})^2 \quad \text{or} \quad I = 1.8 - 0.9 = 0.9 \text{ N-s}$$

- 52.(B) Rate of burning of fuel $\left(\frac{dm}{dt}\right) = 1 \text{ kg/s}$ and velocity of ejected fuel (v) = 60 km/s = $60 \times 10^3 \text{ m/s}$

$$\text{Force} = \text{Rate of change of momentum} = \frac{dp}{dt} = \frac{d(mv)}{dt} = v \frac{dm}{dt} = (60 \times 10^3) \times 1 = 60000 \text{ N}$$

- 53.(C) Thrust is the force with which the rocket move upward given by $F = u \frac{dm}{dt}$

Thus mass of the gas ejected per second to supply the thrust needed to overcome the weight of the rocket is

$$\frac{dm}{dt} = \frac{F}{u} = \frac{m \times a}{u} \quad \text{or} \quad \frac{dm}{dt} = \frac{600 \times 10}{1000} = 6 \text{ kgs}^{-1}$$

- 54.(A) Since 5 kg body explodes into three fragments with masses in the ratio 1 : 1 : 3 thus, masses of fragments will be 1 kg, 1 kg and 3 kg respectively. The magnitude of resultant momentum of two fragments each of mass 1 kg, moving with velocity 21 m/s, in perpendicular direction is $\sqrt{(m_1 v_1)^2 + (m_2 v_2)^2}$

$$m'v' = \sqrt{(21)^2 + (21)^2} = 21\sqrt{2} \text{ kg m/s}$$

According to law of conservation of linear momentum $m_3 v_3 = m'v' = 21\sqrt{2}$ or $3v_3 = 21\sqrt{2}$ or $v_3 = 7\sqrt{2} \text{ m/s}$

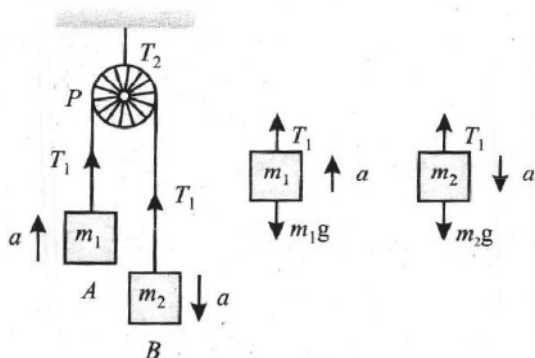
- 55.(A) For equilibrium of system, $F_1 = \sqrt{F_2^2 + F_3^2}$ as $\theta = 90^\circ$

$$\text{In the absence of force } F_1, \text{ Acceleration} = \frac{\text{Net force}}{\text{Mass}} = \frac{\sqrt{F_2^2 + F_3^2}}{m} = \frac{F_1}{m}$$

- 56.(C) As the lift is descending with an acceleration a , so the man also has an acceleration a . Hence the acceleration of the ball with respect to the man in the lift will be $(g - a)$ and with respect to the stationary man on the ground will be g .

- 57.(B) From F.B.D of m_1 : $T_1 - m_1 g = m_1 a$ (i)

$$\text{From F.B.D of } m_2 : m_2 g - T_1 = m_2 a \quad \text{..... (ii)}$$



$$\text{From (i) and (ii), } a = \left(\frac{m_2 - m_1}{m_1 + m_2}\right)g \Rightarrow \frac{g}{8} = \left(\frac{m_2 - m_1}{m_1 + m_2}\right)g \Rightarrow \frac{m_2}{m_1} = \frac{9}{7}$$

- 58.(B) $A + B = 18$ (i)

$$12 = \sqrt{A^2 + B^2 + 2AB \cos \theta} \quad \text{..... (ii)}$$

$$\tan \alpha = \frac{B \sin \theta}{A + B \cos \theta} \Rightarrow \tan 90^\circ = \frac{B \sin \theta}{A + B \cos \theta} \Rightarrow \cos \theta = \frac{-A}{B} \quad \dots \text{ (iii)}$$

Solving (i), (ii) and (iii), we get ; $A = 5\text{ N}$, $B = 13\text{ N}$

59.(B) The friction will provide required centripetal force

From F.B.D of car,

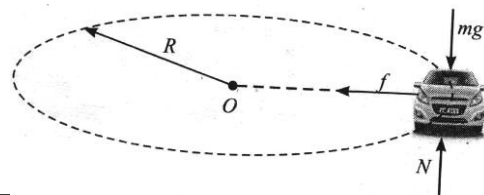
$$f = \frac{mv^2}{R} \quad \dots\dots (i)$$

If the car does not skid, the friction should be static in nature.

$$\therefore f \leq \mu N \quad \Rightarrow \quad \frac{mv^2}{R} \leq \mu mg \quad \Rightarrow \quad v \leq \sqrt{\mu Rg}$$

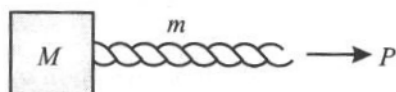
Maximum permitted velocity, $v = \sqrt{\mu Rg}$

$$\therefore v = \sqrt{0.6 \times 150 \times 10} = 30\text{ m/s}$$

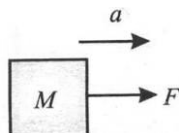


60.(B) The springs are connected in series and the spring balance are massless therefore both the scales read $M\text{ kg}$ each.

61.(A) If we take block and rope as system



$$\text{Acceleration of the system} = \frac{P}{m + M}$$



The force exerted by rope on the mass,

$$F = Ma = M \left(\frac{P}{m + M} \right)$$



62.(B) When the lift is stationary, $W = Mg$

$$\Rightarrow 49 = M(9.8) \Rightarrow M = 5\text{ kg}$$

When the lift is moving downward with an acceleration,

$$Mg - T = Ma \Rightarrow T = mg(g - a)$$

$$T = 5[9.8 - 5] = 24\text{ N}$$

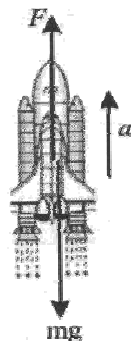
63.(C) From F.B.D of rocket,

$$F - mg = ma$$

$$\Rightarrow F = m(g + a)$$

Hence initial thrust must be

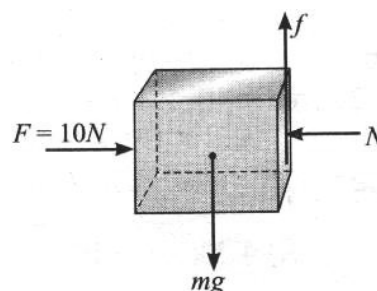
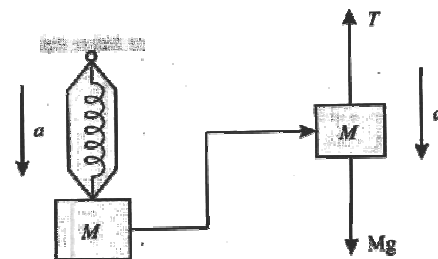
$$F = m[g + a] = 3.5 \times 10^4 (10 + 10) = 7 \times 10^5\text{ N}$$



64.(A) If block does not slide, the friction between the block and wall should be static in nature. This friction force prevents the block from sliding on the wall.

From F.B.D of the block

$$f = mg$$



If block does not slide, $f \leq \mu N$

$$\Rightarrow mg \leq \mu N \Rightarrow mg \leq 0.2 \times 10 \text{ or } mg \leq 2N$$

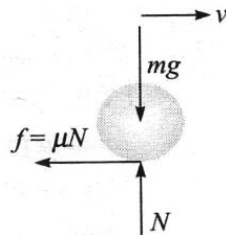
65.(D) Retardation of the marble block because of friction,

$$a = \frac{\mu N}{m} = \frac{\mu mg}{m} = \mu g$$

Now using $v = u + at$, we get;

$$0 = 6 - (\mu \times 10) \times 10$$

Which gives $\mu = 0.06$



66.(C) It is the case of non-uniform circular motion. In this case the car will have radial as well as tangential acceleration.

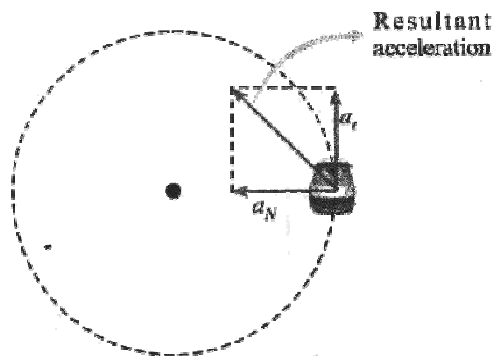
Resultant acceleration is given by

$$|a_{net}| = \sqrt{a_{radial}^2 + a_{tangential}^2}$$

$$\text{Radial acceleration, } a_c = \frac{v^2}{r} = \frac{30 \times 30}{500} = \frac{9}{5} m/s^2,$$

and tangential acceleration, $a_t = 2 m/s^2$

$$\therefore |a_c| = \sqrt{\frac{81}{25} + 4} = 2.7 m/s^2$$



67.(D) u = velocity of bullet

$$\frac{dm}{dt} = \text{Mass fired per second by the gun}$$

$$\frac{dm}{dt} = \text{Mass of bullet } (m_B) \times \text{Bullets fired per sec } (N)$$

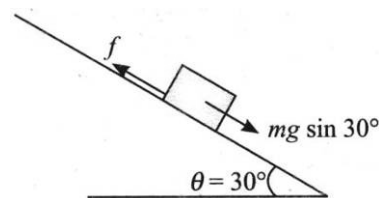
$$\text{Maximum force that man exerts, } F = u \left(\frac{dm}{dt} \right) \therefore F = u \times m_B \times N \Rightarrow N = \frac{F}{m_B \times u} = \frac{144}{40 \times 10^{-3} \times 1200} = 3$$

$$68.(C) a = \frac{m_1 - m_2}{m_1 + m_2} g = \left(\frac{5 - 4.8}{5 + 4.8} \right) 9.8 = 0.2 m/s^2$$

69.(C) The angle of repose, $\alpha = \tan^{-1}(\mu) = \tan^{-1}(0.8) = 37^\circ$

Here, the angle of inclined plane is $\theta = 30^\circ$. It means that block is at rest and therefore the friction should be static in nature. It will balance the component of weight parallel to inclined plane. Static friction = component of weight in downward direction = $mg \sin \theta = 10N$

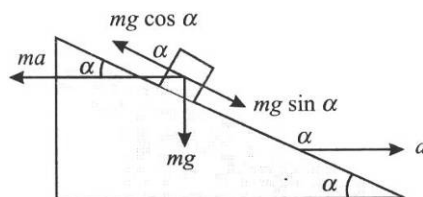
$$\therefore m = \frac{10}{g \times \sin 30^\circ} = 2 kg$$



70.(A) Let the mass of a block is m . It will remain stationary if forces acting on it are in equilibrium i.e.,

$$ma \cos \alpha = mg \sin \alpha \Rightarrow a = g \tan \alpha$$

Here ma = Pseudo force on block, mg = Weight.



71.(B) The force acting on particle, $F = -kx$

Force on particle at 20 cm away, $F = kx$

$$F = 15 \times 0.2 = 3N$$

$$[As \ k = 15 N/m]$$

$$\therefore \text{Acceleration} = \frac{\text{Force}}{\text{Mass}} = \frac{3}{0.3} = 10 m/s^2$$

72.(D) The acceleration of the block in case of smooth surface, $a_1 = \frac{mg \sin \theta}{m} = g \sin \theta$

For smooth plane, using $s = ut + \frac{1}{2}at^2$, we get

$$d = \frac{1}{2}(g \sin \theta)t^2 \quad \dots\dots (i)$$

For rough plane, the acceleration of the block,

$$a_2 = \frac{(mg \sin \theta - \mu_k mg \cos \theta)}{m} = g(\sin \theta - \mu_k \cos \theta)$$

Again using $s = ut + \frac{1}{2}at^2$, we get $d = \frac{1}{2}(g \sin \theta - \mu_k g \cos \theta)(nt)^2$

$$\Rightarrow \frac{1}{2}(g \sin \theta)t^2 = \frac{1}{2}(g \sin \theta - \mu_k g \cos \theta)n^2t^2 \Rightarrow \sin \theta = n^2(\sin \theta - \mu_k \cos \theta)$$

Putting $\theta = 45^\circ$, we get $\sin 45^\circ = n^2(\sin 45^\circ - \mu_k \cos 45^\circ)$

$$\frac{1}{\sqrt{2}} = \frac{n^2}{\sqrt{2}}(1 - \mu_k) \text{ or } \mu_k = 1 - \frac{1}{n^2}$$

73.(C) Retardation due to friction = μg

We know $v^2 = u^2 + 2as$

$$\therefore 0 = (100)^2 - 2(\mu g)s \text{ or } 2\mu gs = 100 \times 100 \text{ or } s = \frac{100 \times 100}{2 \times 0.5 \times 10} = 1000 m$$

74.(D) The acceleration of the body on upper half, $a_1 = \frac{mg \sin \theta}{m} = g \sin \theta$

Let v be the velocity of the block at the end of smooth half.

$$\text{Using } v^2 = u^2 + 2a\frac{l}{2} = 2(g \sin \theta)\frac{l}{2} = gl \sin \theta$$

For lower half, the body comes back to rest at the bottom,

it means the body retards on this section.

The retardation of the body on lower half,

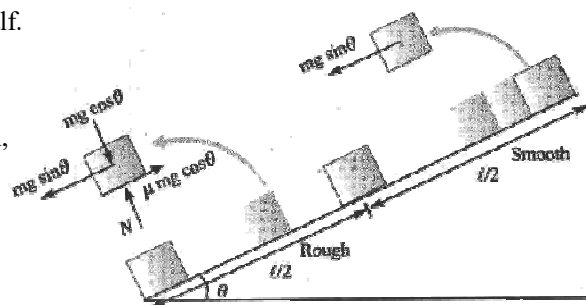
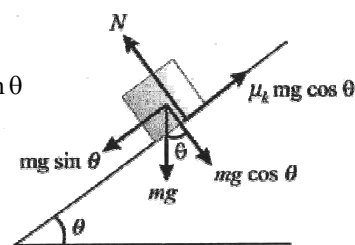
$$a_2 = \frac{(\mu mg \cos \theta - mg \sin \theta)}{m} = g(\mu \cos \theta - \sin \theta)$$

Again using $v^2 = u^2 + 2as$, we get

$$0 = u^2 - 2g(\mu \cos \theta - \sin \theta)\frac{l}{2} \Rightarrow u^2 = 2g(\mu \cos \theta - \sin \theta)\frac{l}{2}$$

But velocity of the body at the starting of this section is equal to final velocity at end of smooth section.

$$\therefore gl \sin \theta = 2g(\mu \cos \theta - \sin \theta)\frac{l}{2} \Rightarrow \mu \cos \theta = 2 \sin \theta \Rightarrow \mu = 2 \tan \theta$$



75.(A) We define average force as : $\left| \vec{F}_{\text{average}} \right| = \left| \frac{\Delta \vec{P}}{\Delta t} \right| = \frac{P_{\text{final}} - P_{\text{initial}}}{\text{duration}} = \left| \frac{(0 - 0.150 \times 20)}{0.1} \right| = 30 \text{ N}$

76.(B) From B to C, $v^2 = u_1^2 - 2 \times 10 \times 2$

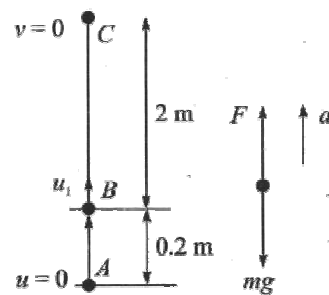
$$\Rightarrow 0 = u_1^2 - 40 \Rightarrow u_1 = \sqrt{40} \text{ m/s}$$

Let from A to B let the acceleration be a .

$$u_1^2 = u^2 + 2a \times 0.2 \Rightarrow 40 = 0^2 + 0.4a \Rightarrow a = 100 \text{ m/s}^2$$

Now $F - mg = ma$

$$\Rightarrow F = m(g + a) = 0.2[10 + 100] = 22 \text{ N}$$



77.(D) Momentum is mv .

$$m = 3.513 \text{ kg}; v = 5.00 \text{ m/s} \quad \therefore mv = 17.57 \text{ ms}^{-1}$$

Because the values will be accurate up to second decimal place only, $17.565 = 17.57$.

78.(B) From the graph. It can be seen that it is a straight line. So the motion is uniform. Because of impulse, the direction of velocity changes as can be seen from the slope of the graph.

$$\text{Initial velocity} = \frac{2}{2} = 1 \text{ m/s}; \text{ Final velocity} = -\frac{2}{2} = -1 \text{ m/s}$$

$$\vec{P}_i = 0.4 \text{ N-s}; \vec{P}_f = -0.4 \text{ N-s}$$

$$\vec{J} = \vec{P}_f - \vec{P}_i = -0.4 - 0.4 = -0.8 \text{ N-s}$$

$$(\vec{J} = 0.8 \text{ N-s})$$

79.(D) $mg \sin \theta = ma \quad \therefore a = g \sin \theta$, where a is along the inclined plane

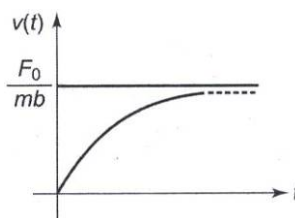
Therefore, the vertical component of acceleration is $g \sin^2 \theta$. Hence, the relative vertical acceleration of A with

respect to B is $g[\sin^2 60 - \sin^2 30] = \frac{g}{2} = 4.9 \text{ m/s}^2$ in vertical direction.

80.(C) $F = ma = F_0 e^{-bt}$

$$\Rightarrow \frac{dv}{dt} = \frac{F_0}{m} e^{-bt} \Rightarrow \int_0^v dv = \frac{F_0}{m} \int_0^t e^{-bt} dt$$

$$\Rightarrow v = \frac{F_0}{m} \left[\frac{e^{-bt}}{-b} \right]_0^t \Rightarrow v = \frac{F_0}{mb} (1 - e^{-bt})$$



81.(C) Block is under limiting friction, so

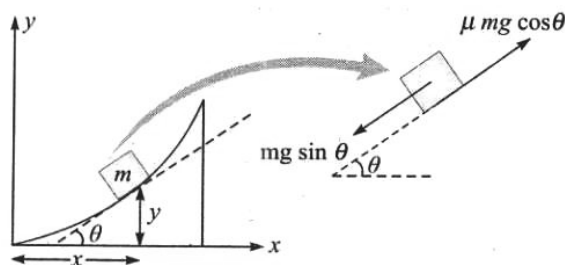
$$mg \sin \theta = \mu mg \cos \theta \Rightarrow \mu = \tan \theta$$

Equation of the surface, $y = \frac{x^3}{6}$

Slope of the curved surface,

$$\tan \theta = \frac{dy}{dx} = \frac{x^2}{2}$$

From equations (i) and (ii), we get



$$0.5 = \frac{x^2}{2} \Rightarrow 0.5 = \frac{x^2}{2} \Rightarrow x^2 = 1 \text{ or } x = 1 \text{ So, } y = \frac{x^3}{6} = \frac{(1)^2}{6} = \frac{1}{6}$$

82.(C) F.B.D of block A and block B

For block A : If block A is at equilibrium, then

$$f_1 = W_A = 20 \text{ N}$$

If friction is static, $f_1 \leq \mu_1 N_1$ or $20 \leq 0.1 \times F$

$$\Rightarrow F \geq \frac{20}{0.1} = 200 \text{ N}$$

Hence limiting value of friction between the blocks,

$$f_1 = 0.1 \times 200 = 20 \text{ N}$$

From F.B.D of block B :

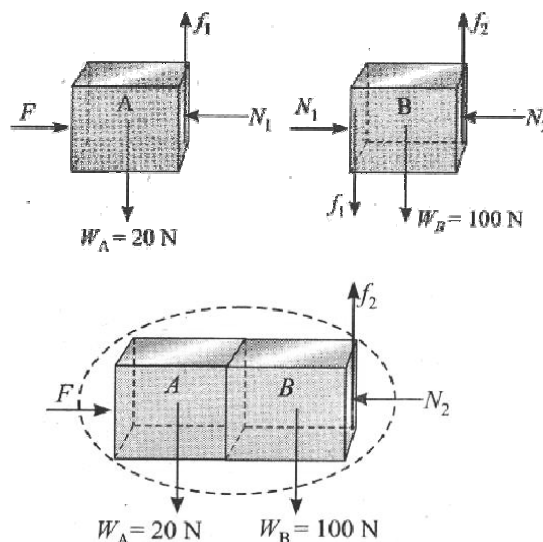
$$f_2 = f_1 + W_B = 20 + 100 = 120$$

Alternative method :

If we take both blocks together as system, then

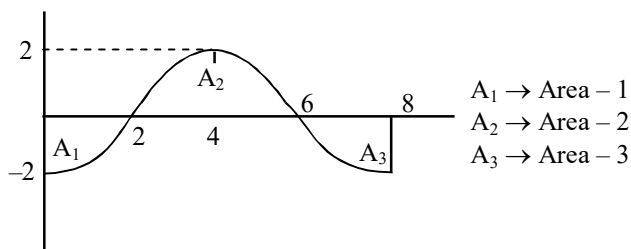
For equilibrium of the system in vertical direction,

$$f_2 = W_A + W_B = 20 + 100 = 120 \text{ N}$$



INTEGER VALUE TYPE

83.(0)

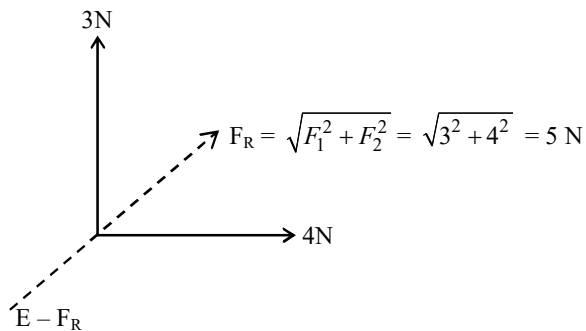


$$A_1 = A_3 : A_2 = A_1 + A_3$$

$$\text{so net Area} = A_2 - (A_1 + A_3) = 0$$

$$F \cdot \Delta T = \Delta P = 0 \text{ so change in linear momentum}$$

84.(5)



So third force should be equal & opposite of F_R so third force will be 5 N.

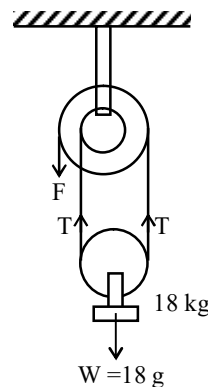
85.(90) From figure

$$2T = 18 \text{ g}; \quad 2T = 180;$$

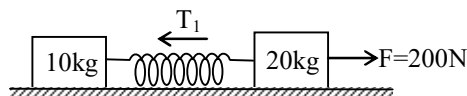
$$T = 90 \text{ Newton}$$

For equilibrium

$$F = T = 90 \text{ N}$$



86.(4)



$$200 - T_1 = 20 a_1 \quad \dots(i)$$

$$T_1 = 10 a_2 = 10 y/2 = 120 \text{ (given } a_2 = 12 \text{ m/s}^2\text{)}$$

$$200 - 120 = 20 a_1$$

$$80 = 20 a_1$$

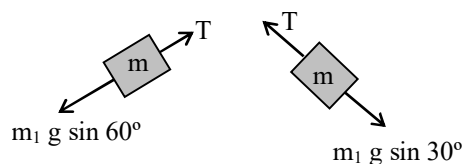
$$a_1 = 4 \text{ m/s}^2$$

87.(175) F. B. D

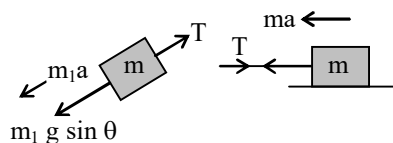
$$m g \sin 60^\circ - T = ma$$

$$T - m g \sin 30^\circ = ma$$

$$a = \frac{g}{2} (\sin 60^\circ - \sin 30^\circ) = 1.75 \text{ m/s}^2 = 175 \text{ cm/s}^2$$



88.(10) F. B. D of Blocks



$$m g \sin \theta - T = ma \quad \dots(i)$$

$$T = ma \quad \dots(ii)$$

$$m g \sin \theta - ma = ma$$

$$m g \sin \theta = 2ma \quad a = \frac{g \sin \theta}{2}$$

$$T = \frac{m g \sin \theta}{2} = 10 \text{ N}$$

89.(5)

$$a_1 = 2a_2$$

$$T_2 = 2T_1$$

$$m_2 a_2 = m_2 g - T_2$$

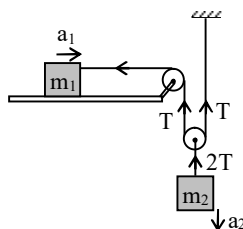
$$m_2 a_2 = m_2 g - 2T_1$$

$$m_1 a_1 = T_1$$

$$m_2 a_2 = m_2 g - 2m_1 a_1 = m_2 g - 2m_1 (2a_2)$$

$$m_2 a_2 + 4m_1 a_2 = m_2 g$$

$$a_2 = \frac{m_2 g}{m_2 + 4m_1} = 5 \text{ m/s}^2$$



90.(5) case (i) when lift accelerating upward

$$N_1 = W + F_s$$

$$N_1 = Mg + Ma \quad \dots(i)$$

case (ii) when accelerating downward

$$N_2 = W - ma = mg - ma$$

$$\text{given } \frac{N_1}{N_2} = \frac{3}{1} = \frac{W + ma}{W - ma} = \frac{mg + ma}{mg - ma}$$

$$\frac{3}{1} = \frac{g + a}{g - a}; a = g/2 = 5 \text{ m/s}^2$$

91.(50) Acceleration of system = $\frac{P}{(M + m)}$

$$\text{Force on } m = m = \frac{Pm}{(M + m)}$$

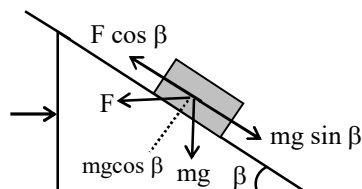
let the reaction of m on M be F . Then

$$F = \frac{Pm}{(M + m)}$$

$$F \cos \beta = mg \sin \beta$$

$$\left(\frac{Pm}{M + m} \right) \cos \beta = mg \sin \beta$$

$$P = (M + m) g \tan \beta = 50 \text{ N}$$



92.(20) $\mu = \frac{1}{4}$

$$N = F \cos 30$$

For equilibrium

$$f_w = \mu F \cos 30$$

$$= \frac{1}{4} \times 100 \times \frac{1}{2}$$

$$f_{\max} = \mu N = \frac{1}{4} F \cos 30 = \frac{1}{4} \times 100 \times \frac{\sqrt{3}}{2}$$

$$= \frac{173.2}{8}$$

$$W = ng - 30\mu$$

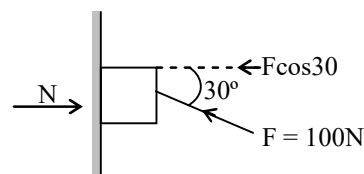
$$F_{\max} = 21.7 \text{ N but}$$

$$F = \text{upward force}$$

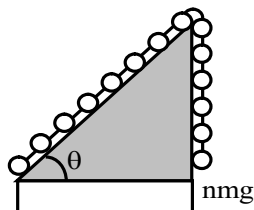
$$= F \sin 30 - mg$$

$$= 50 - 30 = 20 \text{ N}$$

$$\therefore f = 20 \text{ N downward}$$



93.(10)



Suppose out of 16 beads n are hanging vertical downward and so $(16 - n)$ are placed on inclined. Given

$$\sin \theta = \frac{1}{3} \text{ so vertically}$$

$$nmg - T = nma = \frac{nmg}{2} \dots (1)$$

Here $a = g/2$ Given

on inclined

$$T - (16 - n)mg \sin \theta = (16 - n)ma$$

$$T - \frac{(16 - n)mg}{3} = \frac{(16 - n)mg}{2} \dots (2)$$

From eqn (1) & (2) by equating value of T

$$\frac{nmg}{2} = \frac{(16 - n)S}{3}$$

$3n = 80 - 5n$ 10 beads are hanging vertically

$$\therefore n = 10$$

94.(800) Force eqn $m'g - mg = ma = \frac{mg}{6}$

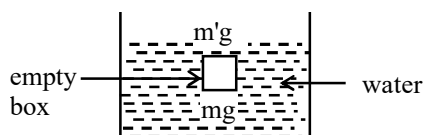
$$\therefore m'g = \frac{7mg}{6}$$

Now after placing sand of mass M

$(m + M)g - m'g = (m + M)a$ Here $a = g/6$

$$mg + Mg - \frac{7mg}{6} = \frac{mg}{6} + \frac{Mg}{6}$$

$$\frac{5Mg}{6} = \frac{mg}{3}; M = \frac{2m}{5} = 0.8 \text{ kg}$$



95.(1) $\uparrow R = 11 \text{ N}$

$\bullet m$

$\downarrow W + ma$

Reading of spring to y shows only

Reaction = 11 N [given $m = 1 \text{ kg}$]

$$w + ma = 11 \text{ N}$$

$$ma = 11 - 10 = 1 \text{ N}$$

$$ma = 1 \text{ N}, a = 1 \text{ m/s}^2$$

96.(8) From graph $F \propto t^2$

or $F = Kt^2 \dots (1)$

$$\therefore K = \frac{F}{t^2} = \frac{50}{25} = 2$$

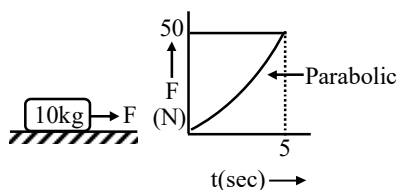
$$K = 2$$

Now $m \frac{dv}{dt} = Kt^2$

$$m \int_0^v dv = k \int_0^t t^2 dt$$

$$mv = \frac{Kt^3}{3}$$

$$v = \frac{Kt^3}{3} = \frac{2 \times 125}{3 \times 10}; v = 8.33 \text{ m/s}$$



97.(25) For equilibrium of A

$$T_1 \cos \alpha = mg + T_2 \cos \beta \quad \dots(1)$$

$$T_1 \sin \alpha = T_2 \sin \beta \quad \dots(2)$$

$$T_2 \cos \beta = mg \quad \dots(3)$$

From equation (3) & (4)

$$T_2 = \sqrt{2} \, mg \quad \tan \beta = 1$$

From equation (1) & (2)

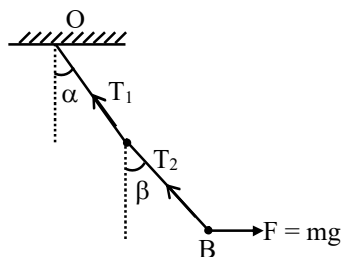
$$\frac{T_1 \sin \alpha}{T_1 \cos \alpha} = \frac{T_2 \sin \beta}{mg + T_2 \cos \beta}$$

$$\tan \alpha = \frac{\sqrt{2}mg \times \frac{1}{\sqrt{2}}}{mg + \sqrt{2}mg \times \frac{1}{\sqrt{2}}} = \frac{1}{2}$$

$$\frac{\tan \alpha}{\tan \beta} = \frac{\frac{1}{2}}{1} = \frac{1}{2} = \tan \beta = 2 \tan \alpha$$

From equation (2) $\sqrt{2} T_1 \sin \alpha = T_2 \sin \beta$

$$T_1 \times \frac{1}{\sqrt{5}} = T_2 \times \frac{1}{\sqrt{2}} \therefore \sqrt{2} T_1 = \sqrt{5} T_2$$



98.(5) For mass A

$$mg \sin 30 + T = ma$$

$$\frac{mg}{2} + T = ma \quad \dots(1)$$

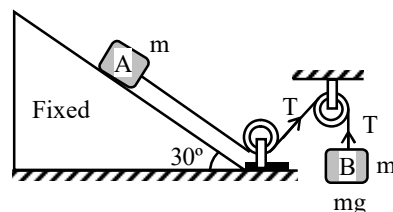
$$\text{For mass B } mg - T = ma \quad \dots(2)$$

$$\frac{mg}{2} + T = ma \quad \dots(1) \text{ on addition}$$

$$\frac{3mg}{2} = 2ma \quad \therefore a = \frac{3g}{4}$$

From eq. (1)

$$\frac{mg}{2} + T = \frac{3mg}{4} \quad \therefore T = \frac{mg}{4}$$



99.(35) In equilibrium

$$T_1 \sin = W$$

$$T_1 = \frac{8}{1/2} = 16\text{N}$$

$$T_1 = 16\text{N}$$

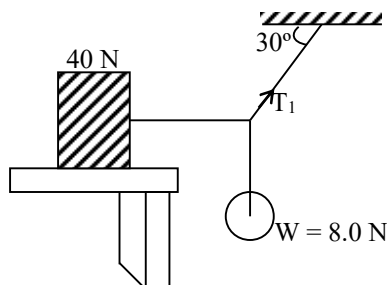
$$T_1 \cos 30 = f$$

$$T_1 \cos 30 = \mu N$$

$$\mu = \frac{T_1 \cos 30}{N}$$

$$= \frac{16 \times \sqrt{3}}{40 \times 2}$$

$$\mu = 0.35$$



100.(4) $f = \mu mg$; $u = 4 \text{ m/s}$
 $f = 0.2 \times 10 \text{ m}$ $u = 0$

$f = 2 \text{ m}$

retardation $= -2 \text{ m/s}^2$

from $v^2 = u^2 + 2as$

$S = \frac{v^2 - u^2}{2a}$; $S = \frac{-16}{-2 \times 2}$; $S = 4 \text{ m}$

101.(5) $m_2 = 6 \text{ kg}$ $m_1 = 2 \text{ kg}$
 $\mu = 0.2$, $g = 10 \text{ /s}^2$

friction f_1 on mass m_1

$f_1 = \mu m_1 g$
 $= 0.2 \times 2 \times 10$

$f_1 = 4 \text{ N}$

force for m_1

$F - T - f_1 = m_1 \frac{a}{2}$

$93 - T - 4 = 2 \times \frac{a}{2}$

$89 - T = a \dots\dots(1)$

from equ 1 & 2

on summation

$65 = 13a$

$\therefore a = 5 \text{ m/s}^2$

friction f_2 on mass m_2

$f_2 = \mu m_2 g$
 $= 0.2 \times 6 \times 10$

$f_2 = 12 \text{ N}$

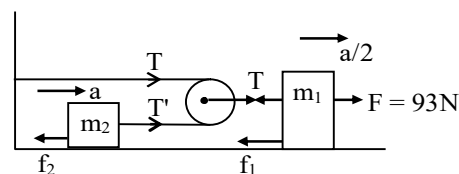
force equ for m_2

$T' - f_2 = m_2 a$

$T' - 12 = 6a$

$T - 2a = 12a \dots\dots(2)$

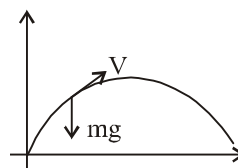
$-T + 89 = a$



Energy & Momentum

1.(A) $x_{cm} = \frac{\int x dm}{dm} = \frac{\int_0^1 x \alpha (1 + bx^2) A dx}{\int_0^1 \alpha (1 + bx^2) A dx} = \frac{3(2+b)}{4(3+b)}$

2.(C)



$P_{in} = \vec{F} \cdot \vec{V}$

$\vec{F} = -mg \hat{j}$

$\vec{V} = V \cos \theta \hat{i} + (V \sin \theta - gt) \hat{j}$

$P_{in} = -mgV \sin \theta + mg^2 t$

3.(D) $Mu = MV_1 + mV_2$, $V_2 - V_1 = eu$

$\Rightarrow Mu = M(V_2 - eu) + mV_2 \Rightarrow V_2 = \frac{Mu(1+e)}{m+M} = \frac{u(1+e)}{m/M+1} t$ $u(1+e) = 2u$ for elastic collision.

4.(B) W r t. heavy ball

Rest



$\leftarrow \frac{u}{\bullet}$

Rest



$\rightarrow \frac{u}{\bullet}$

w.r.t. ground speed is $2u$.

5.(B) Area = Impulse = $\frac{F_0 T}{2} = mu$

6.(A) After collision balls will interchange their velocities

7.(C) $v \cos \phi = eu \cos \theta$

$v \sin \phi = u \sin \theta$

Work Energy Theorem

8.(A) Let V_1 and V_2 be velocity of 1st and 2nd balls after collision

$\Rightarrow mu = mv_1 + mv_2$ and $v_2 - v_1 = eu \Rightarrow v_1 = (1 - e)\frac{u}{2}$ and $v_2 = (1 + e)\frac{u}{2} \Rightarrow \frac{v_1}{v_2} = \frac{1 - e}{1 + e}$

9.(D) Let point of collision is origin and line parallel to BC is x -axis and y -axis along OA .

$\vec{u}_A = -v\hat{j}$

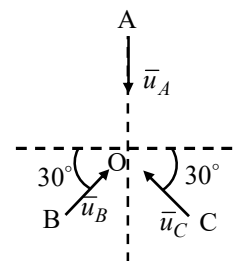
$\vec{u}_B = \frac{v}{2}(\sqrt{3}\hat{i} + \hat{j})$

$\vec{u}_C = \frac{v}{2}(-\sqrt{3}\hat{i} + \hat{j})$ and $\vec{V}_A = 0, \vec{V}_B = -\frac{V}{2}(\sqrt{3}\hat{i} + \hat{j}), \vec{V}_C = ?$

$\Delta \vec{P} = 0 \Rightarrow m\vec{u}_A + m\vec{u}_B + m\vec{u}_C = m\vec{V}_A + m\vec{V}_B + m\vec{V}_C$

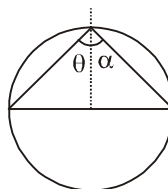
$\Rightarrow -V\hat{j} + \frac{V}{2}(\sqrt{3}\hat{i} + \hat{j}) + \frac{V}{2}(-\sqrt{3}\hat{i} + \hat{j}) = -\frac{V}{2}(\sqrt{3}\hat{i} + \hat{j}) + \vec{V}_C$ (Note initial momentum of A, B and C is zero)

$\Rightarrow \vec{V}_C = \frac{V}{2}(\sqrt{3}\hat{i} + \hat{j}) \Rightarrow V$ along BO .



10.(B) $2May = \frac{1}{2}ky^2 \Rightarrow y = \frac{2Ma}{K}$

11.(B) $\tan \alpha = \frac{\tan \theta}{e}$ as $\alpha = \theta$ so $e = 1$



12.(A) Let V : velocity of block 2 when block 1 breaks off the wall.

$\Rightarrow \frac{1}{2}kx^2 = \frac{1}{2}m_2V^2 \Rightarrow V = \sqrt{\frac{kx^2}{m_2}} \Rightarrow \text{Velocity of centre of mass} = \frac{m_2V}{m_1 + m_2} = \frac{\sqrt{m_2 k} x}{m_1 + m_2}$

13.(D) Negative work done on both masses = increase in potential energy of the spring.

$\Rightarrow -2\omega = \frac{1}{2}kx^2 \Rightarrow \omega = -\frac{1}{4}kx^2$

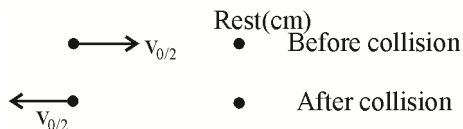
Alternatively: Let both the blocks has been displaced by $s \Rightarrow$ restoring force of the Spring is $K(2s)$

$\Rightarrow \text{Work done on each block} = \int_0^{x/2} H(2s)ds \cos 180^\circ = \frac{kx^2}{4}$

14.(C) $x_{cm} = mu; y_{cm} = 2mu; y = 2x$

15.(C) $V_{cm} = \frac{mV_0}{2m} = \frac{V_0}{2}$

In CM frame :



16.(B) Let σ : mass per unit area of the sheet & Let $2a$: side of corner cut & 2ℓ : side of the square given.

Case I: $x_{cm} = \frac{(\sigma 4\ell^2)(0) + (-\sigma 4a^2)[-(\ell - a)]}{(\sigma 4\ell^2) + (-\sigma 4a^2)} = \frac{a^2(\ell - a)}{\ell^2 - a^2}$

$$\text{Case II: } x_{cm} = \frac{(\sigma 4\ell^2)(0) + 2(-\sigma 4a^2)[-(\ell-a)]}{(\sigma 4\ell^2) + 2(-\sigma 4a^2)} = \frac{2a^2(\ell-a)}{\ell^2-2a^2}$$

$$\text{Case III: } x_{cm} = \frac{(\sigma 4\ell^2)(0) + 2(-\sigma 4a^2)[-(\ell-a)] + (-\sigma 4a^2)(\ell-a)}{\sigma 4\ell^2 - 3\sigma 4a^2} = \frac{a^2(\ell-a)}{\ell^2-3a^2}$$

$$1 \rightarrow x_{cm1} = \frac{a^2(\ell-a)}{\ell^2-a^2}; \quad 2 \rightarrow x_{cm2} = \frac{2a^2(\ell-a)}{\ell^2-2a^2}; \quad 3 \rightarrow x_{cm3} = \frac{a^2(\ell-a)}{\ell^2-3a^2}$$

$$\text{Comparing 1 and 2} \Rightarrow \frac{1}{\ell^2-a^2}, \frac{2}{\ell^2-2a^2} \Rightarrow \ell^2-2a^2, 2\ell^2-2a^2 \Rightarrow x_{cm1} < x_{cm2}$$

$$\text{Comparing 2 and 3} \Rightarrow \frac{2}{\ell^2-2a^2}, \frac{1}{\ell^2-3a^2} \Rightarrow 2\ell^2-6a^2, \ell^2-2a^2$$

$$\Rightarrow \ell^2, 4a^2 \quad \text{Now } \ell > 2a$$

$$\text{Comparing 1 and 3} \Rightarrow \frac{1}{\ell^2-2a^2}, \frac{1}{\ell^2-3a^2} \Rightarrow x_{cm1} < x_{cm3}$$

$\Rightarrow$ correct increasing order is 1, 3, 2

17.(B) Along the wall (i.e. along y-axis) component of velocity remains same & perpendicular to wall it becomes $-e$ times.

$$\Rightarrow \text{Final velocity} = -\hat{i} + 2\hat{j}.$$

$$18.(B) \quad \begin{array}{c} \uparrow F_{\text{thrust}} \\ \bullet \\ \downarrow mg \end{array} \quad a = \frac{F_{th} - mg}{m} = \frac{V_{rel} \frac{dm}{dt} - mg}{m} = 20 \text{ m/s}^2$$

$$19.(C) \quad \text{Minimum elongation in spring to left B (mass M)} = \frac{Mg}{K}$$

$$\Rightarrow \text{Loss in GPE of A} = \text{gain in EPE of spring} \Rightarrow mg \left\{ \frac{Mg}{k} \right\} = \frac{1}{2} K \left(\frac{Mg}{k} \right)^2 \Rightarrow m = \frac{M}{2}$$

$$20.(BD) \quad \text{Just after collision} \quad \begin{array}{c} \xrightarrow{v} \\ \text{[Block]} \text{---} \text{[Spring]} \text{---} \text{[Block]} \end{array}$$

$$\text{During maximum compression } mv = 2mV' \Rightarrow V' = \frac{v}{2}$$

$$\text{K.E.} = 2 \times \frac{1}{2} m \left(\frac{v}{2} \right)^2 = \frac{mv^2}{4}$$

$$\text{Constant energy } \frac{mv^2}{2} = \frac{mv^2}{4} + \frac{1}{2} kx^2$$

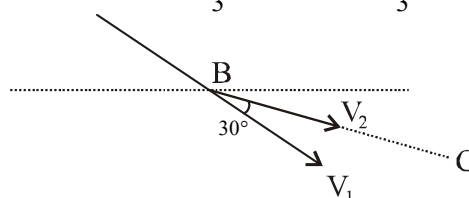
21.(C) Let m_1 and m_2 be masses of the two balls & u & v are initial and final velocities.

$$\Rightarrow m_1 u = m_2 v \text{ and } \frac{1}{2} m_2 v^2 = \frac{1}{3} \cdot \frac{1}{2} m_1 u^2 \Rightarrow v = \frac{1}{3} u \Rightarrow e = \frac{1}{3}$$

22.(B) V_1 = speed just before striking the plane

$$= \sqrt{2hg} = \sqrt{60} \text{ m/s}$$

$$V_2 = V_1 \cos 30^\circ = \sqrt{45}$$

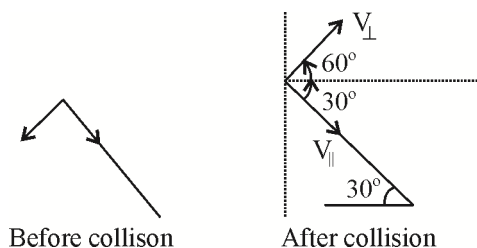


23.(B) $V_3 = \text{velocity just before leaving inclined plane} = \sqrt{V_2^2 + 2gh_2} = \sqrt{45 + 2 \times 10 \times 3} = \sqrt{105}$

24.(C) Component of v_1 parallel to BC will remain unchanged while component perpendicular to BC will reverse its direction

$$V_{||} = V_1 \cos 30^\circ = \sqrt{45} \quad ; \quad V_{\perp} = V_1 \sin 30^\circ = \sqrt{15}$$

$$\Rightarrow V = V_{\perp} \cos 30^\circ - V_{||} \cos 60^\circ = 0$$



25.(C) Net external force passing through the centre of mass of the system = total mass $\times a_{cm}$ and $a_{cm} = V^2/R$.

26.(B) Let x_1 and x_2 be elongations in lower and upper spring

$$\Rightarrow \frac{-x_1 + x}{2} = x_2 \quad \Rightarrow x = 2x_2 + x_1$$

Let T : tension in the string connecting the block.

$$\Rightarrow T = kx_1 \text{ and } T + kx_1 = kx_2 \quad \Rightarrow 2x_1 = x_2 \quad \Rightarrow x_1 = x/5 \text{ and } x_2 = 2x/5$$

Now loss in GPE of block = gain in KE + gain in $EPEs$

$$\Rightarrow mgx = \frac{1}{2}mv^2 + \frac{1}{2}kx_1^2 + \frac{1}{2}kx_2^2 \quad \Rightarrow \frac{1}{2}mv^2 = mgx - \frac{1}{2}k\left\{\frac{x^2}{25} + \frac{4x^2}{25}\right\} = \frac{10mgx - kx^2}{10}$$

27.(D) Kinetic energy is minimum in the reference frame of centre of mass $\Rightarrow V = \frac{m_1 \bar{v}_1 + m_2 \bar{v}_2}{m_1 + m_2}$

28.(C) $(M)u = mv \cos \alpha \Rightarrow u = \frac{mv \cos \alpha}{M}$ (Along horizontal, $\Delta P = 0$)

29.(C) Given, $F = ax + bx^2$

According to work-energy theorem, we know that work done in stretching the rubber band by t is : $|dW| = Fdx$

$$|W| = \int_0^L (ax + bx^2) dx = \left[\frac{ax^2}{2} \right]_0^L + \left[\frac{bx^3}{3} \right]_0^L = \left[\frac{aL^2}{2} - \frac{a \times (0)^2}{2} \right] + \left[\frac{b \times L^3}{3} - \frac{b \times (0)^3}{3} \right]$$

$$\therefore |W| = \frac{aL^2}{2} + \frac{bL^3}{3}$$

30.(B) Let V_1 and V_2 velocities of B and C after first collision towards right $m_B = m$
 $m_A = m_C = 4m$

$$\Rightarrow mV = mV_1 + 4mV_2 \text{ and } V_2 - V_1 = V \Rightarrow V_1 = \frac{-3V}{5} \quad ; \quad V_2 = \frac{2V}{5}$$

and Let V_3 and V_4 be velocity of A and B after 2nd collision towards left.

$$\Rightarrow \frac{3mV}{5} = 4mV_3 + mV_4 \text{ and } V_3 - V_4 = \frac{3V}{5} \Rightarrow V_3 = \frac{6V}{25} \text{ and } V_4 = \frac{-9V}{25}$$

$\Rightarrow B$ moves toward right again but it won't collide with C again.

31.(AD) u_x : velocity at highest point, and v : velocity of other part after the cannon breaks.

$$\Rightarrow mu_x = -\frac{m}{2}u_x + \frac{m}{2}v \Rightarrow V = 3u_x.$$

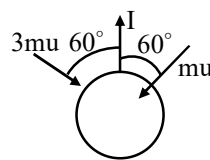
$\Rightarrow$ horizontal distance traveled by 2nd part is 3 times the horizontal distance ($R/2$) traveled by the 1st part after the cannon breaks and KE of 2nd part is 9 times the KE of the first part at highest point.

- 32.(A) Let I : impulse in the string & V : horizontal velocity of combined mass.

$$\Rightarrow I = 3mu \cos 60^\circ + mu \cos 60^\circ = 2mu$$

$$\text{and } 3mu \sin 60^\circ - mu \sin 60^\circ = 12mv \Rightarrow V = \frac{\sqrt{3}u}{12}$$

$$\text{and loss in energy} = \frac{1}{2}m(3u)^2 + \frac{1}{2}mu^2 - \frac{1}{2}m\left(\frac{\sqrt{3}u}{12}\right)^2 = \frac{479}{96}mu^2$$



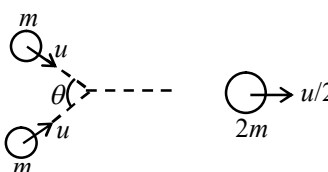
- 33.(ABCD) Given masses are equal and the collision is elastic. If the collision is head-on then A comes to rest and velocity of B relative to A remains the same in magnitude but reverses in direction. And if collision is oblique then angle between their final velocity vectors is 90° . And if they move at an angle of 45° each with the x-axis then their speed is also equal.

- 34.(D) Energy $E = \frac{p^2}{2m}$, where p is momentum, m is the mass moving of the particle. Maximum energy loss occurs when the particles get stuck together as a result of the collision.

Maximum energy loss $(\Delta E) = \frac{p^2}{2m} - \frac{p^2}{2(m+M)}$ where, $(m+M)$ is the resultant mass when the particles get stuck.

$$\text{Also, } p = mv \quad \therefore \Delta E = \frac{m^2v^2}{2m} \left[\frac{M}{m+M} \right] = \frac{mv^2}{2} \left[\frac{M}{m+M} \right]$$

$$\text{Comparing the expression with } \Delta E = f \left(\frac{1}{2}mv^2 \right), f = \frac{M}{m+M}$$

35.(A)  $\Rightarrow 2mu \frac{\cos \theta}{2} = \frac{2mu}{2}$

$$\Rightarrow \frac{\theta}{2} = 60^\circ \Rightarrow \theta = 120^\circ$$

- 36.(D) Work done = increase in potential energy of the hanging part $= \frac{m}{4} \cdot g \cdot \frac{\ell}{8} = \frac{mg\ell}{32}$

- 37.(D) Acceleration of both observers are same so they will observe same acceleration of the block. As initial velocity of the block is differently observed, so displacement of the block is also observed different. So work done by friction, total work done and kinetic energy of the block will be observed differently.

- 38.(D) Extension is maximum when velocities become equal.

$$\Rightarrow m_2v_0 = (m_1 + m_2)V \Rightarrow V = \frac{m_2v_0}{m_1 + m_2}$$

And loss in KE = gain in elastic potential energy.

$$\Rightarrow \frac{1}{2}m_2v_0^2 - \frac{1}{2}(m_1 + m_2)\left(\frac{m_2v_0}{m_1 + m_2}\right)^2 = \frac{1}{2}kx_m^2 \Rightarrow x_m = v_0 \sqrt{\frac{m_1m_2}{(m_1 + m_2)k}}$$

- 39.(C) Let V : common horizontal velocity when the block reaches at the top of the wedge where vertical velocity of the block just becomes zero.

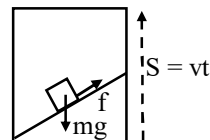
$$\Rightarrow mu = (m + M)V \quad \text{and} \quad \frac{1}{2}mu^2 - \frac{1}{2}(m + M)V^2 = mgh \quad \text{and} \quad M = \eta^m$$

$$\Rightarrow V = \frac{u}{1 + \eta} \quad \& \quad u^2 - (1 + \eta) \frac{u^2}{(1 + \eta)^2} = 2gh \Rightarrow \frac{\eta u^2}{1 + \eta} = 2gh \Rightarrow u = \sqrt{2gh \left(1 + \frac{1}{\eta} \right)}$$

40.(C) $f = mg \sin \theta$

Work done by $f = mg \sin \theta \cdot vt \cdot \cos(90 - \theta)$

(as angle between f and s is $90 - \theta$) $= mg v t \sin^2 \theta$



- 41.(C) Let the particle A and B collide at time t . For their collision, the position vectors of both particles should be same at time t , i.e.

$$\vec{r}_1 + \vec{v}_1 t = \vec{r}_2 + \vec{v}_2 t$$

$$\vec{r}_1 - \vec{r}_2 = \vec{v}_2 t - \vec{v}_1 t = (\vec{v}_2 - \vec{v}_1) t \quad \dots (i)$$

Also $|\vec{r}_1 - \vec{r}_2| = |\vec{v}_2 - \vec{v}_1| t$ or $t = \frac{|\vec{r}_1 - \vec{r}_2|}{|\vec{v}_2 - \vec{v}_1|}$

Substituting this value of t in eqn. (i), we get : $\vec{r}_1 - \vec{r}_2 = (\vec{v}_2 - \vec{v}_1) \frac{|\vec{r}_1 - \vec{r}_2|}{|\vec{v}_2 - \vec{v}_1|}$ or $\frac{\vec{r}_1 - \vec{r}_2}{|\vec{r}_1 - \vec{r}_2|} = \frac{(\vec{v}_2 - \vec{v}_1)}{|\vec{v}_2 - \vec{v}_1|}$

- 42.(C) Here,

Volume of blood pumped by man's hear, $V = 5 \text{ litres} = 5 \times 10^{-3} \text{ m}^3$ ($\because 1 \text{ litre} = 10^{-3} \text{ m}^3$)

Time in which this volume of blood pumps,

$$T = 1 \text{ min} = 60 \text{ s}$$

Pressure at which the blood pumps.

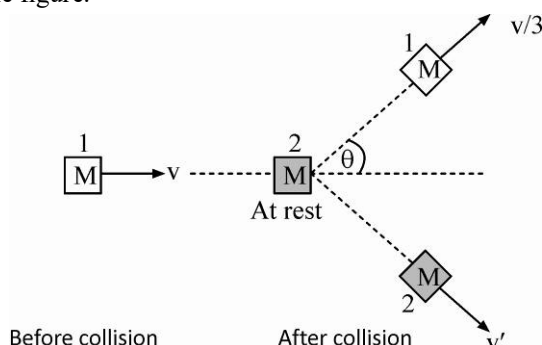
$$P = 150 \text{ mm of Hg} = 0.15 \text{ m of Hg}$$

$$= (0.15 \text{ m}) (13.6 \times 10^3 \text{ kg/m}^3) (10 \text{ m/s}^2) \quad (\because P = h \rho g)$$

$$ABCD = -P(2V) = -2PV \quad \therefore \text{Power of the heart} = \frac{PV}{t}$$

$$= \frac{(20.4 \times 10^3 \text{ N/m}^2)(5 \times 10^{-3} \text{ m}^3)}{60 \text{ s}} = 1.70 \text{ W}$$

- 43.(C) The situation is show in the figure.



Let v' be speed of second block after the collision. As the collision is elastic, so kinetic energy is conserved..

According to conservation of kinetic energy.

$$\frac{1}{2} M v^2 + 0 = \frac{1}{2} M \left(\frac{v}{3} \right)^2 + \frac{1}{2} M v'^2$$

$$v^2 = \frac{v^2}{9} + v'^2$$

or $v'^2 = v^2 - \frac{v^2}{9} = \frac{9v^2 - v^2}{9} = \frac{8}{9} v^2$

$$v' = \sqrt{\frac{8}{9} v^2} = \frac{\sqrt{8}}{3} v = \frac{2\sqrt{2}}{3} v$$

- 44.(C) Constant power acting on the particle of mass m is k watt.

or $p = k$

$$\frac{dW}{dt} = k ; dW = k dt$$

Integrating both sides $\int_0^W dW = \int_0^t k dt \Rightarrow W = kt \dots (i)$

Using work energy theorem, $W = \frac{1}{2}mv^2 - \frac{1}{2}m(0)^2$

$$kt = \frac{1}{2}mv^2 \quad (\text{Using equation (i)})$$

$$v = \sqrt{\frac{2kt}{m}}$$

Acceleration of the particle $a = \frac{dv}{dt}$

$$a = \frac{1}{2} \sqrt{\frac{2k}{m}} \frac{1}{\sqrt{t}} = \sqrt{\frac{k}{2mt}}$$

Force on the particle. $F = ma = \sqrt{\frac{ml}{2t}} \sqrt{\frac{mk}{2}} t^{-1/2}$

- 45.(B) Let $\vec{v}$ be velocity of third piece of mass $2m$. Initial momentum, $\vec{P}_1 = 0$ (As the body is at rest)

Final momentum, $\vec{p}_1 = mv\hat{i} + mv\hat{j} + m\vec{v}'$

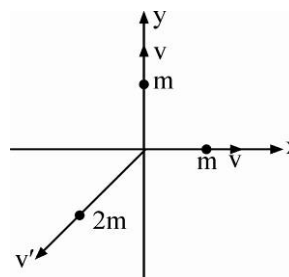
According to law of conservation of momentum

$$\vec{p}_i = \vec{p}_f$$

$$0 = mv\hat{i} + mv\hat{j} + 2m\vec{v}'$$

$$\vec{v}' = -\frac{v}{2}\hat{i} - \frac{v}{2}\hat{j}$$

The magnitude of v' is $v' = \sqrt{\left(-\frac{v}{2}\right)^2 + \left(-\frac{v}{2}\right)^2} = \frac{v}{\sqrt{2}}$



Total kinetic energy generated due to explosion

$$= \frac{1}{2}mv^2 + \frac{1}{2}mv^2 + \frac{1}{2}(2m)v'^2 = \frac{1}{2}mv^2 + \frac{1}{2}mv^2 + \frac{1}{2}(2m)\left(\frac{v}{\sqrt{2}}\right)^2 = mv^2 + \frac{mv^2}{2} = \frac{3}{2}mv^2$$

- 46.(C) Here, $\vec{F} = (3\hat{i} + \hat{j})N$

Initial position, $\vec{r}_1 = (2\hat{i} + \hat{k})m$

Final position, $\vec{r}_2 = (4\hat{i} + 3\hat{j} - \hat{k})m$

Displacement, $\vec{r} = \vec{r}_2 - \vec{r}_1$

$$\vec{r} = (4\hat{i} + 3\hat{j} - \hat{k})m - (2\hat{i} + \hat{k})m = 2\hat{i} + 3\hat{j} - 2\hat{k}m$$

Work done,

$$W = \vec{F} \cdot \vec{r} = (3\hat{i} + \hat{j}) \cdot (2\hat{i} + 3\hat{j} - 2\hat{k}) = 6 + 3 = 9J$$

- 47.(B) Here, $U = \frac{A}{r^2} - \frac{B}{r}$

For equilibrium, $\frac{dU}{dr} = 0 \quad \therefore \quad -\frac{2A}{r^3} + \frac{B}{r^2} = 0 \quad \text{or} \quad \frac{2A}{r^3} = \frac{B}{r^2} \quad \text{or} \quad r = \frac{2A}{B}$

For stable equilibrium, $\frac{d^2U}{dr^2} > 0$

$$\frac{d^2U}{dr^2} = \frac{6AB^4}{16A^4} - \frac{2B^4}{8A^3} = \frac{B^4}{8A^3} > 0$$

So for stable equilibrium, the distance of the particle is $\frac{2A}{B}$

48.(B) $P_a = Fv$

$$\therefore F = ma = m \frac{dv}{dt} \quad \therefore \quad P_0 = mv \frac{dv}{dt} \quad \text{or} \quad P_0 dt = mv dv$$

Integrating both side, we get :

$$\int_0^t P_0 dt = m \int_0^v v dv$$

$$P_0 t = \frac{mv^2}{2}$$

$$v = \left(\frac{2P_0 t}{m} \right)^{1/2} \quad \text{or} \quad v \propto \sqrt{t}$$

49.(B) Power, $P = \vec{F} \cdot \vec{v} = Fv \cos \theta$

Just before hitting the earth $\theta = 0^\circ$. Hence, the power exerted by the gravitational force is greatest at the instant just before the body hits the earth.

50.(A)

51.(D) Here,

Mass per unit length of water $\mu = 100 \text{ kg/m}$

Velocity of water $v = 2 \text{ m/s}$

Power of the engine, $P = \mu v^3 = (100 \text{ kg/m}) (2 \text{ m/s})^3 = 800 \text{ W}$

52.(A) Power = F.V.

where $f = \rho a V^2$

$$\Rightarrow \text{Power} = \rho a V^3$$

Given $m = \rho a$

$$\Rightarrow \text{power} = m V^3 \quad (\text{Rate of imparting of KE})$$

53.(C) Mass of water falling/second = 15 kg/s

$$h = 60 \text{ m}$$

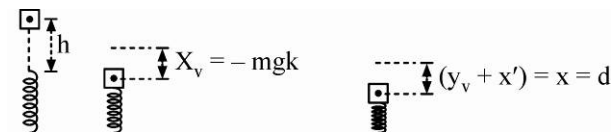
$$g = 10 \text{ m/s}^2, \text{ loss} = 10\%$$

i.e., 90% is used.

$$\begin{aligned} \text{Power generated} &= 15 \times 10 \times 60 \times 0.9 \\ &= 8100 \text{ W} = 8.1 \text{ kW} \end{aligned}$$

54.(A) When a mass falls on a spring from a height h work done by the loss of potential energy of the mass is stored as the potential energy of the spring.

One can write $mg(h+d) = \frac{1}{2}kd^2$



The two energies are equal.

If work done is initial P.E. – final P.E., it is zero. Work done is totally converted (assuming there is no loss).

The work done in compression or expansion is always positive as it is $\propto x^2$. The answer expected is :

$$mg(h+d) - \frac{1}{2}kd^2 \quad \text{or} \quad \frac{1}{2}kd^2 - mg(h+d)$$

As seen from option, but is it not justified. Question could have been more specific like work done by oscillation.

55.(D) $s \frac{t^3}{3}; \frac{ds}{dt} = \frac{2t}{3}; \frac{d^2s}{dt^2} = \frac{2}{3}$

$$\text{Work done, } W = \int F ds = \int m \frac{d^2s}{dt^2} ds = \frac{2}{3} = \int m \frac{d^2s}{dt^2} \frac{ds}{dt} dt = \int_0^2 3 \times \frac{2}{3} \times \frac{2t}{3} dt = \frac{4}{3} \int_0^2 t dt ;$$

$$= \frac{4}{3} \int_0^2 t dt = \frac{4}{3} \left[\frac{t^2}{2} \right]_0^2 = \frac{4}{3} \times 2 = \frac{8}{3} J$$

56.(B) $\vec{P} = \vec{P} \cdot \vec{v} = (64\hat{i} + 45\hat{j} - 3\hat{k}) \cdot (2\hat{j} - 4\hat{j} + 5\hat{k}) ; \quad = 120 - 60 - 15 = 45 \text{ watts}$

57.(C) Velocity after 5 second $v = u - gt = 100 - 10 \times 5 = 50 \text{ m/s}$

By conservation of momentum $1 \times 50 = 0.4 \times (-25) + 0.6 \times v'$

$$60 = 0.6 \times v' \Rightarrow v' = 100 \text{ m/s upwards}$$

58.(C) $\frac{dx}{dt} = 3 - 8t + 3t^2$

$$\left. \frac{dx}{dt} \right|_{t=0} = 3 \quad \left. \frac{dx}{dt} \right|_{t=4} = 19$$

$$\text{Wall} = \Delta KE = \frac{1}{2} \times 3 \times 10^{-3} \times 19^2 - \frac{1}{2} \times 3 \times 10^{-3} \times 3^2 = 528 \text{ mJ}$$

59.(D) Mass of metal ball = 2 kg ;

Speed of metal ball (v_1) = 36 km/h = 10 m/s and mass of stationary ball = 3 kg

Applying law of conservation of momentum $m_1 v_1 + m_2 v_2 = (m_1 + m_2) v$

$$v = \frac{m_1 v_1 + m_2 v_2}{m_1 + m_2} = \frac{(2 \times 10) + (3 \times 0)}{2 + 3} = \frac{20}{5}$$

$$4 \text{ m/s.}$$

$$\text{Therefore loss of energy} = \left[\frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 \right] - \frac{1}{2} (m_1 + m_2) v^2$$

$$= \left[\frac{1}{2} \times 2 \times (10)^2 + \frac{1}{2} \times 3(0)^2 \right] - \frac{1}{2} \times (2 + 3) \times (4)^2 = 100 - 40 = 60 \text{ J}$$

60.(B) $\text{Power} = \frac{\text{work done}}{\text{time taken}} = \frac{W}{t}$

But $W = \text{mass} \times \text{gravity} \times \text{height}$

$$\therefore P = \frac{M \times g \times h}{t}$$

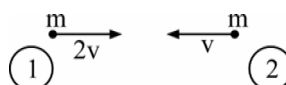
$$\Rightarrow M = \frac{P \times t}{g \times h} = \frac{2000 \times 60}{10 \times 10} = 1200 \text{ kg}$$

i.e. 1200 litres as one litre has a mass of 1 kg.

61.(C) Let the mass of each body be m . Their motion is represented as shown in the figure.

$$\text{From } \vec{V}_{CM} = \frac{m_1 \vec{V}_1 + m_2 \vec{V}_2}{m_1 + m_2}$$

$$\vec{V}_{cm} = \frac{m \times 2v - mv}{m + m} = \frac{v}{2}$$



[The direction of motion of the first particle is taken as positive.]

So the velocity of the centre of mass of the system is $v/2$ in the direction of motion of the particle having larger speed.

62.(C) The bomb initially was at rest therefore

Initial momentum of bomb = 0

Final momentum of system = $m_1 v_1 + m_2 v_2$

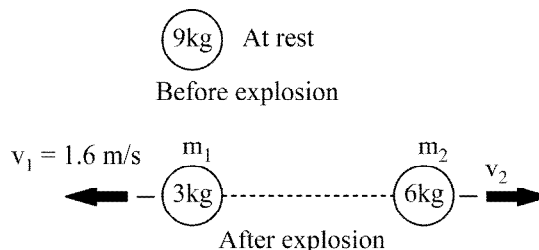
As there is no external force

$$\therefore m_1 v_1 + m_2 v_2 = 0$$

$$\Rightarrow 3 \times 1.6 + 6 \times v_2 = 0$$

Velocity of 6 kg mass, $v_2 = 0.8 \text{ m/s}$ (numerically)

$$\text{Its kinetic energy} = \frac{1}{2} m_2 v_2^2 = \frac{1}{2} \times 6 \times (0.8)^2 = 1.92 \text{ J}$$



63.(D) If the kinetic energy of a system of particles is zero, it means all the particles are at rest. Hence, the momentum is zero. Thus II implies I.

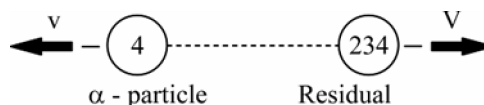
But if the momentum of a system of particles is zero, it does not mean that all the particle are at rest. Hence, the particle may possess kinetic energy. Thus I does not imply II.

64.(A) Initially ^{238}U nucleus was at rest and after decay its parts move in opposite directions.

According to conservation of momentum,

$$4v + 234V = 238 \times 0$$

$$\Rightarrow V = -\frac{4v}{234}$$



65.(B) PE of hanging part = work done

$$\text{Work done} = \left(\frac{4}{2}\right)(0.6)g\left(\frac{0.6}{2}\right) = 2 \times 0.6 \times 0.3 \times 10$$

$$\Rightarrow 1.2 \times 3 = 3.6 \text{ J}$$

- 66.(B) Let mass A moves with velocity v and collides inelastically with mass B, which is at rest.

According to problem, mass A moves in a perpendicular direction and let the mass B moves at angle θ with the initial velocity with velocity v .

Initial horizontal momentum of system

$$(\text{before collision}) = mv \quad \dots (i)$$

Final horizontal momentum of system

$$(\text{after collision}) = mV \cos \theta \quad \dots (ii)$$

From the conservation of horizontal linear momentum, $mv = mV \cos \theta \Rightarrow v = V \cos \theta$

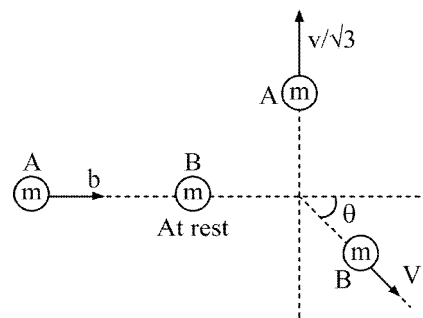
Initial vertical momentum of system (before collision) is zero. Final vertical momentum of system

$$= \frac{mv}{\sqrt{3}} - mV \sin \theta$$

From the conservation of vertical linear momentum $\frac{mv}{\sqrt{3}} - mV \sin \theta = 0$

$$\Rightarrow \frac{v}{\sqrt{3}} = V \sin \theta \quad \dots (iv)$$

By solving (iii) and (iv), we get : $v^2 + \frac{v^2}{3} = V^2 (\sin^2 \theta + \cos^2 \theta) \Rightarrow \frac{4v^2}{3} = V^2 \Rightarrow V = \frac{2}{\sqrt{3}} v$



- 67.(C) CM will not shift as no external force in horizontal direction

- 68.(B) P has to be C.M. of system

- 69.(D) $4V_1 + 12V_2 = 0$

$$\Rightarrow 4V_1 = -12V_2$$

$$V_1 = 3V_2$$

$$\text{Since } V_2 = 4m/s \Rightarrow V_1 = -12m/s$$

$$KE = \frac{1}{2} \times 4 \times (-12)^2 = 288J$$

- 70.(A) $m_1 \Delta x_1 + m_2 \Delta x_2 = 0$ (If No shift of CM)

$$m_1 d + m_2 \Delta x_2 = 0$$

$$\Delta x_2 = \frac{-m_1 d}{m_2}$$

- 71.(D) Let after cutting the disc of radius R , the mass of remaining disc be M' and mass of removed disc be m .

$$\text{Let } (M' + m) = M = \pi(2R)^2 \cdot \sigma = 4\pi R^2 \cdot \sigma$$

(Where σ is mass per unit area)

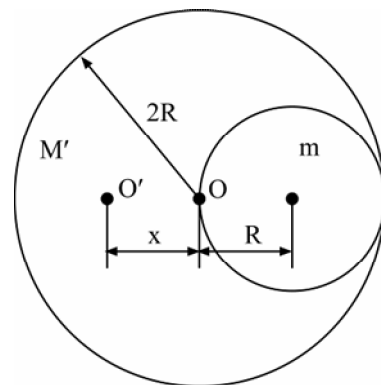
$$\therefore m = \pi R^2 \cdot \sigma \therefore M' = (4\pi R^2 \cdot \sigma) - (\pi R^2 \cdot \sigma) = 3\pi R^2 \cdot \sigma$$

For full disc, the centre of mass is at the centre O. After removing the disc of radius R let centre of mass shift by distance x .

$$\therefore x = \frac{4\pi R^2 \sigma \cdot 0 - \pi R^2 \sigma R}{4\pi R^2 \cdot \sigma - \pi R^2 \cdot \sigma} = -\frac{R}{3}$$

$$\Rightarrow x = -\frac{R}{3} = \alpha R \quad \therefore |\alpha| = \left| \frac{-1}{3} \right|$$

The centre of mass is at $R/3$ to the left on the centre of the original disc.



- 72.(C) By the law of conservation of linear momentum

$$mu = (M + m)v$$

$$0.50 \times 2.00 = (1 + 0.50) v \Rightarrow v = \frac{1.00}{1.50}$$

$$\text{Initial } K.E = (1/2) \times 0.50 \times (2.00)^2 = 1.00 \text{ J}$$

$$\text{Final } K.E = \frac{1}{2} \times 1.50 \times \frac{1.00^2}{(1.50)^2} = \frac{1.00}{3.00} = 0.33 \quad \therefore \quad \text{Loss of energy} = 1.00 - 0.33 = 0.67 \text{ J}$$

- 73.(A) Position of centre of mass.

$$x_{CM} = \frac{\int_0^L \left(\frac{k}{L^n} \cdot x^n \cdot dx \right) x}{\int_0^L \frac{k}{L^n} \cdot x^n \cdot dx} = \frac{\int_0^L x^{n+1} dx}{\int_0^L x^n dx} = \frac{L^{n+2}}{n+2} \cdot \frac{(n+1)}{L^{n+1}} \Rightarrow x_{CM} = \frac{L(n+1)}{(n+2)}$$

$$\text{The variation of the centre of mass with } x \text{ is given by } \frac{dx}{dn} = L \left\{ \frac{(n+2)1 - (n+1)}{(n+2)^2} \right\} = \frac{L}{(n+2)^2}$$

If the rod has the same density as $x = 0$, i.e., $n = 0$, therefore uniform, the centre of mass would have been at $L/2$. As the density increases with length, the centre of mass shifts towards the right. Therefore it can only be option (A).

- 74.(C) As rubber ball is freely falling, the ball is dropped ($u = 0$). If we take downwards direction negative, the velocity at any time t can be given by the relation $v = -gt$. As when coming down, v increase, make collision, the value of v becomes +ve, the speed decreases, momentarily comes to zero and then moves in downwards direction and speed increases. The change from +ve to -ve is almost instantaneous.

Further $h = -\frac{1}{2}gt^2$ is a parabola which opens downwards ; Therefore (C) is correct

- 75.(A) If it is a completely inelastic collision, then $m_1v_1 + m_2v_2 = m_1v + m_2v \Rightarrow v = \frac{m_1v_1 + m_2v_2}{m_1 + m_2}$

Since v_1 and v_2 are in same direction, v will not be zero.

So, total KE cannot be lost.



- 76.(B) Here, mass of the body, $m = 0.4 \text{ kg}$

Since position - time ($x-t$) graph is a straight line, so motion is uniform. Because of impulse direction of velocity change as it can be seen from the slopes of the graph.

From graph,

$$\text{Initial velocity, } u = \frac{(2-0)}{(2-0)} = 1 \text{ ms}^{-1} ; \quad \text{Final velocity, } v = \frac{(0-2)}{(4-2)} = -1 \text{ ms}^{-1}$$

$$\text{Initial momentum, } P_i = mu = 0.4 \times 1 = 0.4 \text{ Ns}$$

$$\text{Final momentum, } P_f = mv = 0.4(-1) = -0.4 \text{ Ns}$$

$$\text{Impulse} = \text{Change in momentum} = p_f - p_i = (-0.4 - 0.4) \text{ Ns} = -0.8 \text{ Ns}$$

$$|\text{Impulse}| = 0.8 \text{ Ns}$$

- 77.(D) Loss of energy is maximum when collision is inelastic as in an inelastic collision there will be maximum deformation.

$$K.E_i = \frac{1}{2}mV^2$$

$$mv = (m + M)V_f \Rightarrow V_f = \left(\frac{m}{m+M} \right) V$$

$$K.E_f - K.E_i = \frac{1}{2}(M + m)V_f^2 - \frac{1}{2}mV^2 = \frac{1}{2}mV^2 \left(\frac{M}{m+M} \right)$$

$$\text{Hence loss in energy is } \frac{1}{2} \left(\frac{Mm}{M+m} \right) V^2 \Rightarrow f = \frac{M}{M+m}$$

78.(C) Applying conservation of linear momentum just before and just after collision,

$$m2v\hat{i} + 2mv\hat{j} = (m + 2m)\vec{v'}$$

$$\vec{v'} = \frac{2}{3}v(i + j)$$

$$\text{Hence } |\vec{v'}| = \frac{2\sqrt{2}}{3}v$$

Initial kinetic energy of the masses just before collision.

$$K_{initial} = \frac{1}{2}m(2v)^2 + \frac{1}{2}2mv^2 = 3mv^2$$

Final kinetic energy of the system just after collision.

$$K_{final} = \frac{1}{2}(3m) \left(\frac{2\sqrt{2}}{3}v \right)^2 = \frac{4}{3}mv^2$$

Loss in kinetic energy.

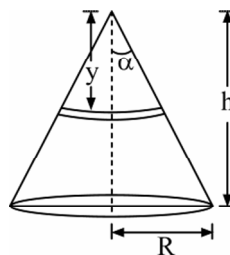
$$\Delta K = K_{initial} - K_{final} = 3mv^2 - \frac{4}{3}mv^2 = \frac{5}{3}mv^2$$

Hence percentage loss in kinetic energy

$$\frac{\Delta K}{K} \% = \frac{\frac{5}{3}mv^2}{3mv^2} \times 100 = 56\%$$

$$79.(B) \quad dm = \pi r^2 dy \rho$$

$$y_{CM} = \frac{\int y dm}{\int dm} = \frac{\int_0^h \pi r^2 dy \times \rho \times y}{\frac{1}{3}\pi R^2 h \rho} = \frac{3h}{4} \left(\text{using } \frac{r}{y} = \frac{R}{h} \right)$$



80.(B) From the conservation of energy, we know that the potential energy at height is equal to the kinetic energy at ground.

Therefore, at height h , the potential energy of ball A is $m_A gh$

Its kinetic energy at ground is

$$\frac{1}{2}m_A v_A^2 \quad \therefore \quad m_A gh = \frac{1}{2}m_A v_A^2 \quad \Rightarrow \quad v_A = \sqrt{2gh}$$

Similarly, $v_B = \sqrt{2gh}$

Therefore, $v_A = v_B$.

81.(D) Using work-energy theorem, $W_{total} = \Delta K$

i.e., work done by stopping forces should be equal to change in the kinetic energy of the car.

$$-F_{stopping} \times \text{stopping distance} = (k_{final} - k_{initial})$$

$$\Rightarrow -F_{stopping} \times s = \left(0 - \frac{1}{2}mv_{initial}^2\right) \Rightarrow F_{stopping} \times s = \frac{1}{2}mv_{initial}^2$$

As stopping force is same for both the cars, hence

$$s \propto v_{initial}^2 \Rightarrow \frac{s_1}{s_2} = \frac{(v_{initial}^2)_1}{(v_{initial}^2)_2} = \frac{u^2}{(4y)^2} = \frac{1}{16}.$$

82.(B) Here the work done in stretching a spring is stored as the potential energy of the spring.

$$\text{Thus } W = \frac{1}{2}kx_2^2 - \frac{1}{2}kx_1^2 = \frac{1}{2}k(x_2^2 - x_1^2) = \frac{1}{2} \times 800 \left[\left(\frac{15}{100}\right)^2 - \left(\frac{5}{100}\right)^2 \right] = 8 \text{ J}$$

83.(B) Kinetic energy, $K = \frac{1}{2}mv^2 = \frac{(mv)^2}{2m} = \frac{p^2}{2m} \Rightarrow p = \sqrt{2mK}$ or $p \propto \sqrt{m} \therefore \frac{p_1}{p_2} = \sqrt{\frac{1}{16}} = \frac{1}{4}$

84.(D) Using work-energy theorem, $W_{total} = \Delta K$ i.e., work done by stopping forces should be equal to change in the kinetic energy of the car.

$$-F_{stopping} \times \text{stopping distance} = (k_{final} - k_{initial})$$

$$-F_{stopping} \times s = -\frac{1}{2}mv_{initial}^2$$

As stopping force is same for both the cars, hence

$$s \propto v_{initial}^2$$

$$\text{Hence, } \frac{s_1}{s_2} = \frac{(v_{initial}^2)_1}{(v_{initial}^2)_2} \quad \text{or} \quad \frac{6}{s_2} = \frac{(v_{initial}^2)_1}{(v_{initial}^2)_2} = \left(\frac{50}{100}\right)^2 \Rightarrow s_2 = 24 \text{ m}$$

85.(C) Here, power $P = F \cdot v = \left(m \frac{dv}{dt}\right) \cdot v \Rightarrow v dv = \frac{P}{m} dt$

$$\text{Now integrating, } \int_0^v v dv = \frac{P}{m} \int_0^t dt \Rightarrow \frac{v^2}{2} = \frac{Pt}{m}$$

$$\text{or } v = \left(\frac{2Pt}{m}\right)^{\frac{1}{2}} \Rightarrow \frac{dx}{dt} = \left(\frac{2P}{m}\right)^{\frac{1}{2}} t^{\frac{1}{2}} \Rightarrow dx = \left(\frac{2P}{m}\right)^{\frac{1}{2}} t^{\frac{1}{2}} dt$$

Integrating, we get :

$$\int_0^x dx = \left(\frac{2P}{m}\right)^{\frac{1}{2}} t^{\frac{1}{2}} dt \quad \text{or} \quad x = \left(\frac{2P}{m}\right)^{\frac{1}{2}} \frac{t^{\frac{3}{2}}}{\frac{3}{2}} \quad \text{or} \quad x \propto t^{\frac{3}{2}}$$

86.(C) Here, $W = \frac{1}{2}k(x_2^2 - x_1^2) = \frac{1}{2} \times 5 \times 10^3 \left[\left(\frac{10}{100}\right)^2 - \left(\frac{5}{100}\right)^2 \right] = \frac{1}{2} \times \frac{5 \times 10^3}{10^4} [100 - 25] = \frac{75}{4} = 18.75 \text{ J}.$

87.(C) $U = \frac{a}{x^{12}} - \frac{b}{x^6}$

Force, $F = -\frac{dU}{dx} = -\frac{d}{dx}\left(\frac{a}{x^{12}} - \frac{b}{x^6}\right) = -\left[\frac{-12a}{x^{13}} + \frac{6b}{x^7}\right] = \left[\frac{12a}{x^{13}} - \frac{6b}{x^7}\right]$

At equilibrium, $F = 0$.

$\therefore \frac{12a}{x^{13}} - \frac{6b}{x^7} = 0$ or $x^6 = \frac{2a}{b}$

$$U_{\text{at equilibrium}} = \frac{a}{\left(\frac{2a}{b}\right)^2} - \frac{b}{\left(\frac{2a}{b}\right)} = \frac{ab^2}{4a^2} - \frac{b^2}{2a} = \frac{b^2}{4a} - \frac{b^2}{2a} = -\frac{b^2}{4a} \Rightarrow U_{(x=\infty)} = 0$$

$$D = [U_{(x=\infty)} - U_{\text{at equilibrium}}] = \left[0 - \left(-\frac{b^2}{4a}\right)\right] = \frac{b^2}{4a}$$

88.(C) According to problem,

Retardation $\propto$ displacement

or $v \frac{dv}{dx} = -kx$ or $\int_{v_1}^{v_2} v dv = -k \int_0^x x dx$

or $\frac{v_2^2}{2} - \frac{v_1^2}{2} = -\frac{kx^2}{2}$ or $\frac{1}{m} \left(\frac{mv_1^2}{2} - \frac{mv_2^2}{2} \right) = \frac{kx^2}{2}$ or $KE_1 - KE_2 = \frac{mk}{2} x^2$

Thus, loss of kinetic energy is proportional to x^2 .

89.(B) Here, $W = \vec{F} \cdot \vec{S} = (5\hat{i} + 3\hat{j} + 2\hat{k}) \cdot (2\hat{i} - \hat{j}) = 10 - 3 = 7 J$

90.(B) Here, applying $v = u + at$, we get ; $v_1 = at_1$ or $a = \frac{v_1}{t_1}$; Also $v = at = \frac{v_1}{t_1} \times t$

Power = $|\vec{F} \cdot \vec{v}| = mav = m \left[\frac{v_1}{t_1} \right] \left[\frac{v_1}{t_1} \times t \right] = \frac{mv_1^2 t}{t_1^2}$

91.(A) When a force of constant magnitude which is perpendicular to the velocity of particle acts on a particle, work done is zero and hence change in kinetic energy is zero.

92.(A) $k_1 x_1 = k_2 x_2 = F$

$$W_1 = \frac{1}{2} k_1 x_1^2 = \frac{(k_1 x_1)^2}{2k_1} = \frac{F^2}{2k_1}$$

Similarly, $W_2 = \frac{F^2}{2k_2} \Rightarrow W \propto \frac{1}{K} \Rightarrow W_1 > W_2 \Rightarrow k_1 < k_2$ Statement-II is true.

Now, $W_1 = \frac{1}{2} K_1 x^2, W_2 = \frac{1}{2} K_2 x^2 ; \therefore K_2 > K_1$ So, $W_2 > W_1$. Hence, statement-I is false.

93.(D) When block of mass M collides with the spring, its kinetic energy gets converted into elastic potential energy of the spring. From the law of conservation of energy.

$$\frac{1}{2} Mv^2 = \frac{1}{2} KL^2 \therefore v = \sqrt{\frac{K}{M}} L$$

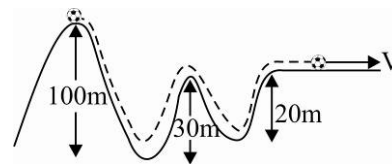
Where v is the velocity of block by which it collides with spring. So, its maximum momentum,

$$P = Mv = M \sqrt{\frac{K}{M}} L = \sqrt{MK} L$$

After collision, the block will rebound with same linear momentum.

94.(A) As per question, restoring force on the particle is $F = 15 \times 0.2 = 3 \text{ N}$. So the acceleration is $3/0.3 = 10 \text{ m/s}^2$.

95.(D) Ball starts from the top of a hill which is 100 m high and finally rolls down to a horizontal which is 20 m above the ground, so from the conservation of energy.



$$mg(h_1 - h_2) = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{2g(h_1 - h_2)} = \sqrt{2 \times 10 \times (100 - 20)} = \sqrt{1600} = 40 \text{ m/s}.$$

96.(A) Restoring force on rubber-band, $F = ax + bx^2$.

Work done in stretching the rubber-band by a small amount dx , $dW = Fdx$

$$\text{Net work done in stretching the rubber-band by } L \text{ is } W = \int_0^L dW = \int_0^L Fdx = \int_0^L (ax + bx^2) dx$$

$$\Rightarrow W = \left[a \frac{x^2}{2} + b \frac{x^3}{3} \right]_0^L = \frac{aL^2}{2} + \frac{bL^3}{3}$$

97.(D) Using work-energy theorem, $W_{total} = \Delta K$ i.e., work done by gravity should be equal to change in the kinetic energy of the particle.

$$W_{gravity} = (K_{final} - K_{initial}) \Rightarrow W_{gravity} = \left(0 - \frac{1}{2} \times \left(\frac{100}{1000} \right) (5)^2 \right) = -1.25 \text{ J}$$

Work done by force of gravity = -1.25 J

98.(C) Total mechanical energy of the particle, $E_T = 2 \text{ J}$.

When kinetic energy is maximum, the potential energy should be minimum.

$$\text{The potential energy of the particle is given by } V(x) = \frac{x^4}{4} - \frac{x^2}{2} \text{ or } \frac{dV}{dx} = \frac{4x^3}{4} - \frac{2x}{2} = x^3 - x = x(x^2 - 1)$$

$$\text{For } V \text{ to the minimum, } \frac{dV}{dx} = 0$$

$$\therefore x(x^2 - 1) = 0, \text{ or } x = 0, \pm 1 \quad ; \quad \text{At } x = \pm 1, V(x) = -\frac{1}{4} \text{ J}$$

$$\therefore (\text{Kinetic energy})_{\max} = E_T - V_{\min} \text{ or } (\text{Kinetic energy})_{\max} = 2 - \left(-\frac{1}{4} \right) = \frac{9}{4} \text{ J}$$

$$\text{or } \frac{1}{2}mv_m^2 = \frac{9}{4} \text{ or } v_m^2 = \frac{9 \times 2}{m \times 4} \text{ or } v_m^2 = \frac{9 \times 2}{1 \times 4} = \frac{9}{2} \Rightarrow v_m = \frac{3}{\sqrt{2}} \text{ m/s}.$$

99.(B) According to work energy theorem "work done by all forces is equal to change kinetic energy".

$$\therefore W_{total} = \Delta K$$

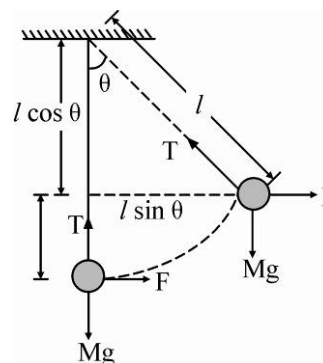
$$W_F + W_{gravity} + W_T = (k_{final} - k_{initial}) = (0 - 0) = 0$$

$$\text{Work done by } F, W_F = F \times l \sin 45^\circ = \frac{Fl}{\sqrt{2}}$$

Work done by gravity,

$$W_{gravity} = -Mg(l - l \cos 45^\circ) = -Mg\ell \left(1 - \frac{1}{\sqrt{2}} \right)$$

$$\text{Work done by tension} = 0 \quad \therefore \frac{Fl}{\sqrt{2}} + \left(-\frac{Mg\ell(\sqrt{2} - 1)}{\sqrt{2}} \right) = 0 \quad \text{or} \quad F = Mg(\sqrt{2} - 1)$$



100.(B) Here the work done by hand should be equal to change in potential energy of the ball i.e., $W_{hand} = \Delta U$

$$\Rightarrow F_{hand} s = mgh \Rightarrow F_{hand} = \frac{mgh}{s} = \frac{0.2 \times 10 \times (2 + 0.2)}{0.2} = 22 \text{ N}$$

101.(D) Let the spring is compressed by x . Loss in KE of block = Gain in PE of spring + Work done against friction

$$\Rightarrow \frac{1}{2} \times 2 \times 4^2 = \frac{1}{2} 10000 x^2 + 15x \quad \text{On solving, we get } x = 5.5 \text{ cm.}$$

102.(B) Let the kinetic energy of the particle at point of projection be K . $K = \frac{1}{2} mu^2$

At highest point, velocity has its horizontal component $u \cos \theta$.

Therefore kinetic energy of a particle at highest point is :

$$K_H = \frac{1}{2} m(u \cos \theta)^2 = \frac{1}{2} mu^2 (\cos \theta)^2 = K \cos^2 \theta \Rightarrow K_H = K \cos^2 60^\circ = \frac{K}{4}$$

103.(D) average velocity $v_{av} = \frac{s}{t} = \frac{100}{10} = 10 \text{ m/s}$

Assuming an athlete has mass about 50 to 100 kg or average mass 75 kg.

His average kinetic energy.

$$k_{av} = \frac{1}{2} mv_{av}^2 = \frac{1}{2} \times 75 \times 100 = 3750 \text{ J}$$

Minimum possible kinetic energy, (for 50 kg)

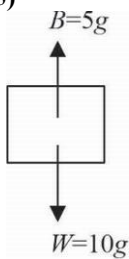
$$k_{\min} = \frac{1}{2} mv_{av}^2 = \frac{1}{2} \times 50 \times 100 = 2500 \text{ J}$$

Minimum possible kinetic energy, (for 100 kg)

$$k_{\max} = \frac{1}{2} mv_{av}^2 = \frac{1}{2} \times 100 \times 100 = 5000 \text{ J}$$

INTEGER VALUE TYPE

104.(245)



$$W_{ext} = (5g)(5) = 25 \times 9.8 = 245 \text{ J.}$$

105.(11) $v \xrightarrow{x} \frac{19v}{20} \quad \frac{1}{2} mv^2 - Fx = \frac{1}{2} m \left(\frac{19v}{20} \right)^2 \quad \dots(i)$

When we have n planks

$$v \xrightarrow{nx} 0. \quad \frac{1}{2} mv^2 - F(nx) = 0 \quad \dots(ii)$$

From (i) and (ii)

$$\frac{1}{2} mv^2 \left(1 - \frac{1}{n} \right) = \frac{1}{2} mv^2 \left(\frac{19}{20} \right)^2$$

$$\frac{1}{n} = \frac{39}{400} \Rightarrow n = \frac{400}{39} = 10.25$$

Thus a minimum of 11 planks are required.

$$106.(30) \text{ Power delivered} = \frac{(1000)(9) \times (10) \times (10)}{5 \times 60} = 3kW.$$

$$\text{Efficiency} = \frac{\text{power delivered}}{\text{Power consumed}} \times 100 = \frac{3}{10} \times 100 = 30\%$$

$$107.(4) \quad e = 1 = \frac{v-1}{2-(-1)} \Rightarrow v = 4 \text{ m/s}$$

108.(6) Conserving momentum

$$(1)(8) + (3)(4) = (1+3)v \Rightarrow v = 5 \text{ m/s}$$

Net work done by friction = Loss in KE

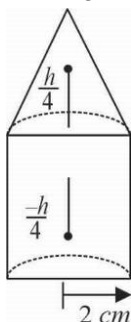
$$= K_i - K_f$$

$$= \left(\frac{1}{2}(1)(8)^2 + \frac{1}{2}(3)(4)^2 \right) - \left(\frac{1}{2}(1+3)(5)^2 \right)$$

$$= 6J$$

$$109.(4) \quad F = \frac{\Delta m}{\Delta t} v = (5)(0.8) = 4N$$

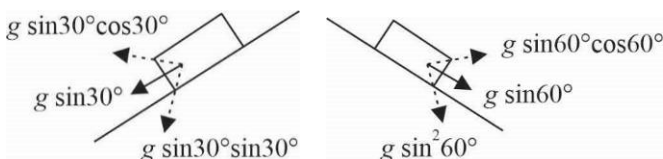
$$110.(5) \quad m_1 = \rho \frac{1}{3} \pi r^2 h = m$$



$$m_2 = \rho \pi r^2 h = 3m$$

$$Y_{cm} = \frac{m \left(\frac{h}{4} \right) + (3m) \left(\frac{-h}{2} \right)}{m + 3m} = \frac{-5h}{16} = -5cm.$$

111.(5)



$$A_y (cm) = \frac{ma_1 + ma_2}{m_1 + m_2} = \frac{a_1 + a_2}{2} = \frac{g}{2} (\sin^2 30^\circ + \sin^2 60^\circ) = \frac{g}{2} = 5 \text{ m/s}^2$$

$$112.(100) y_{cm} (i) = \frac{m(40) + m(0)}{2m} = 20m$$

$$u_{cm} = \frac{m(50) + m(30)}{2m} = 40 \text{ m/s}.$$

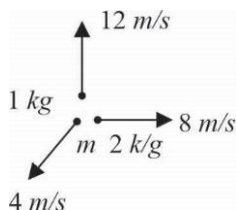
$$a_{cm} = g \text{ (downwards)}$$

$$\text{Max ht} = 20 + \frac{(40)^2}{2 \times 10} = 20 + 80 = 100m$$

$$113.(7) \quad 4(-v) + 0.05(560 - v) = 0$$

$$\Rightarrow \quad v = \frac{28}{4.05} \approx 7 \text{ m/s}$$

114.(5)



Net momentum of first two particles

$$= \sqrt{(12)^2 + (16)^2}$$

$$= 20$$

$$m(4) = 20 \Rightarrow m = 5 \text{ kg}$$

$$115.(5) \quad (2)(0.15) = (2+3)V \Rightarrow V = 0.06 \text{ m/s}$$

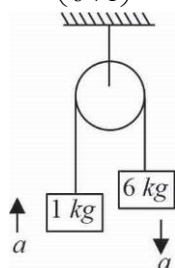
$$\frac{1}{2}(2)(0.15)^2 = \frac{1}{2}(10.8)x^2 + \frac{1}{2}(2+3)(0.06)^2$$

$$\Rightarrow x = 0.05 \text{ m} = 5 \text{ cm}$$

$$116.(9) \quad \text{Mass of the disc removed} = \frac{M}{\pi(28)^2} \times \pi(21)^2 = \frac{9M}{16}$$

$$X_{cm} = \frac{M(0) + \left(\frac{-9M}{16}\right) \times 7}{M - \frac{9M}{16}} = -9 \text{ cm}$$

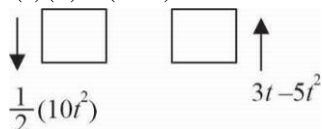
$$117.(5) \quad a = \left(\frac{6-1}{6+1}\right)g = \frac{5}{7}g$$



$$a_{cm} = \frac{6a + 1(-a)}{6+1} = \frac{5}{7}a = \frac{25}{49} \times 9.8$$

$$= 5 \text{ m/s}^2$$

$$118.(30) \quad (1)(9) = (1+2)v \Rightarrow v = 3 \text{ m/s}$$



When displacement of the two blocks become equal the string will become taut again

$$5t^2 = 3t - 5t^2 \Rightarrow t = 0.3 \text{ sec.}$$

Rotational Motion

1.(AD) $\vec{L} = m(\vec{r} \times \vec{v})$

$\vec{L}_{tot} = \vec{L}_1 + \vec{L}_2 = mv(d_2 - d_1) \otimes$

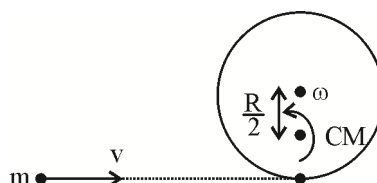
2.(D) $I_B = \frac{1}{2}mr^2$ and I_A about the given axis by parallel axis theorem $= \frac{1}{2}mr^2 + m(2r)^2 = \frac{9}{2}mr^2$

$\Rightarrow I = \frac{1}{2}mr^2 + \frac{9}{2}mr^2 = 5mr^2$

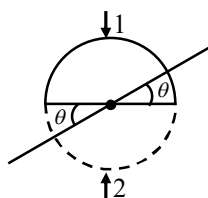
3.(B) Angular momentum conservation about combined COM

$mv \frac{R}{2} = \left(m \left(\frac{R}{2} \right)^2 + mR^2 + m \left(\frac{R}{2} \right)^2 \right) \omega$

$\omega = \frac{V}{3R}$



4.(B)



I_1 about axis $= I_2$ about axis

I of ring about axis $= I_1 + I_2 = 2I_1 = 2I_2$

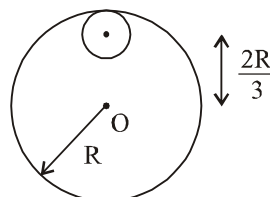
$\Rightarrow I = I = \frac{1}{2} \times I \text{ ring about axis (Diameter)}$

$= \frac{1}{2} \times \left(\frac{1}{2} \times (2M) R^2 \right) = \frac{1}{2} MR^2 \Rightarrow \text{independent of } \theta.$

5.(ABCD) In all the cases net torque about axis is zero.

6.(A) $m' = \frac{9m}{R^2} \times \left(\frac{R}{3} \right)^2 = m$

$I_0 = 9m \frac{R^2}{2} - \left(\frac{m}{2} \left(\frac{R}{3} \right)^2 + m \left(\frac{2R}{3} \right)^2 \right) = 4mR^2$

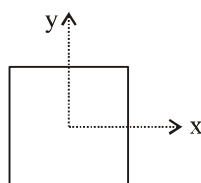


7.(BC) $\vec{\tau} = (\vec{r} \times \vec{F})$ Greater the distance from axis more is torque

8.(ABD) $I_x = 2 \left\{ m \left(\frac{\ell}{2} \right)^2 + \frac{m\ell^2}{12} \right\} = I_y$

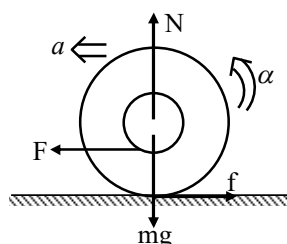
$I_z = I_x + I_y$

$I_{side} = I_x + 4m \left(\frac{\ell}{2} \right)^2 = \frac{5}{3} m\ell^2$



9.(BC) Let F is the force applied to the rope and assuming the cylinder rolls without sliding.

$$\Rightarrow \begin{cases} F - f = ma \\ fR_2 - FR_1 = mk^2 \alpha \\ a = R_2 \alpha \\ N = mf \end{cases} \Rightarrow \begin{cases} a = \frac{F(1 - R_1/R_2)}{m(1 + K^2/R_2^2)} \\ f = \frac{F[K^2 + R_1R_2]}{K^2 + R_2^2} \end{cases}$$



And to avoid sliding $f \leq N\mu$

$$\Rightarrow \frac{F[K^2 + R_1 R_2]}{K^2 + R_2^2} \leq mg\mu_S \Rightarrow F \leq \frac{mg\mu_S[K^2 + R_2^2]}{K^2 + R_1 R_2}$$

$$\Rightarrow F \leq \frac{50 \times 10 \times 0.1[0.3^2 + 0.6^2]}{0.3^2 + (0.3)(0.6)} \Rightarrow F \leq \frac{250}{3} N$$

But applied force is $200 N \Rightarrow$ rolling with sliding and $f = f_K = N\mu_K = mg\mu_K = 50 \times 10 \times 0.08 = 40 N$

$$\left[\alpha = \frac{FR_1 - fR_2}{mk^2} \text{ clockwise} = \frac{200(0.3) - 40(0.6)}{50(0.3)^2} = 8 \text{ rad/s}^2 \right]$$

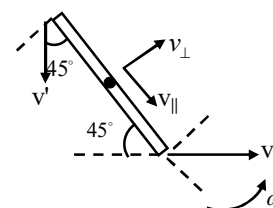
$$a = \frac{F - f}{m} \text{ towards left} = \frac{200 - 40}{50} = 3.2 \text{ m/s}^2$$

10.(BD) By parallel axis theorem $I_z' = I_z + \frac{ma^2}{2}$

Perpendicular axis theorem can't be applied for 3d objects

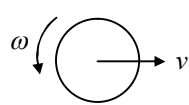
$$I_x = I_y \text{ by symmetry.}$$

$$I_x = I_y \text{ by symmetry.}$$



11.(C) $L = mu_x H = m \frac{u}{\sqrt{2}} \frac{u^2/2}{2g} = \frac{mu^3}{4\sqrt{2}g}$

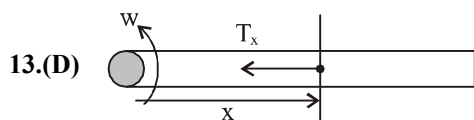
12.(A) $\alpha \Rightarrow a \Rightarrow N\mu = ma \Rightarrow N\mu R = \frac{2}{5} mR^2 \alpha \Rightarrow a = g\mu$



At time $t \Rightarrow \omega = \frac{5g\mu}{2R} t$ and $v + R\omega = v_0 \Rightarrow g\mu t + \frac{5}{2} g\mu t = v_0$

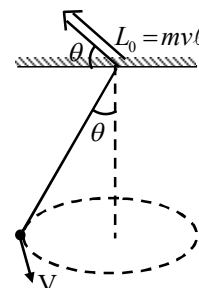
$$\Rightarrow t = \frac{2v_0}{7g\mu}$$

$$s = \frac{1}{2} at^2 = \frac{1}{2} g\mu \frac{4v_0^2}{49g^2\mu^2} = \frac{2v_0^2}{49g\mu} \Rightarrow s = \frac{2v_0^2}{49g(2/7)} = \frac{v_0^2}{7g}$$

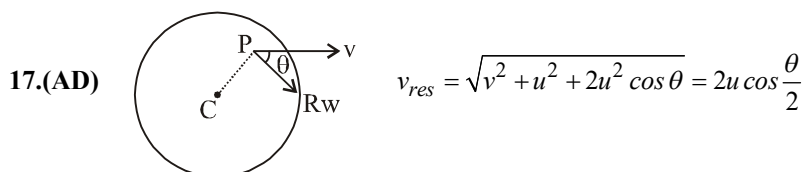
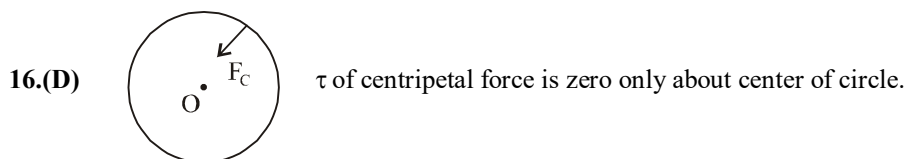


$$T_x = \frac{m}{\ell} (\ell - x) w^2 \left(x + \frac{\ell - x}{2} \right) = \frac{mw^2}{2\ell} (\ell^2 - x^2)$$

14.(BCD) Net force on the particle in uniform circular motion passes through the centre of the circular motion. Angular momentum about O is perpendicular to the plane in which string and velocity vector of the particle lie *i.e.* its makes angle θ with horizontal and horizontal component remains constant in magnitude but vertical component remains constant.



15.(A) $I = mv(3R)(-\hat{k}) + \frac{1}{2}mR^2\omega\hat{k}$ and $V = R\omega = -\frac{5}{2}mR^2\omega\hat{k}$



18.(B) Angular momentum of the system is conserved. $L_{tor} + L_{plate} = \text{constant}$
Moment of inertia of tortoise first increases and then decreases

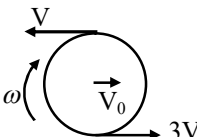
19.(C) Moment of inertia, $I = \frac{1}{2}MR^2 + mx^2$

where m = mass of insect, and x = distance of insect from centre. Clearly, as the insect moves along the diameter of the disc. Moment of inertia first decreases momentum and then increases, angular speed first increases, then decreases.

20.(A) Linear velocity of P is $v + r\omega \Rightarrow$ linear acceleration $= \frac{d}{dt}(v + r\omega) = a + r\alpha$

Angular velocity $= \omega \Rightarrow$ radial acceleration $= \omega^2 r$

$\Rightarrow$ total acceleration of $P = \sqrt{(a + r\alpha)^2 + (\omega^2 r)^2}$

21.(AB) 

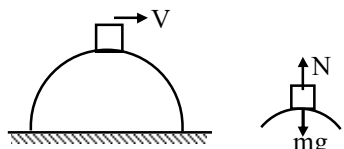
$$\omega = \frac{V + 3V}{2r} = \frac{2V}{r} \text{ also } \omega = \frac{V + V_0}{r} \Rightarrow \frac{V + V_0}{r} = \frac{2V}{r} \Rightarrow V_0 = V$$

22.(D) Let J : impulse generated in the string $\Rightarrow J = \frac{1}{2}mr^2\omega$... for pulley

For block: $-J = mv_2 - mv$... and $v_2 = r\omega$

Where v_2 is velocity of the block just after the string becomes taut.

$\Rightarrow 0 = \frac{mv_2}{2} + mv_2 - mv \Rightarrow v_2 = \frac{2v}{3} = 10/3 \text{ m/s}$

23.(A) 

$$mg - N = \frac{mV^2}{R} \text{ and } N = 0$$

$$\Rightarrow V = \sqrt{Rg}$$

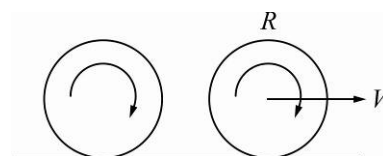
24.(C) From the principle of conservation of angular momentum, we have

$\Sigma l / \omega = \text{constant}$

where, l is moment law of angular momentum,

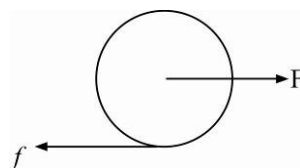
By conservation law of angular momentum, about ground

$mr^2\omega_0 = mvr + mr^2 \times \frac{v}{r} \Rightarrow v = \frac{\omega_0 r}{2}$

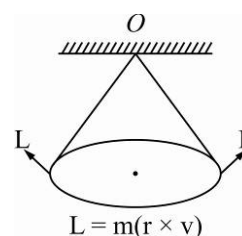


- 25.(A) Since F will produce an anticlockwise torque. So it will cause slipping and hence friction will act towards left. And so will accelerate in left direction.

26.(D) $F - f = ma$ (i)
 $fR = I_C \alpha$ (ii)
 $a = \alpha R$ (iii)
 $\Rightarrow F = ma + \frac{I_C \alpha}{R} \Rightarrow a = \frac{5F}{7M}$
 $\Rightarrow f = \frac{2}{7}F$
 $f \leq f_\ell \Rightarrow \frac{2}{7}F \leq \mu mg$
 $F \leq \frac{7}{2} \mu mg$
 $F_{\max} = \frac{7}{2} \mu mg$



- 27.(C) Net angular momentum will be as shown in figure whose value will be constant ($|L| = mvl$). But its direction will change as shown in the figure.



28.(BD) $K_1 = \frac{1}{2} \left(\frac{1}{2} mr^2 \right) \omega_1^2 = \frac{1}{4} mr^2 \omega_1^2$
 and $\frac{1}{2} mr^2 \omega_1 = \left(\frac{1}{2} mr^2 + mr^2 \right) \omega_2 \dots (\Delta L = 0) \Rightarrow \omega_2 = \omega_1 / 3$
 $\Rightarrow K_2 = \frac{1}{2} \left\{ \frac{1}{2} mr^2 + mr^2 \right\} \omega_2^2 = \frac{1}{2} mr^2 \omega_1^2 = K_1 / 3 \Rightarrow \text{Energy lost} = K_1 - K_2 = 2K_1 / 3$

- 29 (A) Given, force $F = 20t - 5t^2$

$$\alpha = \frac{FR}{I} = \frac{(20t - 5t^2)2}{10} = 4t - t^2$$

$$\Rightarrow \frac{d\omega}{dt} = 4t - t^2 \Rightarrow \int_0^\omega d\omega = \int_0^t (4t - t^2) dt \Rightarrow \omega = 2t^2 - \frac{t^3}{3}$$

When direction is reversed, $\omega = 0$, i.e., $t = 0, 6s$

$$\text{Now, } d\theta = \omega dt \Rightarrow \int_0^6 d\theta = \int_0^6 \left(2t^2 - \frac{t^3}{3} \right) dt \Rightarrow \theta = \left[\frac{2t^3}{3} - \frac{t^4}{12} \right]_0^6$$

$$\Rightarrow \theta = 144 - 108 = 36 \text{ rad} \quad \therefore \text{Number of rotations, } n = \frac{\theta}{2\pi} = \frac{36}{2\pi} < 6$$

- 30.(A) For the mass m ,

$$mg - T = ma$$

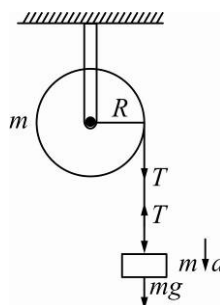
$$\text{As we know, } a = R\alpha \quad \dots(i)$$

$$\text{So, } mg - T = mR\alpha \quad \dots(ii)$$

$$\text{As torque} = T \times R = \frac{mR^2}{2} \alpha$$

From Equations (i) and (ii), we get :

$$\text{Hence, the acceleration body is } \frac{2g}{3}$$



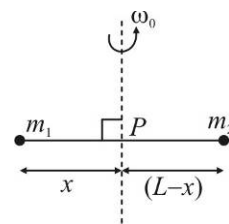
- 31.(B) Moment of inertia of the system about the axis of rotation (through point P) is

$I = m_1 x^2 + m_2 (L-x)^2$. By work energy theorem : work done to set the rod rotating with angular velocity ω_0 = increase in rotational kinetic energy

$$W = \frac{1}{2} I \omega_0^2 = \frac{1}{2} [m_1 x^2 + m_2 (L-x)^2] \omega_0^2. \text{ For } W \text{ to be minimum, } \frac{dW}{dx} = 0$$

$$\text{i.e. } \frac{1}{2} [2m_1 x + 2m_2 (L-x)(-1)] \omega_0^2 = 0 \quad \text{or} \quad m_1 x - m_2 (L-x) = 0 \quad (\dots \omega_0 \neq 0)$$

$$\text{or} \quad (m_1 + m_2)x = m_2 L \quad \text{or} \quad x = \frac{m_2 L}{m_1 + m_2}$$



- 32.(C) Here, speed of the automobile, $v = 54 \text{ km h}^{-1} = 54 \times \frac{5}{18} \text{ ms}^{-1} = 15 \text{ ms}^{-1}$

Radius of the wheel of the automobile, $R = 0.45 \text{ m}$ Moment of inertia of the wheel about its axis of rotation,

$I = 3 \text{ kg m}^2$. Time in which the vehicle brought to rest, $t = 15 \text{ s}$ the initial angular speed of the wheel is

$$\omega_i = \frac{v}{R} = \frac{15 \text{ ms}^{-1}}{0.45 \text{ m}} = \frac{1500}{45} \text{ rad s}^{-1} = \frac{100}{3} \text{ rad s}^{-1}$$

And its final angular speed is $\omega_f = 0$ (as the vehicle comes to rest)

$$\therefore \text{ the angular retardation of the wheel is } \alpha = \frac{\omega_f - \omega_i}{t} = \frac{0 - \frac{100}{3}}{15 \text{ s}} = -\frac{100}{45} \text{ rad s}^{-2}$$

The magnitude of required torque is $\tau = I|\alpha| = (3 \text{ kg m}^2) \left(\frac{100}{45} \text{ rad s}^{-2} \right) = \frac{20}{3} \text{ kg m}^2 \text{ s}^{-2} = 6.66 \text{ kg m}^2 \text{ s}^{-2}$

- 33.(C) For the conservation of angular momentum about origin, the torque $\vec{\tau}$ acting on the particle will be zero.

By definition, $\vec{\tau} = \vec{r} \times \vec{F}$

Here, $\vec{r} = 2\hat{i} - 6\hat{j} - 12\hat{k}$ and $\vec{f} = \alpha\hat{i} + 3\hat{j} + 6\hat{k}$

$$\therefore \vec{\tau} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -6 & -12 \\ \alpha & 3 & 6 \end{vmatrix} = \hat{i}(-36 + 36) - \hat{j}(12 + 12\alpha) + \hat{k}(6 + 6\alpha) = -\hat{j}(12 + 12\alpha) + \hat{k}(6 + 6\alpha)$$

$$\text{But } \vec{\tau} = 0 \quad \therefore \quad 12 + 12\alpha = 0 \text{ or } \alpha = -1 \quad \text{and} \quad 6 + 6\alpha = 0 \text{ or } \alpha = -1$$

- 34.(A) According to law of conservation of angular momentum $mvr = mv'r'$

$$v_0 R_0 = v \left(\frac{R_0}{2} \right); v = 2v_0 \quad \dots (i)$$

$$\therefore \frac{K_0}{K} = \frac{\frac{1}{2} m v_0^2}{\frac{1}{2} m v^2} = \left(\frac{v_0}{v} \right)^2 \quad \text{or} \quad \frac{K}{K_0} = \left(\frac{v}{v_0} \right)^2 = (2)^2 \quad (\text{Using (1)})$$

$$K = 4K_0 = 2m v_0^2$$

- 35.(B) New moment of inertia of the system, $I = I_1 + I_2 + I_3$

The moment of inertia of a shell about its diameter, $I_1 = \frac{2}{3} m r^2$

The moment of inertia of a shell about its tangent is given by

$$I_2 = I_3 = I_1 + m r^2 = \frac{2}{3} m r^2 + m r^2 = \frac{5}{3} m r^2$$

$$\therefore I = 2 \times \frac{5}{3} m r^2 + \frac{2}{3} m r^2 = \frac{12 m r^2}{3} = 4 m r^2$$

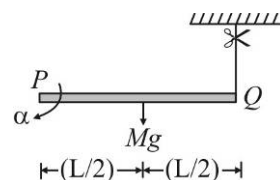
- 36.(C) When the string is cut, the rod will rotate about P . let α be initial angular acceleration of the rod. Then

$$\text{Torque, } \tau = I\alpha = \frac{ML^2}{3}\alpha$$

$$(\text{Moment of inertia of the about one end} = \frac{ML^2}{3})$$

$$\text{Also, } \tau = Mg\left(\frac{L}{2}\right)$$

$$\text{Equating (i) and (ii), we get : } Mg\frac{L}{2} = \frac{ML^2}{3}\alpha \text{ or } \alpha = \frac{3g}{2L}$$



- 37.(B) The kinetic energy of the rolling object is converted into potential energy at height

$$\left[h = \frac{3v^2}{4g} \right]$$

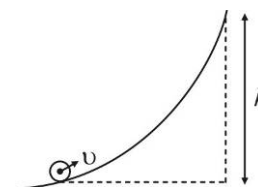
So by the law of conservation of mechanical energy, we have

$$\frac{1}{2}Mv^2 + \frac{1}{2}I\omega^2 = Mgh \quad \left(\because \omega = \frac{v}{R} \right)$$

$$\frac{1}{2}Mv^2 + \frac{1}{2}I\left(\frac{v}{R}\right)^2 = Mg\left(\frac{3v^2}{4g}\right)$$

$$\frac{1}{2}I\frac{v^2}{R^2} = \frac{3}{4}Mv^2 - \frac{1}{2}Mv^2$$

$$\frac{1}{2}I\frac{v^2}{R^2} = \frac{1}{4}Mv^2 \text{ or } I = \frac{1}{2}MR^2$$



- 38.(D) Let M and R be mass and radius of the ring and the disc respectively. Then,

Moment of inertia of ring about an axis passing through its centre and perpendicular to its place is $I_{\text{max}} = MR^2$

Moment of inertia of disc about the same axis is $I_{\text{disc}} = \frac{MR^2}{2}$

As $I = Mk^2$ where k is the radius of gyration

$$\therefore I_{\text{ring}} = Mk_{\text{ring}}^2 = MR^2 \text{ or } k_{\text{ring}} = R$$

$$\text{And } I_{\text{disc}} = Mk_{\text{disc}}^2 = \frac{MR^2}{2} \text{ or } k_{\text{disc}} = \frac{R}{\sqrt{2}}$$

$$\therefore \frac{k_{\text{ring}}}{k_{\text{disc}}} = \frac{R}{R/\sqrt{2}} = \frac{\sqrt{2}}{1} \quad k_{\text{ring}} : k_{\text{disc}} = \sqrt{2} : 1$$

- 39.(C) As no external force acts on the system, therefore centre of mass will not shift.

- 40.(C) As the system is initially at rest, therefore initial angular momentum $L_i = 0$.

According to the principle of conservation of angular momentum, final angular momentum, $L_f = 0$.

$\therefore$ Angular momentum = angular momentum of man is in opposite direction of platform.

$$\text{i.e., } mvR = I\omega \quad \text{or} \quad \omega = \frac{mvR}{I} = \frac{50 \times 1 \times 2}{200} = \frac{1}{2} \text{ rad s}^{-1}$$

$$\text{Angular velocity of man relative to platform is } \omega_r = \omega + \frac{v}{R} = 1 \text{ rad s}^{-1}$$

$$\text{Time taken by the man to complete one revolution is } T = \frac{2\pi}{\omega_r} = \frac{2\pi}{1} = 2\pi \text{ s}$$

41.(A) Given: $\theta(t) = 2t^3 - 6t^2$ $\therefore \frac{d\theta}{dt} = 6t^2 - 12t \Rightarrow \frac{d^2\theta}{dt^2} = 12t - 12$

Angular acceleration, $\alpha = \frac{d^2\theta}{dt^2} = 12t - 12$

When angular acceleration (α) is zero, then the torque on the wheel becomes zero ($\because \tau = I\alpha$)

$\Rightarrow 12t - 12 = 0$ or $t = 1s$

42.(D) As no external torque is applies to the system, the angular momentum of the system remains conserved.

$\therefore L_i = L_f$

According to given problem,

$I_i\omega = (I_i + I_b)\omega_f$

Or $\omega_f = \frac{I_i\omega_i}{(I_i + I_b)}$ (i)

Initial energy, $E_i = \frac{1}{2}I_i\omega_i^2$ (ii)

Final energy, $E_f = \frac{1}{2}(I_i + I_b)\omega_f^2$ (iii)

Substituting the value of ω_f from equation (i) in equation (iii), we get :

Final energy, $E_f = \frac{1}{2}(I_i + I_b)\left(\frac{I_i\omega_i}{I_i + I_b}\right)^2 = \frac{1}{2}\frac{I_i^2\omega_i^2}{(I_i + I_b)}$ (iv)

Loss of energy, $\Delta E = E_i - E_f$

$$= \frac{1}{2}I_i\omega_i^2 - \frac{1}{2}\frac{I_i^2\omega_i^2}{(I_i + I_b)} \quad (\text{Using (ii) and (iv)})$$

$$= \frac{\omega_i^2}{2}\left(I_i - \frac{I_i^2}{(I_i + I_b)}\right) = \frac{\omega_i^2}{2}\left(\frac{I_i^2 + I_bI_i - I_i^2}{(I_i + I_b)}\right) = \frac{1}{2}\frac{I_bI_i}{(I_i + I_b)}\omega_i^2$$

43.(C) The coin will revolve with the record, if force of friction $\geq$ centrifugal force

$\mu mg \geq m r \omega^2$ or $r \leq \frac{\mu g}{\omega^2}$

44.(D) As no external torque is acting about the axis, angular momentum

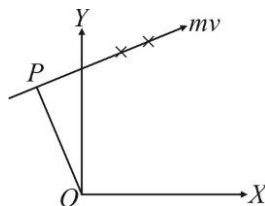
$I_1\omega_1 = I_2\omega_2 \Rightarrow \omega_2 = \frac{I_1\omega_1}{I_2} = \frac{Mr^2\omega}{(M + 2m)r^2} = \frac{M\omega}{(M + 2m)}$

45.(B) Torque is always perpendicular to $\vec{F}$ as well as $\vec{r}$ $\therefore \vec{r} \cdot \vec{\tau} = 0$ as well as $\vec{F} \cdot \vec{\tau} = 0$.

46.(A) Moment of momentum is angular momentum.

OP is the same whether the mass is at A to B .

$\therefore L_A = L_B$



47.(D) Friction acts up the incline but does not work as there is no slipping.

48.(A) $\vec{L} = \vec{r} \times \vec{P}$

$Y = X + 4$ line has been

Shown in the figure.

When $X = 0$,

$$Y = 4, \text{ so } OP = 4.$$

The slope of the line can be obtained by comparing with the equation of line

$$y = mx + c$$

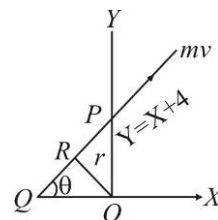
$$m = \tan \theta = 1 \Rightarrow \theta = 45^\circ$$

$$\angle OQP = \angle OPQ = 45^\circ$$

If we draw a line perpendicular to this line.

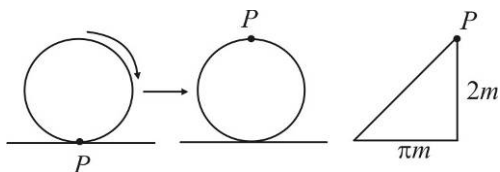
$$\text{Length of the perpendicular} = OR \Rightarrow OR = OP \sin 45^\circ = 4 \frac{1}{\sqrt{2}} = \frac{4}{\sqrt{2}} = 2\sqrt{2}$$

$$\text{So } L = 5 \times 3\sqrt{2} \times 2\sqrt{2} = 60 \text{ units}$$



49.(C)
$$\frac{\frac{1}{2} m K^2 \omega^2}{\frac{1}{2} m K^2 \omega^2 + \frac{1}{2} m (\omega R)^2} = \frac{K^2}{K^2 + R^2}$$

50.(B)



In half rotation point P has moved horizontally $\frac{\pi d}{2} = \pi r = \pi \times 1m = \pi m$. [$\because$ radius = $1m$]

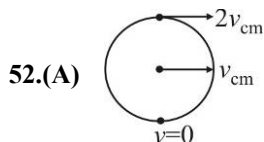
In the same time, it has moved vertically a distance which is equal to its diameter = $2m$

$$\therefore \text{Displacement of } P = \sqrt{\pi^2 + 2^2} = \sqrt{\pi^2 + 4m}.$$

51.(A) A circular disc may be divided into a large number of circular rings. Moment of inertia of the disc will be the summation of the moments of inertia of these rings about the geometrical axis.

Now, moment of inertia of a circular ring about its geometrical axis is MR^2 , where M is the mass and R is the radius of the ring.

Since the density (mass per unit volume) for iron is more than that of aluminium, the proposed rings made of iron should be placed at a higher radius to get more value of MR^2 . Hence to get maximum moment of inertia for the circular disc, aluminium should be placed at interior and iron at the outside.

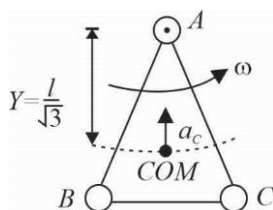


53.(B) As effective distance of mass from BC is greater than the effective distance of mass from AB , therefore $I_2 > I_1$.

54.(D)

INTEGER VALUE TYPE

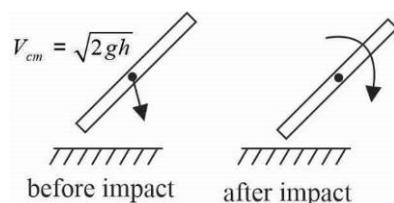
55.(3) $a_c = \omega^2 r$



The center of mass moves along a circular path with constant angular velocity. So, there must be a horizontal centripetal force directed towards the axis at the hinge.

$$F_c = 3m \left(\frac{l}{\sqrt{3}} \right) \omega^2 = \sqrt{3} m l \omega^2$$

56.(4)

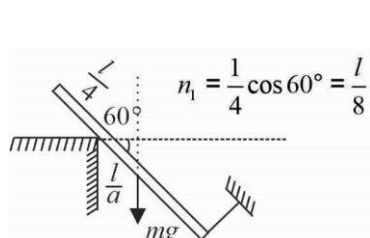


$$L_i = m \sqrt{2gh} \frac{l}{2} \cos 60^\circ \quad L_f = \frac{m l^2}{3} \omega$$

$$L_i = L_f \text{ about point impact}$$

$$m \sqrt{2gh} \frac{l}{4} = \frac{m l}{3} \omega \Rightarrow \omega = \frac{3}{8h} \sqrt{2gh} = \frac{3}{8} \sqrt{\frac{2g}{h}}$$

57.(2)



Net torque about point A = 0

$$T_N = 0 \quad \tau_1 = 0$$

$$T \frac{3l}{4} - Mg \frac{l}{8} = 0$$

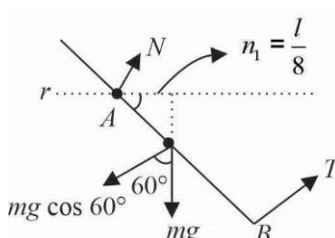
$$T = \frac{Mg \frac{l}{8}}{\frac{3l}{4}} = \frac{Mg}{6}$$

$$T + N = mg \cos 60^\circ$$

$$N = \frac{Mg}{2} - \frac{Mg}{6} = \frac{2Mg}{6} = \frac{Mg}{3}$$

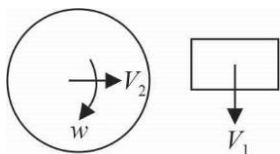
$$f = Mg \sin 60^\circ$$

$$f = \frac{\sqrt{3}Mg}{2}$$



58.(3) Conserving energy

$$-mg(y) + \frac{1}{2}mv_1^2 + \frac{1}{2}mv_2^2 + \frac{1}{2}\left(\frac{2}{5}mR^2\right)\omega^2 = 0.$$



$$v_1 = v_2 + \omega R$$

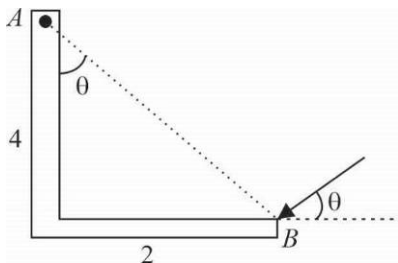
$$v_2 = \omega R$$

Solving we get,

$$v_1 = \sqrt{\frac{10}{27}gy}$$

Hence $x = 3$

59.(2)



In order to have maximum torque about A the force must be perpendicular to AB . Hence $\tan \theta = \frac{2}{4} = \frac{1}{2}$

60.(4) As hinges are smooth the disc continues to rotate at ω so by energy conservation

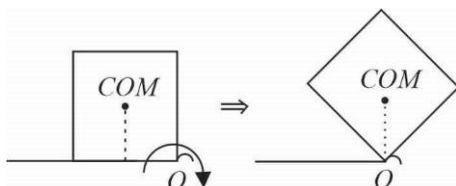
$$\Delta U_{rod} + \Delta U_{disc} + \Delta K_{rod} + \Delta K_{disc} = 0$$

$$mgR(1 - \cos 60^\circ) + mg2R(1 - \cos 60^\circ) + \left(-\frac{1}{2}\frac{m(2R)^2}{3}\omega^2\right) + \left(-\frac{1}{2}m(2R\omega)^2\right) = 0$$

$$\Rightarrow \frac{1}{2}\left(\frac{1}{3}m(2R)^2\right)\omega^2 + \frac{1}{2}m(2R\omega)^2 = (mg(2R) + mgR)\frac{1}{2}$$

$$\Rightarrow 2R^2\omega^2 + \frac{2R^2\omega^2}{3} = \frac{3gR}{2} \Rightarrow \frac{8R\omega^2}{3} = \frac{3g}{2} \Rightarrow \omega = \sqrt{\frac{9g}{16R}}$$

61.(4) $MV\left(\frac{a}{2}\right) = l_0\omega = \left[\frac{Ma^2}{6} + M\left(\frac{a}{\sqrt{2}}\right)^2\right]\omega$



$$= \frac{2}{3}Ma^2\omega$$

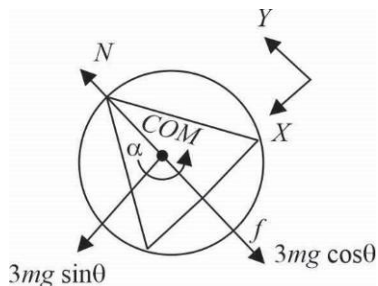
$\therefore$

$$\omega = \frac{3V}{4a}$$

62.(7)

$$\sum F_x = ma$$

$$\Rightarrow 3mg \sin \theta - f = 3ma$$



$$\sum F_y = 0 \quad \Rightarrow \quad N - 3mg \cos \theta = 0$$

$$\sum \tau = fR = I\alpha$$

For no slipping $a = R\alpha$

The moment of inertial of the assembly about its centre of mass is $I = \frac{3}{2}mR^2$

Now, on solving Eqs (i), (ii), (iii) and (iv) simultaneously, we obtain

$$f = mg \sin \theta$$

If μ is the coefficient of friction at the contact surface, then

$$f \leq \mu N$$

$$\text{or} \quad f \leq \mu \times 3mg \cos \theta \quad \text{or} \quad mg \sin \theta \leq 3\mu mg \cos \theta$$

$$\text{or} \quad \mu \geq \frac{1}{3} \tan \theta \quad \therefore \quad \mu_{\min} = \frac{1}{3} \tan \theta = \frac{x}{y} \tan \theta$$

$$x = 1 \quad \therefore x + 2y = 7$$

$$y = 3$$

63.(7) Case 1

$$mgh = \frac{1}{2}mv_1^2 + \frac{1}{2}I\left(\frac{V_1}{R}\right)^2 \Rightarrow V_1 = \sqrt{\frac{2mgh}{m + \frac{I}{R^2}}} = V_0 \quad \dots(i)$$

Case 2

$$mgh = \frac{1}{2}mv_2^2 \Rightarrow V_2 = \sqrt{2gh} = \frac{5V_0}{4} \quad \dots(ii)$$

From (i) & (ii)

$$I = \frac{9}{16}mR^2 \Rightarrow k = \frac{3R}{4}$$

64.(2) At point P,

Decrease in GPH = increase in $(K)_T$ + Increase in $(K)_R$

$$mg \cdot 7R = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 = \frac{1}{2}mv^2 + \frac{1}{2}\left(\frac{2}{5}mr^2\right) \cdot \frac{V^2}{r^2}$$

$$= \frac{7}{10}mv^2 \quad (r \text{ is radius of ball})$$

$$\therefore v^2 = 10gR$$

Centripetal force is $\frac{mv^2}{R} = \frac{m(10gR)}{R} = 10mg = 10(0.1)(10)$
 $= 10N \Rightarrow x = 2$

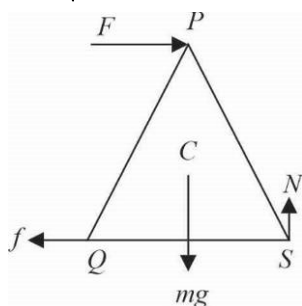
- 65.(6) Let m be the mass of the block, l be length of each side and μ be the coefficient of friction, then the block will not slip if

$$F < \mu mg$$

The block will start toppling when torque of F about $S \geq$ torque of mg about S .

or $F \frac{\sqrt{3}}{2} l \geq mg \cdot \frac{l}{2}$

$$\Rightarrow F \geq \frac{1}{\sqrt{3}} mg$$



Hence, the block will topple without slipping.

$$\text{If } \mu mg \geq \frac{1}{\sqrt{3}} mg \text{ or } \mu \geq \frac{1}{\sqrt{3}}$$

- 66.(3) Torque of weight about hinge P is

$$mg \frac{l}{2} = I_P \alpha \quad \dots(1)$$

$$\text{Now } I_P = \frac{ml^2}{3} + \left\{ \frac{ml^2}{12} + m \left(\frac{\sqrt{3}l}{2} \right)^2 \right\}$$

$$I_P = \frac{5ml^2}{3} \quad \dots(2)$$

$$\therefore \text{Solving 1 \& 2 we get } \alpha = \frac{3g}{10l} \quad \therefore x = 3$$

- 67.(1) $mg \sin \theta - f = ma$

$$fR = \frac{mR^2}{2} \alpha$$

$$a = \alpha R$$



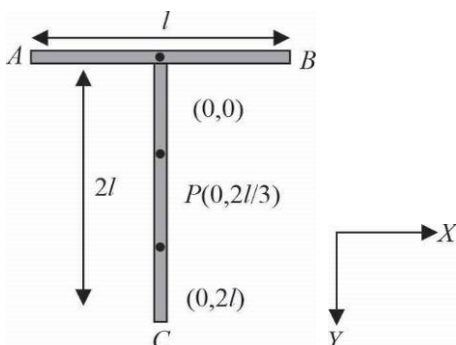
Solving the above equations, we get

$$\frac{mg \sin \theta}{3} = \mu mg \cos \theta \Rightarrow \frac{\tan \theta}{3} = \frac{2-3x}{\sqrt{3}}$$

$$\Rightarrow x = \frac{1}{3}m = \frac{P}{3} \quad \therefore P = 1$$

- 68.(7) For pure translator motion, net torque about center of mass should be zero.
Thus, F is applied at center of mass of system.

$$OP = \frac{m_1 \times O + m_2 \times l}{m_1 + m_2}$$



Where, m_1 and m_2 are masses of horizontal and vertical section of the object. Assuming object is uniform.

$$m_2 = 2m_1 \quad \Rightarrow \quad OP = \frac{2l}{3}$$

$$PC = \left(l - \frac{2l}{3} + 1 \right) = \left(2l - \frac{2l}{3} \right) = \frac{4l}{3}$$

69.(5)
$$I_0 = \frac{MR^2}{2} + 6 \left[\frac{MR^2}{2} + M(2R)^2 \right] = MR^2 \left[\frac{1}{2} + 3 + 24 \right] = \frac{55}{2} MR^2$$

O is the center of mass of the system. Applying parallel axis theorem between O & P .

Moment of inertia about point P

$$I_p = I_0 + 7M(R)^2$$

$$= \frac{55MR^2}{2} + 7MR^2 = \frac{69MR^2}{2}$$

Hence $12x + 9 = 69$

$x = 5$

Gravitation

- 1.(A) Energy required to move a satellite from radius r_1 to r_2 is given by

$$\Delta E = \frac{GMm}{2} \left[\frac{1}{r_1} - \frac{1}{r_2} \right] = \frac{GMm}{2} \left(\frac{1}{2R} - \frac{1}{3R} \right) = \frac{GMm}{12R}$$

- 2.(B) When gravitational force becomes zero, the centripetal force required cannot be provided. So the satellite will move with the velocity as it has at the instant when gravitational force becomes zero, i.e., it moves tangentially to the original orbit.

- 3.(A) Escape velocity $= \sqrt{2gR_e}$. So the escape velocity is independent of m . So it depends upon mass as m^0 .

- 4.(C) The minimum kinetic energy required to project a body of mass m from the earth's surface to infinite is known as escape energy. Therefore,

$$KE = \frac{GM_e m}{R} = mgR \left(\because gR = \frac{GM_e}{R} \right)$$

5.(C) $\frac{T_2}{T_1} = \left(\frac{r_2}{r_1} \right)^{3/2} = \left(\frac{4r}{r} \right)^{3/2} = 8 \quad \therefore T_2 = 8T_1 = 8 \times 5 = 40h$

- 6.(C) If x_1 and x_2 are the distance covered by the two bodies, then $x_1 + x_2 = 9R$

Also $Mx_1 = 5Mx_2 \Rightarrow x_2 = \frac{x_1}{5} \quad \therefore x_1 + \frac{x_1}{5} = 9R \Rightarrow x_1 = 7.5R$

- 7.(C) Work is done against gravitational force, which is a conservative force. Therefore, the escape velocity is independent of the angle of projection.

8.(B) $v_0 = \sqrt{\frac{GM}{R+x}} = \sqrt{\frac{gR^2}{R+x}}$

9.(A) The time period of the satellite is given by $T = 2\pi \sqrt{\frac{(R+h)^3}{GM}}$

Where $R+h$ is the radius of the orbit of the satellite, M is mass of the earth.

10.(B) $\Delta U = \frac{mgh}{1 + \frac{h}{R}} = \frac{1}{2} mgR \quad (\because h = R)$

11.(A) $m\omega^2 R \propto \frac{1}{R^n} \Rightarrow m \left(\frac{4\pi^2}{T^2} \right) R \propto \frac{1}{R^n} \Rightarrow T^2 \propto R^{n+1} \quad \therefore T \propto R^{\left(\frac{n+1}{2} \right)}$

12.(C) $g = \frac{GM}{R^2} = \frac{G}{R^2} \times \frac{4}{3} \pi R^3 \rho = \frac{4}{3} \pi R \rho G \Rightarrow \rho = g$

- 13.(C) At height h above the surface of the earth.

$$g' = g \left(1 - \frac{2h}{R} \right) \Rightarrow \Delta g_1 = \frac{2h}{R} g$$

At depth d below the surface of the earth,

$$g' = g \left(1 - \frac{d}{R} \right) \Rightarrow \Delta g_2 = \frac{d}{R} g$$

Since $\Delta g_1 = \Delta g_2$

$$d = 2h$$

- 14.(D) Work done = Change in GPE = $U_\infty - U_R$

$$W = 0 - \left(\frac{GMm}{R} \right) = \frac{GMm}{R} = \frac{6.67 \times 10^{-11} \times 100 \times 10^{-2}}{10^{-1}} = 6.67 \times 10^{-10} J$$

- 15.(D) Electronic charge = $1.6 \times 10^{-19} C$

It is the fundamental property of an electron. Its value remains the same on the moon as well as on the earth.

$$\therefore \frac{\text{Electronic charge on moon}}{\text{Electronic charge on earth}} = 1$$

- 16.(C) Mass of planet

$$M_P = 10M_e$$

Where M_e is the mass of the earth. Radius of planet $R_P = \frac{R_e}{10}$;

Where R_e is the radius of the earth.

Using formula $v_{planet} = \sqrt{\frac{2GM_P}{R_P}} = \sqrt{\frac{2G(10M_e)}{(R_e/10)}} = 10v_{earth}$

17.(A) $g' = \frac{GM}{(R+h)^2}$, acceleration due to gravity at height h

$$\Rightarrow \frac{g}{9} = \frac{GM}{R^2} \frac{R^2}{(R+h)^2} = g \left(\frac{R}{R+h} \right)^2 \Rightarrow \frac{1}{9} = \left(\frac{R}{R+h} \right)^2 \Rightarrow \frac{R}{R+h} = \frac{1}{3}$$

$$\Rightarrow 3R = R+h \Rightarrow 2R = h$$

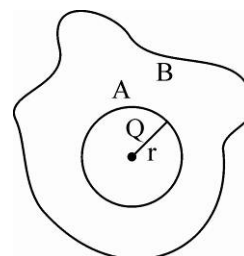
Using formula $v_e = \sqrt{\frac{2GM}{R}}$
 $v = 10$

18.(C) For solving this problem let us use the concept of electric flux. Let A be the Gaussian surface enclosing a spherical charge Q .

Flux enclosed by Gaussian surface

$$\phi = \oint \vec{E} \cdot \vec{ds} = \frac{Q_{enclose}}{\epsilon_0} \Rightarrow E \cdot 4\pi r^2 = \frac{Q}{\epsilon_0}$$

Hence electric field at a distance a from charge is $E = \frac{Q}{4\pi\epsilon_0 r^2}$



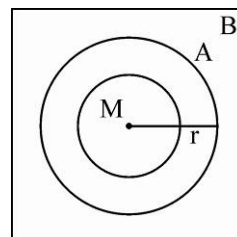
Flux $\phi = \vec{E} \cdot 4\pi r^2 = \frac{Q}{\epsilon_0}$

Every line passing through A , has to pass through B , whether B is a cube or any surface. It is only for Gaussian surface, the lines of field should be normal. Assuming the mass is a point mass.

Gravitational field $g = -\frac{GM}{r^2}$

Flux $\phi_g = |\vec{g} \cdot 4\pi r^2| = \frac{4\pi r^2 \cdot GM}{r^2} = 4\pi GM$

Hence B is a cube. As explained earlier, whatever be the shape, all the lines passing through A are passing through B , although all the lines are not normal.



Statement 2 is correct because when the shape of the earth is spherical, area of the Gaussian surface is $4\pi r^2$.

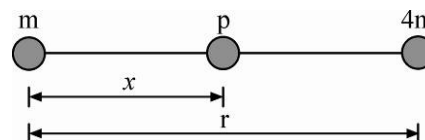
This ensures inverse square law.

19.(D) Let x be the distance of the point P from the mass m where gravitational field is zero.

$$\therefore \frac{Gm}{x^2} = \frac{G(4m)}{(r-x)^2} \text{ or } \left(\frac{x}{(r-x)} \right)^2 = \frac{1}{4}$$

Or $x = \frac{r}{3}$

.....(i)



Gravitational potential at a point P is :

$$V = -\frac{Gm}{x} - \frac{G(4m)}{(r-x)}$$

$$V = -\frac{Gm}{\left(\frac{r}{3}\right)} - \frac{G(4m)}{\left(r - \frac{r}{3}\right)} = -9\frac{Gm}{r} \quad (\text{Using (i)})$$

$$20.(D) \quad W = 0 - \left(-\frac{GMm}{R}\right) = \frac{GMm}{R} = m\left(\frac{GM}{R^2}\right)R = mgR = 1000 \times 10 \times 6400 \times 10^3 = 64 \times 10^9 J = 6.4 \times 10^{10}$$

21.(D) Energy of the satellite on the surface of the planet is

$$E_1 = KE + PE = 0 + \left(-\frac{GMm}{R}\right) = -\frac{GMm}{R} = \frac{1}{2} \frac{GMm}{3R} - \frac{GMm}{3R} = -\frac{GMm}{6R}$$

∴ Minimum energy required to launch the satellite is :

$$\Delta E = E_f - E_i = -\frac{GMm}{6R} - \left(-\frac{GMm}{R}\right) = -\frac{GMm}{6R} + \frac{GMm}{R} = \frac{5GMm}{6R}$$

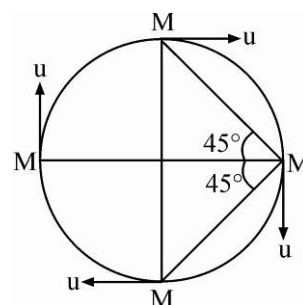
22.(B) Net force on any one particle

$$= \frac{GM^2}{(2R)^2} + \frac{GM^2}{(R\sqrt{2})^2} \cos 45^\circ + \frac{GM^2}{(R\sqrt{2})^2} \cos 45^\circ = \frac{GM^2}{R^2} \left[\frac{1}{4} + \frac{1}{\sqrt{2}} \right]$$

This force will be equal to centripetal force so

$$\frac{Mu^2}{R} = \frac{GM^2}{R^2} \left[\frac{1 + 2\sqrt{2}}{4} \right]$$

$$u = \sqrt{\frac{GM}{R} [1 + 2\sqrt{2}]} = \frac{1}{2} \sqrt{\frac{GM}{R} (2\sqrt{2} + 1)}$$



23.(B) $V = V_1 - V_2$

$$V_1 = -\frac{GM}{2R^3} \left[3R^2 - \left(\frac{R}{2}\right)^2 \right]; \quad V_2 = -\frac{3G\left(\frac{M}{8}\right)}{2\left(\frac{R}{2}\right)} \Rightarrow \quad V = \frac{-GM}{R}$$

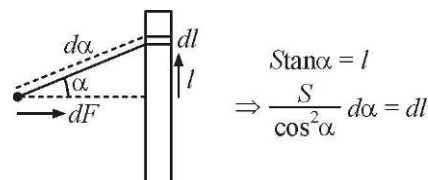
INTEGER VALUE TYPE

$$24.(2) \quad dm = D \times dl = D \times \frac{S d\alpha}{\cos \alpha}$$

$$\text{Gravitational force, } dF = \frac{GMdm}{\left(\frac{S}{\cos \alpha}\right)^2} \cos \alpha$$

$$\text{Total force} \quad F = \int_{-\pi/2}^{\pi/2} \frac{MGD}{S} \cos \alpha d\alpha$$

$$= \frac{2MGD}{S}$$



$$S \tan \alpha = l$$

$$\Rightarrow \frac{S}{\cos^2 \alpha} d\alpha = dl$$

$$25.(8) \quad dF = \frac{GMdm}{(3R)^2}; \quad F = \Sigma dF \cos \theta \quad \Rightarrow \quad F = \frac{GMm2\sqrt{2}}{27R^2} \quad \left(\text{as } \cos \theta = \frac{2\sqrt{2}}{3} \right)$$

$$26.(7) \quad V = \frac{-G4M}{\frac{3}{2}R} - \frac{MG}{2R} = -\frac{19}{6} \frac{GM}{R}$$

27.(2) Let v be the speed of the projectile at highest point and $r_{\max}$ its distance from the centre of the earth. Applying conservation of angular momentum and mechanical energy.

$$mv_0 \sin \alpha + mvr_{\max} \quad \dots(i)$$

$$\frac{1}{2}mv_0^2 - \frac{GM_em}{R_e} = \frac{1}{2}mv^2 - \frac{GM_em}{r_{\max}} \quad \dots(ii)$$

Solving these two equations with given data, we get

$$r_{\max} = \frac{3R_e}{2} \quad \text{or,} \quad \text{the maximum height } h_{\max} = r_{\max} - R_e = \frac{R_e}{2}$$

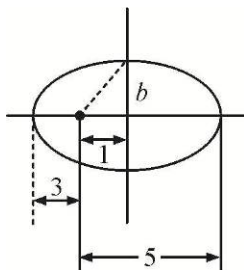
$$28.(6) L_A = 2.2M_s r_1^2 \omega; L_B = 11M_s r_2^2 \omega$$

$$\begin{aligned} \frac{L}{L_B} &= \frac{2.2M_s r_1^2 + 11M_s r_2^2}{11M_s r_2^2} = 1 + \frac{2.2}{11} \left(\frac{r_1}{r_2} \right)^2 \\ &= 1 + \frac{2.2}{11} \times \left(\frac{11}{2.2} \right)^2 \quad (\because m_1 r_1 = m_2 r_2) \end{aligned}$$

$$= 6$$

- 29.(4) From given information, semi-major axis is equal to 4 units. Let e be the eccentricity of ellipse, then $ae = 1 \Rightarrow e = 1/4$

Semi-major axis,



$$\begin{aligned} b &= a\sqrt{1-e^2} = 4\sqrt{1-\frac{1}{16}} \text{ units} \\ &= \sqrt{15} \text{ units} \end{aligned}$$

$$\begin{aligned} \text{So required distance} &= \sqrt{b^2 + 1^2} \\ &= 4 \text{ units} \end{aligned}$$

$$30.(2) F_1 = \frac{DMm}{(2R)^2}, F_2 = F_1 - F' = \frac{GMm}{(2R)^2} - \frac{GMm}{4 \times \left(R + \frac{R}{2} \right)^2}$$

$$\text{Now, } \frac{F_2}{F_1} = \left(\frac{7}{9} \right)$$

- 31.(2) For a particle at a distance r from the centre of earth, force is given by,

$$F = \frac{Gm_1m_2}{r^2}$$

Force becomes one-fourth, when $r = 2R$

(R = radius of earth)

$$\text{Escape velocity, } v_e = \sqrt{\frac{2GM}{R}}$$

Using conservation of energy for the given particle,

$$\frac{1}{2}mv^2 - \frac{GmM}{R} = -\frac{GmM}{2R}$$

This gives, $v = \sqrt{\frac{GM}{R}}$

$\Rightarrow v_e = v\sqrt{2}$

Hence, $N = 2$

32.(7) From FBD of middle mass:

$$\frac{GMm}{9l^2} - \frac{Gm^2}{l^2} - T = ma$$

Since $T = 0$, $\frac{GMm}{9l^2} - \frac{Gm^2}{l^2} = ma \quad \dots(i)$

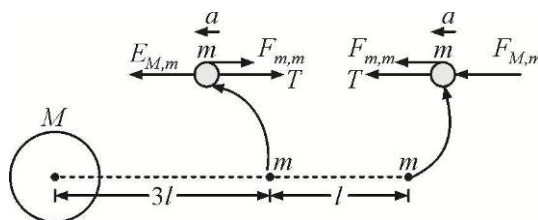
Also, for the right most mass:

$$\frac{GMm}{16l^2} + \frac{Gm^2}{l^2} = ma \quad \dots(ii)$$

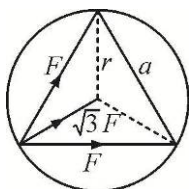
Equating (i) and (ii)

$$\frac{GMm}{9l^2} - \frac{Gm^2}{16l^2} = \frac{2Gm^2}{l^2}$$

Solving $m = \frac{7M}{288}$

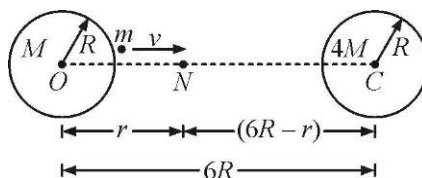


33.(9)



$$\sqrt{3}F = \frac{mv^2}{r} \quad \Rightarrow \quad v = \left[\frac{\sqrt{3}Fr}{M} \right]^{1/2} \quad \text{Where } F = \frac{Gm^2}{a^2}$$

34.(6) Let the projectile of mass m be fired with minimum velocity, v from the surface of sphere of mass M to reach the surface of mass $4M$. Let N be neutral point at a distance r from the centre of the sphere of mass M . At neutral point N .



$$\frac{GMm}{r^2} = \frac{G(4M)}{(6R-r)^2}$$

$$(6R-r)^2 = 4r^2$$

$$6R-r = \pm 2r \quad \text{or} \quad r = 2R \quad \text{or} \quad -6R$$

The point $r = -6R$ does not concern us.

Thus, $ON = r = 2R$

It is sufficient to project the projectile with a speed which would enable it to reach N . Thereafter, the greater gravitational pull of $4M$ would suffice.

The mechanical energy of m at the surface of M is

$$E_i = \frac{1}{2}mv^2 - \frac{GMm}{R} - \frac{G(4M)m}{5R}$$

At the neutral point N , the speed approaches zero

∴ The mechanical energy at N is

$$E_N = -\frac{GMm}{2R} - \frac{G(4M)m}{4R} = -\frac{GMm}{2R} - \frac{GMm}{R}$$

According to law of conservation of mechanical energy,

$$E_i = E_N$$

$$\frac{1}{2}mv^2 - \frac{GMm}{R} - \frac{4GMm}{5R} = -\frac{GMm}{2R} - \frac{GMm}{R}$$

$$v^2 = \frac{2GM}{R} \left[\frac{4}{5} - \frac{1}{2} \right] = \frac{3}{5} \frac{GM}{R} \quad \text{or} \quad v = \left(\frac{3}{5} \frac{GM}{R} \right)^{1/2}$$

35.(6) Here, mass of the particle = M

Mass of the spherical shell = M

Radius of the spherical shell = R

Let O be centre of spherical shell

Gravitational potential at point P due to particle at O is

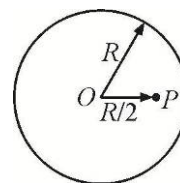
$$V_1 = -\frac{GM}{(R/2)}$$

Gravitational potential at point P due to spherical shell is

$$V_2 = -\frac{GM}{R}$$

Hence, total gravitational potential at the point P is

$$V = V_1 + V_2 = -\frac{GM}{(R/2)} + \left(-\frac{GM}{R} \right) = -\frac{2GM}{R} - \frac{GM}{R} = -\frac{3GM}{R}$$



36.(2) Removed mass

$$m = \frac{M}{\frac{4}{3}\pi R^3} \times \frac{4}{3}\pi \left(\frac{R}{2} \right)^3 = \frac{M}{8}$$

Gravitational potential at centre of cavity is

$$V_P = V_{\text{sphere } P} - V_{mP} \quad \dots(i)$$

Due to a solid sphere gravitational potential at its interior point is given as

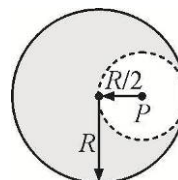
$$V_{\text{in}} = -\frac{GM}{2R^3} (3R^2 - x^2)$$

⇒ at $P \left(x = \frac{R}{2} \right)$ so we have

$$V_{\text{in}} = -\frac{GM}{2R^3} \left(3R^2 - \frac{R^2}{4} \right)$$

$$\Rightarrow V_{\text{in}} = -\frac{11}{8} \frac{GM}{R}$$

Gravitational potential due to removed mass m at centre of cavity is



$$V_{Pm} = -\frac{3}{2} \left(\frac{GM}{(R/2)} \right) = -\frac{3}{2} \frac{G(M/8)}{(R/2)}$$

$$V_{Pm} = -\frac{3}{8} \frac{GM}{R}$$

From equation (i)

$$V_P = \left(-\frac{11}{8} \frac{GM}{R} \right) - \left(-\frac{3}{8} \frac{GM}{R} \right) \Rightarrow V_P = -\frac{GM}{R}$$

37.(5) For EOM elliptical orbit if time period of motion in this orbit is T then we use

$$T^2 \propto \left(\frac{a_m + a_e}{2} \right)^3 \quad \dots(i)$$

Semi-major axis of elliptical orbit for earth if T_e be the period around sun then we use

$$T_e^2 \propto a_e^3 \quad \dots(ii)$$

Here (i)/(ii) gives

$$\frac{T^2}{T_e^2} = \left[\frac{(a_m + a_e)/2}{a_e} \right]^3$$

$$T = \left[\frac{(a_m + a_e)/2}{a_e} \right]^{3/2} T_e$$

$$T = \left[\frac{(1.5 + 2.28)/2}{1.5} \right]^{3/2} \times 365 \text{ days}$$

$$T = \left(\frac{1.89}{1.5} \right)^{3/2} \times 365 \text{ days}$$

Time of motion in half ellipse as described (path EOM) is given as

$$t = \frac{T}{2} = \left[\frac{1.89}{1.5} \right]^{3/2} \times \frac{365}{2} \cong 258 \text{ days}$$

38.(1) If λ be the linear mass density along the length of galaxy then gravitational field at the location of star is given as

$$g_P = \frac{2G\lambda}{r}$$

For circular motion of star, we use

$$mg_P = \frac{mv^2}{r}$$

$$\Rightarrow \frac{2G\lambda}{r} = \frac{v^2}{r}$$

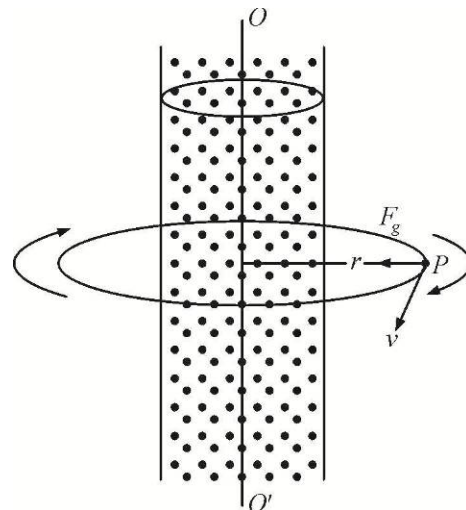
$$\Rightarrow v = \sqrt{2G\lambda} \quad \text{orbital speed of star}$$

Period of revolution of star in circular orbit is

$$T = \frac{2\pi r}{v}$$

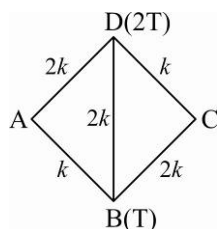
$$\Rightarrow T = \frac{2\pi r}{\sqrt{2G\lambda}}$$

$$\Rightarrow T \propto r$$



Properties of Matter

1-3.



1.(A) 2.(C)

$$\frac{2k(2T - T_A)S}{\ell} = \frac{k(T_A - T)S}{\ell} \Rightarrow T_A = \frac{5}{3}T$$

$$\frac{k(2T - T_C)S}{\ell} = \frac{2k(T_C - T)S}{\ell} \Rightarrow T_C = \frac{4}{3}T$$

3.(D) $H = \frac{2k TS}{\ell}$

4.(B) $\varepsilon = CT \quad \therefore \text{Temperature is halved} \quad \therefore \text{emissivity is halved} \quad \therefore H = \frac{dQ}{dt} = \left(\frac{1}{2}\right)^5 \times 160 = 5 \text{ W}$

5.(B) Specific heat increases with temp.

For every 1°C drop in temperature of the hotter block more than 1°C will be gained by the colder block

Final temperature will be greater than 50°C

6.(C) The vertical component of tension of both the string should balance weight of the frame

$\therefore$ Tension will be least in (ii)

7.(B) $\therefore$ Area of wire B is greater than wire A.

For stress (F/A) to be equal for both means tension in wire having greater cross section should be more.

8.(A) $T_1 = mg \Rightarrow \Delta \ell_1 = \ell \frac{mg/A}{Y}$

$T_2 = \frac{mg}{\cos \theta} \Rightarrow \Delta \ell_2 = \ell \frac{mg/A \cos \theta}{Y} \quad \therefore \Delta \ell_2 = \frac{\Delta \ell_1}{\cos \theta}$

9.(B) $\lambda T = \text{constant}$

$Q \propto AT^4$

$Q_A : Q_B : Q_C = \frac{4\pi R^2}{(3000)^4} : \frac{4\pi (2R)^2}{(4000)^4} : \frac{4\pi (3R)^2}{(5000)^4}$

10.(A) By Hooke's law,

Young's modulus (Y) = $\frac{\text{Tensile stress}}{\text{Tensile strain}}$

So, $Y = \frac{F/A}{\Delta L/L} = \frac{FL}{A\Delta L}$

If the rod is compressed, then compressive stress and strain appear. Their ratio Y is same as that for tensile case.

Given, length of a steel wire (CL) = 10 cm

Temperature (t) = 100°C

As length is constant.

$\therefore \text{Strain} = \frac{\Delta L}{L} \alpha \Delta \theta \quad \left[\text{as } \Delta L = (L\Delta \theta) \right]$

Now, pressure = stress = $Y \times \text{strain}$

[Given $Y = 2 \times 10^{11} \text{ N/m}^2$ and $\alpha = 1.1 \times 10^{-5} \text{ K}^{-1}$]

$= 2 \times 10^{11} \times 1.1 \times 10^{-5} \times 100 = 2.2 \times 10^8 \text{ Pa}$

11.(A) $L_{\text{vap}} = \frac{1000 \times 900}{200} \text{ J/gm} = 4.5 \times 10^3 \text{ J/gm} = 4.5 \times 10^6 \text{ J/kg}$

- 12.(AD) ∴ Change in volume is zero for ideal liquid
 ∴ Bulk modulus becomes infinite
 For very small shear stress ≈ 0
 Shear strain is very high (shape changes)
 ∴ Shear modulus = 0

- 13.(AD) ∴ Stress = F/A {Same for both} & Strain = $\frac{\text{stress}}{Y}$ {Different for both}

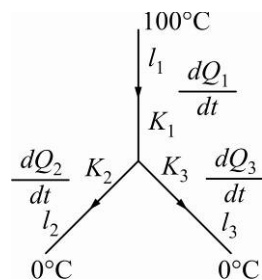
14.(C) Then $\frac{\Delta Q}{\Delta t} = \frac{KA(T_1 - T_2)}{x}$

According to thermal conductivity, we get $\frac{dQ_1}{dt} = \frac{dQ_2}{dt} + \frac{dQ_3}{dt}$

$$\frac{0.92(100 - T)}{46} = \frac{0.26(T - 0)}{13} + \frac{0.12(T - 0)}{12}$$

$$\Rightarrow T = 40^\circ\text{C}$$

$$\therefore \frac{dQ_1}{dt} = \frac{0.92 \times 4(100 - 40)}{46} = 4.8 \text{ cal/s}$$



- 15.(D) Strain is same in both (A) and (B).

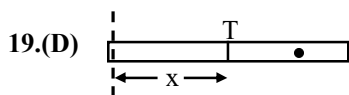
$$\text{Energy density} = \frac{1}{2}(\text{Stress}) \times (\text{Strain}) = \frac{1}{2}Y(\text{Strain})^2$$

Since Y is different, so energy density is different

- 16.(D) As soon as water temp drops below 4°C colder water goes to the top surface and will freeze first.

17.(C) $\Delta \ell$ in upper half = $\int_{\ell/2}^{\ell} \frac{mgx}{\ell} \frac{dx}{AY} = \frac{3mg\ell}{8AY}$

18.(A) $T = m\omega^2 \frac{\ell}{2}$



$$T = \frac{m(\ell - x)}{\ell} \omega^2 \left(\frac{\ell + x}{2} \right) = \frac{m\omega^2}{2\ell} (\ell^2 - x^2)$$

$$\Rightarrow \Delta \ell = \int_0^{\ell} \frac{m\omega^2}{2\ell} \frac{(\ell^2 - x^2)}{AY} dx = \frac{m\omega^2}{2AY} \left(\ell^3 - \frac{\ell^3}{3} \right) = \frac{m\omega^2 \ell^2}{3AY} = \frac{mg\ell}{2AY} \Rightarrow \omega = \sqrt{\frac{3g}{2\ell}}$$

- 20.(C) The various forces acting on the cylinder are shown in the figure.

Let x_0 be the extension in spring

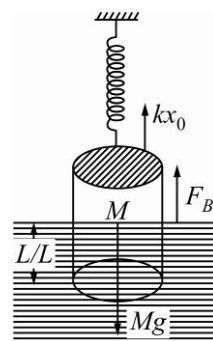
At equilibrium,

$$\text{Upward forces} = \text{Downward forces} \quad kx_0 + F_B = mg$$

where, k is spring constant, F_B is force due to buoyancy which is equal to weight of liquid displaced.

$$F_B = \sigma \cdot \frac{L}{2} \cdot A \cdot g$$

$$\therefore kx_0 + \sigma \frac{L}{2} Ag = Mg \Rightarrow x_0 = \frac{Mg - \frac{\sigma LA g}{2}}{k} = \frac{Mg}{k} \left(1 - \frac{\sigma LA}{2M} \right)$$



- 21.(C) According to Newton's law of cooling, the rate of loss of heat is directly proportional to the difference of temperature of the body and surroundings.

$$\frac{dQ}{dt} = k(T_2 - T_1) \quad \dots(i)$$

and $dQ = msdT_2$

$$\therefore \frac{dQ}{dt} = \frac{msdT_2}{dt} \quad \dots(ii)$$

From equation (i) and (ii), we get :

$$-\frac{msdT_2}{dt} = k(T_2 - T_1) \Rightarrow \frac{dT_2}{T_2 - T_1} = -\frac{k}{ms} dt = -K dt \quad \left[\because K = \frac{k}{ms} \right]$$

On integrating, we get :

$$\log_0(T_2 - T_1) = -kt + C \quad \text{or} \quad T_2 = T_1 + C'e^{-kt} \quad [\text{where, } C' = e^C]$$

22.(B) $T = m\omega^2 \ell \Rightarrow \omega = \sqrt{\frac{T}{m\ell}} \quad \text{and} \quad T = \text{breaking stress} \times \text{area}$

$$= 4.8 \times 10^7 \times 10^{-6} = 48 N \quad \Rightarrow \quad \omega = \sqrt{\frac{48}{10(0.3)}} = 4 \text{ rad/s}$$

23.(C) Heat required to change the temperature of vessel by a small amount dT , $-dQ = mC_p dT$

$$\text{Total heat required } -Q = m \int_{20}^4 32 \left(\frac{T}{400} \right)^3 dT = \frac{100 \times 10^{-3} \times 32}{(400)^3} \left[\frac{T^4}{4} \right]_{20}^4 \Rightarrow Q = 0.001996 \text{ kJ}$$

Work done required to maintain the temperature of sink to T_2 ,

$$W = Q_1 - Q_2 = \frac{Q_1 - Q_2}{Q_2} Q_2 = \left(\frac{T_1}{T_2} - 1 \right) Q_2 \Rightarrow W = \left(\frac{T_1 - T_2}{T_2} \right) Q_2$$

For $T_2 = 4 \text{ K}$: $W_1 = \frac{300 - 20}{20} \times 0.001996 = 0.028 \text{ kJ}$

and For $T_2 = 4 \text{ K}$: $W_2 = \frac{300 - 4}{4} \times 0.001996 = 0.148 \text{ kJ}$

As temperature is changing from 20 K to 4 K, work done required will be more than W_1 but less than W_2 .

24.(A) As $F = -kx$

$$k \propto \frac{1}{x} \propto \frac{1}{l}$$

For spring, $k \propto \frac{1}{l} \quad \therefore \quad \frac{k_A}{k} = \frac{l}{l_A} \Rightarrow k_A = \frac{l_A + l_B}{l_A} k = \frac{5}{2} k$

25.(B) Frequency, $f = \frac{v}{2l} = \frac{1}{2l} \sqrt{\frac{T}{\mu}} = \frac{1}{2l} \sqrt{\frac{T}{Ad}}$

Where, T is tension in the wire and μ is the mass per unit length of wire.

Also, Young's modulus, $Y = \frac{Tl}{A\Delta l} \Rightarrow \frac{T}{A} = \frac{Y\Delta l}{l}$

Putting this value in expression of frequency, we have $f = \frac{1}{2} \sqrt{\frac{Y\Delta l}{ld}}$ Given, $l = 1.5 \text{ m}$, $\frac{\Delta l}{l} = 0.01$

$$d = 7.7 \times 10^3 \text{ kg/m}^3, Y = 2.2 \times 10^{11} \text{ N/m}^2$$

Putting these values we, have $f = \frac{1}{2l} \sqrt{\frac{2.2 \times 10^{11} \times 0.01}{7.7 \times 10^3}} \Rightarrow f = \sqrt{\frac{2}{7}} \times \frac{10^3}{3} \Rightarrow f = 178.2 \text{ Hz}$

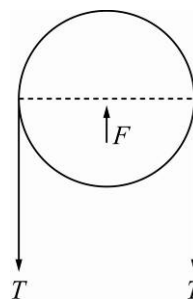
- 26.(D) Elongation due to change in temperature, $\Delta l = L\alpha\Delta T$

Which is compensated by elastic strain, when temperature becomes normal,

$$i.e., \quad \Delta l = \frac{TL}{YS}$$

$$\text{Thus,} \quad \frac{TL}{YS} = L\alpha\Delta T \quad \Rightarrow \quad T = VS\alpha\Delta T$$

At equilibrium, force exerted by one half on other, $F = 2T = 2YS\alpha\Delta T$



- 27.(D) As volume is same.

$$\therefore A_1 l_1 = A_2 l_2$$

$$\Rightarrow l_2 = \frac{A_1 l_1}{A_2} = \frac{A_1 \times l_1}{3A} = \frac{l_1}{3} \quad \Rightarrow \quad \frac{l_1}{l_2} = 3$$

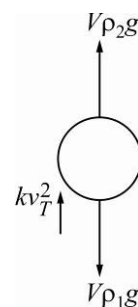
$$\therefore \Delta x_1 = \frac{F_1}{A\gamma} l_2 \quad \text{and} \quad \Delta x_2 = \frac{F_2}{3A\gamma} l_2$$

$$\text{Here,} \quad \Delta x_1 = \Delta x_2$$

$$\therefore \frac{F_2}{3A\gamma} l_2 = \frac{F_1}{A\gamma} l_1 \quad \Rightarrow \quad F_2 = 3F_1 \times \frac{l_1}{l_2} = 3F_1 \times 3 = 9F$$

- 28.(A) The force acting on the ball are gravity force, buoyancy force and viscous force. When ball acquires terminal speed, it is in dynamic equilibrium, let terminal speed

$$\text{of ball } v_T. \text{ So, } V\rho_2 g + kv_T^2 = V\rho_1 g \quad \text{or} \quad v_T = \sqrt{\frac{V(\rho_1 - \rho_2)g}{k}}$$



- 29.(D) Let ρ_0 and ρ_T be densities of glycerin at $0^\circ C$ and $T^\circ C$ respectively. then, $\rho_T = \rho_0(1 - \gamma\Delta T)$ Where γ is the coefficient of volume expansion of glycerine and ΔT is rise in temperature.

$$\frac{\rho_T}{\rho_0} = 1 - \gamma\Delta T \quad \text{or} \quad \gamma\Delta T = 1 - \frac{\rho_T}{\rho_0}$$

$$\text{Thus,} \quad \frac{\rho_0 - \rho_T}{\rho_0} = \gamma\Delta T$$

$$\text{Here,} \quad \gamma = 5 \times 10^{-4} K^{-1} \quad \text{and} \quad \Delta T = 40^\circ C = 40K$$

$$\therefore \text{The fractional change in the density of glycerin} = \frac{\rho_0 - \rho_T}{\rho_0} = \gamma\Delta T = (5 \times 10^{-4} K^{-1})(40K) = 0.020$$

- 30.(D) Let L and A be length and area of cross section of each wire. In order to have the lower ends of the wires to be at the same level (i.e. same elongation is produced in both wires), let weights W_s and W_b are added to steel and brass wires respectively. Then by definition of young's modulus, the elongation produced in

$$\text{the steel wire is } \Delta L_s = \frac{W_s L}{Y_s A} \quad \left(\text{as } Y = \frac{W/A}{\Delta L/L} \right)$$

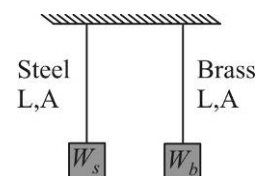
$$\text{And that in the brass wire is } \Delta L_b = \frac{W_b L}{Y_b A}$$

$$\text{But } \Delta L_s = \Delta L_b \quad (\text{given})$$

$$\therefore \frac{W_s L}{Y_s A} = \frac{W_b L}{Y_b A} \quad \text{or} \quad \frac{W_s}{W_b} = \frac{Y_s}{Y_b}$$

$$\text{As } \frac{Y_s}{Y_b} = 2 \quad (\text{given}) \quad \therefore \frac{W_s}{W_b} = \frac{2}{1}$$

31.(B)



- 32.(D) According to Wein's displacement law $\lambda_m T = \text{constant} \dots (i)$

For star P , intensity of violet colour is maximum

For star Q , intensity of red colour is maximum.

For star R , intensity of green colour is maximum.

Also, $\lambda_r > \lambda_g > \lambda_v$

Using equation (i), $T_r < T_g < T_v$

$$T_Q < T_R < T_P$$

- 33.(A) Depth of ocean $d = 2700 \text{ m}$

Density of water, $\rho = 10^3 \text{ kg m}^{-3}$

Compressibility of water, $K = 45.4 \times 10^{-11} \text{ Pa}^{-1}$

Excess pressure at the bottom, $\Delta P = \rho g d = 10^3 \times 10 \times 2700 = 27 \times 10^6 \text{ Pa}$

We know, $B = \frac{\Delta P}{(\Delta V/V)}$

$$\left(\frac{\Delta V}{V} \right) = \frac{\Delta P}{B} = K \cdot \Delta P \left(\because K = \frac{1}{B} \right) = 45.4 \times 10^{-11} \times 27 \times 10^6 = 1.2 \times 10^2$$

- 34.(D) Here,

Specific heat of water, $s_w = 1 \text{ cal g}^{-1} \text{ } ^\circ\text{C}^{-1}$

Latent heat of steam, $L_s = 540 \text{ cal g}^{-1}$

Heat lost by $m \text{ g}$ of steam at 100°C to change into water at 80°C is

$$Q_1 = mL_s + ms_w \Delta T_w = m \times 540 + m \times 1 \times (100 - 80) = 540m + 20 = 560m$$

Heat gained by 20 g of water to change its temperature from 10°C TO 80°C is

$$Q_2 = m_w s_w \Delta T_w = 20 \times 1 \times (80 - 10) = 1400$$

According to principle to calorimetry $Q_1 = Q_2$

$$\therefore 560m = 1400 \text{ or } m = 2.5 \text{ g}$$

Total mass of water present $= (20 + m) \text{ g} = (20 + 2.5) \text{ g} = 22.5 \text{ g}$

- 35.(A) Let T_s be the temperature of the surroundings.

According to Newton's law to cooling

$$\frac{T_1 - T_2}{t} = K \left(\frac{T_1 + T_2}{2} - T_s \right)$$

For first 5 minutes, $T_1 = 70^\circ \text{C}, T_2 = 60^\circ \text{C}, t = 5 \text{ minutes}$

$$\therefore \frac{70 - 60}{5} = K \left(\frac{70 + 60}{2} - T_s \right) \quad \frac{10}{5} = K (65 - T_s)$$

For next 5 minutes, $T_1 = 60^\circ \text{C}, T_2 = 54^\circ \text{C}, t = 5 \text{ minutes}$

$$\therefore \frac{60 - 54}{5} = K \left(\frac{60 + 54}{2} - T_s \right) \quad 6 \times \frac{6}{5} = K (57 - T_s)$$

Divide equation (i) by equation. (ii) we get : $\frac{5}{3} = \frac{65 - T_s}{57 - T_s}$

$$285 - 5T_s = 195 - 3T_s \quad 2T_s = 90 \text{ or } T_s = 45^\circ \text{C}$$

- 36.(D) According to Wien's displacement law

$$\lambda_m T = \text{constant}$$

$$\lambda_m = \frac{\text{constant}}{T}$$

So when a piece of iron is heated, λ_m decreases i.e. with rise in temperature the maximum intensity of radiation emitted gets shifted towards the shorter wavelength. So the colour of the heated object will change that of longer wavelength (red) to that of shorter (reddish yellow) and when the temperature is sufficiently high and all wavelengths are emitted, the colour will become white.

- 37.(C) Young's modulus, $Y = \frac{FL}{A\Delta L} = \frac{4FL}{\pi D^2 \Delta L}$ or $\Delta L = \frac{4FL}{\pi D^2 Y}$

When F is the force applied, L is the length, D is the diameter and ΔL is the extension of the respectively.

As each wire is made up of same material therefore their Young's modulus is same for each wire. For all the four wires, Y, F (= tension) are the same.

$$\therefore \Delta L \propto \frac{L}{D^2}$$

$$\text{In (a)} \quad \frac{L}{D^2} = \frac{200\text{cm}}{(0.2\text{cm})^2} = 5 \times 10^3 \text{ cm}^{-1}$$

$$\text{In (b)} \quad \frac{L}{D^2} = \frac{300\text{cm}}{(0.3\text{cm})^2} = 3.3 \times 10^3 \text{ cm}^{-1}$$

$$\text{In (c)} \quad \frac{L}{D^2} = \frac{50\text{cm}}{(0.05\text{cm})^2} = 20 \times 10^3 \text{ cm}^{-1}$$

$$\text{In (d)} \quad \frac{L}{D^2} = \frac{100\text{cm}}{(0.1\text{cm})^2} = 10 \times 10^3 \text{ cm}^{-1}$$

Hence, ΔL is maximum in (c).

- 38.(B) As $Y = \frac{FL}{A\Delta L} = \frac{4FL}{\pi D^2 \Delta L}$

$$\Delta L = \frac{4FL}{\pi D^2 Y}$$

$$\therefore \frac{\Delta L_s}{\Delta L_c} = \frac{F_s}{F_c} \frac{L_s}{L_c} \frac{D_c^2}{D_s^2} \frac{Y_c}{Y_s}$$

Where subscripts S and C refer to copper and steel respectively.

$$\text{Here, } F_s = (5m + 2m)g = 7mg$$

$$F_c = 5mg$$

$$\frac{L_s}{L_c} = q, \frac{D_s}{D_c} = p, \frac{Y_s}{Y_c} = s$$

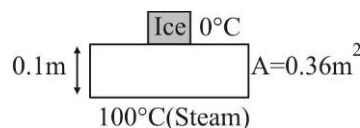
$$\therefore \frac{\Delta L_s}{\Delta L_c} = \left(\frac{7mg}{5mg} \right) (q) \left(\frac{1}{p} \right)^2 \left(\frac{1}{s} \right) = \frac{7q}{5p^2 s}$$

- 39.(D) According to Stefan's law, $Q = \sigma AT^4$ or $T = \left(\frac{Q}{\sigma A} \right)^{1/4} = \left(\frac{Q}{\sigma 4\pi R^2} \right)^{1/4}$

- 40.(A) Heat flows through the slab in t s is

$$Q = \frac{KA(T_1 - T_2)t}{L} = \frac{K \times 0.36 \times (100 - 0) \times 3600}{0.1}$$

$$= \frac{K \times 0.36 \times 100 \times 3600}{0.1} \quad \dots (i)$$



So ice melted by this heat is $m_{ice} = \frac{Q}{L_f}$... (ii) or $Q = m_{ice}L_f = 4.8 \times 3.36 \times 10^5$

From (i) and (ii), we get : $\frac{K \times 0.36 \times (100 - 0) \times 3600}{0.1} = 4.8 \times 3.36 \times 10^5$

$$K = \frac{4.8 \times 3.36 \times 10^5 \times 0.01}{0.36 \times 100 \times 3600} = 1.24 \text{ J/m/s/}^\circ\text{C}$$

41.(B) The amount of heat flows in time t through a cylindrical metallic rod of length L and uniform area of cross-section $A (= \pi R^2)$ with its ends maintained at temperatures T_1 and T_2 ($T_1 > T_2$) is given by

$$Q = \frac{KA(T_1 - T_2)t}{L} \quad \dots (i)$$

Where K is the thermal conductivity of the material of the rod.

Area of cross-section of new rod

$$A' = \pi \left(\frac{R}{2} \right)^2 = \frac{\pi R^2}{4} = \frac{A}{4} \quad \dots (ii)$$

As the volume of the rod remains unchanged $\therefore AL = A'L'$

Where L' is the length of the new rod or $L' = L \frac{A}{A'} \quad \dots (iii)$
 $= 4L$ (Using (ii))

Now, the amount of heat flows in same time t in the new rod with its ends maintained at the same temperatures T_1 and T_2 is given by

$$Q' = \frac{KA'(T_1 - T_2)t}{L'} \quad \dots (iv)$$

Substituting the values of A' and L' from equations (ii) and (iii) in the above equation, we get

$$Q' = \frac{K(A/4)(T_1 - T_2)t}{4L} = \frac{1}{16} \frac{KA(T_1 - T_2)t}{L} = \frac{1}{16} Q$$

42.(C) Power P radiated by the sun with its surface temperature $(t + 273)K$ is given by Stefan's Boltzmann law.

$$P = \sigma e 4\pi r^2 (t + 273)^4$$

Where r is the radius of the sun and the sun is treated as a black body where $e = 1$.

The radiant power per unit area received by the surface at a distance R from the centre of the sun is given by

$$S = \frac{P}{4\pi R^2} = \frac{\sigma 4\pi r^2 (t + 273)^4}{4\pi R^2} = \frac{r^2 \sigma (t + 273)^4}{R^2}$$

43.(A) According to Wein's displacement law,

$$\lambda_{\max} T = \text{constant}$$

$$\therefore \frac{\lambda_{\max 1}}{\lambda_{\max 2}} = \frac{T_2}{T_1} \quad \text{or} \quad \lambda_{\max 2} = \frac{\lambda_{\max 1} \times T_1}{T_2} = \frac{5000 \times 1500}{2500} = 3000 \text{ \AA}$$

44.(B) Radiating power of a black body $= E_0 = \sigma(T^4 - T_0^4)A$

Where σ is known as the Stefan-Boltzmann constant, A is the surface area of a body,

T is the temperature of the black body and T_0 is the temperature of the surrounding.

$$\therefore 60 = \sigma(1000^4 - 500^4) \quad \dots (i) \quad \left[T = 727^\circ C = 727 + 273 = 1000K, T_0 = 227^\circ C = 500K \right]$$

In the second case, $T = 1227^\circ C = 1500K$ and let E' be the radiating power.

$$\therefore E' = \sigma(1500^4 - 500^4) \quad \dots (ii)$$

$$\text{Form (i) and (ii) we have } \frac{E'}{60} = \frac{1500^4 - 500^4}{1000^4 - 500^4} = \frac{15^4 - 5^4}{10^4 - 5^4} = \frac{50000}{9375} \quad \therefore E' = \frac{50000}{9375} \times 60 = 320W.$$

45.(B) Heat flow rate $\frac{dQ}{dt} = \frac{KA(T_1 - T_2)}{L} = Q$

When linear dimensions are double.

$$A_1 \propto r_1^2, L_1 = L \quad A_2 \propto 4r_1^2, L_2 = 2L \quad \text{so } Q_2 = 2Q_1.$$

46.(B) According to Wein's law,

$$\lambda_m T = \text{constant} \quad \therefore \lambda' = (2/3)\lambda_m$$

47.(A) The rate of cooling is directly proportional to the temperature difference of the body and the surroundings. So, cooling will be fastest in the first case and slowest in the third case.

INTEGER VALUE TYPE

48.(82) As shown in figure for equilibrium of mass M ,

$$Mg = 2T \sin \theta \quad \dots (i)$$

But from figure,

$$\sin \theta = \tan \theta = (x/L) \quad \dots (ii)$$

and by definition of Young's modulus,

$$T = \frac{YA}{L} \Delta L = \frac{YA}{L} [(L^2 + x^2)^{1/2} - L] \approx \frac{YAx^2}{2L^2} \quad \dots (iii)$$

So, substituting the values of $\sin \theta$ and T from equations (ii) and (iii) in equation (i), we get

$$Mg = 2 \times \frac{YAx^2}{2L^2} \times \frac{x}{L}, \quad \text{e.g.,} \quad M = \frac{YAx^3}{gL^3} \quad \dots (iv)$$

Now as here $2L = 1m, x = 1cm = 10^{-2}m$

$$\text{and } A = \pi r^2 = \pi[(0.8/2) \times 10^{-3}]^2$$

$$= \pi \times (4 \times 10^{-4})^2 m^2$$

$$\text{So, } M = \frac{2 \times 10^{11} \times \pi (4 \times 10^{-4})^2 \times (10^{-2})^2}{9.8 \times (1/2)^3} kg$$

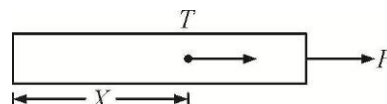
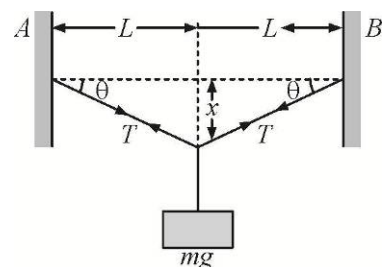
49.(6) As discussed earlier, the tension T in the string at a distance x from its free end is given as.

$$T = \frac{F}{l} x$$

$$\text{Hence, } p = \frac{T}{A} = \frac{F}{Al} x$$

Substituting (p) in the formula

$$U = \frac{1}{2Y} \int p^2 dV$$



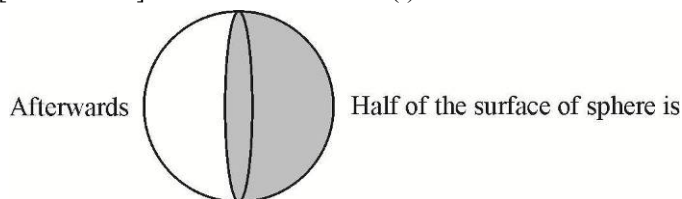
We have, $U = \frac{1}{2Y} \int_0^l \frac{F^2}{A^2 l^2} x^2 dV$

Where $dV = A dx$

This gives $U = \frac{F^2 l}{6AY}$

50.(3) Initially $T = 300$ $T_s = 500$

So, $300 = \sigma e A [500^4 - 300^4]$... (i)



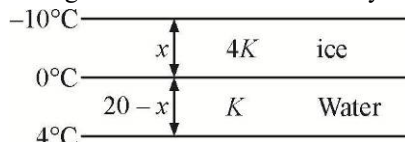
Completely blackened

So, $600 = \frac{\sigma e A}{2} [500^4 - 300^4] + \frac{\sigma A}{2} [500^4 - 300^4]$... (ii)

Dividing equations (2) by equation (1)

$$2 = \frac{\left(\frac{e}{2} + \frac{1}{2}\right)}{e} \Rightarrow 2e = \frac{e}{2} + \frac{1}{2} \Rightarrow e = \frac{1}{3}$$

51.(2) The rate of heat flow is the same through water and ice in the steady state so.

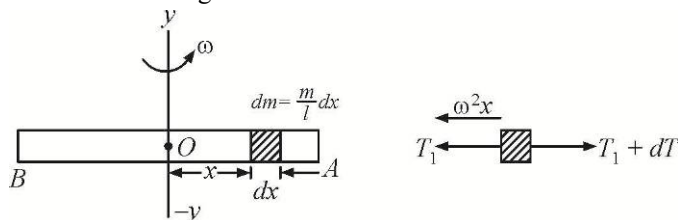


$$\frac{KA(4-0)}{20-x} = \frac{4KA[0-(-10)]}{x}$$

$$\Rightarrow x = \frac{200}{11} m$$

52.(80) The tension in the rotating rod is the maximum at the mid-point. One can assert this intuitively prior to any calculations. Do you feel it? If the stress at the mid-point equals σ_b (the breaking strength for copper), the rod breaks.

For the sake of computation, consider an element of length dx of the rod, as shown in the figure. Let the tension in the rod vary from T to $T + dT$ when the position of the cross-section changes from x to $x + dx$. The forces on the element dx are also shown in the figure.



Now apply Newton's second law of motion to this element.

$$-dT = \left(\frac{m}{l} dx\right) \cdot \omega^2 x$$

Tensions at the end A and also at end B of the rod are zero. Why? If the tension at the mid-point be T_0 ,

$$-\int_{T_0}^0 dT = \frac{m}{l} \omega^2 \int_0^{1/2} x dx$$

$$\text{or } T_0 = \frac{m\omega^2 l}{8}$$

Hence, for the rod not to break

$$T_0 \leq \sigma_b A \quad (\sigma_b = \text{breaking strength})$$

$$\text{or } \frac{m\omega^2 l}{8} \leq \sigma_b A \quad \text{or } \omega^2 \leq \frac{8\sigma_b A}{ml} \quad \text{or } \omega \leq \sqrt{\frac{8\sigma_b A}{ml}}$$

$$\Rightarrow \omega_{\max} = \sqrt{\frac{8\sigma_b A}{(\rho A l)l}} = \sqrt{\frac{8\sigma_b}{\rho l^2}}$$

$$\text{or } (2\pi n)_{\max} = \sqrt{\frac{8\sigma_b}{\rho l^2}} \quad (n = \text{number of revolutions per second})$$

$$\begin{aligned} \text{or } n_{\max} &= \left(\sqrt{\frac{2\sigma_b}{\rho}} \right) \times \frac{1}{\pi l} \\ &= \left(\sqrt{\frac{2 \times 3.0 \times 10^7 \text{ N/m}^2}{8.9 \times 10^3 \text{ kg/m}^3}} \right) \cdot \frac{1}{3.14 \times 1.0 \text{ m}} \\ &= 80 \text{ rps} \end{aligned}$$

53.(23) Figure below shows the forces on the elemental length ds of the ring. The element ds subtends an angle $d\theta$ at the centre of the ring.

Apply Newton's second law to this element,

$$2T \sin \frac{d\theta}{2} = \left(\frac{m}{2\pi} d\theta \right) \omega^2 r$$

$$\text{or } 2T \frac{d\theta}{2} = \left(\frac{m}{2\pi} d\theta \right) \omega^2 r \quad \left(\sin \frac{d\theta}{2} \approx \frac{d\theta}{2} \right)$$

$$\text{or } T = \frac{mr\omega^2}{2\pi}$$

If the area of cross-section of the wire be a and the density of lead ρ , the mass of the ring $m = (2\pi r \cdot a) \rho$.

On substituting for m in the above equation,

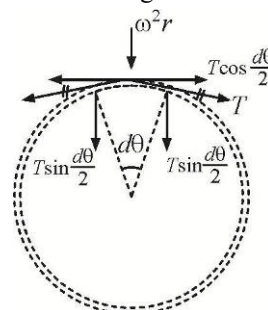
$$T = \left(\frac{r\omega^2}{2\pi} \right) \times 2\pi r a \rho$$

It is obvious that the ring would rupture if the tensile strength developed in it due to rotation exceeds the breaking strength of lead.

The ring would not rupture if $\sigma < \sigma_m$,

$$\text{or } \left(\frac{r\omega^2}{2\pi} \right) 2\pi r \rho < \sigma_m \quad \text{or } \omega < \sqrt{\frac{\sigma_m}{r^2 \rho}}$$

$$\text{or } 2\pi n < \sqrt{\frac{\sigma_m}{r^2 \rho}}$$



$$\text{or } n < \left(\sqrt{\frac{\sigma_m}{\rho}} \right) \cdot \frac{1}{2\pi r} \Rightarrow n_{\max} = \left(\sqrt{\frac{\sigma_m}{\rho}} \right) \cdot \frac{1}{2\pi r}$$

$$= \sqrt{\frac{1.5 \times 10^7 \text{ N/m}}{1.13 \times 10^4 \text{ kg/m}^3}} \cdot \frac{1}{2 \times 3.14 \times 25 \times 10^{-2} \text{ m}}$$

$$= 23 \text{ rps}$$

- 54.(3) (a)** Consider an element of length dr at a distance r from the axis of rotation as shown in figure. The centripetal force acting on this element will be $dT = dm r \omega^2 = (\rho A dr) r \omega^2$. As this force is provided by tension in the rod (due to elasticity), so the tension in the rod at a distance r from the axis of rotation will be due to the centripetal force due to all elements between $x = r$ to $x = L$ i.e.,

$$T = \int_r^L \rho A \omega^2 r dr = \frac{1}{2} \rho A \omega^2 [L^2 - r^2] \quad \dots(i)$$

$$\text{So, } T = \frac{1}{2} \cdot 10^4 \cdot \pi \cdot 10^{-2}$$

$$(400)^2 \left[\left(\frac{1}{2} \right)^2 - r^2 \right] = 8\pi \times 10^6 \left[\frac{1}{4} - r^2 \right] N$$

Now if dy is the elongation in the element of length dr at position r then strain

$$\frac{dy}{dr} = \frac{\text{stress}}{Y} = \frac{T}{AY} = \frac{1}{2} \frac{\rho \omega^2}{Y} [L^2 - r^2]$$

So the elongation of the whole rod

$$\Delta L = \frac{\rho \omega^2}{2Y} \int_0^L (L^2 - r^2) dr = \frac{1}{3} \frac{\rho \omega^2 L^2}{Y} = \frac{1}{3} \frac{10^4 \times (400)^2 (0.5)^3}{2 \times 10^{11}} = \frac{1}{3} \times 10^{-3} \text{ m}$$

- 55.(1)** Increased in length of the iron wire

$$I \alpha \Delta t = I \times 12 \times 10^{-6} \times 40$$

$$= 48I \times 10^{-5} \text{ m}$$

Increase in length of the brass tube

$$= I \alpha \Delta t = I \times 12 \times 10^{-6} \times 40$$

$$= 72I \times 10^{-5} \text{ m}$$



Since the increase in length of the brass tube is greater than that of the iron wire, the latter will be tightened. Let T be the tension in the wire. Then

$$\text{Stress} = \frac{T}{\pi r^2} = \frac{T}{\pi (3 \times 10^{-4})^2} = \frac{T}{9\pi \times 10^{-8}}$$

$$\text{Strain} = \frac{(72I - 48I) \times 10^{-5}}{I} = 24 \times 10^{-5}$$

$$\text{Since stress} = Y \times \text{strain}, \frac{T}{9\pi \times 10^{-8}} = 21 \times 10^{10} \times 24 \times 10^{-5}$$

$$T = 21 \times 10^{10} \times 24 \times 10^{-5} \times 9\pi \times 10^{-8} = 14.2 \text{ N}$$

- 56.(0)** Let t be the common temperature above zero Celsius.

$$\text{Heat lost by calorimeter and water added} = 400 \times 10^{-3} \times 1000(5 - t) + 200 \times 10^{-3} \times 1000(10 - t)$$

$$\text{Heat gained by ice} = 400 \times 10^{-3} \times 5000(60 + t) + 400 \times 10^{-3} \times 80000 + 400 \times 10^{-3} \times 1000t$$

Since Heat lost = heat gained

$$\therefore 400(5-t) + 200(10-t) = 200(60+t) + 400 \times 80 + 400t$$

$$\text{or, } t = -24.33^\circ\text{C}$$

This is contradictory to the assumption that t is above zero.

Hence, the final temperature is either 0°C or less than 0°C .

Let the temperature be $t^\circ\text{C}$ (less than 0°C).

$$\text{Then } 400 \times 10^{-3} \times 1000 \times 5 + 200 \times 10^{-3} \times 100 \times 10 + 600 \times 10^{-3} \times 80000 +$$

$$600 \times 10^{-3} \times 500 \times t = 400 \times 500 (60 - t)$$

$$\text{or, } 400 \times 5 + 200 \times 10 + 600 \times 80 + 300t = 200(60 - t)$$

$$\text{or, } 20 + 20 + 480 + 3t = 120 - 2t$$

$$\text{or, } t = -80^\circ\text{C}$$

This is absurd because the lowest temperature is -60°C and we discard this second assumption also.

So the final temperature is 0°C

Let x grams of water be converted into ice.

$$\text{Heat lost} = 400 \times 10^{-3} \times 1000 \times 5 + 200 \times 10^{-3} \times 1000 \times 10 + x \times 10^{-3} \times 8000 = 4000 + 80x$$

$$\text{Heat gained} = 400 \times 10^{-3} \times 500 \times 60$$

$$\text{or, } 4000 + 80x = 12000$$

$$\text{or, } x = 100 \text{ g}$$

Hence, the final result is 500 g of ice and 500 g of water at 0°C .

- 57.(51)** Assume that a body cools down from temperature θ_i to θ_f in t seconds, and θ_0 is the temperature of surroundings. Applying Newton's law of cooling

$$\frac{d\theta}{dt} = -K(\theta - \theta_0); \quad \theta_0 = 45^\circ\text{C}; \quad \int \frac{d\theta}{\theta - 45} = -\int dt \quad (K \text{ is constant})$$

From $t = 0 \text{ s}$ to $t = 30 \text{ s}$, θ changes from 60°C to 55°C

$$\int_{55}^{60} \frac{d\theta}{\theta - 45} = -K \int_0^{30} dt \quad \Rightarrow \quad \ln\left(\frac{60-45}{55-45}\right) = -K(30-0) \quad \dots(i)$$

$$\int_{55}^{50} \frac{d\theta}{\theta - 45} = -K \int_{30}^t dt \quad \Rightarrow \quad \ln\left(\frac{50-45}{55-45}\right) = -K(t-30) \quad \dots(ii)$$

Divide equation (ii) by equation (i) to set:

$$\frac{t-30}{30} = \frac{\ln \frac{50-45}{55-45}}{\ln \frac{55-45}{60-45}}$$

$$\Rightarrow \frac{t-30}{30} = \frac{\ln 2}{\ln 3/2} \quad \Rightarrow \quad t = 30 \left(1 + \frac{\ln 2}{\ln 3/2} \right) = 81.28 \text{ s}$$

Time from $\theta = 55^\circ\text{C}$ to $\theta = 50^\circ\text{C}$ is $(t - 30) = (81.28 - 30) = 51.28 \text{ s}$

- 58.(8)** Let us learn the application theory this illustration

Let m = mass of water, f = fraction which freezes

L_1 = latent heat of vapourization

L_2 = latent heat of fusion

Mass of water frozen = mf

Heat lost by freezing water = mfL_2

Mass of vapour formed = $m(1-f)$

Heat gained by vapours = $m(1-f)L_1$

Now heat loss = heat gain:

$$mfL_2 = m(1-f) \times 7L_2$$

$$f = 7 - 7f \quad \text{or} \quad f = 7/8$$

59.(0) Here in the given mixture initially we have

Mixture = 2 kg copper at 100°C + 0.75 kg of ice at 0°C

Now heat released by copper to attain 0°C is

$$Q_1 = 2 \times 378 \times 100 = 7.56 \times 10^4 \text{ Joule}$$

Heat gained by ice to melt is

$$Q_2 = -0.75 \times 3.36 \times 10^5 = -25.2 \times 10^4 \text{ Joule}$$

Thus, mixture contains

Mixture = 2 kg copper at 0°C + 0.75 kg

Water at 0°C + $7.56 \times 10^4 \text{ J} - 25.2 \times 10^4 \text{ J}$

= 2 kg copper at 0°C + 0.75 kg water at $0^\circ\text{C} - 17.64 \times 10^4 \text{ J}$

Here the available heat is negative which implies that whole ice will not melt here we can freeze the available water at 0°C upto the extent when the available negative heat is balanced. $17.64 \times 10^4 \text{ J}$ heat is released when m mass of water is frozen thus m is given as

$$m = \frac{17.64 \times 10^4}{3.36 \times 10^5} = 0.525 \text{ kg}$$

Thus final mixture contains

Mixture = 2 kg copper at 0°C + 0.525 kg

Ice at 0°C + 0.225 kg water at 0°C

Thus, it is obvious that the equilibrium temperature is 0°C .

60.(3) When the block is in equilibrium in the water and spring is stretched by a distance x and the spring force balances the effective weight of block i.e., weight of block minus the buoyant force on block. Thus for equilibrium of block in water, we have

kx + weight of liquid displaced = weight of block

$$200x + \frac{2}{5000} \times 1000 \times 10 = 2 \times 10 \quad \text{or,} \quad x = \frac{20-4}{200} = \frac{16}{200} = 0.08 \text{ m} = 8 \text{ cm}$$

Thus in equilibrium, energy stored in spring is

$$U = \frac{1}{2} kx^2 = \frac{1}{2} \times 200 \times (0.08)^2 = 0.64 \text{ Joule}$$

When the support is broken, the mass fall down to the bottom of vessel and the potential energy stored in spring and the gravitational potential energy of block is released and used in heating the water and block. Thus we have

$$0.64 + mg(0.4) = m_w \times s_w \times \Delta T + m_B \times s_B \times \Delta T$$

$$\text{or,} \quad 0.64 + 2 \times 10 \times 0.4 = [0.3 \times 4200 + 2 \times 250] \Delta T \quad \text{or,} \quad \Delta T = \frac{8.64}{1760} = 0.0049^\circ\text{C}$$

61.(26) Here container and water are at 20°C , thus when iron block is dropped into water it will loose energy to it and its temperature will fall. If equilibrium temperature is T_0 , then we have

Heat lost by iron block = heat gained by water plus container

$$m_i \times s_i (100 - T_0) = m_w \times s_w (T_0 - 20) + m_c \times s_c (T_0 - 20)$$

$$\text{or,} \quad T_0 = \frac{m_i s_i (100) + m_w s_w (20) + m_c s_c (20)}{m_i s_i + m_w s_w + m_c s_c}$$

$$= \frac{0.2 \times 470 \times 100 + 0.2 \times 4200 \times 20 + 0.5 \times 910 \times 20}{0.2 \times 470 + 0.2 \times 4200 + 0.5 \times 910}$$

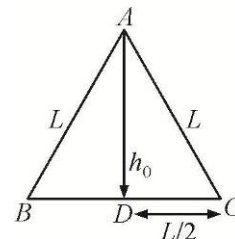
$$= \frac{9400 + 16800 + 9100}{94 + 840 + 455} = \frac{35300}{1329} = 26.56^\circ\text{C}$$

62.(4) Let L be the length of each rod at 0°C and h_0 be the altitude AD at 0°C . Then (figure)

$$h_0 = \left(L^2 - \frac{L^2}{4} \right)^{1/2} \quad \dots(i)$$

When the rods are heated to a temperature t , the altitude becomes

$$h = \left[L^2 (1 + \alpha_1 t)^2 - \frac{L^2}{4} (1 + \alpha_2 t)^2 \right]^{1/2}$$



Since α_1 and α_2 are much less than unity, we can write $(1 + \alpha_1 t)^2 = 1 + 2\alpha_1 t$ and $(1 + \alpha_2 t)^2 = 1 + 2\alpha_2 t$

$$\text{Thus } h = \left[L^2 (1 + 2\alpha_1 t) - \frac{L^2}{4} (1 + 2\alpha_2 t) \right]^{1/2} \quad \dots(ii)$$

Equating (i) and (ii), we get ($\because$ given that $h = h_0$)

$$\left(L^2 - \frac{L^2}{4} \right)^{1/2} = \left[L^2 (1 + 2\alpha_1 t) - \frac{L^2}{4} (1 + 2\alpha_2 t) \right]^{1/2}$$

Which simplifies to give $\frac{\alpha_1}{\alpha_2} = \frac{1}{4}$

Gaseous State & Thermodynamics

- 1.(D) In (i) heat supplied is used to increase internal energy as well as to perform work on surroundings
 In (ii) heat supplied is completely used to increase internal energy as well as surroundings performs some work on the gas which also increase internal energy
 In (iii) heat supplied is used to increase only internal energy $C_I > C_{III} > C_{II}$
- 2.(B) Pressure is independent of the motion of the gas vessel with constant velocity
- 3.(A) Gas performs work in expanding against the atmosphere as well as spring.
- 4.(D) At higher temperature additional degrees of freedom come into play $\therefore C_V > \frac{5}{2}R$
- 5.(AB) $\frac{dU}{dT} = \frac{3nR}{2}$ $\frac{dQ}{dT} = \frac{5}{2}nR$
- 6.(D) Kinetic energy is same for molecules of all gases at given temperature.
- 7.(B) 8.(D) 9.(D) 10.(B)
- 11.(D) According to first law of thermodynamics, we get
 (i) Change in internal energy from A to B i.e., ΔU_{AB}
- $$\Delta U_{AB} = nC_V (T_B - T_A) = 1 \times \frac{5R}{2} (800 - 400) = 1000 R$$
- (ii) Change in internal energy from B to C

$$\Delta U_{BC} = nC_V(T_C - T_B) = 1 \times \frac{5R}{2}(600 - 800) = -500R$$

(iii) Change in internal energy from C to A

$$i.e., \Delta U_{CA}$$

$$\Delta U_{CA} = nC_V(T_A - T_C) = 1 \times \frac{5R}{2}(400 - 600) = -500R$$

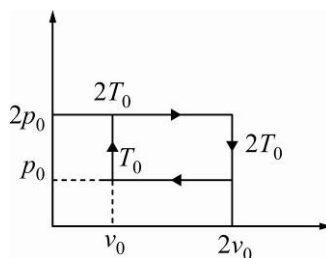
12-13. 12.(A) Work done = $\frac{1}{2}(2V_0)(P_0 + 3P_0) = 4P_0V_0$ 13.(D) Specific heat of isothermal process is infinite

14.(B) Heat supplied

$$nC_V(2T_0 - T_0) + nC_P(4T_0 - 2T_0)$$

$$= \frac{n3RT_0}{2} + \frac{n5R}{2}2T_0 = \frac{13}{2}nRT_0$$

$$= \frac{13}{2}P_0V_0$$



15.(B) Thermal energy corresponds to internal energy

$$\text{Mass} = 1 \text{ kg}$$

$$\text{Density} = 4 \text{ kg m}^{-3} \Rightarrow \text{Volume} = \frac{\text{Mass}}{\text{Density}} = \frac{1}{4} \text{ m}^3$$

$$\text{Pressure} = 8 \times 10^4 \text{ Nm}^{-2}$$

$$\text{As internal energy} = \frac{f}{2}nRT = \frac{f}{2}pV$$

$$\text{Degree of freedom for diatomic} = 5 \quad \therefore \quad \text{Internal energy} = \frac{5}{2}p \times V$$

Putting the values of $p = 8 \times 10^4 \text{ Nm}^{-2}$ and $V = \frac{1}{4} \text{ m}^3$, we get :

$$\text{Internal energy} = \frac{5}{2} \times 8 \times 10^4 \times \frac{1}{4} = 5 \times 10^4 \text{ J}$$

16.(A) $T = V + T_0$

$$PV = RT$$

$$\Rightarrow PV = R(V + T_0)$$

$$P = R + \frac{RT_0}{V}$$

17.(C) Pressure applied by piston $\frac{Mg}{A} = p_0$

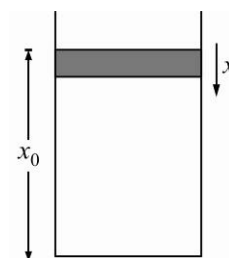
As no exchange of heat, so process is adiabatic.

$$p_0V_0^\gamma = pV^\gamma, p_0Ax_0^\gamma = pA(x_0 - x)^\gamma$$

$$p = \frac{p_0x_0^\gamma}{(x_0 - x)^\gamma}$$

Let piston be displaced by x .

$$Mg - \left(\frac{p_0x_0^\gamma}{(x_0 - x)^\gamma} \right) A = F_{\text{restoring}}$$



$$p_0 A \left(1 - \frac{x_0^\gamma}{(x_0 - x)^\gamma} \right) = F_{\text{restoring}} \quad [\because x_0 - x = x_0]$$

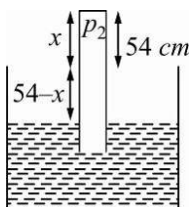
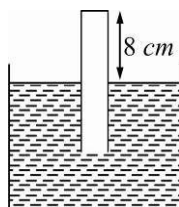
$$F = -\frac{\gamma p_0 A x}{x_0} \quad \therefore \quad f = \frac{1}{2\pi} \sqrt{\frac{\gamma p_0 A}{x_0 M}} = \frac{1}{2\pi} \sqrt{\frac{\gamma p_0 A^2}{M V_0}}$$

18.(A)

19.(B)

20.(C)

21.(A)



$$V_1 = A \times B$$

$$[\because A = \text{area of cross-section}]$$

$$p_2 = p_{\text{atm}} - \rho g (54 - x) = \rho g (22 + x)$$

$$V_2 = A \cdot x$$

At constant temperature,

$$p_1 V_1 = p_2 V_2$$

$$\rho g 76 \cdot A 8 = \rho g (22 + x) A x$$

$$x^2 + 22x - 78 \times 8 = 0 \quad \Rightarrow \quad x = 16 \text{ cm}$$

22.(D)

23.(A) $P^2 = CT$

Also $PV = nRT \Rightarrow T = \frac{PV}{nR} \Rightarrow P = KV$

$$nC dT = nC_V dT + \int P dV \quad [\because Q = \Delta U + W]$$

So $\int P dV = \int_{V_1}^{V_2} K V dV = \frac{K}{2} [V_2^2 - V_1^2] = \frac{P_2 V_2 - P_1 V_1}{2} = \frac{nR \Delta T}{2}$

$$\Rightarrow nC \Delta T = nC_V \Delta T + \frac{nR \Delta T}{2} \Rightarrow C = C_V + \frac{R}{2} = 2R \Rightarrow Q = nC \Delta T = 2R(2T_0 - T_0) = 2RT_0$$

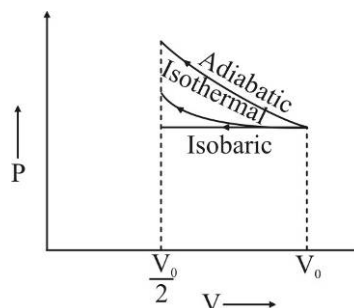
24.(B) Given $P = KU \Rightarrow \Delta P = K(nC_V \Delta T)$

Also $PV = nRT$

$$P = \left(\frac{nR}{V} \right) T \Rightarrow \Delta P = \left(\frac{nR}{V} \right) \Delta T$$

So we can say volume remains constant and temperature is increasing.

25.(C) The P - V diagram of an ideal gas compressed from its initial V_0 to $\frac{V_0}{2}$ by several processes is shown in the figure. Work done on the gas = area under P - V curve is maximum for adiabatic process, so work done on the gas is maximum for adiabatic process.



- 26.(D) According to an ideal gas equation, the molecular weight of an ideal gas is :

$$M = \frac{\rho RT}{P} \quad \left(\text{as } P = \frac{\rho RT}{M} \right)$$

Where P , T and ρ are the pressure, temperature and density of the gas respectively and R is the universal gas constant.

$$\therefore \text{The molecular weight of A is } M_A = \frac{\rho_A RT_A}{P_A}$$

$$\text{And that of B is } M_B = \frac{\rho_B RT_B}{P_B}$$

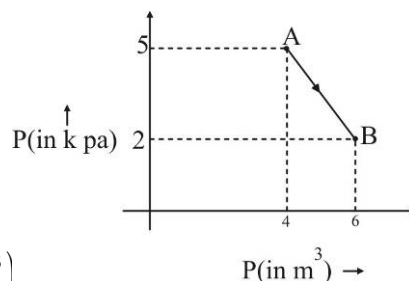
$$\text{Hence, their corresponding ratio is } \frac{M_A}{M_B} = \left(\frac{\rho_A}{\rho_B} \right) \left(\frac{T_A}{T_B} \right) \left(\frac{P_B}{P_A} \right)$$

$$\text{Here, } \frac{\rho_A}{\rho_B} = 1.5 = \frac{3}{2}, \frac{T_A}{T_B} = 1 \text{ and } \frac{P_A}{P_B} = 2 \quad \therefore \frac{M_A}{M_B} = \left(\frac{3}{2} \right) (1) \left(\frac{1}{2} \right) = \frac{3}{4}$$

$$27.(A) \quad \beta = \frac{1-\eta}{n} \Rightarrow \beta = 9 \quad \therefore Q = 10\beta = 90J$$

- 28.(D) We know, $\Delta U = nC_v \Delta T$

$$\begin{aligned} &= n \left(\frac{5R}{2} \right) (T_B - T_A) \quad [\text{For diatomic gas, } C_v = \frac{5R}{2}] \\ &= \frac{5nR}{2} \left(\frac{P_B V_B}{nR} - \frac{P_A V_A}{nR} \right) [\because PV = nRT] \\ &= \frac{5}{2} (P_B V_B - P_A V_A) = \frac{5}{2} (2 \times 10^3 \times 6 - 5 \times 10^3 \times 4) = \frac{5}{2} (-8 \times 10^3) \end{aligned}$$



- 29.(A) As initial and final points are same so $\Delta U_{ABC} = \Delta U_{AC}$

AB is isochoric process

$$\Delta W_{AB} = 0$$

$$\Delta Q_{AB} = \Delta U_{AB} = 400J$$

BC is isobaric process.

$$\Delta Q_{BC} = \Delta U_{BC} + \Delta W_{BC}$$

$$100 = \Delta U_{BC} + 6 \times 10^4 (4 \times 10^{-3} - 2 \times 10^{-3})$$

$$100 = \Delta U_{BC} + 12 \times 10$$

$$\Delta U_{BC} = 100 - 120 = -20J$$

$$\text{As, } \Delta U_{ABC} = \Delta U_{AC}$$

$$\Delta U_{AB} + \Delta U_{BC} = \Delta Q_{AC} - \Delta W_{AC}$$

$$400 - 20 = \Delta Q_{AC} - \left(2 \times 10^4 \times 2 \times 10^{-3} + \frac{1}{2} \times 2 \times 10^{-3} \times 4 \times 10^4 \right) = 460J$$

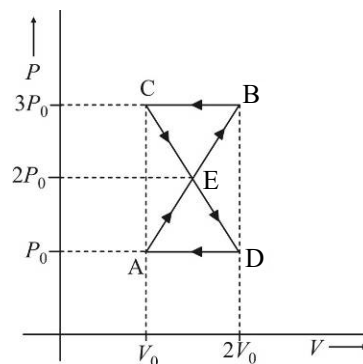
- 30.(D) In a cyclic process work done is equal to the area under the cycle and is positive if the cycle is clockwise and negative if anticlockwise.

As is clear from figure,

$$W_{AEDA} = + \text{area of } \triangle AED = + \frac{1}{2} P_0 V_0$$

$$W_{BCEB} = - \text{Area of } \triangle BCE = - \frac{1}{2} P_0 V_0$$

The net work done by the system is



$$W_{net} = W_{AEDA} + W_{BCEB} = +\frac{1}{2}P_0V_0 - \frac{1}{2}P_0V_0 = \text{zero}$$

31.(A) According to ideal gas equation $PV = nRT$

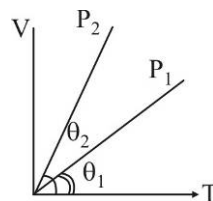
$$\text{Or } V = \frac{nRT}{P}$$

For an isobaric process, $P = \text{constant}$ and $V \propto T$
therefore, $V-T$ graph is a straight line passing through origin. Slope of this line is inversely proportional to P .

In the given figure,

$$(\text{slope})_2 > (\text{slope})_1$$

$$\therefore P_2 < P_1$$



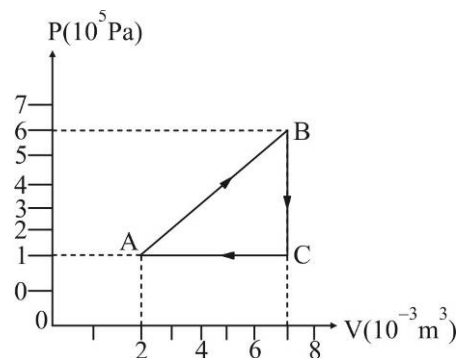
32.(D) In a cyclic process, work done is equal to the area under the cycle and is positive if the cycle is clockwise and negative if the cycle is anticlockwise.

$\therefore$ The network done by the gas is

$$W = \text{area of the cycle } ABCA$$

$$= \frac{1}{2} \times (7-2) \times 20^{-3} \times (6-2) \times 10^5$$

$$= \frac{1}{2} \times 5 \times 10^{-3} \times 4 \times 10^5 = 10 \times 10^2 J = 1000 J$$



33.(C) As here volume of the gas remains constant, therefore the amount of heat energy required to raise the temperature of the gas is $\Delta Q = nC_v \Delta T$

$$\text{Here of moles, } n = \frac{1}{4}$$

$$C_v = \frac{3}{2}R \quad (\because \text{He is a monatomic})$$

$$\Delta T = T_2 - T_1 \quad \therefore \quad \Delta Q = \frac{1}{4} \times \frac{3}{2} R (T_2 - T_1) = \frac{3}{8} N_a k_B (T_2 - T_1) \quad \left(\because k_B = \frac{R}{N_a} \right)$$

34.(D) According to first law of thermodynamics $\delta Q = \delta U + \delta W$

Along the path adc

$$\text{Change in internal energy, } \delta U_1 = \delta Q_1 - \delta W_1 = 50J - 20J = 30J$$

Along the path abc

Change in internal energy,

$$\delta U_2 = \delta Q_2 - \delta W_2$$

$$\delta U_2 = 36J - \delta W_2$$

As change in internal energy is path independent.

$$\therefore \delta U_1 = \delta U_2 \quad \therefore \quad 30J = 36J - \delta W_2$$

$$\delta W_2 = 36J - 30J = 6J$$

35.(A) In a cyclic process, $\Delta U = 0$.

In a cyclic process work done is equal to the area under the cycle and is positive if the cycle is clockwise and negative if anticlockwise.

$$\therefore \Delta W = - \text{area of rectangle } ABCD = -P(2V) = -2PV$$

According to first law of thermodynamics

$$\Delta Q = \Delta U + \Delta W \text{ OR } \Delta Q = \Delta W \quad (\text{As } \Delta U = 0)$$

i.e., done supplied to the system is equal to the work done

So heat absorbed, $\Delta Q = \Delta W = -2PV$ $\therefore$ Heat rejected by the gas = $2PV$ **36.(D)**

37.(A) Change in internal energy is path independent and depends only of the initial and final states.

As the initial and final states in the three processes are same. Therefore,

$$\Delta U_1 = \Delta U_2 = \Delta U_3$$

Work done, W = are under P - V graph as area under curve 1 > area under curve 2 > are under curve 3

$$\therefore W_1 > W_2 > W_3$$

According to first law of thermodynamics, $Q = W + \Delta U$

$$\text{As } W_1 > W_2 > W_3 \text{ and } \Delta U_1 = \Delta U_2 = \Delta U_3 \quad \therefore Q_1 > Q_2 > Q_3$$

38.(D)

39.(A) According to first law of thermodynamics $\Delta Q = \Delta U + \Delta W$... (I)

Were, ΔQ = Heat supplied to the system

ΔU = Increase in internal energy of the system

ΔW = Work done by the system

For an adiabatic process $\Delta Q = 0$

From equation (i), we get $\Delta U = -\Delta W$

For an isothermal process $\Delta U = 0$

From equation (i), we get $\Delta Q = \Delta W$

Hence, option (A) is true.

40.(C) Let C_V and C_P be molar specific heats of the ideal gas at constant volume and constant pressure, respectively,

$$\text{then } C_P = Mc_P \text{ and } C_V = Mc_V$$

$$C_P - C_V = R \quad \therefore Mc_P - Mc_V = R$$

$$\Rightarrow c_P - c_V = R/M$$

41.(C) heat energy given $dQ = dU + dW$ where dU is the change in internal energy and dW is the work done.

$$\text{Given } dQ = 2\text{kcal} = 2000 \times 4.2\text{J} \text{ and } dw = 500\text{J} \quad \therefore dQ = 2000 \times 4.2 = dU + 500 \Rightarrow dU = 7900\text{J}$$

42.(C) Efficiency of an engine, $\eta = 1 - \frac{T_2}{T_1}$

Where T_1 is the temperature of the source and T_2 is the temperature of the sink.

$$\therefore \frac{1}{6} = 1 - \frac{T_2}{T_1} \text{ or, } \frac{T_2}{T_1} = \frac{5}{6} \quad \dots (i)$$

When the temperature of the sink is decreased by 62°C (or 62), efficiency becomes double.

Since, the temperature of the source remain unchanged

$$\therefore 2 \times \frac{1}{6} = 1 - \frac{(T_2 - 62)}{T_1} \text{ or, } \frac{1}{3} = 1 - \frac{(T_2 - 62)}{T_1}$$

$$\text{Or, } \frac{2}{3} = \frac{T_2 - 62}{T_1} \text{ or, } 2T_1 = 3T_2 - 186 \text{ or, } 2T_1 = 3\left[\frac{5}{6}T_1\right] - 186 \quad [\text{Using (i)}]$$

$$\therefore \left[\frac{5}{2} - 2\right]T_1 = 186 \text{ or, } \frac{T_1}{2} = 186 \text{ or, } T_1 = 372\text{K} = 99^\circ\text{C}$$

43.(A) As $C_p / C_v = \gamma$, $\therefore \frac{C_p - C_v}{C_v} = \gamma - 1$ or $\frac{C_p - C_v}{\gamma - 1} = \frac{R}{\gamma - 1}$.

Change internal energy, $\Delta U = nC_v \Delta T = \frac{nR \Delta T}{(\gamma - 1)} = \frac{nPdV}{(\gamma - 1)} = \frac{nP(2V - V)}{\gamma - 1}$. For one mole, $n = 1$, $\therefore \Delta U = PV / (\gamma - 1)$.

44.(D) For adiabatic change, $PV^\gamma = \text{constant}$

$$\Rightarrow P \left(\frac{RT}{\rho} \right)^\gamma = \text{constant} \Rightarrow P^{1-\gamma} T^\gamma = \text{constant} \Rightarrow P \propto T^{\frac{\gamma}{\gamma-1}}$$

Therefore, the value of constant $C = \frac{\gamma}{(\gamma-1)}$. For monoatomic gas, $\gamma = \frac{5}{3}$. Therefore $C = \frac{5/3}{(5/3)-1} = \frac{5/3}{2/3} = \frac{5}{2}$.

45.(C) Work done = area under curve ACBDA

46.(D) $C_p - C_v = R$ for all gases.

47.(C) According to Dalton's law of partial pressure, we have $P = P_1 + P_2 + P_3$

INTEGER VALUE TYPE

48.(2) $C = \sqrt{\frac{3RT}{M}}$

In first case, $V = \sqrt{\frac{3RT}{M}}$

And in second case, $V_2 = \sqrt{\frac{3R(2T)}{M/2}}$
 $= \sqrt{\frac{4 \times (3RT)}{M}} = 2V$

49.(3) As we know, $pV^\gamma = \text{constant}$

Using equation $pV = nRT$, we get

$$p^{1-\gamma} T^\gamma = \text{constant} \quad \text{or} \quad p \propto T^{\left[\frac{\gamma}{\gamma-1} \right]}$$

Here $p \propto T^3$

or $\frac{\gamma}{\gamma-1} = 3$ or $\gamma = \frac{3}{2}$

Therefore, $2\gamma = 2 \times \frac{3}{2} = 3$

50.(1) The average number of degree of freedom per molecule.

$$f = \frac{\text{Total number of degree of freedom}}{\text{Total number of molecule}}$$

$$\therefore f = \frac{n_1 N_A f_1 + n_2 N_A f_2}{n_1 N_A + n_2 N_A} \quad \text{where } N_A = \text{Avogadro constant}$$

Here, $f_1 = 3$ and $f_2 = 5$

$$\therefore f = \frac{n_1 f_1 + n_2 f_2}{n_1 + n_2} = \frac{3n_1 + 5n_2}{n_1 + n_2}$$

Also, $\gamma = 1 + \frac{2}{f} = 1.5 \Rightarrow f = 4$ $\therefore \frac{3n_1 + 5n_2}{n_1 + n_2} = 4$

or $3n_1 + 5n_2 = 4n_1 + 4n_2$ or $n_1 = n_2$ $\therefore k = 1$

51.(50) From first law,

$$\Delta Q = \Delta V + \Delta W$$

We have, $\Delta Q = -2000J$

$$\Delta W = -2500J$$

Now, $\Delta Q = \Delta U + \Delta W$

$$-2000 = \Delta U - 2500$$

or $\Delta U = 500J$

$$\text{Therefore } \frac{x}{10} = \frac{500}{10} = 50J$$

$$52.(6) \quad P = \alpha V^2 \quad \therefore \frac{P_1}{P_2} = \left(\frac{V_1}{V_2} \right)^2$$

$$P_2 = \alpha V_2^2$$

$$= 5 \times (2)^2 = 20 \text{ atm}$$

$$\text{Work done by the gas} = \int_{V_i}^{V_f} P dv$$

$$= \int_{V_i}^{V_f} \alpha V^2 dv = \alpha \int_{V_i}^{V_f} V^2 dv = \frac{\alpha}{3} [V_f^3 - V_i^3]$$

$$\text{Putting the value of } \alpha, V_i \text{ and } V_f, \text{ we get } W = \frac{1}{3} (5 \text{ atm} / m^6) (1.013 \times 10^5 \text{ Pa} / \text{atm}) (2^3 - 1^3) = 1.18 \times 10^6 J$$

$$\therefore n = 6$$

53.(5) Let us take one mole of an ideal gas, the equation of state becomes, $PV = RT$. We are given, $PT = \text{constant} = K$ (say).

$$\text{or } P = \frac{K}{T}$$

we can now write,

$$\left(\frac{K}{T} \right) V = RT \quad \text{or } V = \frac{R}{K} T^2$$

Differentiating the above equation, we get

$$dV = \frac{2RT}{K} dT \quad \text{or } dw = PdV$$

$$= \frac{K}{T} \left(\frac{2RT}{K} \right) dT$$

From, first law of thermodynamics,

$$dQ = dV + dW$$

$$CdT = CvdT + 2RdT$$

$$C_v = C - 2R$$

$$= 4.5R - 2R = 2.5R = \frac{5}{2} R$$

$$\text{But we define } C_v = \frac{f}{2} R$$

Hence, degree of freedom = 5

54.(4) Given, $Q_1 = 50 \text{ Kcal}$ per cycle.

And $\eta = 20\%$

$$\text{We have } W = \eta Q_1 = 20\% \times 50 = 10 \text{ kcal}$$

Heat injected to sink in one cycle

$$= Q_1 - W$$

$$= 50 - 10 = 40 \text{ Kcal} \quad \therefore y = 4$$

55.(3) $P \propto V$ since graph is a straight line.

$$PV^{-1} = \text{constant}$$

For the process, $PV^x = \text{constant}$, molar heat capacity is given by C

$$C = \frac{R}{\gamma - 1} + \frac{R}{1 - x}$$

Here, $\gamma = 1.4$ & $x = -1$

$$\text{So, } C = \frac{R}{1.4 - 1} + \frac{R}{1 + 1} = 3R = nR \quad \therefore n = 3$$

56.(9) Here $\Delta Q = 15000 \text{ J}$ (Given)

In an isobaric process,

$$\Delta Q = nC_p \Delta T$$

$$\Delta U = nC_v \Delta T$$

$$\text{or } \frac{\Delta U}{\Delta Q} = \frac{C_v}{C_p} = \frac{1}{\gamma} \Rightarrow \Delta U = \frac{\Delta Q}{\gamma} \quad \text{or } \Delta U = \frac{3}{5} \times 15 = 9 \text{ kJ}.$$

57.(6) For process I : $dQ = dU + dW$

$$\Rightarrow 4 = dU + 5 \Rightarrow dU = -1$$

$$\text{For process II : } 5 = dU + dW \Rightarrow 5 = -1 + dW \Rightarrow dW = 6 \text{ J}$$

58.(3) $PV^\gamma = \text{constant}$

$$PV^{4/3} = \text{constant}$$

$$\ln V^4 + \ln P^3 = \text{constant} \quad \therefore 4 \left(\frac{dV}{V} \right) = -3 \left(\frac{dP}{P} \right)$$

$$\Rightarrow 4PdV + 3VdP = 0$$

$$\Rightarrow 3(PdV + VdP) + PdV = 0 \quad \dots(i)$$

For adiabatic process $dQ = 0$

$$\Rightarrow dU + dW = 0$$

$$\Rightarrow dU + PdV = 0 \quad \dots(ii)$$

Comparing equation (i) and (ii)

$$dU = 3(PdV + VdP)$$

$$U = K + 3PV$$

So, $n = 3$

59.(3) Let x moles of diatomic gas be associated. x moles of diatomic gas becomes $2x$ moles of a monoatomic gas after dissociation.

Internal energy of n moles of an ideal gas $= \frac{3}{2}nRT$ for monoatomic gas

Internal energy of n moles of an ideal gas $= \frac{5}{2}nRT$ for diatomic gas

So, (Internal energy of $2x$ moles of a monoatomic gas + internal energy of $(4 - x)$ moles of a diatomic gas) –

Internal energy of 4 moles of a diatomic gas $= \frac{3}{2}RT$ (given)

$$\Rightarrow (2x) \left(\frac{3}{2}RT \right) + (4 - x) \left(\frac{5}{2}RT \right) - 4 \left(\frac{5}{2}RT \right) = \frac{3}{2}RT$$

on solving, we get $x = 3$ moles

60.(5) Area of the bounded region = $|w|$

$$= \frac{\pi(4-3)(4-2)}{2} + \frac{\pi(2-1)(3-2.5)}{2} = \frac{2.5\pi}{2} = \frac{5\pi}{4} \text{ atm L} \quad \therefore n = 5$$

61.(10) $R = \frac{2}{3} C_v \quad \dots(i)$

We know, $C_p - C_v = R$

or $\gamma - 1 = \frac{R}{C_v}$

or $R = C_v(\gamma - 1) \quad \dots(ii)$

Comparing (i) and (ii)

We get $\gamma = \frac{5}{3}$

As we know, $P^{1-\gamma} T^\gamma = \text{constant}$

$\therefore P \propto T^{\left(\frac{\gamma}{\gamma-1}\right)}$ and $\gamma = \frac{5}{3} \quad \therefore P \propto T^{5/2} \quad \therefore 4x = 4 \times \frac{5}{2} = 10$

62.(39) $Q_{AB} = \Delta U_{AB} + W_{AB}$

$W_{AB} = 0$ (Process is isochoric)

$\Delta U_{AB} = \frac{1}{2} nR\Delta T \quad \therefore Q_{AB} = 2.5P_0V_0$

In process BC,

$Q_{BC} = W_{BC} = 2P_0V_0 \ln 2 = 1.386P_0V_0$

$Q_{net} = Q_{BC} + Q_{AB} = 3.886P_0V_0 \quad ; \quad 10n = 38.86 = 39$

Liquids

1.(A) As density increases with depth, the change in pressure per unit depth increases.

$\therefore$ Slope of P vs h should increase with h .

2.(B) $\rho gh = \frac{4T}{r} \quad \Rightarrow \quad T = \frac{\rho ghr}{4}$

3.(B) Equating pressure of liquid at level B

$\rho g \times (5 \text{ cm}) = 500 \text{ kg/m}^3 (g)(20 \text{ cm}) \quad \Rightarrow \quad \rho = 2000 \text{ kg/m}^3$

4.(C) $V = \sqrt{2gh} = \sqrt{2 \times 10 \times 4} = 4\sqrt{5} \text{ m/s}$

5.(A) $A_1V_1 = A_2V_2 \quad \Rightarrow \quad \frac{\pi}{4}(2.5)^2 V_1 = \frac{\pi}{4}(3.75)^2 V_2 \quad \Rightarrow \quad \frac{V_1}{V_2} = \frac{9}{4}$

6.(AB) ℓ decreases (wooden block will displace liquid equal to its weight only)

h also decreases (initially coin displaced liquid equal to its weight but it displaces liquid equal to its volume)

7.(C) $\frac{mg}{a} = \rho gh \quad \Rightarrow \quad h = \frac{m}{\rho a}$

8.(BD) Rate of liquid out flow at maximum height = rate of inflow

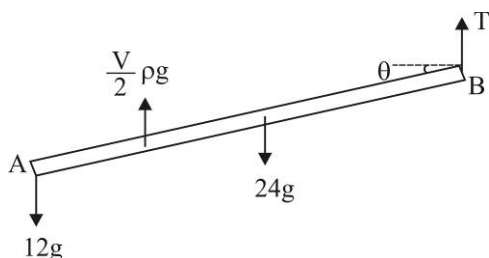
$av = \phi$

$a\sqrt{2gh} = \phi \quad \Rightarrow \quad h \propto \phi^2, \quad h \propto \frac{1}{a^2}$

9.(C) $V_T \propto r^2(d - \rho) \quad \therefore \rho$ is negligible, $V_T \propto r^2d$

$$\therefore \frac{r_1}{r_2} = \frac{1}{2} \quad \therefore \frac{d_1}{d_2} = 8 \quad \therefore \frac{V_{T_2}}{V_{T_1}} = \frac{r_2^2 d_2}{r_1^2 d_1} = \frac{1}{2}$$

10.(C)



About B:

$$24g \frac{\ell}{2} \cos \theta + 12g \ell \cos \theta = \frac{V}{2} \rho g \frac{3\ell}{4} \cos \theta$$

$$\Rightarrow V = 6.4 \times 10^{-2} \text{ m}^3$$

11.(C) Initial pressure inside bubble = $P_1 + \frac{4T}{r}$, Final pressure inside bubble = $P_2 + \frac{4T}{r/2}$

For Isothermal compression,

$$\left(P_1 + \frac{4T}{r}\right) \frac{4}{3} \pi r^3 = \left(P_2 + \frac{8T}{r}\right) \frac{4}{3} \pi \left(\frac{r}{2}\right)^3 \Rightarrow 8\left(P_1 + \frac{4T}{r}\right) = P_2 + \frac{8T}{r} \Rightarrow P_2 = 8P_1 + \frac{24T}{r}$$

12.(C) $V = \sqrt{\frac{2\Delta P}{\rho}} = \sqrt{\frac{2(\rho_1 g h_1 + \rho_2 g h_2)}{\rho_2}} = \sqrt{2(0.9 \times 400 + 1 \times 100)980} \text{ cm/s} = \sqrt{920 \times 980} \text{ cm/s}$

13.(B) As pressure varies linearly $\therefore$ Mean pressure = pressure at half depth = $hdg/2$

14.(C) $S_1 = 2(4\pi r_1^2)$ & $S_2 = 4\pi(2)^{2/3} r_1^2$ $\therefore$ Ratio to surface energy = $\frac{S_2}{S_1} = 1 : 2^{1/3}$

15.(D) The block loses $1/3^{\text{rd}}$ of weight $\therefore$ Specific gravity of block = 3

16.(B) $V_T = \frac{2r^2(d - \rho)g}{9\eta}$ $\therefore V_T \propto r^2$

17.(B) Acceleration of ball upwards = $\frac{V(d - D)g}{VD} = \left(\frac{d}{D} - 1\right)g$

Velocity when it reaches water surface = $\sqrt{2h\left(\frac{d}{D} - 1\right)g}$

Height attained above water surface = $\frac{2h\left(\frac{d}{D} - 1\right)g}{2g} = h\left(\frac{d}{D} - 1\right)$

18.(A) Density of ice < density of water form < density of sugar solution

As density of water form is less than the density of sugar solution $\Rightarrow$ total volume increases.

19.(D) $\therefore$ For same material

$$V_T \propto r^2$$

$\therefore$ Ratio of radius = 3 : 2

$\therefore$ Ratio of volume = 27 : 8

20.(D) $p_1 < p_2$

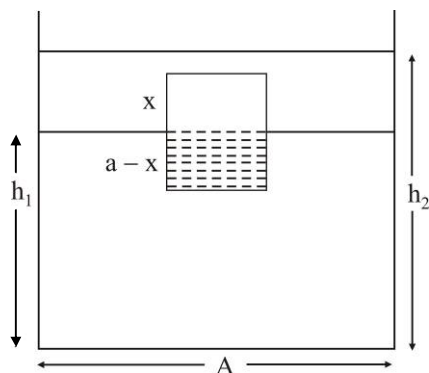
As liquid 1 is above liquid 2 : $\rho_3 < \rho_2$

Else ball would sink in liquid 2 : $\rho_1 < \rho_3$

Ball sank in liquid 1 $\therefore \rho_1 < \rho_3 < \rho_2$

21.(B) $V = \sqrt{\frac{2\Delta P}{\rho}} = \sqrt{\frac{2 \times (2 \times 10^5)}{1000}} = 20 \text{ m/s}$

22.(A)



$$\rho_{oil} < \rho_{ice} < \rho_{water}$$

Initially volume of ice inside water $= a^2(a-x)$,

assuming the ice is cube of side a .

From buoyancy: $a^3 \rho_{ice} g = a^2(a-x) \rho_{water} g + a^2 x \rho_{oil} g$

$$\Rightarrow a \rho_{ice} = (a-x) \rho_{water} + x \rho_{oil}$$

$$\begin{aligned} \text{Volume of water formed} &= \frac{\text{mass of ice}}{\text{density of water}} = \frac{\rho_{ice} a^3}{\rho_{water}} \\ &= \frac{a^2}{\rho_{water}} [(a-x) \rho_{water} + x \rho_{oil}] = a^2(a-x) + \frac{a^2 x \rho_{oil}}{\rho_{water}} \end{aligned}$$

Volume of water formed = initial volume of ice inside water + $\frac{a^2 x \rho_{oil}}{\rho_{water}} \Rightarrow h_1$ is increased.

$$\text{Also volume of water formed} = a^2(a-x) + \frac{\rho_{oil} x a^2}{\rho_{water}}.$$

$$= a^3 - a^2 x \left[1 - \frac{\rho_{oil}}{\rho_{water}} \right] < a^3 \quad (\text{as } \rho_{oil} < \rho_{water}) \Rightarrow h_2 \text{ is decreased.}$$

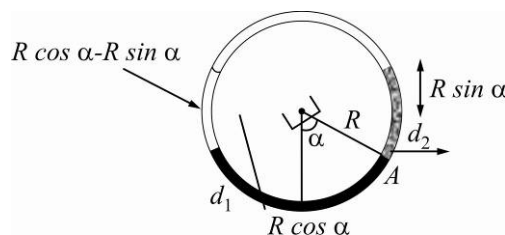
23.(C) Equating pressure at A , we get

$$R \sin \alpha d_2 + R \cos \alpha d_2 + R(1 - \cos \alpha) d_1$$

$$= R(1 - \sin \alpha) d_1$$

$$(\sin \alpha + \cos \alpha) d_2 = d_1 (\cos \alpha - \sin \alpha)$$

$$\Rightarrow \frac{d_1}{d_2} = \frac{1 + \tan \alpha}{1 - \tan \alpha}$$



24.(A) The bubble will detach, if

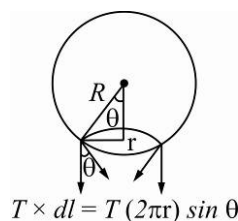
Buoyant force $\geq$ Surface tension force

$$\text{i.e., } \frac{4}{3} \pi R^3 \rho_w g \geq \int T \times dl \sin \theta$$

$$T \times dl = T(2\pi r) \sin \theta$$

$$(\rho_w) \left(\frac{4}{3} \pi R^3 \right) g \geq (T)(2\pi r) \sin \theta$$

$$\sin \theta = \frac{r}{R}$$



$$T \times dl = T(2\pi r) \sin \theta$$

$$\text{Solving, } r = \sqrt{\frac{2\rho_w R^4 g}{3T}} = R^2 \sqrt{\frac{2\rho_w g}{3T}}$$

25.(D) If a layer of thickness dr is evaporated, then change in surface energy

$$= \text{Change in surface area} \times T = [4\pi(r+dr)^2 - 4\pi r^2] T = [4\pi r^2 + 4\pi dr^2 + 8\pi r dr - 4\pi r^2]$$

As dr is very small, so dr^2 neglected. So, change in surface energy $= 8\pi r dr T$

Energy required to evaporate layer of thickness $dr = (4\pi r^2 dr) \rho L$

The process of evaporation just starts, if change in surface energy is just sufficient to evaporate the water layer.

$$\therefore (4\pi r^2 dr) \rho L = (8\pi r dr) T \Rightarrow r = \frac{2T}{\rho L}$$

26.(C) As terminal velocity, $u = \frac{2}{9} r^2 \frac{(\rho - \rho_0)}{\eta}$

Then, $v \propto \frac{\rho - \rho_0}{\eta}$

$$\therefore \frac{v_2}{v_1} = \frac{\rho - \rho_{02}}{\rho - \rho_{01}} \times \frac{\eta_1}{\eta_2} = \frac{7.8 - 1.2}{7.8 - 1} \times \frac{8.5 \times 10^{-4} \times 10}{13.2}$$

$$6.25 \times 10^{-4} \text{ cm s}^{-1}$$

27.(C) From Bernoulli's theorem, $\rho gh = \frac{1}{2} \rho (v_2^2 - v_1^2)$

$$\Rightarrow gh = \frac{1}{2} v_1^2 \left[\left(\frac{v_2}{v_1} \right)^2 - 1 \right] \Rightarrow gh = \frac{1}{2} v_1^2 \left[\left(\frac{A_1}{A_2} \right)^2 - 1 \right] \quad [\because A_1 v_1 = A_2 v_2]$$

$$\Rightarrow \left(\frac{A_1}{A_2} \right)^2 = 1 + \frac{2gh}{v_1^2} \Rightarrow \left(\frac{D_1}{D_2} \right)^4 = 1 + \frac{2gh}{v_1^2} \Rightarrow D_2 = \frac{D_1}{\left(1 + \frac{2gh}{v_1^2} \right)^{1/4}}$$

$$= \frac{8 \times 10^{-3}}{\left(1 + \frac{2 \times 10 \times 0.2}{(0.4)^2} \right)^{1/4}} = 3.6 \times 10^{-3} \text{ m}$$

28.(D) $A_1 V_1 = A_2 V_2$ So $V_2 = \frac{V \pi (2R)^2}{\pi R^2} = 4V$

29.(C) Because of upward acceleration pressure will increase.

30.(B) To get maximum upward acceleration pressure will increase

31.(B) $\frac{F_u}{W} = \frac{\rho_1}{\rho_s} = \frac{2}{3}$ or $\rho_s = \frac{3}{2} \rho_\ell$

32.(C) $v \rho g = v_1 \rho_1 g + v_2 \rho_2 g$, $v_1 \rho_1 + v_2 \rho_2 = v \rho$ and $v_1 + v_2 = v$

$$(v - v_2) \rho_1 + v_2 \rho_2 = v \rho, v_2 (\rho_2 - \rho_1) = v (\rho - \rho_1), \frac{v_2}{v} = \frac{\rho - \rho_1}{\rho_2 - \rho_1} = \frac{\rho_1 - \rho}{\rho_1 - \rho_2}$$

33.(B) $\frac{H}{10}$ part of ice slab will be out of water. Volume of ice slab outside the water = $\frac{A(0.5)}{10} m^3$ (A = area of slab)

$$A \times \left(\frac{0.5}{10} \right) \times 10^3 g = 100 g, A = \frac{10}{5} = 2 m^2$$

34.(B) $P_2 + a \rho L = P_1$, $P_1 - P_2 = a \rho L$ $h_1 - h_2 = \frac{aL}{g}$

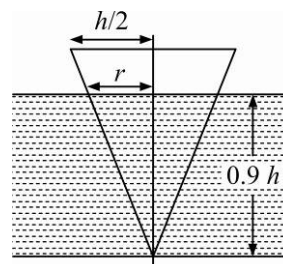
35.(D) Since the force due to fluid acts normal to the surface, hence about point O, no net torque is acting on the gate hence F = O.

36.(A) $\frac{h}{h/2} = \frac{0.9h}{r} \Rightarrow r = \frac{0.9h}{2}$

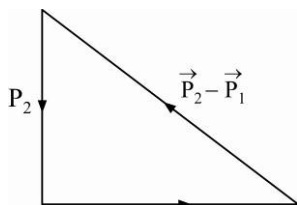
$$\text{Upthrust} = \frac{1}{2} \pi r^2 \cdot (0.9h) g \rho_1 = \frac{1}{3} \pi \left(\frac{h}{2} \right)^2 h \cdot \rho \cdot g \quad \dots (i)$$

$$\text{When totally submerged } w + \frac{1}{3} \pi \left(\frac{h}{2} \right)^2 gh \rho_s = \frac{1}{3} \pi \left(\frac{h}{2} \right)^2 g \rho_\ell h$$

On solving (i) and (ii) and (iii) W = 568 m



- 37.(B) Rate of change of momentum $|\vec{P}_2 - \vec{P}_1| = \sqrt{2}sv^2\rho$



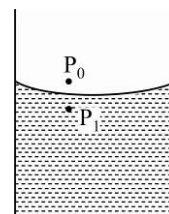
38.(C)

- 39.(A) Pressure difference in both cases should be same.

$$10\rho g = L \cos 45^\circ \rho g \text{ or } L = 10\sqrt{2}cm$$

- 40.(D) Shape will be hemispherical.

- 41.(B) Shape of meniscus is convex, if seen from inside the water, hence pressure inside the water is $P_1 = P_2 - \frac{2T}{r}$ where r = radius of curvature of meniscus.



- 42.(B) Excess pressure $\Delta P = \frac{4T}{R}$

$\Delta P \cdot R = 4T = \text{constant}$, R is increasing linear with time t . Hence $\Delta Pt = \text{constant}$.

- 43.(A) Since viscous force depends upon velocity, viscous force on sphere will increase till it achieves terminal velocity.

44.(C)

- 45.(D) $\Delta A = 4\pi(R_2^2 - R_1^2) = 12\pi R^2$ $W = \Delta E = 2S\Delta A = 24\pi R^2 S$

- 46.(B) Pressure in the concave surface is $\frac{2T}{r}$ greater than convex surface, hence P inside = $P + hdg + \frac{2T}{r}$.

- 47.(D) $6 \times 10^{-2} \times (2\pi r) = 75 \times 10^{-4}$ $(2\pi r) = \frac{75}{6} \times 10^{-2} = 12.5 \times 10^{-2} m$

INTEGER VALUE TYPE

- 48.(15) $\rho_1 h_1 = \rho_2 h_2$

$$1000 \times h_1 = 1300 \times h_2 \quad ; \quad h_1 = \frac{1300}{1000} \times 5 = 6.5 cm$$

$$\begin{aligned} \text{Difference} &= h_1 - h_2 = 6.5 - 5 \\ &= 1.5 cm \\ 10h &= 15 cm \end{aligned}$$

- 49.(10) Let mass of cube = m

$$(m + 200)g = \ell^3 \rho g \quad \text{--- (1)}$$

$$mg = (\ell - 2)\ell^2 \rho g \quad \text{--- (2)}$$

Subtract (2) from (1)

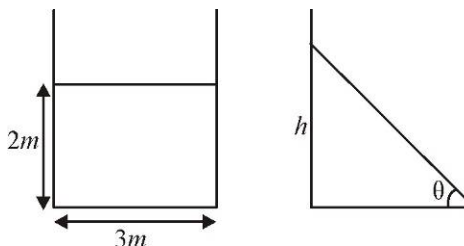
$$200 = \rho[\ell^3 - (\ell - 2)\ell^2]$$

$$\ell = 10 cm \quad \ell = 1g/cc$$

- 50.(7) Volume will be same

$$2 \times 3 = \frac{1}{2} \times h \times 3 \Rightarrow h = 4m$$

$$\tan \theta = \frac{4}{3} = \frac{a}{g} \Rightarrow a = \frac{4}{3}g$$



$$51.(9) \quad (2.4 \times 10^{-4}) \times 1.5 = 20 \times 2 \times 10^{-2} \times 10^{-4} \times V$$

$$V = \frac{2.4 \times 1.5 \times 10^2}{40} \text{ m/s} = 9 \text{ m/s}$$

$$52.(2) \quad \text{Reading will be total mass} = 2 \text{ kg}$$

$$53.(8) \quad \text{Pressure at } A = P_o + \rho gh$$

Also pressure at A

$$= P_o + \int \rho a dx$$

$$a = \omega^2 x$$

$$P_A = P_o + \left[\frac{\rho \omega^2 x^2}{2} \right]_0^R$$

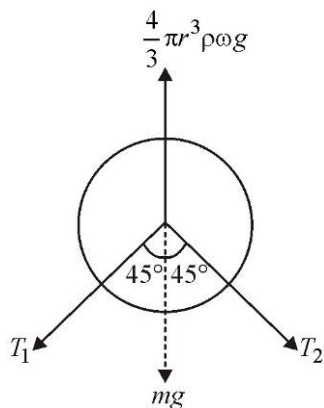
$$P_A = P_o + \frac{\rho \omega^2 R^2}{2}$$

$$\therefore P_o + \rho gh = P_o + \frac{\rho \omega^2 R^2}{2}$$

$$h = \frac{\omega^2 R^2}{2g} \quad ; \quad \omega = 2\pi(4 \text{ rad/s})$$

$$\therefore h = 8 \text{ cm}$$

$$54.(250) \quad T_1 = T_2$$



$$Mg + T\sqrt{2} = \frac{4}{3}\pi R^3 \rho wg$$

$$T_1 = 250\sqrt{2} \text{ N}$$

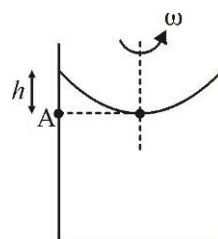
$$55.(4) \quad A(dp) = (eAgdx)xw^2$$

$$\int dp = ew^2 \int_0^L x dx$$

$$\Delta P = \frac{ew^2 L^2}{2} \Rightarrow egH = \frac{ew^2 L^2}{2}$$

$$H = \frac{w^2 L^2}{2g} = \frac{4g}{L} \cdot \frac{L^2}{2g} = 2L = 4 \text{ m}$$

$$56.(5) \quad V_1 = \pi R^2 \cdot R - \frac{2}{3} \pi R^3 \quad [e \text{ is density of liquid}]$$



$$= \frac{\pi R^3}{3}$$

$$F_1 \text{ evl } g = \frac{\pi R^3 eg}{3}$$

$$-F_1 + F_2 = F_b$$

$$F_2 = \frac{4}{3} \pi R^3 eg + \frac{\pi R^3 eg}{3}$$

$$= \frac{5\pi R^3 eg}{3}$$

$$\frac{F_2}{F_1} = 5$$

$$57.(50) \quad h = h_o = \frac{2T \cos \theta}{\text{erg}} = 57.6 \text{ cm} > 50 \text{ cm}$$

So it will rise only 50 cm

58.(5) Let the volume of the cylinder be V when the cylinder is floating upthrust = weight hence

$$\rho = \left(\frac{3}{4} V\right) g = mg \Rightarrow V = \frac{4m}{3\rho}$$

Let the acceleration of the particle vessel be A (upwards). In the reference from of vessel the acceleration of the cylinder is A/3.

$$\therefore mg + mA - \text{upthrust} = m \left(\frac{A}{3}\right)$$

$$mg + mA - \rho V g' = m \left(\frac{A}{3}\right)$$

Where $g' = g + A = \text{effective value of } g \text{ for upthrust}$

$$\therefore mg + mA - \rho V (g + A) = m \left(\frac{A}{3}\right)$$

$$mg - \frac{4}{3} m(g + A) = m \cdot \frac{2}{3} A$$

$$A = \left(-\frac{9}{2}\right) \text{ upwards}$$

59.(5) Due, to rotation, let the shift of the liquid be x cm.

Let the cross section area of the tube be A.

In the right limb for compressed air.

$$P_1 V_1 = P_2 V_2$$

$$P_0 A \times 6 = P_2 A (6 - x)$$

$$P_2 = \frac{6P_1}{6 - x}$$

Force at the corner C of right limb climb due to liquid above.

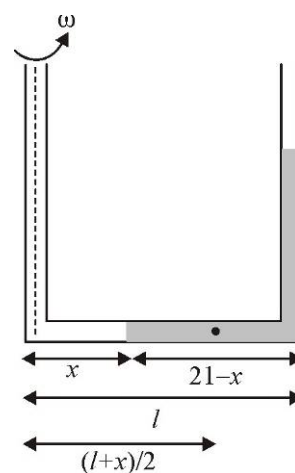
$$P_1 = \left[\frac{6P_1}{6 - x} + x\rho y \right] A$$

Mass of the liquid in the horizontal arm, $M = \rho(l - x)A$ it is rotated above left limb. Then the centripetal force is

$$F_2 = m\omega_0^2 r = \rho(l - x)A\omega_0^2 \frac{(l - x)}{2} (l^2 - x^2)$$

But $F_1 = F_2$

$$\frac{\rho A \omega_0^2 (l^2 - x^2)}{2} = \left[\frac{\delta \rho_0}{\delta - x} + x\rho g \right] A$$



$$= \frac{10^5 \times 100(2l^2 - x^2) \times 10^{-4}}{2}$$

$$= \left[\frac{\delta \times 10500}{(\delta - x)} + x \times 10^5 \times 10 \times 10^{-2} \right]$$

On solving, we get $x = 0.01\text{m} = 1\text{cm}$

The length of the air column is $= 6 - 1 = 5\text{cm}$

60.(2) Let y be the length of the plank inside water.

$$y = \frac{0.5}{\cos \theta}$$

Let A be the cross section area of the plank, then the force on it is

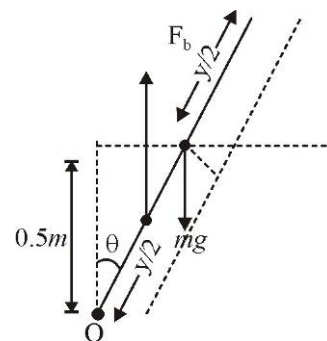
$$F_b = V\rho\omega g = (Ay)\rho\omega g$$

Since the plank is in rotational equilibrium, so $\sum \vec{\tau}_0 = 0$

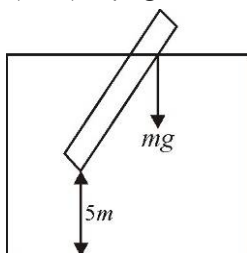
$$\text{or } mg \times \frac{l}{2} \sin \theta - F_b \cdot \frac{y}{2} \sin \theta = 0 \quad \text{or } mgl - F_b \times y = 0$$

$$\text{or } (Al \times 0.5)gl - (Ay)y = 0 \Rightarrow 0.5l^2 = y^2$$

$$\text{or } 0.5(1)^2 = \left(\frac{0.5}{\cos \theta} \right)^2 \quad \text{or } \cos^2 \theta = \frac{1}{2}$$



61.(1) $(6-x)^2 A\rho\omega g \cdot 2 = 18A\rho g$

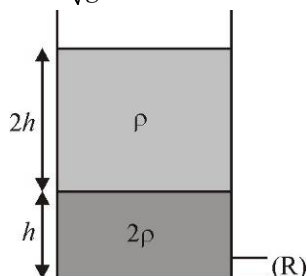


62.(2) Pressure at 1, $P_1 = P_{atm} + P_g(2x)$

Apply Bernoulli's theorem between point (1) & (2)

$$(P_{atm} + 2\rho gh) + \rho g(3h - h) = P_{atm} + \frac{1}{2}(2\rho)v^2$$

$$v = 2\sqrt{gh}$$



Simple Harmonic Motion

$$1.(C) \quad \frac{1}{2}KA^2 = \frac{1}{2}mu^2 \quad \left\{ \text{where } A = \frac{a+b}{2} \right\}$$

$$\Rightarrow \quad \frac{m}{k} = \left(\frac{a+b}{2u} \right)^2 \quad \therefore \quad T = 2\pi\sqrt{\frac{m}{k}} = \frac{\pi(a+b)}{u}$$

$$2.(C) \quad x = \frac{F}{K} = \frac{3}{10} = 0.3m \quad \therefore \quad \text{length} = 2.3m$$

$$3.(D) \quad \frac{1}{2}mA^2\omega^2 = 0.9 \quad ; \quad \frac{1}{2}(0.2)A^2(50)^2 = 0.9$$

4.(D)

5.(C) The particle oscillates but period decreases with time

$$6.(B) \quad v_1 = \frac{1}{2\pi}\sqrt{\frac{K_1}{m}} \quad ; \quad v_2 = \frac{1}{2\pi}\sqrt{\frac{K_2}{m}} \quad ; \quad v = \frac{1}{2\pi}\sqrt{\frac{K_1+K_2}{m}} \quad \therefore \quad v = \sqrt{v_1^2 + v_2^2}$$

$$7.(C) \quad \text{Time taken to reach half the mean position} = \frac{T}{12} = \frac{6}{12} = \frac{1}{2} \text{ sec.}$$

8.(B) The equation of motion of a point at $x = 75 \text{ cm}$ is $y = 0.04 \sin(600\pi t + 1.5\pi)$ or $0.04 \sin(600\pi t - 1.5\pi)$

According to as wave traveling towards left or right respectively.

$$\Rightarrow \quad \text{At } t = 0.015, \quad y(\text{at } x = 75 \text{ cm}) = 0.04 \sin(7.5\pi) \quad \text{or} \quad 0.04 \sin(4.5\pi) \quad \Rightarrow \quad y = 0.04 \quad \text{or} \quad -0.04m$$

$$9.(A) \quad T.E. = \frac{1}{2}KA^2 = \frac{1}{2}(m\omega^2)A^2 = \frac{1}{2}\left(\frac{m4\pi^2}{T^2}\right)A^2$$

$$10.(A) \quad \Delta\phi = \frac{2\pi}{\lambda}(\Delta x) = k\Delta x$$

$$\frac{\pi}{3} = 3(\Delta x)$$

$$\Delta x = \frac{\pi}{9}m.$$

11.(B) Comparing given equation with $y = A \sin(kx - \omega t)$ $c = 2m/s$. Position $= c \times t = 2 \times 5 = 10m$.

12.(B) Intensity of reflected wave $= 100 - 36 = 64\% \Rightarrow$ amplitude $(\propto \sqrt{I})$ of reflected wave $= 0.8A$

13.(A) Equation of reflected wave is $y = 0.8A \sin\left(ax - bt + \frac{3\pi}{2}\right)$, due to reflection at denser medium a phase difference of π occurs.

$$\text{Equation to resulting wave is } Y = A \sin(ax + bt + \pi/2) + 0.8A \sin\left(ax - bt + \frac{3\pi}{2}\right)$$

$$\Rightarrow \quad y = A \cos(ax + bt) - 0.8A \cos(ax - bt) = 0.8A [\cos(ax + bt) - \cos(ax - bt)] + 0.2A \cos(ax + bt)$$

$$= -1.6A \sin ax \sin bt + 0.2A \cos(ax + bt) \quad \Rightarrow \quad C = 0.2$$

$$14.(C) \quad \sin ax = \pm 1 \Rightarrow ax = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \dots \Rightarrow x = \frac{3\pi}{2a} \text{ for 2nd antinodes.}$$

15.(A) Let the block is displaced down by x so that x_1 and x_2 be further elongations in left and right springs respectively.

$$\Rightarrow \quad x = \frac{x_1 + x_2}{2} \quad \& \quad kx_1 = kx_2 \quad \Rightarrow \quad x = x_1 = x_2$$

$$\Rightarrow \text{restoring force} = 2kx \quad \Rightarrow \quad T = 2\pi\sqrt{\frac{m}{2k}}$$

$$16.(C) \quad T = 2\pi\sqrt{\frac{m}{K}}, K_{eq} = 3K$$

$$17.(D) \quad T = 2\pi\sqrt{\frac{\ell}{g}} \Rightarrow T^2 = 4\pi^2 \frac{\ell}{g} \quad \text{So graph is parabolic}$$

$$18.(D) \quad f = ma \quad (\text{for block})$$

$$\Rightarrow \mu mg = m\omega^2 A \quad (A \text{ is maximum amplitude})$$

$$\Rightarrow \mu g = (2\pi\nu)^2 A \quad \Rightarrow \quad A = \frac{\mu g}{4\pi^2\nu^2}$$

$$19.(A) \quad h_p = \frac{dy}{dt} \Big|_{\max} = 2\pi ab$$

$$\frac{dx}{dt} = \frac{b}{c}$$

$$\frac{dy}{dx} \Big|_{\max} = 2\pi ac$$

$$\therefore 2\pi ab = \frac{ab}{c} \Rightarrow c = \frac{1}{\pi a}$$

$$20.(D) \quad a = -\omega^2 y$$

$$21.(B) \quad (KE)_{\text{mean}} = \frac{1}{2} KA^2$$

$$(PE)_{\text{at } A/2} = \frac{1}{2} K \left(\frac{A}{2}\right)^2$$

$$\frac{KE_{\text{mean}}}{(PE)_{\text{at } A/2}} = \frac{4}{1}$$

$$22.(C) \quad \text{Amplitude of damped oscillator is given by } A = A_0 e^{\frac{bt}{2m}}$$

$$\text{After 5 s, } 0.9 A_0 = A_0 e^{\frac{b(5)}{2m}} \Rightarrow 0.9 = e^{\frac{-b(15)}{2m}} \quad \dots(i)$$

$$\text{After, 10 s } A = A_0 e^{\frac{-b(15)}{2m}} \Rightarrow A = A_0 (e^{\frac{-5b}{2m}})^3$$

$$\text{From equation (i) and (ii), we get: } A = 0.729 A_0 \quad \text{Hence, } \alpha = 0.729$$

$$23.(D) \quad \text{For damped harmonic motion}$$

$$ma = -kx - mbv \quad \text{or} \quad ma + mbv + kx = 0$$

$$\text{Solution of above equation is } x = A_0 e^{\frac{-b\tau}{2}} \sin \omega\tau; \text{ with } \omega^2 = \frac{k}{m} - \frac{b^2}{4}$$

$$\text{where, amplitude drops exponentially with time. i.e., } A_\tau = A_0 e^{\frac{-b\tau}{2}}$$

Average time τ is that duration when, amplitude drops by 63% i.e., becomes A_0/e .

$$\text{Thus, } A_\tau = \frac{A_0}{e} = A_0 e^{\frac{-b\tau}{2}} \quad \text{Or} \quad \frac{b\tau}{2} = 1 \quad \text{or} \quad \tau = \frac{2}{b}$$

$$24.(B) \quad \text{For a string fixed at both end, the resonant frequencies are } \nu_n = \frac{nv}{2L} \text{ when } n = 1, 2, 3, \dots$$

$$\text{The difference between two consecutive resonant frequencies is } \Delta\nu_n = \nu_{n+1} - \nu_n = \frac{(n+1)v}{2L} - \frac{nv}{2L} = \frac{v}{2L}$$

$$\text{Which is also the lowest resonant for the given string} = 420\text{Hz} - 315\text{Hz} = 105\text{Hz}$$

25.(B) If A and ω be amplitude and angular frequency of vibration, then

$$\alpha = \omega^2 A \quad \dots (i)$$

And $\beta = \omega A \quad \dots (ii)$

Dividing equation (i) by equation (ii), we get, $\frac{\alpha}{\beta} = \frac{\omega^2 A}{\omega A} = \omega$

$$\therefore \text{Time period vibration is } T = \frac{2\pi}{\omega} = \frac{2\pi}{(\alpha/\beta)} = \frac{2\pi\beta}{\alpha}$$

26.(D) in SHM, velocities of a particle at distances x_1 and x_2 from mean position are given by

$$V_1^2 = \omega^2 (a^2 - x_1^2) \quad \dots (i)$$

$$V_2^2 = \omega^2 (a^2 - x_2^2) \quad \dots (ii)$$

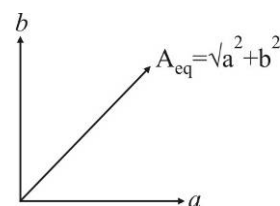
From equations (i) and (ii), we get:

$$V_1^2 - V_2^2 = \omega^2 (x_2^2 - x_1^2) \quad \omega = \sqrt{\frac{V_1^2 - V_2^2}{x_2^2 - x_1^2}} \quad \therefore T = 2\pi \sqrt{\frac{x_2^2 - x_1^2}{V_1^2 - V_2^2}}$$

27.(A) Here, $y_1 = a \sin \omega t$

$$y_1 = b \cos \omega t = b \sin \left(\omega t + \frac{\pi}{2} \right)$$

Hence, resultant motion is SHM with amplitude $\sqrt{a^2 + b^2}$.



28.(C) Here, $X = A \cos \omega t$

$$\therefore \text{Velocity, } v = \frac{dX}{dt} = \frac{d}{dt}(A \cos \omega t) = -A \omega \sin \omega t$$

$$\text{Acceleration, } a = \frac{dv}{dt} = \frac{d}{dt}(-A \omega \sin \omega t) = -A \omega^2 \cos \omega t$$

Hence the variation of a with t is correctly shown by graph (c).

$$29.(C) \quad y = \sin \omega t - \cos \omega t = \sqrt{2} \left[\frac{1}{\sqrt{2}} \sin \omega t - \frac{1}{\sqrt{2}} \cos \omega t \right] = \sqrt{2} \sin \left(\omega t - \frac{\pi}{4} \right)$$

It represents a SUM with time period, $T = \frac{2\pi}{\omega}$.

$$y = \sin^3 \omega t = \frac{1}{4} [3 \sin \omega t - \sin 3 \omega t]$$

It represents a period motion with time period $T = \frac{2\pi}{\omega}$ but not SHM.

$$y = 5 \cos \left(\frac{3\pi}{4} - 3\omega t \right) = 5 \cos \left(3\omega t - \frac{3\pi}{4} \right) \quad [\because \cos(-\theta) = \cos \theta]$$

It represents a SHM with time period, $T = \frac{2\pi}{3\omega}$.

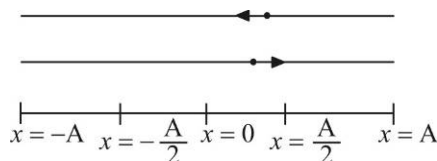
$y = 1 + \omega t + \omega^2 t^2$ represents a non-periodic motion. Also it is not physically acceptable as the $y \rightarrow \infty$ as $t \rightarrow \infty$.

- 30.(C) The time taken by the particle to travel from $x = 0$ to $x = \frac{A}{2}$ is $\frac{T}{12}$.

The time taken by the particle to travel from $x = A$ to $x = \frac{A}{2}$ is $\frac{T}{6}$.

$$\text{Time difference} = \frac{T}{6} + \frac{T}{6} = \frac{T}{3}$$

$$\text{Phase difference, } \phi = \frac{2\pi}{T} \times \text{time difference} = \frac{2\pi}{T} \times \frac{T}{3} = \frac{2\pi}{3}$$



- 31.(A) $x = a \sin^2 \omega t = a \left(\frac{1 - \cos 2\omega t}{2} \right)$ ($\because \cos \theta = 1 - 2\sin^2 \theta$) $= \frac{a}{2} - \frac{a \cos 2\omega t}{2}$

$$x - \frac{a}{2} = -\frac{a}{2} \cos 2\omega t \Rightarrow X = -\frac{a}{2} \cos 2\omega t$$

The given motion is SHM motion with mean at $x = \frac{a}{2}$ and time period $T = \frac{2\pi}{2\omega} = \frac{\pi}{\omega}$

- 32.(A) The spring has a length l . when m is placed over it the equilibrium position becomes O' if is pressed from O' (the equilibrium position) to O'' , $O'O''$ is the amplitude.

$$OO' = \frac{mg}{k} = \frac{2 \times 10}{200} = 0.10 m. \quad mg = kx_0.$$

If the restoring force $mA\omega^2 > mg$, then the mass will move

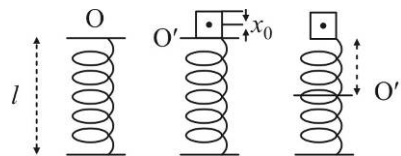
Up with acceleration, detached from the pan.

$$\text{i.e.} \quad A > \frac{g}{k/m} \Rightarrow A > \frac{20}{200} > 0.10 m$$

The amplitude $> 10 \text{ cm}$.

i. e. the minimum is just greater than 10 cm .

(the actual compression will include x_0 also. But when talking of amplitude, it is always from the equilibrium position with respect to which the mass is oscillating).



- 33.(C) Kinetic energy + potential energy = total energy when kinetic energy is maximum, potential energy is zero and vice versa.

$$\therefore \quad \text{Maximum potential energy} = \text{total energy. } 0 + K_0 = K_0$$

(K.E. + P.E = total energy).

- 34.(C) Let k be the force constant of spring. If k' is the force constant of each part, then

$$\frac{1}{k} = \frac{4}{k'} \Rightarrow k' = 4k$$

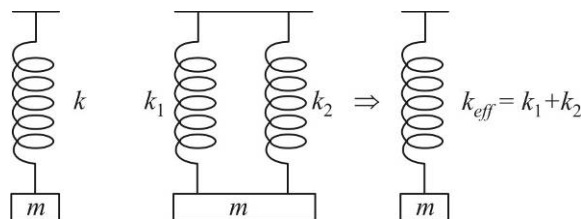
$$\therefore \quad \text{Time period} = 2\pi\sqrt{\frac{m}{4k}} = \frac{1}{2} \times 2\pi\sqrt{\frac{m}{k}} = \frac{T}{2}.$$

- 35.(B) The time period of a spring mass system as shown in figure 1 is given by $T = 2\pi\sqrt{m/k}$, where k is the spring constant.

$$\therefore \quad t_1 = 2\pi\sqrt{m/k_1} \quad \dots \text{ (i)}$$

$$\text{And } t_2 = 2\pi\sqrt{m/k_2} \quad \dots \text{ (ii)}$$

Now, when they are connected in parallel as shown in Figure 2(a), the system can be replaced by a single spring of Spring constant, $k_{\text{eff}} = k_1 + k_2$.



[Since $mg = k_1x + k_2x = k_{eff} x$]

$$\therefore t_0 = 2\pi\sqrt{m/k_{eff}} = 2\pi\sqrt{m/(k_1 + k_2)}$$

$$\text{From (i), } \frac{1}{t_2^2} = \frac{1}{4\pi^2} \times \frac{k_1}{m} \quad \text{From (ii), } \frac{1}{t_2^2} = \frac{1}{4\pi^2} \times \frac{k_2}{m} \quad \text{From (iii), } \frac{1}{t_0^2} = \frac{1}{4\pi^2} \times \frac{k_1 + k_2}{m}$$

$$\Rightarrow \frac{1}{t_1^2} + \frac{1}{t_2^2} = \frac{1}{4\pi^2 m} (k_1 + k_2) = \frac{1}{t_0^2} \quad \therefore t_0^{-2} = t_1^{-2} + t_2^{-2}$$

36.(B) The amplitude and velocity resonance occurs at the same frequency.

At resonance, i.e., $\omega_1 = \omega_0$ and $\omega_2 = \omega_0$ the amplitude and energy of the particle would be maximum.

37.(C) $x = a \sin \omega t$

$$y = a \sin(\omega t + \pi/2) = a \cos \omega t$$

$$\text{or } \frac{x}{y} = \frac{\sin \omega t}{\cos \omega t} = \tan \omega t \quad \text{or } \frac{x}{y} = \frac{x}{\sqrt{a^2 - x^2}} \quad \text{or } y^2 = a^2 - x^2 \quad \text{or } x^2 + y^2 = a^2.$$

It is an equation of a circle.

38.(D) Maximum velocity, $v_{\max} = A\omega$

$$\text{According to question, } \frac{V_{\max}}{2} = \frac{A\omega}{2} = \omega\sqrt{A^2 - y^2}$$

$$\frac{A^2}{4} = A^2 - y^2 \Rightarrow y^2 = A^2 - \frac{A^2}{4} \Rightarrow y = \frac{\sqrt{3}A}{2}.$$

39.(D) Period of oscillation $T = 2\pi\sqrt{\frac{\ell}{g}}$. therefore T will decrease when acceleration (g) increases, and g will increase when the rocket moves up with a uniform acceleration.

40.(A) Time period = 4 sec. in one simple harmonic oscillation, the same kinetic and potential energies are repeated two times. So the difference will be 2 seconds.

41.(D) The effective value of acceleration due to gravity is $\sqrt{(a^2 + g^2)}$

42.(C) For simple harmonic motion velocity $v = \omega\sqrt{a^2 - x^2}$ at displacement x .

$$10 = \omega\sqrt{a^2 - 16} \quad \dots \text{ (i)}$$

$$8 = \omega\sqrt{a^2 - 25} \quad \dots \text{ (ii)}$$

$$\frac{100}{\omega^2} = a^2 - 16 \quad \dots \text{ (iii)}$$

$$\frac{64}{\omega^2} = a^2 - 25 \quad \dots \text{ (iv)}$$

$$\therefore \text{Equation (iii) - (iv) gives } \frac{36}{\omega^2} = 9 \Rightarrow \omega = 2 \text{ rad/s} \quad \text{or} \quad T = \frac{2\pi}{\omega} = \frac{2\pi}{2} = \pi \text{ sec}$$

43.(C) P.E. $V = \frac{1}{2} m \omega^2 x^2$

$$\text{And } K.E. \quad T = \frac{1}{2} m \omega^2 (a^2 - x^2) \quad \therefore \quad \frac{T}{V} = \frac{a^2 - x^2}{x^2}$$

INTEGER VALUE TYPE

44.(6) $V_{equilibrium} = \omega(\text{Amplitude})$

$$\text{So, } \frac{x}{1} = \frac{2}{1} \Rightarrow x = 6$$

45.(20) $mg = mk^2 A$
 $\Rightarrow k = 20$

46.(2) Moment of inertia $= \frac{2mL^2}{3}$

$$r_{com} = \frac{L}{\sqrt{2}}$$

$$T = 2\pi \sqrt{\frac{\sqrt{2}l}{3g}} = 2\pi \sqrt{\frac{\sqrt{2} \times 60\sqrt{2}}{30}} = 4\pi$$

47.(125) $mg + \frac{mg}{4} = kA \quad \Rightarrow A = \frac{5mg}{4K} = \frac{5 \times 1 \times 10}{4 \times 10} = 1.25m = 125 \text{ cm.}$

48.(1) $K.E_{equilibrium} = 4 = \frac{1}{2} \times m \times v_{eq}^2$

$$v_{eq} = 2m/s = \omega \times A \quad \Rightarrow \omega = 2 = \frac{2\pi}{T} \Rightarrow T = \pi \text{ so, } K=1$$

49.(2) Momentum conservation:

$$\frac{m}{4} v = mv_1 \Rightarrow v_1 = \frac{v}{4}$$

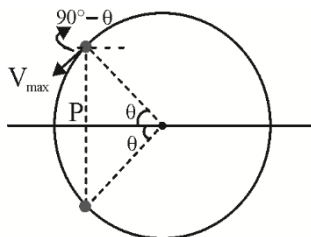
Energy conservation:

$$\frac{1}{2} m \left(\frac{v}{4} \right)^2 = \frac{1}{2} m \left(\frac{v}{8} \right)^2 \times 2 + \frac{1}{2} KA^2$$

$$\text{So, } A = v \sqrt{\frac{m}{32k}} = 1 \sqrt{\frac{64}{32 \times 0.5}} = 2m$$

50.(9) Time-period $= T = 0.5 + 1.5 = 2 \text{ sec}$

$$\omega = \frac{2\pi}{T} = \pi \text{ rad/s}$$



$$\therefore \omega \times 0.5 = 2\theta \text{ so, } \theta = \frac{\pi}{4}$$

$$v_{at P} = v_{max} \sin \theta \Rightarrow 3 = \frac{v_{max}}{\sqrt{2}} \Rightarrow v_{max} = 3\sqrt{2} \approx 4.2 \text{ m/s} . \text{ So Maximum kinetic energy} = 9 \text{ J}.$$

$$51.(75) \quad m\omega^2 x = mg$$

$$x = \frac{g}{\omega^2}$$

$$\text{Height above lowest point} = A + \frac{g}{\omega^2} = 5 \text{ cm} + \frac{10}{\left(2\pi \times \frac{10}{\pi}\right)^2} \times 100 \text{ cm} = 7.5 \text{ cm} = 75 \text{ mm}.$$

$$52.(2) \quad \omega T = 2\pi$$

$$\omega \frac{T}{8} = \frac{2\pi}{8} = \frac{\pi}{4}$$

$$v_{t=\frac{T}{8}} = v_{max} \cos \frac{\pi}{4} = \frac{1}{\sqrt{2}} v_{max}$$

$$\text{ratio} = \sqrt{2} = 1.414 \text{ so square of ratio is } 2.$$

$$53.(60) \quad x_1 = A \sin \omega t; \quad x_2 = A \sin(\omega t + \phi)$$

$$|x_2 - x_1|_{max} = \sqrt{A^2 + A^2 - 2A^2 \cos \phi} = 2A \sin \frac{\phi}{2} = A$$

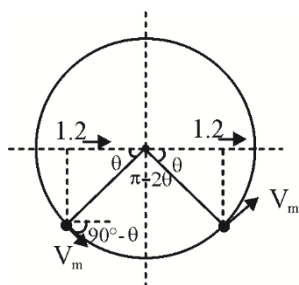
$$\sin \frac{\phi}{2} = \frac{1}{2} \Rightarrow \frac{\phi}{2} = 30^\circ \Rightarrow \phi = 60^\circ$$

$$54.(32) \quad \text{maximum displacement} = A\sqrt{2}$$

$$\text{Maximum average velocity} = 4 \frac{A\sqrt{2}}{T}$$

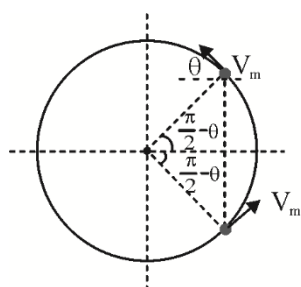
$$\text{So, } k = 4\sqrt{2} = 5.656 \text{ so } K^2 = 32.$$

$$55.(2)$$



$$V_m \cos(90^\circ - \theta) = 1.2$$

$$\text{i.e. } V_m \sin \theta = 1.2$$



$$V_m \cos \theta = 1.6$$

$$\text{So, } V_m = 2$$

$$v_m \cos \theta = 1.6$$

$$\text{So, } v_m = 2$$

$$56.(10) \quad a\alpha - \omega^2 \alpha$$

$$\text{So, } \omega = \sqrt{100} = 10 \text{ rad/s}$$

$$57.(2) \quad \left(\frac{2\pi}{T_1} - \frac{2\pi}{T_2} \right) \times 18 = 2\pi$$

$$\frac{1}{T_1} - \frac{1}{T_2} = \frac{1}{18}$$

$$\frac{1}{T_1} = \frac{1}{2} + \frac{1}{18} = \frac{9+1}{18}$$

$$T_1 = 1.8 \text{ sec (round to 2)}$$

$$58.(5) \quad F \cdot \frac{F}{K} - \frac{1}{2} K \left(\frac{F}{K} \right)^2 = \frac{1}{2} m V_{\max}^2$$

$$V_{\max} = F \sqrt{\frac{m}{K}}$$

$$\text{So, } V_{\max} \propto \frac{1}{\sqrt{K}}$$

If K becomes 4 times, $V_{\max}$ becomes 0.5 times. So Answer is 5.

Wave Motion

$$1.(D) \quad \phi_1 = \pi \quad \{ \because \text{both points are in adjacent loops} \}$$

$$\phi_2 = (x_2 - x_1) K = \frac{7}{6} \pi \quad \therefore \quad \frac{\phi_1}{\phi_2} = \frac{6}{7}$$

$$2.(BC) \quad C = \sqrt{\frac{\gamma P}{\rho}}$$

C increases if P increases. If humidity increases, ρ decreases $\therefore C$ increases

3.(D) Adiabatic compression of atmosphere. No transfer of heat

4.(B) Equation of reflected wave

$$y = 0.4 \sin(2\pi t + x\pi + \pi)$$

$$y = -0.4 \sin 2\pi \left(t + \frac{x}{2} \right)$$

5.(C) $v' = \frac{vc}{c-v}$ (while train is approaching) $v'' = \frac{vc}{c+v}$ (train is moving away)

6.(BCD) Phase of particles are different amplitude of all particles is same particles execute SHM
Wave velocity depends on properties of the medium.

7.(ABD) Between two consecutive nodes all particles vibrate in phase with same frequency. Their amplitude depends on the position of the particle energy between two nodes is constant.

8.(AB) The frequency of sound $v' = v \left(\frac{C + V_m - V_o}{C + V_m - V_s} \right)$
 $v' = v$

9.(C) $V = \sqrt{\frac{T}{\mu}} = 100 \text{ m/s}$

$$f_1 = \frac{V}{2\ell} = \frac{100}{2 \times 1} = 50 \text{ Hz}$$

$$f_2 = 100 \text{ Hz and } f_3 = 150 \text{ Hz.}$$

$$\eta_1 = \frac{V}{4\ell} = \frac{100}{4 \times 1} = 25 \text{ Hz. ; } \eta_2 = \frac{3V}{4\ell} = 75 \text{ Hz and } \eta_3 = \frac{5V}{4\ell} = 125 \text{ Hz}$$

10.(C) $1000 = \frac{(2n-1) \times 320}{4\ell} = \frac{(2n+1) \times 320}{4\ell_1} \Rightarrow \ell_1 - \ell = 0.16 \text{ m} = 16 \text{ cm}$

$$\text{Volume of water drained out} = 16 \times 100 = 1600 \text{ cm}^3$$

11.(A) $\Delta P = BKS_0$ and $B = rP$

$$\Delta P = (rP) \frac{2\pi}{\lambda} S_0$$

$$S_0 = \frac{\Delta P}{rP} \cdot \frac{\lambda}{2\pi} = \frac{1}{100r} \cdot \frac{\lambda}{2\pi}$$

$$S_0 = \frac{1}{100} \times \frac{P}{C^2 P} \times \frac{2\ell}{3.2\pi}$$

$$S_0 = \frac{1}{40} \text{ m} = 2.5 \text{ cm}$$

12.(A) $v_1 = \frac{V_1}{2\ell} ; v_2 = \frac{V}{\ell}$

$$\frac{v_1}{v_2} = \frac{V_1}{2V_2} = \frac{\sqrt{T_1}}{2\sqrt{T_2}} \Rightarrow T_1 = 4T_2 \Rightarrow x = \frac{L}{5}$$

13.(C) $\frac{V_{H_2}}{V_{He}} = \sqrt{\frac{\gamma_1}{M_1} \times \frac{M_2}{\gamma_2}} = \frac{\sqrt{42}}{5}$

14.(D) $v_1 = \frac{1}{2\ell} \sqrt{\frac{T_1}{\mu_1}} = 600 ; v_2 = \frac{1}{4\ell} \sqrt{\frac{T_2}{\mu_2}} = \frac{1}{12\ell} \sqrt{\frac{T_1}{\mu_1}} = 200 \text{ Hz.}$

15.(A) $\frac{\lambda}{2} = \frac{10}{5} \Rightarrow \lambda = 4 \text{ m} \therefore v = \frac{20}{4} = 5 \text{ Hz}$ 16.(D) $\frac{2\pi}{3}(\Delta x) = \pi \Rightarrow \Delta x = \frac{3}{2} \text{ cm}$

17.(A) Phase difference between the waves = $\frac{\pi}{2}$ $\therefore$ Resultant Amplitude = $\sqrt{4^2 + 3^2} = 5$

18.(B) $V_{max}(\text{Particle}) = 2\pi f y_0$

$V_{wave} = \lambda f \quad \therefore \quad \lambda = \frac{\pi y_0}{2}$

19.(A)

20.(D) As intensity, $I = \frac{1}{2} \rho \omega^2 A^2 v$

Here, ρ = density of medium,

A = amplitude

ω = angular frequency

and v = velocity of wave

$\therefore$ Intensity depend upon amplitude, frequency as well as velocity of wave.

Also, $I_1 = I_2$

21.(A) $y(x, t) = e^{-(ax^2 + bt^2 + 2\sqrt{ab}xt)} = e^{-(\sqrt{a} + \sqrt{b}t)^2}$

It is a function of type $y = f(\omega t + kx)$

So, $y(x, t)$ represents wave traveling along $-x$ direction.

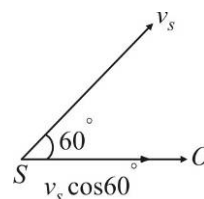
$\therefore$ Speed of wave = $\frac{\omega}{k} = \frac{\sqrt{b}}{\sqrt{a}} = \frac{\sqrt{b}}{\sqrt{a}}$

22.(D) Here,

Frequency of source, $\nu_0 = 100\text{Hz}$

Velocity of source, $\nu_s = 19.4\text{ms}^{-1}$

Velocity of sound in air, $\nu = 330\text{ms}^{-1}$



As the velocity of source along the source observer line is $\nu_s \cos 60^\circ$ and the observer is at rest, so the apparent frequency observed by the observer is :

$$\begin{aligned} \nu &= \nu_0 \left(\frac{\nu}{\nu - \nu_s \cos 60^\circ} \right) = (100\text{Hz}) \left(\frac{330\text{ms}^{-1}}{330\text{ms}^{-1} - (19.4\text{ms}^{-1}) \left(\frac{1}{2} \right)} \right) = (100\text{Hz}) \left(\frac{330\text{ms}^{-1}}{330\text{ms}^{-1} - 9.7\text{ms}^{-1}} \right) \\ &= (100\text{Hz}) \left(\frac{330\text{ms}^{-1}}{320.3\text{ms}^{-1}} \right) = 103\text{Hz} \end{aligned}$$

23.(A) For closed organ pipe, fundamental

Frequency is given by $\nu_c = \frac{\nu}{2l'}$. For open organ pipe, fundamental frequency is given by $\nu_o = \frac{\nu}{2l'}$

2nd overtone of open organ pipe $\nu' = 3\nu_o$; $\nu' = \frac{3\nu}{2l'}$, According to question, $\nu_c = \nu' \frac{\nu}{4l} = \frac{3\nu}{2l'}$ $l' = 6l$

Here, $= 20\text{cm}$, $l' = ?$ $\therefore \quad l' = 6 \times 20 = 120\text{cm}$

24.(A) $n_1 = \frac{1}{2l_1} \sqrt{\frac{T}{\mu}} \quad \dots \text{ (i) } ; \quad n_2 = \frac{1}{2l_2} \sqrt{\frac{T}{\mu}} \quad \dots \text{ (ii) } ; \quad n_3 = \frac{1}{2l_3} \sqrt{\frac{T}{\mu}} \quad \dots \text{ (iii) }$

$$n = \frac{1}{2l} \sqrt{\frac{T}{\mu}} \quad \dots \text{(iv)}$$

From equation (iv), we get $\frac{1}{n} = \frac{2l}{\sqrt{\frac{T}{\mu}}}$

As $l = l_1 + l_2 + l_3 \quad \therefore \frac{1}{n} = \frac{2(l_1 + l_2 + l_3)}{\sqrt{\frac{T}{\mu}}} = \frac{2l_1}{\sqrt{\frac{T}{\mu}}} + \frac{2l_2}{\sqrt{\frac{T}{\mu}}} + \frac{2l_3}{\sqrt{\frac{T}{\mu}}} = \frac{1}{n_1} + \frac{1}{n_2} + \frac{1}{n_3}$ [Using (i), (ii) and (iii)]

25.(D) Fundamental frequency of the closed organ pipe is $\nu = \frac{v}{4L}$

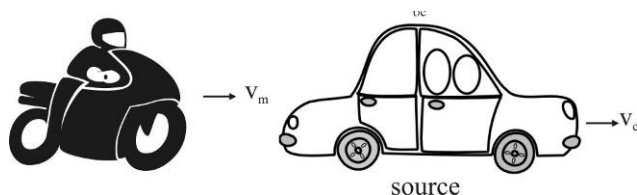
Here, $v = 340 \text{ ms}^{-1}$, $L = 85 \text{ cm} = 0.85 \text{ m} \quad \therefore \nu = \frac{340 \text{ ms}^{-1}}{4 \times 0.85 \text{ m}} = 100 \text{ Hz}$

The natural frequencies of the closed organ pipe will be

$$\begin{aligned} \nu_n &= (2n-1)\nu = \nu, 3\nu, 5\nu, 7\nu, 9\nu, 11\nu, 13\nu, \dots \\ &= 100 \text{ Hz}, 300 \text{ Hz}, 500 \text{ Hz}, 700 \text{ Hz}, 900 \text{ Hz}, 1100 \text{ Hz}, 1300 \text{ Hz}, \dots \end{aligned}$$

Thus, the natural frequencies lies below the 1250 Hz is 6.

26.(C)



Here, speed of motorcyclist, $v_m = 36 \text{ km hour}^{-1} = 36 \times \frac{5}{18} = 10 \text{ ms}^{-1}$

Speed of car, $v_c = 18 \text{ km hour}^{-1} = 18 \times \frac{5}{18} \text{ ms}^{-1} = 5 \text{ ms}^{-1}$

Frequency of source, $\nu_0 = 1392 \text{ Hz}$. Speed of sound, $v = 343 \text{ ms}^{-1}$

The frequency of the honk heard by the motorcyclist is

$$\nu' = \nu_0 \left(\frac{v + v_m}{v + v_c} \right) = 1392 \left(\frac{343 + 10}{343 + 5} \right) = \frac{1392 \times 353}{348} = 1412 \text{ Hz}$$

27.(B) Pressure change will be minimum at both ends.

28.(C) The standard equation of a wave travelling along +ve x-direction is given by : $y = A \sin(kx - \omega t)$

Where A = amplitude of the wave, k = angular wave number and ω = angular frequency of the wave

Given: $A = 1 \text{ m}$, $\lambda = 2\pi \text{ m}$, $\nu = \frac{1}{\pi} \text{ Hz}$

As $k = \frac{2\pi}{\lambda} = \frac{2\pi}{2\pi} = 1 \quad \Rightarrow \quad \omega = 2\pi\nu = 2\pi \times \frac{1}{\pi} = 2$

$\therefore$ The equation of the given wave is $y = 1 \sin(1x - 2t) = \sin(x - 2t)$

29.(C) Let ν be frequency of the unknown source. As it gives 4 beats per second when sounded with a source of frequency 250 Hz,

$\therefore \nu = 250 \pm 4 = 246 \text{ Hz or } 254 \text{ Hz}$

- 30.(D)** Let $L (=100\text{ cm})$ be the length of the wire and L_1, L_2 and L_3 are the length of the segments as shown in the figure.

Fundamental frequency, $\nu \propto \frac{1}{L}$

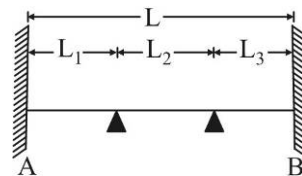
As the fundamental frequencies are in the ratio of 1:3:5,

$$\therefore L_1 : L_2 : L_3 = \frac{1}{1} : \frac{1}{3} : \frac{1}{5} = 15 : 5 : 3$$

Let x be the common factor, then $15x + 5x + 3x = L = 100$

$$23x = 100 \text{ or } x = \frac{100}{23}$$

$$\therefore L_1 = 15 \times \frac{100}{23} = \frac{1500}{23} \text{ cm}, L_2 = 5 \times \frac{100}{23} = \frac{500}{23} \text{ cm} \text{ and } L_3 = 3 \times \frac{100}{23} = \frac{300}{23} \text{ cm}$$



- 31.(C)** Let l be the length of the string fundamental frequency is given by $\nu = \frac{1}{2l} \sqrt{\frac{T}{\mu}}$ or $\nu \propto \frac{1}{l}$ ($\because T$ and μ are constants)

Here, $l_1 = \frac{k}{\nu_1}, l_2 = \frac{k}{\nu_2}, l_3 = \frac{k}{\nu_3}$ and $l = \frac{k}{\nu}$ But $l = l_1 + l_2 + l_3 \therefore \frac{1}{\nu} = \frac{1}{\nu_1} + \frac{1}{\nu_2} + \frac{1}{\nu_3}$

- 32.(D)** Given: $y_1 = 4 \sin 600\pi t, y_2 = 5 \sin 608\pi t$
 $\therefore \omega_1 = 600\pi$ or $2\pi\nu_1 = 600\pi$ or $\nu_1 = 300$ $A_1 = 4$
 And $\omega_2 = 608\pi$ or $2\pi\nu_2 = 608$ or $\nu_2 = 304$ $A_2 = 5$
 Number of beats heard per second $= \nu_2 - \nu_1 = 304 - 300 = 4$

$$\frac{I_{\max}}{I_{\min}} = \frac{(A_1 + A_2)^2}{(A_1 - A_2)^2} = \frac{(4+5)^2}{(4-5)^2} = \frac{81}{1}$$

- 33.(B)** The given wave equation is $y = 3 \sin \frac{\pi}{2} (50t - x)$; $y = 3 \sin \left(25\pi t - \frac{\pi}{2} x \right)$

The standard wave equation is $y = A \sin (\omega t - kx)$... (i)

Comparing (i) and (ii), we get : $\omega = 25\pi, k = \frac{\pi}{2}$... (ii)

Wave velocity, $\nu = \frac{\omega}{k} = \frac{25\pi}{(\pi/2)} = 50 \text{ ms}^{-1}$

Particle velocity, $\nu_p = \frac{dy}{dt} = \frac{d}{dt} \left(3 \sin \left(25\pi t - \frac{\pi}{2} x \right) \right) = 75\pi \cos \left(25\pi t - \frac{\pi}{2} x \right)$

Maximum particle velocity, $(\nu_p)_{\max} = 75\pi \text{ ms}^{-1}$ $\therefore \frac{(\nu_p)_{\max}}{\nu} = \frac{75\pi}{50} = \frac{3}{2}\pi$

- 34.(A)** $y_1 = a \sin (\omega t + kx + 0.57)$ $\therefore$ phase, $\phi_1 = \omega t + kx + 0.57$

$y_2 = a \cos (\omega t + kx) = a \sin \left(\omega t + kx + \frac{\pi}{2} \right)$ $\therefore$ Phase $\phi_2 = \omega t + kx + \frac{\pi}{2}$

Phase difference, $\Delta\phi = \phi_2 - \phi_1 = \left(\omega t + kx + \frac{\pi}{2} \right) - (\omega t + kx + 0.57) = \frac{\pi}{2} - 0.57 = (1.57 - 0.57) \text{ radian} = 1 \text{ radian}$

- 35.(C)** Here, $\nu_{\text{air}} = 350 \text{ m/s}, \nu_{\text{brass}} = 3500 \text{ m/s}$

When a sound wave travels from one medium to another medium its frequency remains the same

$$\therefore \text{frequency, } v = \frac{v}{\lambda} \Rightarrow \frac{v_{air}}{\lambda_{air}} = \frac{v_{brass}}{\lambda_{brass}}$$

$$\lambda_{brass} = \lambda_{air} \times \frac{v_{brass}}{v_{air}} = \lambda_{air} \times \frac{3500}{350} = 10\lambda_{air}$$

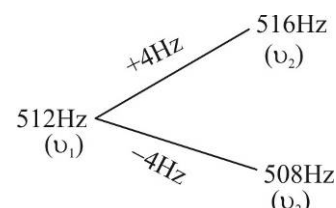
$$36.(B) \text{ As } v = \frac{1}{2L} \sqrt{\frac{T}{\mu}} \therefore \frac{\Delta v}{v} = \frac{1}{2} \frac{\Delta T}{T}$$

$$\frac{\Delta T}{T} = 2 \frac{\Delta v}{v} = 2 \times \frac{6}{600} = 0.02$$

37.(D) Let the frequencies of tuning fork and piano string be v_1 and v_2 respectively.

$$\therefore v_2 = v_1 \pm 4 = 512 \text{ Hz} \pm 4$$

$$= 516 \text{ Hz or } 508 \text{ Hz}$$



Increase in the tension of a piano string increases its frequency.

If $v_2 = 516 \text{ Hz}$, further increase in v_2 resulted in decreases in the beat frequency. This is given in the question.

When the beat frequency decreases to 2 beats per second. Therefore, the frequency of the piano string before increasing the tension was 508 Hz.

$$38.(A) l_1 = 0.516 \text{ m}, l_2 = 0.491 \text{ m}, T = 20 \text{ N.}$$

Mass per unit length, $\mu = 0.001 \text{ kg/m}$

$$\text{Frequency, } v = \frac{1}{2l} \sqrt{\frac{T}{\mu}} ; \quad v_1 = \frac{1}{2 \times 0.516} \sqrt{\frac{20}{0.001}} \text{ and } v_2 = \frac{1}{2 \times 0.491} \sqrt{\frac{20}{0.001}}$$

$$\therefore \text{Number of beats} = v_1 - v_2 = 7$$

$$39.(C) \text{ Amplitude} = 2 \text{ cm} = 0.02 \text{ m}, v = 128 \text{ m/s}$$

$$\lambda = \frac{4}{5} = 0.8 \text{ m}; v = \frac{128}{0.8} = 160 \text{ Hz}$$

$$\omega = 2\pi v = 2\pi \times 160 = 1005; l = \frac{2\pi}{\lambda} = \frac{2\pi}{0.8} = 7.85 \quad \therefore y = 0.02 \sin(7.85x - 1005t).$$

$$40.(A) y = 0.25 \sin(10\pi x - 2\pi t)$$

$$y_{\max} = 0.25$$

$$k = \frac{2\pi}{\lambda} = 10\pi \Rightarrow \lambda = 0.2 \text{ m} ; \quad \omega = 2\pi f = 2\pi \Rightarrow f = 1 \text{ Hz}$$

The sign is negative inside the bracket, therefore the wave travels in the positive x-direction.

$$41.(B) Y_1 = 4 \sin 500\pi t, Y_2 = 2 \sin 506\pi t$$

$$\omega_1 = 500\pi, \omega_1 = 2\pi v, v_1 = 250, v_2 = 253$$

$$v = v_2 - v_1 = 253 - 250 = 3 \text{ beats/s}$$

$$\text{Number of beats per minute} = 3 \times 60 = 180.$$

$$42.(C) d_1 = 2 \text{ m}, d_2 = 3 \text{ m}$$

$$\text{Intensity} \propto \frac{1}{(\text{distance})^2}$$

$$I_1 \propto \frac{1}{2^2} \text{ and } I_2 \propto \frac{1}{3^2} \quad \therefore \frac{I_1}{I_2} = \frac{9}{4}.$$

43.(B) 1st the car in the source and at the cliff, one observer f' . $\therefore f' = \frac{v}{v-v_s} f$

2nd cliff is now source, it emits frequency f' and the observer is not the driver who observes f'' .

$$\therefore f'' = \left[\frac{v+v_o}{v} \right] f' \text{ or } 2f = \left[\frac{v+v_o}{v-v_s} \right] f$$

$$\Rightarrow 2v - 2v_o = v + v_o \Rightarrow v = v_o + 2v_o \quad [\text{as } v_s = v_o]$$

$$\Rightarrow v_o = \frac{v}{3}$$

44.(B) Apparent frequency $f' = \frac{v+v_o}{v} f = \frac{v+(1/5)v}{v} f = 1.2f$. Wavelength does not change by motion of observer.

45.(C) The equation of progressive wave travelling in positive x -direction is given by $y = a \sin \frac{2\pi}{\lambda} (vt - x)$.

$$\text{Here } a = 0.2m, v = 360m/sec, \lambda = 60m = 0.2 \sin \frac{2\pi}{60} (360t - x) = 0.2 \sin \left[2\pi \left(6t - \frac{x}{60} \right) \right]$$

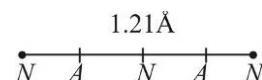
46.(B) Resultant amplitude $= 2a(1 + \cos \phi) = a$ $\therefore (1 + \cos \phi) = 1/2; \cos \phi = -1/2; \phi = \frac{2\pi}{3}$.

47.(A) As $n \propto (1/l)$ and $l = l_1 + l_2 + l_3$ $\therefore \frac{1}{n} = \frac{1}{n_1} + \frac{1}{n_2} + \frac{1}{n_3}$.

48.(C) Distance between a node and adjoining anti-node $= \lambda/4$

From figure, distance between two atoms.

$$= 4 \times \frac{\lambda}{4} = 1.21\text{\AA} \text{ or, } \lambda = 1.21\text{\AA}.$$



49.(D) For the cylindrical tube open at both ends, $f = v/2l$.

When half of the tube is in water, it behaves as a closed pipe of length $l/2$.

$$\therefore f' = \frac{v}{4(l/2)} = \frac{v}{2l}. \therefore f' = f$$

50.(A) Wavelength (λ) = 5000 Å and velocity (v) = $1.5 \times 10^4 m/s$ wavelength of the approaching star,

$$(\lambda') = \lambda \frac{c-v_s}{c} \text{ or } \frac{\lambda'}{\lambda} = 1 - \frac{v}{c} \quad \text{or} \quad \frac{v}{c} = 1 - \frac{\lambda'}{\lambda} = \frac{\lambda - \lambda'}{\lambda} = \frac{\Delta\lambda}{\lambda}$$

$$\text{Therefore } \Delta\lambda = \lambda \times \frac{v}{c} = 5000\text{\AA} \times \frac{1.5 \times 10^6}{3 \times 10^8} = 25\text{\AA} \quad (\text{where } \Delta\lambda \text{ is the change in the wavelength})$$

INTEGER VALUE TYPE

$$51.(150) 1000 = \frac{C}{4L_0} \Rightarrow \frac{C}{4000} = L_0 \text{ (Fundamental)}$$

$$\text{So, } \Delta L = 2L_0 = \frac{C}{2000}$$

$$\text{Volume to be drained } \Delta L \times 10 = 150 \text{ cm}^3.$$

$$52.(11) A = \frac{\Delta P}{SC^2 \cdot K} \text{ in 3rd harmonic, } \frac{3\lambda}{2} = L \Rightarrow \lambda = 2.6\pi$$

$$\text{solving } \Delta P = 10^3 N$$

$$S = 1.3 \text{ Kg/m}^3$$

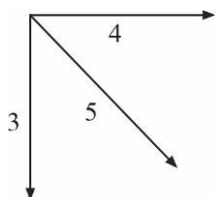
$$C = 300 \text{ m/s}$$

$$A = 1.11 \text{ cm} = 11 \text{ mm (nearest)}$$

$$53.(300) \quad f_0 = \frac{1}{2Lr} \sqrt{\frac{T}{3\pi}} \quad ; \quad \frac{600}{f} = \sqrt{4} \Rightarrow f = 300 \text{ Hz}$$

$$54.(10) \quad 5 \frac{\lambda}{2} = 10 \Rightarrow \lambda = 4 \text{ m} \quad ; \quad f = \frac{C}{\lambda} = \frac{40}{4} = 10 \text{ Hz}$$

$$55.(5) \quad A = \sqrt{A_1^2 + A_2^2} = \sqrt{3^2 + 4^2} = 5$$



$$56.(576) \text{ Frequency reaching wall} = 512 \times \frac{34}{32}$$

$$\text{Frequency reaching girl} = 512 \times \frac{34}{32} \times \frac{36}{34} = 576 \text{ Hz}$$

$$57.(101) \quad f = 100 \times \frac{330}{330 - 3.46 \times \frac{\sqrt{3}}{2}} \approx 101 \text{ Hz}$$

$$58.(60) \quad \frac{\lambda_{\text{closed}}}{4} = 10 \text{ cm} \quad ; \quad \frac{3\lambda_{\text{open}}}{2} = L_{\text{open}}$$

$\lambda_{\text{closed}} = \lambda_{\text{open}}$ so, solving we get $L_{\text{open}} = 60 \text{ cm}$.

$$59.(7) \quad (2n+1) \frac{C}{4L} \leq 1300 \Rightarrow n = 0, 1, 2, 3, 4, 5, 6$$

$$60.(186) \quad f_{\text{obs}} = 1830 \times \frac{300 + 10}{300 + 5} = 1860 \text{ Hz} = 186 \text{ deca Hz}$$

$$61.(254) \quad f_{\text{unknown}} = 254 \text{ or } 246 \text{ Hz}$$

Second harmonic = 508 Hz or 492 Hz

Out of these two 508 Hz can only make 5 beats / sec with 513.

$$62.(1) \quad y_1 = a \sin(\omega t + kx + 0.57)$$

$$y_2 = a \cos(\omega t + kx) = a \sin(\omega t + kx + \pi/2)$$

$$\text{So, } \Delta\phi = \frac{\pi}{2} - 0.57 \approx 1 \text{ radian}$$

$$63.(496) \quad f_{\text{piano}} = 504 \text{ or } 496 \text{ Hz}$$

If tension increases, frequency increases.

So, 496 Hz must be answer

$$64.(5) \quad f = \frac{1}{2L} \sqrt{\frac{T}{\mu}} \quad ; \quad \text{Calculating we get 5 beat / sec.}$$

$$65.(180) \quad f_1 = 250 \text{ Hz}$$

$$f_2 = 253 \text{ Hz}$$

$$f_{\text{beat}} = 3 / \text{sec}$$

So, in 1 minute = 180